

NCR 5XXX Device Support Manual

BASE24-atm[®]



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What's New

The following table highlights the major changes that have been made in the most recent update to the *NCR 5XXX Device Control Manual*. The first column of the table lists the sections and appendixes in which major changes have been made. The second column of the table describes the major changes for each section or appendix. These changes update the manual to release 1.0, version 06.0v10.

September 2012

Section/ Appendix	Major Changes
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- | | |
|---|---|
| G | Clarifies the description of the MOBILE-OPERATOR-ID-VT <i>identifier</i> param, where <i>identifier</i> is a value in the range of 00–29 representing a voucher type. This param specifies a mobile operator six-digit IIN, two-digit digit product ID, and the mobile operator name (up to 20 characters) associated with the mobile telephone operator. |
|---|---|

January 2012

Section/ Appendix	Major Changes
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- | | |
|---|---|
| 1 | Adds supporting references for NCR 66XX-series models |
|---|---|

September 2011

Section/ Appendix	Major Changes
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- | | |
|---|---|
| 1 | Updates description for NCR 5XXX-series models |
| E | Adds the “Enhanced Signature-Based Processing” topic, renames “Signature-Based Processing” topic to “Basic Signature-Based Processing”, updates existing topics “Configuration Considerations”, “Load Commands”, and “Sending a Command Using the DCT”. |

December 2009

Section/ Appendix	Major Changes
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- | | |
|---|--|
| 4 | <p>Adds load group value 21 (support for withdrawals and inquiries on a default account) to the GROUP NUMBER field on the NCR 5XXX Configuration File Menu screen.</p> <p>Removes load group value 480 (Apra Edge XML message support) from the GROUP NUMBER field on the NCR 5XXX Configuration File Menu screen. This corrects a documentation change made with version 8.</p> <p>Updates the screen number on page 1 of the NCR 5XXX State Table Screens.</p> |
| 5 | <p>Adds financial receipt field type codes CA, DC, DE, DO, EA, and OA.</p> |

Preface

The *BASE24-atm NCR 5XXX Device Support Manual* provides an overview of the NCR 5XXX Device Handler process and NCR-related software options required by the BASE24-atm product. This manual discusses the interaction between the BASE24 product and the NCR 5XXX-series ATMs or ATMs emulating the NCR 5XXX-series ATMs. It also examines the role of the BASE24 product in the features, specifications, and functions of the NCR 5XXX-series ATMs.

Audience

This manual is intended as a reference for financial institution personnel responsible for updating and maintaining NCR 5XXX-series ATMs or ATMs emulating the NCR 5XXX-series ATMs for use with the BASE24-atm product.

Prerequisites

This manual discusses a number of general topics relating to NCR 5XXX-series ATMs. These discussions have been included primarily to provide a stronger insight into the way the BASE24-atm NCR 5XXX Device Handler process handles NCR 5XXX-series ATMs. This manual is not intended as a substitute for the vendor literature relating to NCR 5XXX-series ATMs. It is recommended that individuals making decisions on BASE24-atm 5XXX-series ATM processing be familiar with the vendor literature available for the specific type of 5XXX-series ATM under consideration.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional

information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information.

This manual contains references to the following BASE24 and NET24-XPNET publications:

- The ***BASE24-atm NCR 5XXX SSB Manual*** contains information on the self-service banking (SSB) modules that can be added to the BASE24-atm product, including field descriptions for SSB-specific screens, receipt templates for new and modified transactions, and SSB-specific configuration.
- The ***NET24-XPNET Management Programming Manual*** and the ***NET24-XPNET Network Control Command Reference Manual*** contain information about protocols.
- The ***BASE24 Device Control Manual*** contains detailed information on the operational control of the NCR 5XXX-series ATMs by BASE24. This includes instructions for such actions as downloading information and performing status inquiries.
- The ***BASE24 Base Files Maintenance Manual*** contains screen images and field descriptions of the Acquirer Processing Code File (APCF), Card Prefix File (CPF), Institution Definition File (IDF), Issuer Processing Code File (IPCF), External Message File (EMF), and the Surcharge File (SURF).
- The ***BASE24-atm Files Maintenance Manual*** contains screen images and field descriptions of the BASE24-atm Terminal Data files (ATD), Receipt File (RCPT), Terminal Receipt File (TRF), and the Transaction Log File (TLF).
- The ***BASE24-atm Settlement and Reporting Manual*** describes the settlement and cutover procedures, and contains samples and field descriptions of the reports produced by BASE24-atm.
- The ***BASE24-atm Transaction Processing Manual*** contains descriptions of the supported transactions and response codes, and the Standard Internal Message (STM) used by the BASE24-atm product.
- The ***BASE24-atm EMV Support Manual*** contains information about EMV screens and tokens.
- The ***BASE24 Logical Network Configuration File Manual*** describes the Logical Network Configuration File (LCONF) and its assigns and params.
- The ***BASE24 Transaction Security Manual*** contains information on message authentication.
- The ***BASE24 Text Command Reference Manual*** contains information on the BASE24 operator functions available from a network control facility.

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- The ***BASE24 Log Message Reference Manual*** describes event messages, and how to access documentation for event messages.
 - The ***BASE24 User Security Manual*** contains general information about logging on to the BASE24 user access system, using the default function keys, and creating screen access security.
 - The ***BASE24 External Message Manual*** describes the standard external message used by ISO hosts.
 - The ***BASE24 ISO Host Interface Manual*** contains information on message processing between the BASE24 system and ISO hosts.
 - The ***BASE24 Tokens Manual*** describes the tokens that are added to the standard internal message to carry additional information.
 - The ***BASE24 Transaction Security Services Processing Guide*** describes the Transaction Security Services process requirements for implementing the Automated Key Distribution System add-on product. This manual also describes the first phases of automated key distribution and the Transaction Security Services user interface.

In addition to BASE24 and NET24-XPNET publications, the following NCR manuals are referenced in this manual:

- The ***NCR NDC Programmer's Overview Manual*** contains a general overview of the NCR configuration file and the state tables, FIT tables, and configuration parameters.
- The ***NCR NDC Programmer's Reference Manual*** contains specific information on the NCR configuration file for NCR 50XX-series terminals.
- The ***NCR NDC+ Programmer's Reference Manual*** contains specific information on the NCR configuration file for NCR 56XX-series terminals.
- The ***NCR NDC+ Supervisor's Reference Manual*** describes the Rear Supervisor/Settlement transaction and the screens used by the transaction, as well as terminal configuration options.
- The ***NCR 56XX ATM Dial-Up Interface Specification*** contains information about the Dial-Up Configuration File (DIALUP).
- The ***NCR Aptra Edge Remote Configuration Guide*** contains information on formatting instructional data.
- The ***NCR Aptra Edge Standard NDC Message Format Guidelines*** contains information on next states and associated screen verbiage.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

“Conventions Used in this Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” introduces the reader to the NCR 5XXX-series ATM models that are available and describes how the BASE24-atm product provides support for NCR 5XXX-series ATMs in the following areas: ATM features, protocols used, device control functions, transaction security, cutover and settlement, and financial and administrative transactions.

Section 2, “ATM Configuration,” contains information about NCR 5XXX-series ATM configuration requirements, NCR 5XXX-series ATM configuration files, and the BASE24 files used in configuring NCR 5XXX-series ATMs.

Section 3, “BASE24-atm Device Handler Process,” contains an overview of the BASE24-atm NCR 5XXX Device Handler process, the processing steps the Device Handler process takes when it is first started, an overview of the network commands that can be sent to the Device Handler process, and information on how to access the event messages generated by the Device Handler process.

Section 4, “Configuration File Maintenance,” describes the NCR 5XXX Configuration File (CNFG5070) maintenance that can be performed using the BASE24-atm product. It includes the different screens used to change information in the ATM configuration file.

Section 5, “ATM Printer Support,” describes standard printer support for NCR 5XXX-series ATMs. Receipts are generated when a financial transaction is performed, when a cardholder performs a statement print transaction at the ATM, and when an administrative transaction is performed by institution personnel. Examples of receipts generated at NCR 5XXX-series ATMs are shown as they would appear to the cardholder, as well as how you can customize the financial and administrative receipts. Also included are the printing requirements for printing receipts, and the options for not printing receipts.

Appendix A, “Statement Print Data Compression,” describes data compression techniques used by the BASE24 system to send and receive messages from the host system and to send messages to an ATM. Data compression techniques are used to increase efficiency of information transfer in order to save time and expenses associated with transmission of repeated information such as blanks used to format and position printed text.

Appendix B, “Digital Audio Service,” describes the NCR 5XXX Device Handler support of the NDC+ Digital Audio Service (DAS).

Appendix C, “Dispensing Bills and Noncurrency Items,” describes the bill dispensing methods used by the NCR 5XXX Device Handler process. It also describes the BASE24-atm Non–Currency Dispense add-on product.

Appendix D, “NCR 5XXX Dial-Up Device Handler Extension,” describes the NCR 5XXX Dial-Up Device Handler extension.

Appendix E, “Distributing Master Keys and Transaction Keys,” describes the Automated Key Distribution System add-on product and the Dynamic Key Management add-on product. The Automated Key Distribution System add-on product enables you to remotely load master keys to an ATM. The Dynamic Key Management add-on product enables you to automatically load PIN and message authentication code (MAC) keys to an ATM.

Appendix F, “Multiple Currency Dispense Add-On Feature,” describes the BASE24-atm Multiple Currency Dispense add-on feature. This feature, part of the BASE24-atm Multiple Currency add-on product, enables you to dispense multiple currencies from a single ATM.

Appendix G, “BASE24-atm Mobile Top-Up Extension,” describes the BASE24-atm Mobile Top-Up Device Handler extension. This extension, which is used in conjunction with the BASE24-atm Non–Currency Dispense add-on product, enables an institution to offer mobile telephone top-up minutes as an option at an ATM.

Appendix H, “NCR 5XXX Aptra Edge Standard NDC Device Handler Extension,” describes the BASE24-atm NCR 5XXX Aptra Edge Standard NDC Device Handler extension. This add-on product, enables an institution to support the Aptra Edge Standard NDC message format and standard NDC+ terminals using a single Device Handler process.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, AT-DA031-01 for the *BASE24-atm NCR 5XXX Device Support Manual*) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Sep-2012 for September, 2012) indicates the issue of the manual. The software release information (for example, R6.0v10 for release 6.0, version 10) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

Conventions Used in This Manual

This section explains how fields and blank characters are documented in this manual.

Field Descriptions

Each field appearing on a file maintenance screen is listed by name and then described. Field descriptions briefly summarize the contents, purpose, and permissible values, as shown in the following example taken from the BASE24-atm Terminal Data files (ATD).

Example

FRMT — Indicates to the host whether or not the Device Handler process uses the columnar or the statement print data compression statement print format. Valid values are as follows:

- 0 = Columnar format (statement print data compression not supported)
- 1 = BASE24-atm statement print data compression format supported (data compression supported)

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	ATDS1.[atds1-dev-info-ncr-5xxx].STMT-PRNTR-CNTL. FRMT

Explanation

Each field description is completed by one or more of the following items of information:

Item	Description
Example	Illustrates a possible entry for the field to further clarify the value or values that can be entered.
Field Length	Specifies the size of the field and the type of characters that can be entered. This length refers to the field and valid values for the file maintenance screen, not the field in the DDLs. Possible values are alphabetic, alphanumeric, hexadecimal, and numeric. The term <i>alphanumeric</i> includes all alphabetic, numeric, and special characters that can be entered from a keyboard without using a control sequence. The term <i>hexadecimal</i> includes all numbers and the letters A through F. When a field value cannot be modified by the operator, the field length is System protected.
Occurs	Indicates the number of times the field can be displayed on the screen. This information is provided only when the field can be displayed multiple times.
Required Field	Specifies whether a value has to be entered in the field. Possible values are Yes and No. Some fields are required only under certain conditions. In this case, the entry is Yes, followed by the conditions that determine when the field is required.
Default Value	Specifies the value that is automatically placed in the field when the screen is first displayed or when the F8 key is pressed to clear the screen.
Data Name	Provides the DDL name associated with the field appearing on the screen. Data names are included in the documentation to assist in communicating screen and field issues to your technical staff. Note that screen data is not always stored in the BASE24 database as it appears on the screen or as it is described in the field description. If you need information on how screen data is actually stored, consult the DDLs.

Blank Characters

Field descriptions for internal and external messages, tokens, files maintenance screens, and reports often include lists of valid values. When the value contains a blank character, this manual uses the symbol *b* to indicate the blank character.

Unlabeled Fields on Screens

Angle brackets (< >) indicate an unlabeled field on a screen. An unlabeled field is a field that is present on a screen but is not preceded by an identifying literal label. For the purposes of documenting the field, a label has been assigned and appears inside the angle brackets. A multiple line unlabeled field is displayed as a shaded area and also has a label in angle brackets.

In the field descriptions for the screen, the unlabeled field appears according to its place on the screen and is identified by the same label.

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1: Introduction

This section provides an introduction to BASE24-atm support of NCR 5XXX and 66XX-series ATMs. It contains information about the following items:

- NCR 5XXX and 66XX-series ATM models supported by the BASE24-atm product
- BASE24-atm support capabilities
- NCR 5XXX-series ATM features supported by the BASE24-atm product
- Protocols supported by the BASE24-atm product for NCR 5XXX and 66XX-series ATMs
- Transaction security
- Cutover and settlement
- Financial and administrative transactions supported by the BASE24-atm product for the NCR 5XXX and 66XX-series ATMs, and print options for these transactions
- Device configuration and control functions supported by the BASE24-atm product

NCR 5XXX and 66XX-Series Models

The BASE24-atm product supports most models of NCR terminal devices in the 5XXX and 66XX product lines. Within these NCR product lines, there are full-service terminals that allow cardholders to perform a wide variety of banking services as well as those that perform limited services such as inquiry and receipt printing. Additional services are supported in the BASE24 system through the BASE24-atm self-service banking (SSB) add-on product. For more information on SSB, refer to the ***BASE24-atm NCR 5XXX SSB Manual***.

The major distinction between the 50XX and the 56XX series is that the later models are based upon the IBM personal computer (PC) architecture and OS/2 operating system as opposed to the less capable chip architecture of the 50XX device line.

Because of the unique design of BASE24-atm support, terminals from the 50XX, 56XX, and 66XX groups are supported such that you can mix devices from all groups in any combination without concern for keeping terminals from like groups together.

For specific model applications and design specifications, readers should consult the appropriate NCR documentation supporting the devices in question.

All NCR 5XXX and 66XX-series ATMs meet the applicable requirements of the Bank Protection Act and Regulation P, as well as the applicable UL Standard 291 for ATMs.

In addition, the BASE24-atm NCR 5XXX Device Handler process supports the 58XX-series *personaS* ATM, as well as the basic transaction set for any other ATM supporting native NCR emulation, such as the Nautilus Hyosung ATM.

BASE24-atm Support

The BASE24-atm product provides the following support specifically tailored to the NCR 5XXX-series terminals:

- Supported features
- Supported protocols
- Transaction security

Supported Features

The BASE24-atm NCR 5XXX Device Handler process is compatible with NCR NDC ATM software up to release 8.0. The Device Handler process is also compatible with NCR NDC+ software up to release 7.0, Aprta Edge release 2.04.00, and Aprta Advance release 3.00 software. NCR 56XX-series terminals that have a Document Processing Module (DPM) and that are using the BASE24-atm self-service banking (SSB) product require release 5.02.

The following NCR 5XXX and 66XX-series ATM features are supported by BASE24-atm:

- Receipt printer
- Journal printer
- Electronic journal as a backup to the journal printer (only with NDC+ release 4.0 and higher)
- Dual mode journal printing (only with NDC+ release 6.01 and higher)
- Statement printer
- Thermal statement printer (only with NDC+ release 5.02 and higher)
- Statement print cancel
- Data compression on statement print transactions
- Depository
- Track 1 name display
- Color support
- 999 states
- Rear Supervisor/Settlement Transaction Facility

- Data Encryption Standard (DES) PIN encryption
- Diebold PIN encryption
- ANSI X9.8 standard PIN block format
- Local PIN verification
- Message authentication
- Downloading the configuration files
- Card reader
- Dip and Smart Hardware (DASH) card reader
- Voice guidance
- Touch screen (56XX-series only)
- Digital camera system (only with NDC+ release 5.02 and higher)
- Digital Audio Service (DAS) (only with NDC+ release 6.00 and higher)
- Enhanced Electronic Journal Backup support (only with NDC+ release 6.00 and higher)
- Dial-up protocol
- Bunch Note Acceptor support (only with NDC and Shared NDC+ BNA)

Supported Protocols

The BASE24-atm product can support communications with NCR 5XXX and 66XX-series terminals using the following protocols:

- SDLC XF Multipoint
- CSS Multipoint Supervisor Burroughs
- CSS Bisynch Multipoint Supervisor
- X.25
- PLU
- TCP/IP

Note: Protocol requirements for the Automated Key Distribution System may vary. Refer to the “Protocol Requirements” topic in Appendix E of this manual for more information.

For more information about protocols, refer to the *NET24-XPNET Management Programming Manual* and the *NET24-XPNET Network Control Command Reference Manual*.

Transaction Security

The BASE24-atm product offers transaction security options to protect an institution and its cardholders from fraud and includes such functions as PIN encryption, PIN verification, and message authentication.

The BASE24-atm product supports the PIN Change transaction for the NCR ATMs, which enables the cardholder to perform a PIN change at the ATM. The BASE24-atm product also supports the EMV PIN Unblock transaction for the NCR ATMs, which enables the cardholder to reset the bad PIN tries accumulator for the EMV card's offline PIN at the ATM. If local PIN verification is used, the PIN Change transaction and the EMV PIN unblock transaction are not supported.

The BASE24-atm product supports transaction security using both hardware and software. Hardware security options include interfaces to various models of Atalla and Thales e-Security (Racal) hardware. Software security options include several methods of DES PIN encryption and PIN verification. Refer to the *BASE24 Transaction Security Manual* for more information on transaction security.

Local PIN Verification

The BASE24-atm product supports local PIN verification in which NCR 5XXX and 66XX-series devices, which have PIN verification capability, perform PIN verification at the device for financial and administrative transactions without the involvement of the BASE24 network.

To perform local PIN verification, the NCR terminal must be equipped with the proper hardware. Only those PINs with an offset encoded on the card can be verified at the ATM. You must consider which cards can be verified at the ATM because some interchanges do not support a preverified PIN.

Local PIN verification requires a setting in the BASE24-atm Terminal Data files (ATD) as well as updates to the financial institution table (FIT) and the PIN Entry state in the NCR 5XXX Configuration File (CNFG5070). The configuration file must be downloaded after these changes are made to implement local PIN verification. Refer to [section 4](#) for information on how to access the CNFG5070

using the BASE24 product. Refer to the appropriate NCR vendor documentation for complete information on setting up the configuration file to support local PIN verification.

Note: The BASE24-atm system does not record the number of bad PIN attempts by a cardholder when the ATM performs PIN verification.

Message Authentication

Message authentication ensures that transaction messages are received exactly as created by the legitimate originator. Message authentication protects financial transaction messages against accidental or deliberate alteration.

The message authentication code (MAC) is generated by the originator of the message, based upon message elements identified in advance by the originator and recipient, and included with the message.

The BASE24-atm product supports the generation and verification of MAC values in hardware.

Message authentication is supported for the following message types:

- Transaction Request
- Transaction Reply
- Message Authentication Field Selection Load

Based on a table in the N50NAMS, message authentication can optionally be performed on the following message types:

- State Table Load
- FIT Load
- Solicited Status Message
- Dispenser Currency Cassette Mapping Table (DCCMT) Load

Message authentication can be performed in two ways—using full message authentication or selective message authentication. Full message authentication includes the whole message starting from the first field following the protocol-dependent header up to, but excluding, the field separator preceding the MAC field. Selective message authentication involves calculating the MAC based on fields specified in a field selection table in the N50NAMS.

Message authentication is also supported by a series of field settings in the device-specific portion of the ATD. These fields provide message authentication options to the institution. For complete descriptions of these fields, refer to the *BASE24-atm Files Maintenance Manual*.

For complete information on the MAC calculation for NCR 5XXX-series terminals, refer to the appropriate NCR vendor documentation.

Device Handler Process Keys

There are two types of keys that are used in PIN encryption and message authentication between terminals and their BASE24-atm Device Handler processes: transaction keys and master keys.

Transaction Keys

A transaction key is distributed electronically by the BASE24 system to its terminals using the same lines over which all other communications travel. There are separate transaction keys for PIN encryption and message authentication.

Master Keys

Since transaction keys are distributed to BASE24-atm terminals using the same lines over which all other communications travel, a means of sending transaction keys in a secure manner is required. This security is provided by a separate master key known to both the terminal and the BASE24 system. The master key is used to encrypt transaction keys for movement over the communication lines. The same master key is used to encrypt the PIN encryption transaction key and message authentication transaction key.

Master keys are by nature *long-term* and, once in place, usually remain unchanged unless they are compromised. The master key of each terminal must be physically entered at the ATM, unless the Automated Key Distribution add-on product is used.

Transmitting Transaction Keys to the Terminal

There are several instances in which Device Handler processes transmit new transaction keys to their terminals. They are identified in the following list:

- During load sequences of the ATMs.
- When commands are issued from the BASE24 Device Control Terminal (DCT) or from a network control facility.

Each time a new transaction key is sent to a terminal, the Device Handler process checks the MAC SUPPORT TYPE field on ATD NCR 5XXX screen 9. The Device Handler process provides software message authentication and security module message authentication.

Software Message Authentication. If the MAC SUPPORT TYPE field on ATD screen 9 is set to a value of 1, the Device Handler process generates a new transaction key, encrypts the key under the hard-coded master key, and sends the key to the terminal.

Security Module Message Authentication. If the MAC SUPPORT TYPE field on ATD screen 9 is set to a value of 3, the Device Handler process requests a new transaction key from the security module. The Device Handler process stores the encrypted transaction key in the ATD so that the key can be used to generate MACs for messages outbound for the terminal. In addition, MACs in messages received from the terminal can be verified.

Cutover and Settlement

The role of the BASE24-atm product in settlement consists of providing institutions with detailed transaction reports that reflect cardholder and terminal activity on both a daily and monthly basis. The BASE24-atm system defines the business day, captures and keeps track of each day's transaction totals, provides a daily file of transactions for processing by a host, and reports the funds due from and owed to the various participants in the logical network for balancing purposes.

A term used in connection with the BASE24-atm settlement process is cutover. Cutover refers to closing the current posting day and opening the next. It is the point in time when transactions stop being logged to the current day's business and start being logged to the next day's business by a specified terminal.

During institution cutover, the Settlement Initiator process reads the BASE24-atm Terminal Data files (ATD) to find ATMs that are not yet cut over. The Settlement Initiator process performs a force cutover on those terminals, and logs the totals to the Transaction Log File (TLF). An institution can purchase the Forced Cutover Audit module to write the terminal totals to the journal receipt at the ATM.

For a complete description of BASE24-atm settlement and cutover and the Forced Cutover Audit module, refer to the ***BASE24-atm Settlement and Reporting Manual***.

Transaction Support

The BASE24-atm product supports a number of different financial and administrative transactions for NCR 5XXX-series ATMs. They are listed below. For complete descriptions of supported transactions, refer to the ***BASE24-atm Transaction Processing Manual***. Additional transactions are provided with the BASE24-atm self-service banking (SSB) add-on product and are described in the ***BASE24-atm NCR 5XXX SSB Manual***.

Financial Transactions

The following financial transactions are supported by the BASE24-atm product for NCR ATMs.

- Withdrawal from checking account
- Withdrawal from savings account
- Fast cash withdrawal
- Cash advance from credit account
- Deposit to checking account
- Deposit to savings account
- Transfer from checking to savings
- Transfer from savings to checking
- Transfer from credit account to checking
- Transfer from checking to checking
- Transfer from savings to savings
- PIN change
- EMV PIN unblock
- Log only #1
- Log only #2
- Log only #3
- Log only #4
- Payment enclosed
- Payment from checking to credit account

- Payment from savings to credit account
- Balance inquiry from checking
- Balance inquiry from savings
- Inquiry into credit account availability
- Message enclosed from cardholder to financial institution
- Statement print
- Non-currency dispense transaction
- Mobile top-up from checking
- Mobile top-up from savings
- Mobile top-up from credit

Transaction Receipt Printing

All of the transactions listed above can have receipts printed for the cardholder's records. There are several ways that you can determine whether or not the receipt will print, and if so, how the receipt will look. These options are discussed more fully in [section 5](#).

Optional Receipts

You can determine whether or not a receipt will be printed by transaction type. You can also allow the cardholder to determine if a receipt will be printed on all transactions except for the statement print transaction and administrative transactions or allow the cardholder to continue the transaction without a receipt in the case of a printer failure.

Configurable Receipts

The BASE24-atm product comes with standard formats, or templates, for how financial and administrative receipts will look. You can use these templates or modify them to print the information that you want the cardholder to receive. Header and trailer templates allow you to customize the receipt to print your institution's name and marketing information. You can even print your graphic logo, if the printer is capable.

PAN Truncation

You can determine if the account numbers on a receipt are printed in full or are truncated so only some digits of the number appear. This option allows you to reduce the amount of fraud caused by carelessly discarded receipts falling into the wrong hands.

Data Compression

The BASE24-atm product supports data compression on statement print transactions. Data compression involves reducing the amount of transaction data that is carried in a statement print transaction message. Reducing the amount of transmitted data results in a more efficient use of processing resources. For more information on data compression for statement print transactions, refer to [appendix A](#).

Mobile Top-Up

If you have licensed the BASE24-atm Mobile Top-Up Device Handler extension, the NCR 5XXX Device Handler process supports printing two receipts for a mobile top-up transaction. The Device Handler process sends top-up information (that is, information provided by the mobile telephone operator) and funds information (that is, standard financial receipt information) to the ATM. Refer to appendix G for more information about the BASE24-atm Mobile Top-Up Device Handler extension.

Transaction Surcharge

An optional surcharge can be added to certain transactions performed at your terminals. This surcharge is in addition to any fees that the cardholder's institution imposes; in other words, surcharges are for transaction acquirers, not issuers. A surcharge can be positive or negative. A positive surcharge amount debits the cardholder account, while a negative surcharge amount (i.e., an incentive) credits the cardholder account.

You determine whether the surcharge is added, and the amount added, depending on such variables as whether the transaction is on-us or not-on-us, the account type of the transaction, the transaction amount, the card prefix, and the card type. Terminals can also be grouped for surcharging purposes; for example, all of your

terminals in airports can have the same surcharge profile. You also determine whether or not a surcharge is assessed when a partial amount is dispensed and the rest of the transaction is reversed.

Surcharging is customized through the Card Prefix File (CPF) and the Surcharge File (SURF), both of which are documented in the ***BASE24 Base Files Maintenance Manual***, and the ATD, which is documented in the ***BASE24-atm Files Maintenance Manual***. The surcharge amount is carried in the Surcharge Data token, which is appended to the BASE24-atm Standard Internal Message (STM). Tokens are described in the ***BASE24 Tokens Manual***.

The following financial transactions can have a surcharge added by the BASE24-atm system:

- Withdrawal from checking account
- Withdrawal from savings account
- Fast cash
- Cash advance from credit account
- Noncurrency dispense
- Deposit
- Deposit with cash back
- Split deposit

Interactive Request Notification

The standard surcharging feature uses an interactive request notification sequence to notify the cardholder of a transaction surcharge. That is, the notification and acceptance of the surcharge is processed before the Authorization process sends the transaction to the issuer for authorization. The cardholder account is impacted only after the cardholder accepts the fee.

Interactive request notification is the default processing for the surcharging feature. To use this processing, the ATM-FEE-INTRACTV-RESP-NTFY param must be set to a value of N. Interactive request notification also is used if the ATM-FEE-INTRACTV-RESP-NTFY param is not present.

Interactive Response Notification

Surcharge interactive response notification processing routes the transaction request, including fees, to the issuer for authorization before notifying the cardholder of a fee. The cardholder account is impacted before the cardholder accepts the fee. If the cardholder does not accept the fee, the transaction is reversed.

To use surcharge interactive response notification processing, the ATM-FEE-INTRACTV-RESP-NTFY param must be set to a value of Y.

The interactive response notification of fees can include both issuer fees and acquirer fees (i.e., surcharges). BASE24 maintains information in the Surcharge File (SURF) for acquirer fees only; it does not maintain information in its database for issuer fees. The setting of issuer fees must be done at a host system or an interchange. Issuer fee information is carried in the Issuer Fee/Rebate token. This token is described in the *BASE24 Tokens Manual*.

Administrative Transactions

You can perform administrative transactions in one of the following ways:

- You can use a machine-readable administrative card at the terminal.
- You can use the Rear Supervisor/Settlement transaction to initiate administrative transactions from the rear of the ATM without the use of an administrative card.
- You can perform administrative services using the Device Control Terminal (DCT).

The administrative transactions available through the BASE24-atm product include the following transactions:

- Balance with current cash
- Balance with standard replenishment
- Terminal subtotals
- Increase cash
- Decrease cash

The terminal balancing transactions such as balance with current cash and balance with standard replenishment, balance terminals using standard replenishment or current cash balancing methods.

The terminal subtotal transactions allow activity totals accumulated since the last balancing transaction (and up to the time the terminal subtotal transaction is initiated) to be printed on journal and receipt printers. Terminal totals are not cleared by these transactions.

Increase and decrease cash transactions (called the midpoint cash adjustment transaction), allow the amount of cash contained in a hopper to either be increased or decreased without balancing the terminal.

For more information on administrative transactions, refer to the ***BASE24-atm Settlement and Reporting Manual***.

Device Configuration and Control

The appropriate software to run NCR 5XXX-series ATMs is supplied by NCR and allows the ATMs to operate without additional support.

The BASE24-atm product has the diagnostic capabilities to identify ATM problems and minimize down time using a CRT terminal called the Device Control Terminal (DCT).

BASE24-atm support provides the ability to customize and control the NCR ATMs with greater flexibility. You can make changes to the NCR Configuration File (CNFG5070) using the NCR 5XXX Commands screen—selection 45 from the DCT Menu—and download the changes to the ATM.

Terminal Balancing

A terminal must be balanced on a regular basis once it is operational. This can be accomplished using the DCT functions or using administrative transactions performed at the ATM. The DCT allows a terminal to be closed, balanced, and opened again and cash totals to be increased or decreased. Other functions such as adding to the cash supply and servicing the terminal require administrative transactions to be performed at the terminal site.

Event Messages

The Device Handler process generates event messages periodically and as needed to notify an institution of the state of an ATM. The messages range in severity from informational to critical and can be printed or displayed using the EMSperus utility.

The NCR 5XXX Device Handler process also can capture native ATM status messages and display them in their native format to the EMSperus utility. If the ATM-DH-DISPLAY-NATIVE-STATUS-MSG param is present and set to the value Y, the NCR 5XXX Device Handler process appends the native status message it received from the ATM to existing status event messages sent to the EMSperus utility. The Device Handler process does this for all NDC+ unsolicited status messages and any NDC+ solicited status messages with a status descriptor of 8 or F it receives. If the param is set to a value other than Y or N, EMS raises an event stating that the value was invalid and defaults to the value N. If the param is set to the value N, the EMS system does not capture native ATM messages.

Native message fields b through g are required to display native solicited status messages. Native message fields b through e are required to display native unsolicited status messages.

The format of the data appended to the existing event message is [NATIVE]NNDC*native-message*, where *native-message* is the message received from the ATM.

PIN Handling

The Device Handler process is responsible for the interface between the BASE24 system and the NCR ATMs attached to it relative to PIN handling. Based on a combination of device capabilities and your preferences, the Device Handler process passes on or decrypts PINs upon entry into the system.

Discretionary Data Suppression

The BASE24-atm product protects consumer data in acquirer and issuer transactions as well as consumer data carried internally in the BASE24-atm Standard Internal Message (STM). If the ATM-DISCR-DATA-STM-SUPPRS param is present and set to the value Y, this discretionary data from Track 1 or Track 2 can be suppressed.

Basic Device Control Functions

The BASE24-atm product provides several ways for you to control NCR terminals. One of these ways is to use the DCT. The DCT allows you to perform all of the functions necessary to control a terminal.

The basic device control functions common to all types of terminals include the following functions:

- Balancing terminals
- Inquiring into the status of the terminals
- Opening and closing the terminals
- Starting and stopping terminals
- Displaying subtotals
- Increasing and decreasing cash in the hoppers

Each of the functions listed above is associated with the DCT, since each is critical to the operation of ATMs. The central location of the DCT allows these functions to be used more efficiently. For example, when institutions need to balance their terminals, they can close an ATM from the same screen from which they execute the balancing command.

BASE24-atm processes also can be reinitialized or warmbooted from the DCT. This is important because Device Handler processes must be reinitialized when changes to files are made that affect the Device Handler processes. For example, updates to the CPF are not utilized until the Device Handler process is reinitialized. Refer to the ***BASE24 Device Control Manual*** for more information about device control functions.

2: ATM Configuration

BASE24-atm support of NCR 5XXX-series ATMs includes flexible configuration options. NCR ATMs can be configured in numerous ways, depending on the institution's preferences. For example, you can allow some ATMs to offer certain transactions, while prohibiting ATMs in other locations from offering the same transactions. Each ATM can have its own transaction set, based on your selections. You can also change the wording on screen displays for ATMs in different locations.

This section contains information about the following items:

- NCR Configuration File
- BASE24 files used by the NCR 5XXX Device Handler process during transaction processing

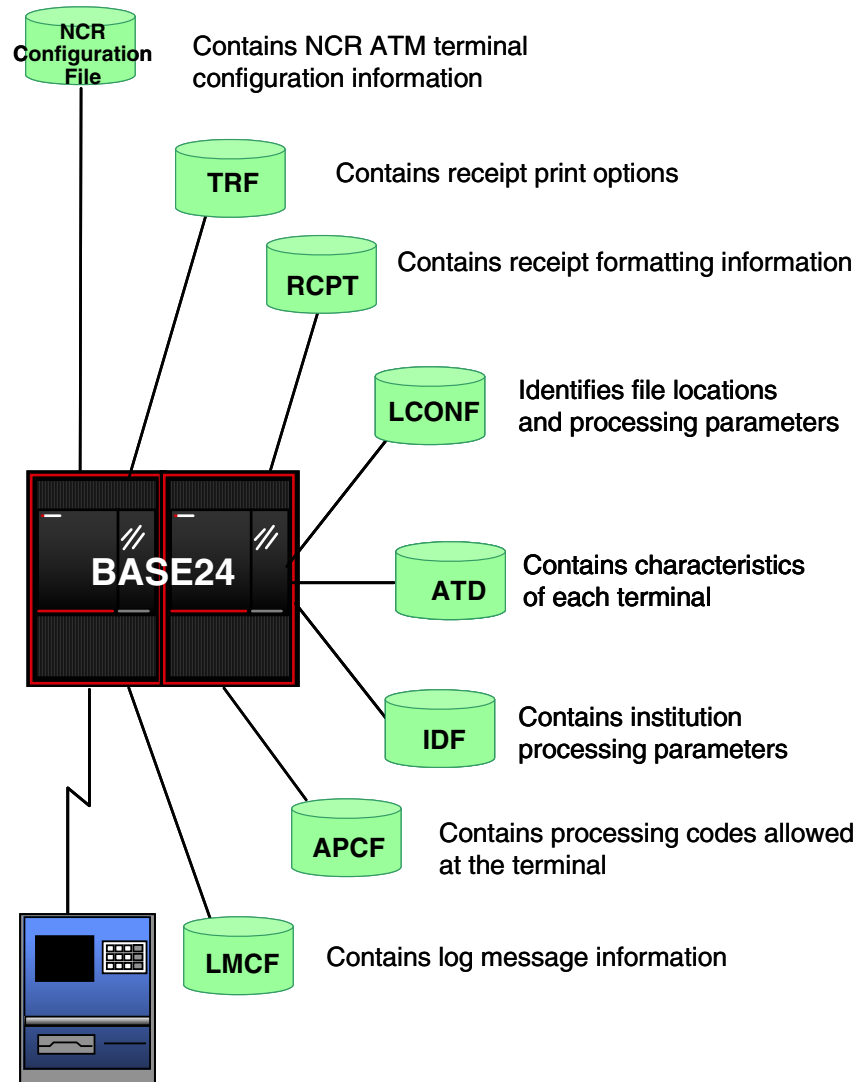
NCR 5XXX-Series ATM Configuration

The NCR 5XXX-series ATM is configured using software contained in the ATM itself. The NCR internal software contains the control logic and associated parameters that describe how the ATM is to process transactions using screen data and timer data.

In addition to the NCR internal software, a number of different BASE24 files are used in configuring the terminals. Because of the many combinations of options that are available during configuration, the files are discussed in this section one at a time. The various combinations that can be derived from these individual options is left up to you. The following files are used with NCR ATMs:

- NCR Configuration File (CNFG5070)
- Receipt File (RCPT)
- Terminal Receipt File (TRF)
- Logical Network Configuration File (LCONF)
- BASE24-atm Terminal Data files (ATD)
- Institution Definition File (IDF)
- Acquirer Processing Code File (APCF)
- Log Message Configuration File (LMCF)

The following illustration shows the files required to configure an NCR terminal on a BASE24-atm system.



NCR Configuration File

NCR ATMs are configured using software and files supplied by NCR and contained within the ATMs themselves.

The BASE24-atm system supports the NCR 5XXX ATM configuration using the NCR Configuration File (CNFG5070). Changes can be made to the CNFG5070 on the BASE24-atm system using the NCR 5XXX Commands screen and can be downloaded to the ATM from the BASE24 Device Control Terminal (DCT). The CNFG5070 is described in more detail in [section 4](#). Refer to the ***BASE24 Device Control Manual*** for information on the procedures used when downloading files.

The CNFG5070 is organized by groups. When a change is made, you can download a data group to the ATM without physically going to the terminal. The BASE24-atm product supplies several groups of data in the CNFG5070, which are listed in [section 4](#). Groups 106 through 112 are described in more detail in the ***BASE24-atm NCR 5XXX SSB Manual***, and groups 300 through 303 and 311 through 313 are described in [section 5](#) of this manual.

Receipt File

The Receipt File (RCPT), which is used to configure the printed receipts, provides you with the ability to establish ATM receipt formats or templates that are unique to the institution. In addition, this customization option supports the display of an informational or marketing message on the cardholder receipt, the use of flexible monetary symbols, multiple languages, and the option to print the account description and account type on the receipt. The configurable receipt option includes approved and denied financial transaction receipts as well as administrative receipts. The RCPT also is used to configure Digital Audio Service (DAS) templates. Refer to appendix B for more information about configuring DAS templates.

Receipt configuration is supported by receipt group, transaction type, card issuer FIID, *from* and *to* account type, and printer type. Receipt groups allow the use of a single receipt format for a group of ATM terminals rather than configuring receipts for an individual terminal. Receipt groups are specified for each terminal on ATD screen 8. For further information on establishing receipt groups in the ATD, refer to the ***BASE24-atm Files Maintenance Manual***. The information on a receipt can also be tailored for specific card issuers; for instance, you can have different marketing information for on-us and not-on-us transaction receipts. Having different templates for different printers allows you to truncate the PAN on a receipt to reduce the chance of fraud, while allowing the entire PAN to be printed on the journal so that you have a complete record of the transaction. The receipt can also be customized to allow separate templates for different *from* and *to* account types. For example, you can truncate savings account numbers differently than credit card numbers.

There are several ways you can configure the BASE24 product to require receipts or allow transactions without a printed receipt. The first option allows you to decide by transaction type whether receipts are required, optional (i.e., allow the transaction if the receipt printer is not operational), or not printed. These settings are contained in the Terminal Receipt File (TRF) using the terminal receipt profile configured for the ATM on ATD screen 8. You can also allow the cardholder to determine if a receipt is printed on all transactions (except for the statement print transaction and administrative transactions), or allow the cardholder to continue the transaction without a receipt in the case of a printer failure.

For a description of how the Device Handler process uses the RCPT in determining receipt text and format, examples of default receipt templates for various transactions, and how to configure whether receipts are printed for a transaction using the TRF, refer to [section 5](#). For a description of how the Device Handler process uses the RCPT in specifying voice text and format, refer to appendix B.

Terminal Receipt File

The Terminal Receipt File (TRF) contains receipt print options for approved and denied transactions. The TRF contains one record for each terminal receipt profile, message category (the second digit in the message number), and ISO processing code combination in the BASE24-atm system. The terminal receipt profile defines the type of transactions that require a receipt at the terminal. The terminal receipt profile is defined in the BASE24-atm Terminal Data files (ATD).

Refer to section 5 for more information about how the TRF is used in receipt processing. For more information about TRF screens, refer to the ***BASE24-atm Files Maintenance Manual***.

Logical Network Configuration File

The Logical Network Configuration File (LCONF) contains the information that links physical file names to the file names as they are referenced in BASE24 code and information that sets processing values for a logical network.

The information that links physical file names to the file names as they are referenced in code are known as assigns. The information that sets processing values are known as params.

The LCONF assigns and params used by NCR devices are listed below, including a short description of each. The following add-on products, features, and extensions use additional assigns and params that are described only in appendixes of this manual:

Add-On Product, Feature, or Extension	Appendix
BASE24-atm Non-Currency Dispense	Appendix C
NCR 5XXX Dial-Up Device Handler Extension	Appendix D
Automated Key Distribution System	Appendix E
Dynamic Key Management	Appendix E
Multiple Currency Dispense	Appendix F
BASE24-atm Mobile Top-Up Device Handler Extension	Appendix G

The BASE24-atm self-service banking (SSB) product uses additional assigns and params that are described in the ***BASE24-atm NCR 5XXX SSB Manual***. Refer to the ***BASE24 Logical Network Configuration File Manual*** for detailed information on LCONF assigns and params.

Assigns

The assigns used by the Device Handler process to locate the appropriate files are listed below. Add-on products, features, and extensions use additional assigns that are described only in appendixes of this manual.

APCFEMT — The fully qualified file name of the Acquirer Processing Code File (APCF) extended memory table (APCFEMT), which contains APCF records.

ATM-ATD-DYN-GNRL — The fully qualified file name of the BASE24-atm Terminal Data Dynamic File—general data (ATDD1). This file contains dynamic core terminal data (i.e., data common to all types of devices that is subject to change), device information (i.e., data specific to a device type), and device-dependent data (i.e., data specific to each terminal defined). Key STM fields for the last transaction processed for a device are also stored in this file. The NCR 5XXX Device Handler process uses this assign to read and update dynamic terminal data as transactions are processed.

ATM-ATD-DYN-SCRATCH-PAD — The fully qualified file name of the BASE24-atm Terminal Data Dynamic File—scratch pad (ATDD2). This file contains dynamic token data for reversal processing. This assign is generally optional but required for ATDD2 processing.

ATM-ATD-STATIC-GNRL — The fully qualified file name of the BASE24-atm Terminal Data Static File—general data (ATDS1). This file contains static data, which is information that does not change during the course of transaction processing. The NCR 5XXX Device Handler process uses this assign to retrieve static terminal data, which is maintained in extended memory.

ATM-DH-ATD-EXTMEM-SWAPVOL — The fully qualified name of the Terminal Data File extended memory swap volume in which the NCR 5XXX Device Handler process temporarily stores data when building extended memory for a terminal's BASE24-atm Terminal Data files records. If this assign is not present in the LCONF or contains an invalid value, the HP NonStop operating system assigns a swap volume location.

Example: \B24.\$DATA

ATM-DH-CNF5070 — The fully qualified file name of the configuration file for NCR ATMs, such as \B24.\$DATA.PRO1DATA.CNFG5070. This assign is used by the NCR 5XXX Device Handler process when downloading data to terminals. This value can be overridden by a corresponding value in the device-dependent information in the individual ATD record for the ATM.

ATM-NIDFEMT — The fully qualified file name of the Note ID File (NIDF) extended memory table, which contains NIDF records.

ATM-TRFEMT — The fully qualified name of the Terminal Receipt File extended memory table (TRFEMT), which contains TRF records.

ERFEMT — The fully qualified file name of the Exchange Rate File (ERF) extended memory table. This assign is used by the Currency Conversion utilities.

IDF — The physical file name of the Institution Definition File (IDF).

IDFEMT — The fully qualified file name of the Institution Definition File (IDF) extended memory table (IDFEMT), which contains IDF records.

LMCF — The physical file name of the Log Message Configuration File (LMCF). This file is used during the initialization and reinitialization of the Device Handler process.

RCPTENT — The fully qualified file name of the Receipt File extended memory table (RCPTENT), which contains Receipt File (RCPT) records.

SECURE-DEV1 — The name of the primary Transaction Security Services process used for PIN management, message authentication processing, and security module key management.

SECURE-DEV2 — The name of the secondary Transaction Security Services process used for PIN management, message authentication processing, and security module key management.

SECURE-DEV-xx — The name of one of 20 Transaction Security Services processes accessed on a rotating basis, where *xx* is any two characters.

Params

The following params are used for setting processing values at the logical network level. Add-on products, features, and extensions use additional params that are described only in appendixes of this manual.

ATM-DH-AUTH-TIMER — The number of seconds the Device Handler process waits for a reply from the BASE24-atm Authorization process before denying a request.

ATM-DH-CONS-MAC-ERR-LMT — A code indicating the number of consecutive message authentication code (MAC) errors that can occur before the Device Handler process sends a shutdown message to the ATM. If the Dynamic Key Management add-on product has been purchased, the NCR 5XXX Device Handler process loads new MAC and PIN keys to the ATM and attempts to reopen the ATM.

ATM-DH-DYN-TBL-RETN-PRD-LEASE — The number of seconds the Device Handler process waits for the next leg of a multiple-leg transaction before the dynamic data for a terminal connected to a leased line can be removed from the dynamic data table. A value of 0 disables this processing.

ATM-DH-DISPLAY-NATIVE-STATUS-MSG — A value indicating whether the native status message data is displayed in the event message. If this parameter does not exist or has a value other than Y, the native status message data is not displayed in the event message.

ATM-DH-TERM-IN-CPLF-RETN-PRD — The number of minutes that the BASE24-atm Check Proof List files (CPLF) records and its terminal are retained in extended memory. Valid values are 1 through 60. A value of 5 is the default.

ATM-DH-MAX-TERMS — The maximum number of BASE24-atm Terminal Data files (ATD) records that each BASE24-atm NCR 5XXX Device Handler process can hold in extended memory at one time. Thus, this param indicates the maximum number of terminals each NCR 5XXX Device Handler process can service.

ATM-DH-MAX-TERMS-IN-CPLF-TBL — The maximum number of unique terminals that can exist at one time in the CPLF extended memory table. A symbolic name (for example, P1A^N50) is required in this field. A value of 1/5 of the value specified in the ATM-DH-MAX-TERMS param is the default.

ATM-DH-MAX-TERMS-IN-DYN-TBL — The maximum number of BASE24-atm Terminal Data Dynamic File—general data (ATDD1) records that a BASE24-atm NCR 5XXX Device Handler process can store in extended memory at one time. Thus, this param indicates the amount of extended memory each Device Handler process must reserve for ATDD1 records and the maximum number of transactions each Device Handler process can process at one time.

ATM-DH-MAX-TERMS-IN-DYN-TBL — The maximum number of BASE24-atm Terminal Data Dynamic File—general data (ATDD1) records that a BASE24-atm NCR 5XXX Device Handler process can store in extended memory at one time. Thus, this param indicates the amount of extended memory each Device Handler process must reserve for ATDD1 records and the maximum number of transactions each Device Handler process can process at one time.

ATM-DH-MAX-TERMS-IN-STATIC-TBL — The maximum number of BASE24-atm Terminal Data Static File—general data (ATDS1) records that a BASE24-atm NCR 5XXX Device Handler process can store in extended memory at one time. Thus, this param indicates the amount of extended memory each Device Handler process must reserve for ATDS1 records and the maximum number of transactions each Device Handler process can process at one time.

ATM-DH-N50-DPM-DFLAGS — A code indicating whether the NCR 5XXX Device Handler process checks the Document Processing Module (DPM) magnetic flux and courtesy amount verification (CAV) fields in requests. During normal processing, the NCR 5XXX Device Handler process checks both of these fields.

ATM-DH-OPT-LVL — A value indicating whether the Device Handler process writes the ATD to disk upon completion of normal transactions.

ATM-DH-STALE-RQST-TIMEOUT — The number of seconds an incoming transaction can remain in the queue of the Device Handler process before being discarded by the Device Handler process as too old to process.

ATM-DISCR-DATA-STM-SUPPRS — A flag indicating whether the discretionary data carried in the STM should be suppressed for transactions that are authorized by BASE24 or a host system. If this parameter does not exist or has a value other than Y, discretionary data is not suppressed in the STM.

ATM-FCA-CUTOVER-AUDIT — A flag indicating whether the Forced Cutover Audit function is turned on. This param specifies the use of the Forced Cutover Audit function for the entire logical network. It is used only if the Forced Cutover Audit add-on product has been purchased and installed.

ATM-FEE-INTRACTV-RESP-NTFY — A flag indicating whether surcharge interactive response notification processing requirements (i.e., performing interactive notification sequences for acquirer and issuer fees and rebates) are used.

ATM-REAR-ADMN-TRAN-PIN-ALWD — A flag indicating whether all rear administrative transactions that contain PIN data should be allowed. If this parameter has a value of Y, rear administrative transactions are allowed. If this parameter has a value of N, rear administrative transactions are declined.

DATE-DISPLAY-TYPE — A code indicating the format of the month and day portion of the dates, such as DDMM or MMDD, on reports and BASE24-atm Device Handler-formatted receipts. The DATE-DISPLAY-TYPE param does not affect dates that include the year such as dates in the form YYMMDD.

SECURE-DEV-FAIL-NOTIFY — A code indicating the interval that messages are logged due to transactions that are denied because of a security device failure. This param is accessed by security utility procedures during the initialization or reinitialization of the Device Handler process.

SECURE-DEV-TIM-LMT — A code indicating the number of seconds to wait for a response from a security device. This param is accessed by security utility procedures during the initialization or reinitialization of the Device Handler process.

SECURE-DEV-TYP — A code indicating the type of security module that is being used by the logical network. This param is accessed by security utility procedures during the initialization or reinitialization of the Device Handler process.

BASE24-atm Terminal Data Files (ATD)

The BASE24-atm Terminal Data files (ATD) contain one record for each ATM in the BASE24 system. A file contains either dynamic or static data associated with the device.

The BASE24-atm product uses the ATM Terminal Data Dynamic File—general data (ATDD1) to help maintain ATM control. Each record in this file contains information reflecting the current status of an ATM. The status of every terminal is monitored continuously so that the BASE24-atm system operator is notified when a problem is detected. The system operator, in turn, can notify the terminal-owning institution about any errors at its ATMs that may require service intervention. The institution can determine from the type of error reported whether the problem is of a nature that the ATM vendor should be called or if it can be corrected by the institution staff.

The ATM Terminal Data Static File—general data (ATDS1) for an ATM defines much of the functionality for the individual ATM. Whenever the BASE24 system receives a message from an ATM, or a message is sent by the BASE24 system to an ATM, the static ATD record for that ATM is accessed. A separate ATD record must exist for each ATM controlled by the BASE24-atm system.

You can access ATD records using BASE24-atm files maintenance functions. ATD information is accessed by 15 screens. To view these screens for NCR 5XXX-series ATMs, set the DEVICE TYPE field on ATD screen 1 to 22. The first six screens contain general information that is common to all ATMs, regardless of manufacturer. The next three screens contain information specific to NCR 5XXX-series ATMs. The next three screens contain information specific to the BASE24-atm self-service banking (SSB) add-on product, and are only used if you have installed the SSB product. Screen 13 contains configurable dispense information specific to NCR 5XXX-series ATMs. Screen 21 contains EMV information. Screen 22 contains AKDS (Automated Key Distribution System) information that is common to all ATMs regardless of the manufacturer. For information on the values of the individual fields in the ATD and how the ATD can be maintained, refer to the ***BASE24-atm Files Maintenance Manual***. Refer to the ***BASE24-atm NCR 5XXX SSB Manual*** for information about the SSB screens. Refer to the ***BASE24-atm EMV Support Manual*** for information about the EMV screen.

General ATD Information

The basic information contained in ATD records for an NCR 5XXX ATM includes the following items:

- Reference information
- Transaction processing information
- Status information
- Transaction security information
- AKDS (Automated Key Distribution System) configuration data

Reference information includes information such as the physical location of the terminal, the owner of the terminal, and the Device Handler and Authorization processes involved in the specified transaction processing for that ATM.

Transaction information includes information on the kind of transactions allowed at the terminal. The Authorization process determines whether a match exists between the sharing groups indicated in the ATD record for the ATM and the sharing groups set in the IDF for the institution that issued a card accepted by the ATM. This check indicates the type of transaction agreement, if any, that has been made between the ATM owner and the institution issuing the card.

Transaction activity can be monitored through a series of timer settings established in the ATD for peak and nonpeak periods and holiday schedules. Receipt groups for configurable receipts are defined in the ATD.

Information on status changes to the ATM such as the state of the ATM and the amount of transaction activity for the ATM is also kept in the ATD.

Note: Using DCT screen 45, configuration parameters must be retrieved after starting and before loading a terminal corresponding to a newly defined ATD record.

Transaction security information relating to the encryption and decryption of data, local PIN verification, and message authentication is defined in the ATD.

AKDS configuration information relating to the name of the Host Public Key Owner (also known as the Key Signer), the ATM manufacturer, and the EPP format, is also defined in the ATD.

Specific NCR Information in the ATD

The ATD contains information specific to NCR 5XXX-series ATMs. The NCR 5XXX-specific screens contain information such as the type of line protocol used with the terminal, cardholder balance information, and the transaction security options used at the ATM.

Message Authentication

Message authentication ensures that transaction messages are received exactly as created by the legitimate originator. Message authentication protects financial transaction messages against accidental or deliberate alteration.

The BASE24-atm product supports message authentication for NCR terminals through controls in the ATD and, optionally, in a field selection table that is stored in the N50NAMS.

BASE24-atm Terminal Data Files (ATD)

Message authentication information in the ATD includes fields that specify whether message authentication is supported, the character set of the data to be authenticated (i.e., ASCII or EBCDIC), whether the solicited status message is to be authenticated, and whether full or partial message authentication is to be used.

Message authentication can be performed in two ways—full message authentication or selective message authentication. Full message authentication includes the whole message starting from the first field following the protocol-dependent header up to, but excluding, the field separator preceding the MAC field. Selective message authentication involves calculating the MAC based on fields specified in a field selection table in the N50NAMS. This table can also specify that full message authentication be performed.

If the ATD indicates that full message authentication is to be used, then message authentication is performed on all of the fields in the message. If the ATD indicates that selective field message authentication is to be used, then selective message authentication is performed only according to the values established in the field selection table.

For example, full message authentication takes place under one or both of the following conditions:

- The ATD indicates full message authentication.
- The ATD indicates selective message authentication and the field selection table for the message indicates full message authentication.

If the ATD indicates selective message authentication and the field selection table for a specific message type indicates selected fields, then only selected fields are taken in sequence from the message to build a single data string. Field and group separators are not included and empty fields are omitted.

For more information about the message authentication information in the ATD, refer to the *BASE24-atm Files Maintenance Manual*.

Field Selection Tables

Field selection tables are used to determine which messages are authenticated and which fields are included in the MAC calculation. These tables are stored in the N50NAMS. The data in these tables is downloaded to the terminal in the Message Authentication Field Selection Load message when hardware message authentication is supported. These are default tables which you can change. For more information about message authentication, refer to the appropriate NCR vendor documentation and the *BASE24 Transaction Security Manual*.

Field Selection Table	Authentication
Transaction request table (field h)	The following fields are authenticated: e, h, j, k, and l.
Transaction reply table (field i)	The following fields are authenticated: f, g, h, i, j, l, and p.
Solicited status table (field j)	Full message authentication of all fields in the message.
Other message table (field k)	Full message authentication of all fields in the FIT Load message and the State Table Load message.
Track 1 table (field l)	Full message authentication of all subfields in the field in the message.
Track 2 table (field m)	Full message authentication of all subfields in the field in the message.
Track 3 table (field n)	Full message authentication of all subfields in the field in the message.

Note: Sentinels are included in the MAC calculation when they are contained in specified subfields. If the first field of track data is selected, the start sentinel is included. If the last field of track data is selected, the end sentinel is included.

Institution Definition File

The Institution Definition File (IDF) is used to define the processing parameters for each institution participating in the logical network. For example, the IDF is read by the Device Handler process during an administrative transaction in order to retrieve the dates the Device Handler process requires for cutting over its terminals. In addition, the Device Handler process reads the IDF extended memory table if terminals are configured, through the ATD, to support inactivity timers. The IDF also contains routing tables used for transaction routing within the BASE24 system, date parameters, transaction sets, and ATM processing control parameters.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

Acquirer Processing Code File

The Acquirer Processing Code File (APCF) contains the processing codes allowed at the terminal. The key to the APCF is the acquirer transaction profile, the message category (the second digit of the message type), and the ISO processing code (a six-character code containing the ISO transaction code, a *from* account type, and a *to* account type). The acquirer transaction profile, which is defined in the ATD or IDF, can be assigned to an individual terminal or to a group of terminals. Acquirer transaction profiles defined in the ATD override the default acquirer transaction profiles defined in the IDF. Each acquirer transaction profile defines a set of transactions supported for an individual acquiring terminal or group of terminals.

For more information on the APCF, refer to the ***BASE24 Base Files Maintenance Manual***.

Log Message Configuration File

The Log Message Configuration File (LMCF) contains one record for each event message that has been altered by a BASE24 network operator. Because LMCF records must be read into memory by the processes using them, updates to the LMCF do not take effect until the processes are reinitialized.

An LMCF record is accessed using BASE24-atm operator functions. The file itself contains changes made to event messages, in the form of records. The routing code, severity, and text of each event message can be modified using BASE24 operator functions.

3: BASE24-atm Device Handler Process

Device Handler processes are the primary interfaces between ATMs and the BASE24 product. All messages that are sent by the BASE24 system to an ATM first must be received by the Device Handler process for the ATM and then forwarded to the designated ATM. Likewise, all messages that are sent by an ATM to the BASE24 system must first be received by the Device Handler process for the ATM.

Because each type of ATM has its own message format and way of processing transactions, separate Device Handler processes must exist for each type of terminal supported by the BASE24 product. For example, Diebold 910 ATMs must have their own type of Device Handler process, which would differ from the Device Handler process used for Diebold 10XX/478X ATMs or NCR 5XXX-series ATMs.

The BASE24-atm self-service banking (SSB) add-on product provides additional functionality by modifying the Device Handler process. These modifications are described in the ***BASE24-atm NCR 5XXX SSB Manual***.

This section provides the following information:

- An overview of the BASE24-atm NCR 5XXX Device Handler process.
- Descriptions of processing steps taken by the Device Handler process when it is first started.
- A list of BASE24 commands received by the Device Handler process to perform specialized types of activities.
- Information on how to access the event messages generated by the Device Handler process.

BASE24-atm NCR 5XXX Device Handler Process Overview

NCR 5XXX-series terminals, or other vendor's ATMs emulating the NCR 5XXX-series terminals, are linked to the BASE24-atm system by a Device Handler process tailored to the NCR 5XXX-series terminals. Every transaction initiated at an NCR terminal connected to the BASE24-atm system is passed to this Device Handler process when it enters the BASE24-atm system. The message coming from the terminal is in the native mode of the device and the Device Handler process converts this message into the BASE24-atm Standard Internal Message (STM) for sending to a BASE24-atm Authorization process. The BASE24-atm product supports both financial and administrative transactions using NCR ATMs.

Transactions processed at different NCR terminals can be handled differently depending on how the individual terminal is configured in the BASE24-atm Terminal Data files (ATD) and the Acquirer Processing Code File (APCF) extended memory table. Each terminal has a record in the ATD, which includes different sharing arrangements and acquirer transaction profiles. The Device Handler process reads the ATD whenever it needs information about a terminal. It uses the acquirer transaction profile specified in the ATD as part of the key to the APCF extended memory table. Values in the APCF indicate if and how the transaction is allowed.

The Institution Definition File (IDF) extended memory table is read by the Device Handler process to obtain the dates the Device Handler process requires for cutting over its terminals and the default acquirer transaction profile, if needed. It is also used to determine the holidays established by an institution for use in supporting the inactivity timer option. Inactivity timers enable an institution to monitor transaction activity during specified periods.

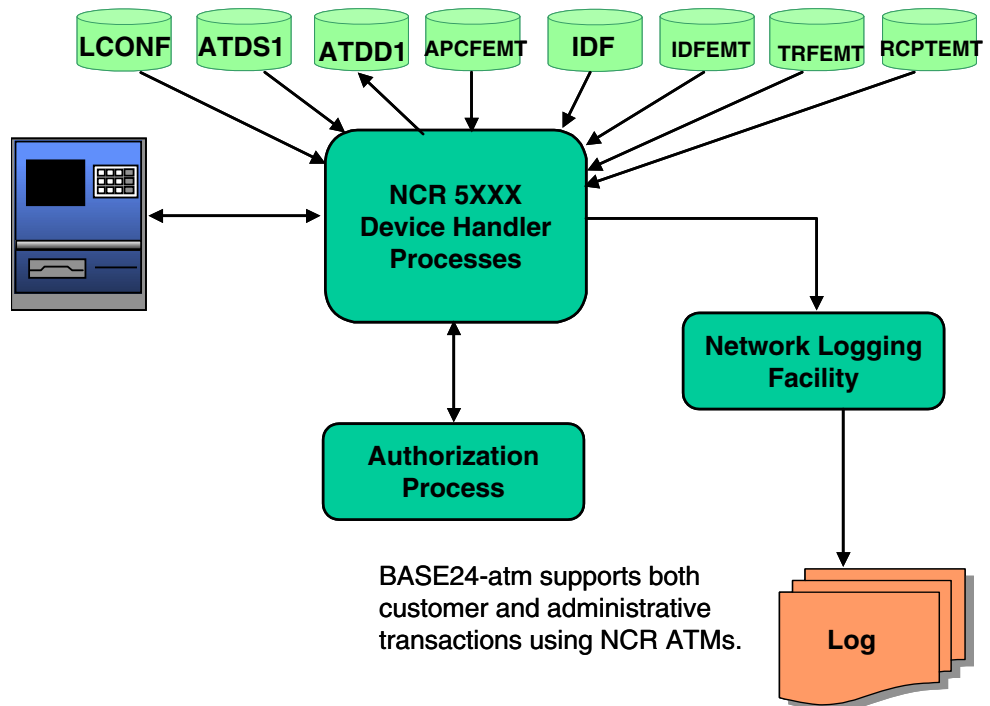
The Terminal Receipt File (TRF) extended memory table is read by the Device Handler process to determine whether a cardholder receipt and journal receipt are required for the transaction at the terminal.

The Receipt File (RCPT) extended memory table is read by the Device Handler process to determine the text and format of financial and administrative receipts as well as the format for any DAS templates present.

The Device Handler process also uses the Logical Network Configuration File (LCONF). The Device Handler process reads the LCONF to obtain the names of the RCPT extended memory table, ATD, and IDF as well as a number of other file names and processing params.

The Device Handler process creates event messages when problems occur with the terminal and sends them to the BASE24 network logging facility. The network logging facility then writes the event messages to output devices (e.g., printers, disk files, or terminals).

The following graphic shows the role of the Device Handler process during transaction processing.



Initialization Steps

The Device Handler process performs the following when it is first started or initialized:

1. Initializes its timer table.

When a Device Handler process creates timers, they are placed in the Device Handler timer table. Device Handler processes maintain three types of timers:

- The Authorization Response timer times each authorization response. If an authorization response is not returned to the Device Handler process within this time limit, the request is automatically denied. (Set by the ATM-DH-AUTH-TIMER param in the LCONF.)
- The Download timer (only kept by Device Handler processes that are capable of performing downloads) times each station download record. If no response is received from the station within this time limit, the Device Handler process automatically aborts the download. (Set to 45 seconds.)
- The Inactivity timer indicates a period of time during which some activity should occur at the terminal. If no activity is detected during the period, the Device Handler process logs a message. This timer is set on ATD screen 7, with different settings for workday and weekend/holiday peak usage time periods. This timer allows a maximum number of consecutive inactivity timeouts before terminal monitoring is discontinued. This maximum timeout value is also specified on ATD screen 7.

2. Initializes the global extended message table structure.
3. Reads the name of the LCONF into memory.

The startup message from the XPNET process contains the name of the LCONF, which the Device Handler process reads into memory. This name allows the Device Handler process to access the LCONF for the assigns and params it uses in processing.

4. Reads the following assigns from the LCONF:

APCFEMT
IDFEMT
RCPTMT
ATM-TRFEMT
ATM-ATD-DYN-GNRL
ATM-ATD-DYN-SCRATCH-PAD

ATM-ATD-STATIC-GNRL
 ATM-DH-ATD-EXTMEM-SWAPVOL
 ATM-DH-CNF5070
 LMCF
 IDF

5. Initializes the network logging facility process for writing event messages.
6. Reads the LMCF for its modified event messages.

The LMCF contains a record for each modified event message in the system. The Device Handler process opens the LMCF, retrieves its own modified event messages, and closes the file.

7. Loads the following params from the LCONF.

ATM-DH-AUTH-TIMER
 ATM-DH-DISPLAY-NATIVE-STATUS-MSG
 ATM-DH-DYN-TBL-RETN-PRD-DIAL-UP
 ATM-DH-DYN-TBL-RETN-PRD-LEASE
 ATM-DH-MAX-TERMS
 ATM-DH-MAX-IN-DYN-TBL
 ATM-DH-MAX-IN-STATIC-TBL
 ATM-DH-OPT-LVL
 ATM-DH-STALE-RQST-TIMEOUT
 DATE-DISPLAY-TYPE
 ATM-DH-CONS-MAC-ERR-LMT
 ATM-DH-NON-VALUE-THRESHOLD
 ATM-DISCR-DATA-STM-SUPPRS
 ATM-FEE-INTRACTV-RESP-NTFY

- a. Reads and edits the ATM-DH-MAX-IN-DYN-TBL, ATM-DH-MAX-IN-STATIC-TBL, and ATM-DH-MAX-TERMS params in the LCONF.

If the ATM-DH-MAX-IN-DYN-TBL param is invalid or cannot be found, the Device Handler process uses the lesser of the default value of 25 or the value of the ATM-DH-MAX-TERMS param.

If the ATM-DH-MAX-IN-STATIC-TBL param is invalid or cannot be found, the Device Handler process uses the lesser of the default value of 25 or the value of the ATM-DH-MAX-TERMS param.

If the ATM-DH-MAX-TERMS param is invalid or cannot be found, the Device Handler process uses a default value of 100.

- b. Reads and edits the ATM-DH-OPT-LVL param in the LCONF.

If the ATM-DH-OPT-LVL param is not 0 or 1, or cannot be found, the Device Handler process uses a default value of 0 (two writes to the ATD).

- c. Reads and edits the ATM-DH-AUTH-TIMER param in the LCONF.
If the ATM-DH-AUTH-TIMER param is not within a range of 10 to 180, or if the param cannot be found, the Device Handler process uses a default value of 45 seconds.
- d. Reads and edits the ATM-DH-STALE-RQST-TIMEOUT param in the LCONF.
If the ATM-DH-STALE-RQST-TIMEOUT param is not within a range of 10 to 180, or if the param cannot be found, the Device Handler process uses a default value of 45 seconds.
- e. Reads and edits the DATE-DISPLAY-TYPE param in the LCONF.
If the DATE-DISPLAY-TYPE param is not set to 1 or 2, or cannot be found, the Device Handler process logs a message to the network logging facility and uses a default value of 2 (date in U.S. format).
- f. Reads and edits the ATM-DH-CONS-MAC-ERR-LMT param in the LCONF.
If the ATM-DH-CONS-MAC-ERR-LMT param is invalid or cannot be found, the Device Handler process logs a message to the network logging facility and uses a default value of 3.
- g. Reads and edits the ATM-DH-NON-VALUE-THRESHOLD param in the LCONF.
If the ATM-DH-NON-VALUE-THRESHOLD param cannot be found, the ASCII to integer conversion fails, or the converted value is less than 5, the Device Handler process logs a message and uses a default value of 5.
- h. Reads and validates the ATM-FEE-INRTRACTV-NTFY param.
If the ATM-FEE-INRTRACTV-NTFY param is invalid or cannot be found, the Device Handler process uses a default value of N.
- i. Reads and validates the ATM-DH-DYN-TBL-RETN-PRD-DIAL-UP param.
If the ATM-DH-DYN-TBL-RETN-PRD-DIAL-UP param is invalid or cannot be found, the Device Handler process uses a default value of 90.
- j. Reads and validates the ATM-DH-DYN-TBL-RETN-PRD-LEASE param.
If the ATM-DH-DYN-TBL-RETN-PRD-LEASE param is invalid or cannot be found, the Device Handler process uses a default value of 60.

- k. Reads and validates the ATM-DH-DISPLAY-NATIVE-STATUS-MSG param.

If the ATM-DH-DISPLAY-NATIVE-STATUS-MSG param is invalid or cannot be found, the Device Handler process raises an event message and uses a default value of N.

8. Retrieves the params and assigns for any add-on products, such as BASE24-atm SSB, BASE24-atm Non-Currency Dispense, the BASE24-atm Mobile Top-Up Device Handler extension, the NCR 5XXX Dial-Up Device Handler Extension, the Automated Key Distribution System, Dynamic Key Management, and Forced Cutover Audit.
9. Closes the LCONF.
10. Opens the ATD files with read/write access using the name specified in the ATD file assigns in the LCONF.
11. Opens the IDF with read access using the name specified in the IDF assign in the LCONF.
12. Initializes an extended memory area for finding and keeping ATD records.

The Device Handler process uses the ATM-DH-MAX-TERMS param in the LCONF to build a table of the ATD key record positions.

The Device Handler process uses the ATM-DH-MAX-IN-DYN-TBL param in the LCONF to determine how much memory is needed to keep copies of the ATD records in extended memory. This param indicates the maximum number of transactions that can be processed by a Device Handler process at one time.

13. Allocates and initializes extended memory for the various EMTs. The called procedure also deallocates the old extended memory, if needed.
14. Calls a security utility procedure that reads either the SECURE-DEV1 and SECURE-DEV2 assigns or just the SECURE-DEV-xx assign from the LCONF and attempts to open the security device, if a security device is being used.

If the security devices cannot be opened, the Device Handler process creates an event message indicating that one or both of the security devices could not be opened and sends it to the logging facility.

The SECURE-DEV-FAIL-NOTIFY, SECURE-DEV-TIM-LMT, and SECURE-DEV-TYP params are also accessed in the LCONF at this time by the security utility procedure.

15. Reads all ATD records for this process name.

Warmboot Procedures

The following paragraphs describe how the Device Handler process handles warmboot commands.

WARMBOOT Command Procedures

The WARMBOOT command is a general use operator command sent using the Device Control Terminal (DCT) screen or a network control facility. General use operator commands enable the operator to control or change the way the Device Handler process is operating.

When Device Handler processes are warmbooted, they do not go through their entire initialization routine. Instead, they only perform the following steps:

1. Reread the LCONF and retrieve all of their assigns and params (including module-specific assigns and params).
2. Reinitialize the LMCF.
3. Release the current extended memory.
4. Close and reopen the files they use.
5. Reallocate new extended memory based on the current settings in the LCONF.
6. Reinitialize the ATD.
7. Close and reopen the security module.

For detailed information on the WARMBOOT command and other operator commands, refer to the ***BASE24 Text Command Reference Manual***. For more information on the DCT, refer to the ***BASE24 Device Control Manual***.

WARMBOOT ATD Procedures

When the Device Handler process receives a WARMBOOT ATD command, it warmboots static terminal data held in extended memory. The module rereads its ATM Terminal Data Static File—general data (ATDS1) into extended memory. Any time static data is changed for an NCR 5XXX ATM on the BASE24-atm Terminal Data files (ATD) screens, it is not used in processing until the associated Device Handler process receives this command or is reinitialized.

Device Handler Text Commands

Text commands are used to send information from a network control facility to a process. These commands serve as a tool that operators can use to perform network management functions such as changing certain processing parameters, warmbooting a specific process, monitoring a specific process, and resolving problems encountered by a specific process. The following are some of the text commands that can be used with the Device Handler process:

- The OPEN command causes the Device Handler process to send the appropriate native-mode message to open the ATM.
- The CLOSE command causes the Device Handler process to send the appropriate native-mode message to close the ATM.
- The LOAD commands direct the Device Handler process to download some form of configuration information to the ATM.
- The RESET command causes the Device Handler process to interrupt any current transaction or load in progress for the given ATM. Operators should not enter this command unless requested to do so by ACI personnel.
- The WARMBOOT command directs the Device Handler process to warmboot.
- The WARMBOOT ATD command causes the Device Handler process to warmboot static terminal data held in extended memory.
- The ALTERATTRIBUTE command causes the Device Handler process to warmboot (reallocate) a specified shared extended memory table after it has been rebuilt.
- The STATUSEXTMEM command causes the Device Handler process to display statistics about its BASE24-atm Terminal Data files (ATD) extended memory segment on the network log.

For a list of the steps the Device Handler process follows when a WARMBOOT command is issued, refer to the [“Warmboot Procedures”](#) topic found previously in this manual.

For detailed descriptions and the syntax of all BASE24 text commands, refer to the ***BASE24 Text Command Reference Manual***.

Event Message Generation

The Device Handler process generates event messages that keep operators aware of activity occurring in the system. Event messages can be used to assist operators with their daily tasks or to notify operators of problems occurring in the system. Event messages provide operators with insight as to how the system is running.

In addition, operators can view event message documentation electronically. This event message documentation provides operators with greater flexibility by allowing them to print out event message documentation as needed at their location or to electronically manipulate the data to suit their needs. The event message documentation provides the message number, severity, and text for each event message generated by the Device Handler process. It also provides the cause of each message and the recommended action to take.

Event message documentation specific to the NCR 5XXX Device Handler process can be found in the \$vol.baxxlogm.at5070 file where *xx* is the current BASE24 release (for example, 60 for BASE24 Release 6.0, regardless of the version). Each file on the baxxlogm subvolume contains event messages for a specific BASE24 module. The files are organized according to product module ID (PMID). One PMID exists for each BASE24 module.

Operators also can view the contents of ATM status messages received by the Device Handler process in their native format. The ATM-DH-DISPLAY-NATIVE-STATUS-MSG param in the LCONF indicates whether the Device Handler process sends this information to the EMS system so it can be reviewed.

For a complete explanation of event messages and for information on how to access the documentation for event messages generated by the Device Handler process, refer to the ***BASE24 Event Message Reference Manual***.

4: Configuration File Maintenance

The NCR 5XXX Configuration File (CNFG5070), which is resident in the BASE24-atm system, can be accessed and downloaded to the ATM from the DCT using screen 45.

With access to the CNFG5070, you can make changes to ATM screen displays, state table entries, FIT table data, configuration parameters, and timers. You then download these changes to the ATM using the DCT.

This section details the BASE24-atm screens you can use to make changes to the CNFG5070 from the DCT.

Refer to the *NCR NDC Programmer's Overview Manual* for a general overview of the configuration file and the state tables, FIT tables, and parameters. Specific information on the configuration file is in the *NCR NDC Programmer's Reference Manual* for NCR 50XX-series terminals, and the *NCR NDC+ Programmer's Reference Manual* for NCR 56XX-series terminals. Information on downloading the CNFG5070 to NCR 5XXX-series ATMs can be found in the *BASE24 Device Control Manual*.

NCR 5XXX Configuration File Menu Function Keys

The function keys specific to the NCR 5XXX Configuration File Menu are described below. A separate set of function keys for use with the NCR 5XXX Commands screen can be found in the *BASE24 Device Control Manual*.

The first column of information below shows the BASE24 keys. The second column describes the function that can be accomplished with these function keys.

Key	Description
F1	Select — Checks the selected field for errors and, if the selection is valid, executes the selected command.
F7	Go to New Page — Displays a different screen in the same file in which you are working. The screen to be displayed must be identified in the NEW PAGE field at the bottom of the screen.
F10	Print Screen — Sends the screen currently being displayed to the spooler location specified on the Logon screen. The Logon screen is explained in the <i>BASE24 User Security Manual</i> .
F16	Exit — Exits the BASE24 system or the file currently being accessed. This function key also allows you to move between logical networks and files. This functionality is explained in the <i>BASE24 User Security Manual</i> .
Shift-F16	Logoff — Logs you off of the BASE24 system. When this function is used while you are accessing BASE24 screens, a blank Logon screen is displayed. If the Logon screen is displayed when this function is used, the Logon screen continues to be displayed or a TACL prompt is displayed, depending upon how the terminal is set up.

NCR 5XXX Configuration File Menu

The NCR 5XXX Configuration File Menu is accessed using the NCR 5XXX Commands screen. To do so, you select the ACCESS CONFIGURATION FILE option on the NCR 5XXX Commands screen of the DCT.

You select files maintenance functions from the NCR 5XXX Configuration File Menu by placing the cursor next to the function and pressing the **F1** key.

Following the selection of a function, a work screen for that particular function is displayed. Each of these work screens is described on the following pages.

The NCR 5XXX Configuration File Menu is used to select the configuration file name and group number to be accessed. The default file name of CNFG5070 and default group number of 0 are initially displayed on the screen. You can change these as required. The NCR 5XXX Configuration File Menu is shown below, followed by field descriptions, a map of the screens that you can access from this menu, and function key definitions.

```

BASE24-DCT   NCR 5XXX CONFIG FILE   LLLL           YY/MM/DD  HH:MM  01 OF 01

      CONFIGURATION FILE NAME

      GROUP NUMBER      0

      -  ENHANCED CONFIGURATION PARAMETERS
      -  STATE
      -  SCREEN
      -  FINANCIAL INSTITUTION TABLES
      -  SUPERVISOR CARD DATA
      -  PRINTER CONTROL SCREENS (Kxx SCREENS)
      -  CONFIGURATION IDENTITY NUMBER
      -  HOPPER MAPPING TABLE
      -  EMV CONFIGURATION MENU
      -  WINCOR EMV AID TABLE

      -  PRINT

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
F1-SELECT F7-GOTO F10-PRINT F12-HELP F16-EXIT SF16-LOGOFF

```

CONFIGURATION FILE NAME — The fully qualified file name of the CNFG5070. This name must be entered using the Hewlett-Packard (HP) NonStop file naming convention of *\$volume.subvolume.filename*.

GROUP NUMBER — The number used to identify a particular load group (i.e., a collection of states, screens, tables, and parameters) that is downloaded to the terminal. Each group provides certain functionality. The group number for each terminal is assigned in the TERMINAL GROUP field on screen 7 of the BASE24-atm Terminal Data files (ATD). Groups 106 through 113 and 706 through 810 are described in more detail in the *BASE24-atm NCR 5XXX SSB Manual*. Groups 300 through 303 and 311 through 313 are described in [section 5](#) of this manual. The usage of groups 305, 306, 310, 315, 354, 355, 364, 365, 501, and 502 are described in appendix D of this manual. Valid values are as follows:

- 0 = The basic set of states and screens that control all ATM transactions; this group is always downloaded to the terminal.
- 1 = Fast Cash transactions.
- 2 = Statement Print transactions. The statement is printed on the 80-column receipt printer—not the 40-column journal printer.
- 3 = Both Fast Cash, Statement Print, and Cash Check transactions.
- 4 = Screens for terminals that require the cards to be passed through a card reader without being retained.
- 5 = Entry of decimal amounts on all transactions.
- 12 = SSB peak load group example.
- 17 = Supports card then cash, on group 0 transaction set only.
- 18 = Supports parallel dispense (eject card and cash at the same time), on group 0 transaction set only.
- 19 = Supports card, then cash on Fast Cash transactions. Must be loaded on top of a Fast Cash load group (for example, group 1).
- 20 = ANSI PIN block support group.
- 21 = Supports withdrawals and inquiries on default account. May be loaded on top of load group 0, 1, 17, 18, or 19.
- 24 = Encrypted PIN block support group.
- 100 = Local PIN verification group.
- 106 = SSB Basic check depository transaction.
- 107 = SSB Fast Cash transactions and basic check depository transactions.
- 108 = SSB Statement Print transactions and basic check depository transactions.
- 109 = SSB Fast Cash plus Statement Print transactions and basic check depository transactions.
- 110 = SSB Split deposit transaction group.
- 111 = SSB Super Teller group function.
- 113 = SSB Check cashing card group function.
- 200 = Multiple account selection by qualifier group.
- 206 = Multiple account selection by qualifier and SSB basic check depository transactions. This group must be loaded on top of groups 0, 200, and 106.
- 210 = Multiple account selection by qualifier and SSB split deposit transactions. This group must be loaded on top of groups 0, 200, 110.

- 211 = Multiple account selection by qualifier and the SSB Super Teller functions.
- 250 = Multiple account selection by qualifier and noncurrency dispense transactions. This group must be loaded on top of groups 0, 500, and 200.
- 256 = Multiple account selection by qualifier with noncurrency dispense and SSB check depository transactions.
- 261 = Multiple account selection by qualifier with noncurrency dispense and SSB Super Teller functions.
- 300 = The basic group for the optional receipts function. This group is merged with Group 0, described above, as well as the appropriate group below (from group 301 to group 313), as needed.
- 301 = The group for the optional receipts function for when the printer currently is operational and the cardholder selects optional receipts once for all chained transactions, on all approved or approved and denied transactions.
- 302 = The group for the optional receipts function for when the printer currently is inoperable and the cardholder selects optional receipts once for all chained transactions.
- 303 = The group for the optional receipts function for when the printer currently is operational and the cardholder selects optional receipts once for all chained transactions, only if the printer becomes inoperable.
- 304 = The group for the standard surcharge function (i.e., request notification processing, leased line).
- 305 = The group for dial-up ATM noninteractive surcharge processing for withdrawal and envelope deposit transactions.
- 306 = The group for dial-up ATM noninteractive surcharge processing for fast cash transactions.
- 307 = The group for dial-up ATM interactive surcharge processing.
- 308 = The group for dial-up ATM noninteractive surcharge processing (includes support for withdrawal, envelope deposit, SSB check deposit, and deposit with cash back transactions).
- 309 = The group for dial-up ATM noninteractive surcharge processing (includes support for withdrawal, envelope deposit, SSB check deposit, deposit with cash back, and split deposit transactions).
- 310 = The group for surcharging with surcharge interactive response notification processing.
- 311 = The group for the optional receipts function for when the printer currently is operational and the cardholder selects optional receipts before each transaction, on all approved or approved and denied transactions.
- 312 = The group for the optional receipts function for when the printer currently is inoperable and the cardholder selects optional receipts before each transaction.
- 313 = The group for the optional receipts function for when the printer currently is operational and the cardholder selects optional receipts before each transaction, only if the printer becomes inoperable.

- 314 = The group for surcharge processing and logging canceled transactions to the TLF.
- 315 = The group for dial-up noninteractive surcharging and logging canceled transactions to the TLF.
- 317 = The group for dial-up interactive surcharging and logging canceled transactions to the TLF.
- 318 = The group for dial-up noninteractive surcharging and logging canceled transactions to the TLF (includes support for withdrawal, envelope deposit, SSB check deposit, and deposit with cash back transactions).
- 319 = The group for dial-up noninteractive surcharging and logging canceled transactions to the TLF (includes support for withdrawal, envelope deposit, SSB check deposit, and deposit with cash back transactions).
- 320 = The group for Dynamic Currency Conversion support.
- 321 = The group for Dynamic Currency Conversion interactive response notification surcharge processing. This group must be loaded on top of groups 310 and 320.
- 322 = The group for Dynamic Currency Conversion support with dial-up. This group must be loaded on top of group 320.
- 450 = Digital Audio Service (DAS) group.
- 500 = Non-Currency Dispense group.
- 501 = The group for dial-up ATM Non-Currency Dispense processing.
- 502 = The group for dial-up ATM Non-Currency Dispense noninteractive surcharge processing.
- 503 = The group for Non-Currency Dispense with Mobile Top-Up processing.
- 504 = The group for Non-Currency Dispense with Mobile Top-Up processing and logging canceled transactions to the TLF.
- 550 = The group for Non-Currency Dispense with Mobile Top-Up definitions-based processing.
- 600 = EMV Support group.
- 601 = EMV fast cash load group.
- 602 = EMV PIN unblock load group.
- 603 = EMV DASH (Dip And Smart Hardware) card reader support load group.
- 604 = The EMV surcharge load group.
- 605 = The EMV Non-Currency Dispense load group.
- 609 = The EMV group for nonstandard machines (i.e., Wincor).
- 610 = The EMV Dial-Up Device Handler group for surcharge interactive response notification processing.
- 611 = The EMV Dial-Up Device Handler group for noninteractive surcharge processing.
- 612 = The EMV Dial-Up Device Handler group for noninteractive surcharge processing for fast cash transactions.
- 613 = The EMV Dial-Up Device Handler group for processing Non-Currency Dispense transactions.

- 614 = The EMV Dial-Up Device Handler group for processing noninteractive Non-Currency Dispense transactions.
- 620 = The EMV Dynamic Currency Conversion support group. This group must be loaded on top of groups 600 and 320.
- 622 = The EMV group for Dynamic Currency Conversion support with dial-up. This group must be loaded on top of groups 320, 600, and 620.
- 624 = EMV Encrypted PIN Block support load group.
- 630 = Basic Shared EMV support.
- 631 = Shared EMV encrypted PIN change load group.
- 634 = Shared EMV surcharging load group.
- 635 = Shared EMV non-currency dispense load group.
- 706 = NCR Bunch Note Acceptor support load group.
- 710 = NCR SSB Base Check Processing Module (CPM) and Bunch Note Acceptor (BNA) and split deposit load group.
- 711 = NCR SSB Base CPM and BNA Super Teller load group.
- 800 = Shared Bunch Note Acceptor (BNA) cash deposit transaction load group for support of different currencies and different denominations in a single deposit transaction. Must be loaded on top of groups 0 and 106.
- 801 = Shared CPM check processing support load group. Must be loaded on top of groups 0 and 106.
- 802 = Shared CPM Bunch Note Acceptor (BNA) load group for support of both cash deposit and check processing transactions. Must be loaded on top of groups 0 and 106.
- 803 = Shared CPM Optical Character Recognition (OCR) support load group. Must be loaded on top of groups 0 and 106.
- 810 = Shared CPM split deposit load group. Must be loaded on top of groups 0, 106, 110, and 801.
- 901 = The group for the Multiple Currency Dispense add-on feature.

Field Length: 1–4 numeric characters

Required Field: No

Default Value: 0

ENHANCED CONFIGURATION PARAMETERS — Accesses the Enhanced Configuration Parameters screen from which you can change a variety of optional features and timers.

Field Length: Cursor placement only

STATE — Accesses the State Table Data Menu from which you can specify a state screen. Each state typically has a screen number and next state to perform, as well as other parameters. The screens supply sequential instructions that provide the terminal with the information to perform cardholder and administrative transactions.

Field Length: Cursor placement only

SCREEN — Accesses the Screen Data Format screen from which you can customize the screen display information and cardholder keyboard.

Field Length: Cursor placement only

FINANCIAL INSTITUTION TABLES — Accesses the Financial Institution Table screen where you can specify parameters to be used depending on the terminal PIN verification method to be used.

Field Length: Cursor placement only

SUPERVISOR CARD DATA — Accesses the Supervisor Card Data from which you can configure the predefined supervisor transaction data to support the Rear Supervisor/Settlement transaction.

Field Length: Cursor placement only

PRINTER CONTROL SCREENS (Kxx SCREENS) — Accesses the Kxx Screen Data Format screen from which you can customize various printer settings.

Field Length: Cursor placement only

CONFIGURATION IDENTITY NUMBER — Accesses the Configuration Identity Number screen from which you can establish the current configuration version present at the terminal.

Field Length: Cursor placement only

HOPPER MAPPING TABLE — Accesses the Hopper Mapping Table screen from which you can determine the currency denominations and currency types present in each hopper.

Field Length: Cursor placement only

EMV CONFIGURATION MENU — Lists four options from which a screen can be accessed in order to configure various tables. Refer to the ***BASE24-atm EMV Support Manual*** for more information about this screen.

Field Length: Cursor placement only

WINCOR EMV AID TABLE — Accesses the Wincor EMV AID Table screen from which you can identify the applications that can be processed with the NDC/Diebold 91x message format. Refer to the ***BASE24-atm EMV Support Manual*** for more information about this screen.

Field Length: Cursor placement only

PRINT — Accesses the Print Menu from which you can print all or part of the configuration file load information in the CNFG5070.

Field Length: Cursor placement only

Configuration File Function Keys

The function keys specific to the Enhanced Configuration Parameters, State, Screen Data Format, Financial Institution Table, Supervisor Card Data, Configuration Identity Number, and Printer Control screens are described below.

The first column of information below shows the BASE24 keys. The second column describes the functions that can be accomplished with these function keys.

Key	Description
F1	Validate Data — Checks the data that has been entered on the screen for errors, and enters the data in the file.
F2	Read Record — Reads data in the file.
F3	Add Record — Adds data to the file.
F4	Delete Record — Deletes data from the file.
F5	Update Record — Changes data already in a file.
F6	Read Next Record — Reads the next record in the file.
F7	Go to New Page — Displays a different screen in the same file in which you are working. The screen to be displayed must be identified in the NEW PAGE field at the bottom of the screen.
F8	Clear Screen — Clears any value in the record in which you are working to spaces or replaces them with default values. This key impacts all screens associated with the file currently being accessed. This includes the current screen, any previous screens, and any remaining screens. Once this function has been used, the update function cannot be used until after the read function has been completed.
F9	Next Screen — Displays the next screen of the file in which you are working. If you are on the last screen of the file, this function key takes you to the first screen of the file.
F10	Print Screen — Sends the screen currently being displayed to the spooler location specified on the Logon screen. The Logon screen is explained in the <i>BASE24 User Security Manual</i> .

Key	Description
F11	Previous Screen — Displays the previous screen of the file in which you are working. If you are on the first screen of the file, this function key takes you to the last screen of the file.
F12	Display Help Screen — Displays the Help screen. The Help screen displayed depends on the screen currently being viewed. The Help screen contains information about BASE24 function keys or menu options.
F13	Change Current Logical Net — Changes the logical network to the one specified in the NEW LOGICAL NETWORK ID field.
F16	Exit or Go to File — This key has several purposes. It is used to exit the BASE24 screens or the file or table currently being accessed. It also allows the user to move between logical networks and files or tables. This functionality is explained in the <i>BASE24 User Security Manual</i> .
Shift-F16	Log Off BASE24 — Logs the user off the BASE24 screens. When this function is used while a user is accessing a BASE24 screen, a blank Logon screen is displayed. If the Logon screen is displayed when this function is used, the Logon screen continues to be displayed or a TACL prompt is displayed, depending upon how the terminal is set up. Note: A hyphen connecting two keys indicates the keys are pressed simultaneously (for example, Shift-F16 indicates the Shift and F16 keys are pressed simultaneously).

Enhanced Configuration Parameters Screens

NCR terminals use internal “Direct Connect” ATM software, which facilitates the use of the Enhanced Configuration Parameters screens. The Enhanced Configuration Parameters screens are used to change a variety of optional features and timers to suit the particular requirements of the financial institution. For more detailed information on enhanced configuration parameters load information, refer to the *NCR NDC Programmer’s Reference Manual* for NCR 50XX-series terminals, and the *NCR NDC+ Programmer’s Reference Manual* for NCR 56XX-series terminals.

Enhanced Configuration Parameters Screen 1

Enhanced Configuration Parameters screen 1 contains information about optional features for NCR 5XXX terminals. This screen is displayed when you place the cursor to the left of the Enhanced Configuration Parameters option and press the **F1** key on the NCR 5XXX Configuration File Menu.

Enhanced Configuration Parameters screen 1 is shown below, followed by descriptions of its fields.

```

BASE24-DCT   ENHANCED CONFIG PARAM   LLLL           YY/MM/DD hh:mm  01 OF 02

GROUP:       0

OPTIONS:
CAMERA CONTROL          1 (ON)
READY/SUPPLY            1 (READY, AMT 8 DGTS)
AUTO VOICE              1 (ON)
ROLL WIDTH              40
TRACK 1 FORMAT          1 (ISO, BCKWD SRCH)
SPEC CMD RJCT           0 (NO)
TRAN STATUS             1 (YES)
J-PRNTR BKUP TIM LMT    999 (NOT USED)
ENV DISP FLT            999 (NOT USED)
ENHANCED TI             999 (NOT USED)
KEYBOARD AUDIO ECHO     0 (NO)
RPT EJ DUAL-MD ERRS     0 (NONE)
INCLUDE PAN             0 (YES)
INCLUDE EMV ERR INFO    999 (NOT USED)
ACQUIRER ID            0
DATE FORMAT             1 (MMDDYY)
LEFT PRINT COL          1
J-PRNT BNA CNTS         999 (NOT USED)
BNA MSG                 999 (NOT USED)
J-PRNTR BKUP PRNT LMT  999 (NOT USED)
MEI FLASH RATE          999 (NOT USED)
REPORT DAS ERRORS       0 (NO)
EJ BKUP MODE            0 (STANDARD)
CARDS ACCEPTED          0 (MAG ONLY)
EMV ACTIONED            0 (YES)

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:    NEW LOGICAL NETWORK ID:
F9-NEXT PAGE      F11-PREVIOUS PAGE     F12-HELP

```

GROUP — The number used to identify a particular load group (i.e., a collection of states, screens, tables, and parameters) that is downloaded to the ATM. Each group provides certain functionality (e.g., group 0 is the basic group, group 1 provides Fast Cash support, etc.). The group number of each ATM is assigned in the **TERMINAL GROUP** field on ATD screen 7. Refer to the topic [“NCR 5XXX Configuration File Menu”](#) for a list of valid values for this field.

Field Length: 1–4 numeric characters
 Required Field: No
 Default Value: 0

ACQUIRER ID — The eight-digit acquirer identifier. This field is reserved for future use.

Field Length: 8 numeric characters
Required Field: No
Default Value: 00000000

OPTIONS

The following fields describe optional features available on NCR 5XXX terminals.

CAMERA CONTROL — Activates or deactivates the automatic camera. If the automatic camera is on, a record of the person performing the transaction is made after a transaction reply has been received by the ATM. This field can be zero filled to the left. Valid values are as follows:

- 1 = Automatic camera on
- 2 = Automatic camera off

Field Length: 1–3 numeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: 1

READY/SUPPLY — Defines the active supply and ready status options for the ATM. This field can be zero filled to the left. Valid values are as follows:

- 1 = Indicates that a separate READY (B) status is sent in response to a transaction reply command. When this value is specified, the BASE24 system expects a completion message after a transaction, and the BASE24 product specifies the next state for the terminal. This option supports amount fields up to a length of eight digits.
- 3 = Specifies the option in which the BASE24 system expects a completion message after a transaction and the terminal proceeds to the next state automatically. This option supports amount fields up to a length of eight digits.
- 9 = Specifies the same processing as value 1, except this value supports amount fields up to a length of 12 digits.
- 11 = Specifies the same processing as value 3, except this value supports amount fields up to a length of 12 digits.

Field Length: 1–3 numeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: 1

AUTO VOICE — Controls the automatic voice responder. This field can be zero filled to the left. Valid values are as follows:

- 1 = Automatic voice on
- 2 = Automatic voice off

Field Length: 1–3 numeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: 1

DATE FORMAT — Controls the date format for supervisor/journal messages and camera messages. This field can be zero filled to the left. Valid formats are as follows:

- 1 = MMDDYY
- 2 = DDMMYY
- 3 = YYMMDD
- 4 = YYDDMM

Field Length: 1–3 numeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: 1

ROLL WIDTH — Defines the number of receipt and journal printer columns used for printing messages from the host. This field can be zero filled to the left. Valid values are 1 through 40.

Note: If you change the default value of this field, you must make sure that the receipt templates in the RCPT do not cause a receipt to exceed the new number of characters per line.

Field Length: 1–3 numeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: 40

LEFT PRINT COL — Defines the left-most column to be used for receipt and journal messages from the host. This field can be zero filled to the left. Valid values are 1 through 40.

Note: If you change the default value of this field, you must make sure that the receipt templates in the RCPT do not cause a receipt to exceed the new number of characters per line.

Field Length: 1–3 numeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: 1

TRACK 1 FORMAT — Defines acceptable formats for displaying the Track 1 name field on the CRT. Track 1 data usually contains the cardholder's name. The option selected determines if the first name/title separator (i.e., a space in ISO format or a full stop in Visa format) appears in the first name field. Options 3 and 4 allow the format, which is either ISO or Visa, to be selected from data on the ATM card. This field can be zero filled to the left. Valid values are as follows:

- 1 = ISO format, backward search
- 2 = Visa format, backward search
- 3 = Auto select, backward search
- 4 = Auto select, forward search

Field Length: 1–3 numeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: 1

SPEC CMD RJCT — Determines whether the ATM responds with a specific command reject status or with a general command reject status. This field can be zero filled to the left. Valid values are as follows:

- 0 = Do not transmit specific command reject status responses; transmit general command reject status responses only.
- 1 = Transmit specific command reject status responses.

Note: ACI recommends the transmission of specific command reject status responses because they provide more detailed information about the cause of the logged rejection.

Field Length: 1–3 numeric characters, followed by a system-protected text description
Required Field: No
Default Value: 0

J-PRNT BNA CNTS — Indicates whether Bunch Note Acceptor vaulted note counts are automatically printed to the journal printer. Valid values are as follows:

- 000 = Do not automatically print vaulted note counts to the journal printer.
- 001 = Automatically print vaulted note counts to the journal printer.
- 999 = Do not use this option.

Field Length: 1–3 numeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: 999

TRAN STATUS — Controls whether the last transaction status data is appended to the transaction request message from the terminal to the Device Handler process. If this option is enabled, the transaction request message contains the last transaction status code of the previous transaction performed by the terminal, appended to the request message. This field can be zero filled to the left. Valid values are as follows:

- 0 = Do not append the information
- 1 = Append the information

The last transaction status code is a one-byte code that identifies the last known status for the previous transaction. Depending on the status, the type of transaction, and the ATM model, the BASE24-atm product can generate the

following reversals: a suspect reversal, a full reversal, or a partial reversal. Implementing this option can reduce the number of suspect reversals that are generated.

The table below identifies the last transaction status codes that can be appended to the last message and describes the action that the BASE24 product takes based on each code.

Code	Last Transaction Status	BASE24 Action
0	None sent	Generates a suspect reversal.
1	Good termination sent	Generates no reversal.
2	Error status sent	Generates a full or partial reversal. If the previous transaction was a withdrawal, the BASE24 product generates a full or partial reversal depending on the amount dispensed. Deposit, payment enclosed, and message enclosed transactions are fully reversed.
3	Transaction request rejected	Generates a full reversal for all transactions except statement print or balance inquiry transactions.

Field Length: 1–3 numeric characters, followed by a system-protected text description

Required Field: No

Default Value: 1

BNA MSG — Indicates whether Bunch Note Acceptor transaction counts are included in the transaction request message sent to the BASE24 system. Valid values are as follows:

000 = Do not include transaction counts; accept up to 90 notes; do not retract notes.

001 = Include transaction counts; accept up to 90 notes; do not retract notes.

002 = Do not include transaction counts; accept more than 90 notes; do not retract notes.

003 = Include transaction counts; accept more than 90 notes; do not retract notes.

004 = Do not include transaction counts; accept up to 90 notes; retract notes.

- 005 = Include transaction counts; accept up to 90 notes; retract notes.
- 006 = Do not include transaction counts; accept more than 90 notes; retract notes left by customer.
- 007 = Include transaction counts; accept more than 90 notes; retract notes left by customer.

Field Length: 1–3 numeric characters, followed by a system-protected text description

Required Field: Yes

Default Value: 000

J-PRNTR BKUP TIM LMT — Defines the maximum number of hours that journal printer backup is allowed before the ATM stops writing to the electronic journal. Valid values are 0 through 255 and 999. A value of 999 indicates that this option is not used.

Field Length: 1–3 numeric characters, followed by a system-protected text description

Required Field: Yes

Default Value: 999

J-PRNTR BKUP PRNT LMT — Defines the maximum number of print operations, in hundreds, that can be buffered while the journal printer is not operational. Valid values are 0 through 225 and 999. A value of 999 indicates that this option is not used.

Field Length: 1–3 numeric characters, followed by a system-protected text description

Required Field: Yes

Default Value: 999

ENV DISP FLT — Determines whether envelope dispenser status messages are sent, remote status indicators are set, and the remote relay is activated. Valid values are as follows:

- 000 = Do not send envelope dispenser status messages, set the remote status indicators, or activate the remote relay.
- 001 = Send envelope dispenser status messages, but do not set the remote status indicators or activate the remote relay.

- 002 = Set the remote status indicators and activate the remote relay, but do not send envelope dispenser status messages.
- 003 = Send envelope dispenser status messages, set the remote status indicators, and activate the remote relay.
- 999 = Do not use this option.

Field Length: 1–3 numeric characters, followed by a system-protected text description

Required Field: Yes

Default Value: 999

ENHANCED TI — Indicates whether the Enhanced TI/Sensor Status Unsolicited message is sent from the terminal when tampering is suspected on devices not supported in the existing TI/Sensor Status Unsolicited message. Valid values are as follows:

- 000 = Do not send the Enhanced TI/Sensor Status Unsolicited message. Continue to send the existing TI/Sensor Status Unsolicited message when appropriate.
- 001 = Send the Enhanced TI/Sensor Status Unsolicited message when appropriate. Do not send the existing TI/Sensor Status Unsolicited message.
- 999 = Do not use this option.

Field Length: 1–3 numeric characters, followed by a system-protected text description

Required Field: Yes

Default Value: 999

MEI FLASH RATE — Defines the flash rate for the media entry indicators. This field can be zero filled to the left. Valid values are as follows:

- 000 = 4.0 hertz
- 001 = 2.0 hertz
- 002 = 1.0 hertz
- 003 = 0.5 hertz
- 004 = 0.25 hertz
- 005 = Continuously on
- 999 = Do not use this option

Field Length: 1–3 numeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: 999

KEYBOARD AUDIO ECHO — Indicates whether the keyboard audio echo feature is to be used. Valid values are as follows:

000 = No, do not use the keyboard audio echo feature.
001 = Yes, use the keyboard audio echo feature.
999 = Do not use this option.

Field Length: 3 numeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: 0

REPORT DAS ERRORS — Indicates whether unsolicited Digital Audio Service (DAS) errors are sent to the Device Handler process. Valid values are as follows:

000 = No, do not send unsolicited DAS errors.
001 = Yes, send unsolicited DAS errors.
999 = Do not use this option.

Field Length: 3 numeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: 999

RPT EJ DUAL-MD ERRS — Indicates whether electronic journal dual mode errors are logged. Valid values are as follows:

000 = Do not report electronic journal dual mode errors
001 = Report electronic journal errors only
002 = Report both electronic journal and hard-copy back-up errors
999 = Do not use this option

Field Length: 3 numeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: 999

EJ BKUP MODE — Indicates the mode of operation used at the ATM to support the logging, recalling and copying of multiple electronic journal back-up files held on the system disk. Standard mode allows you to recall only the most recent back-up files. Enhanced mode allows you to select and recall ten back-up files. Valid values are as follows:

000 = Standard mode
001 = Enhanced mode
999 = Do not use this option

Field Length: 3 numeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: 0

INCLUDE PAN — Indicates whether the cardholder's primary account number (PAN) is sent to the security camera (Digital Camera System cameras only). Valid values are as follows:

000 = Yes, include the PAN.
001 = No, do not include the PAN.
999 = Do not use this option.

Field Length: 3 numeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: 0

CARDS ACCEPTED — Indicates the type of cards accepted at the ATM. Valid values are as follows:

0 = Accept cards with only a magnetic stripe.

- 1 = Accept cards with both a magnetic stripe and an integrated circuit chip.
- 2 = Accept cards with an integrated circuit chip only when the Card Read state is not 0 (i.e., accept only when a magnetic stripe has already started the transaction).
- 999 = This option is not used.

Field Length: 3 numeric characters, followed by a system-protected text description

Required Field: No

Default Value: 0

INCLUDE EMV ERR INFO — Indicates whether EMV handling commands are *actioned* or *not actioned*. For more information on *actioned* and *not actioned*, see the *NCR NDC+ Programmer's Reference Manual* for NCR 5XXX terminals.

Valid values are as follows:

- 0 = Do not include EMV error information. EMV command and response data are not to be included in EMV application error messages, both solicited and unsolicited.
- 1 = Include EMV error information. EMV command and response data are included in EMV application error messages, both solicited and unsolicited.
- 999 = Do not use this option.

Field Length: 3 numeric characters, followed by a system-protected text description

Required Field: No

Default Value: 999

EMV ACTIONED — Indicates whether EMV commands are actioned. Valid values are as follows:

- 0 = EMV commands are actioned.
- 1 = EMV commands are not actioned.

Field Length: 3 numeric characters, followed by a system-protected text description

Required Field: Yes

Default Value: 0

Enhanced Configuration Parameters Screen 2

Enhanced Configuration Parameters screen 2 contains a variety of timers used by NCR 5XXX devices. This screen is displayed when you press the **F9** key on Enhanced Configuration Parameters screen 1.

Enhanced Configuration Parameters screen 2 is shown below, followed by descriptions of its fields.

```

BASE24-DCT   ENHANCED CONFIG PARAM   LLLL           YY/MM/DD  hh:mm  02 OF 02

GROUP:       0

TIME-OUTS: (IN SECONDS - EXCEPT WHERE SPECIFIED DIFFERENTLY)
  KEYBOARD ENTRY           30   CARDHOLDER RESPONSE TO TIME-OUT SCREEN  30
  CLOSE STATE SCREEN       30   COMMUNICATIONS RESPONSE           30
  ENVELOPE INSERTION       30   CASH RETRACT                     30
  POLL/SELECT              30   PRESENT                          30
  NIGHT SAFE DEPOSITORY    30   CARD REMOVAL                     30
  ADDITIONAL PRESENT       30   STATEMENT PRESENT                30
  DOOR ACCESS              30   FORM INSERTION                   30
  VANDAL GUARD CLOSE       30   SOH CYCLICAL DISPLAY / FAULT DISPLAY 999
  EJECT DOCUMENT           30   SOH NEXT/PREVIOUS                999
  CAPTURE DOCUMENT         999   REAR SETTLEMENT                  999
  CAPTURE STATEMENT        999   CAM DLY PRD (1/10th Secs)        999
  PICT FILE RETENT (Days)  999   CASH ACCEPTANCE                  999
  CASH REJECTION           999   CARD REMOVAL FROM DASH CARD READER 999

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
  F9-NEXT PAGE      F11-PREVIOUS PAGE      F12-HELP

```

Time-Outs

The following configuration options set timers that are used by the ATM to control transactions.

KEYBOARD ENTRY — The number of seconds the cardholder has to respond, using the terminal keyboard, to a screen prompting him or her for information before the terminal enters the timeout state (timer 00). Valid values are 1 through 204 seconds.

Field Length: 1–3 numeric characters
 Required Field: Yes
 Default Value: 30

CARDHOLDER RESPONSE TO TIME-OUT SCREEN — The number of additional seconds the cardholder has for keyboard entry, form insertion, or envelope insertion when the cardholder responds to the timeout state screen by requesting more time (timer 01). Valid values are 1 through 204 seconds.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

CLOSE STATE SCREEN — The number of seconds the ATM remains in the Close state (state J) after a card has been ejected (timer 02). At the end of the time interval, the terminal proceeds to the next state number provided the card has been taken or captured. If the card has not been removed, exiting from the Close state is delayed until the Card Removal timer expires or the card is taken. Valid values are 1 through 204 seconds.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

COMMUNICATIONS RESPONSE — The maximum number of seconds within which the BASE24 system must return a response to an ATM request (timer 03). If the timer expires before the ATM receives a response from the BASE24 system, the terminal exits the transaction request state and advances to the timeout state specified in the Transaction Request state table. Valid values are 1 through 204 seconds.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

ENVELOPE INSERTION — The number of seconds during which the cardholder must insert a deposit envelope in the depository slot before the terminal enters the timeout state (timer 04). The keyboard beeper sounds continuously while this timer is active. Valid values are 1 through 204 seconds.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

CASH RETRACT — The number of seconds cash is to remain accessible to the cardholder before a retract operation is attempted (timer 05). This parameter is only valid for models with cash presenters. Valid values are 1 through 204 seconds.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

POLL/SELECT — The number of seconds the ATM allows between poll or select sequences from BASE24 (timer 06). If the ATM has not been polled or selected within the interval of time specified, the ATM proceeds to offline mode (out-of-service to cardholders). Valid values are 1 through 204 seconds.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

PRESENT — The number of seconds the Withdrawal Completion screen is displayed (timer 07). When the timer expires, the ATM enters the next state specified in the Transaction Reply state table. Valid values are 1 through 204 seconds.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

NIGHT SAFE DEPOSITORY — The number of seconds the night safe depository door is unlocked to accept a night safe deposit (timer 08). If the bag-detection mechanism detects no bag and the timer expires, the terminal enters the timeout state. If the bag-detection mechanism is not functioning, the terminal does not enter the timeout state, which gives the cardholder more time to complete a deposit. Valid values are 1 through 204 seconds.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

CARD REMOVAL — The number of seconds the cardholder has to remove the card before the card is captured (timer 09). Valid values are 1 through 204 seconds.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

ADDITIONAL PRESENT — The number of seconds to add to the present time interval to achieve Diebold compatibility (timer 10). NCR 5XXX-series ATMs can be configured to provide some of the same functions as Diebold ATMs. This field allows additional seconds to be added to the timer to emulate Diebold ATMs. Valid values are 1 through 204 seconds.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

STATEMENT PRESENT — The number of seconds the cardholder has to remove the printed statement before the ATM enters the next state specified in the transaction reply (timer 96). Valid values are 1 through 204 seconds.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

DOOR ACCESS — The number of seconds the vestibule door is unlocked when the cardholder's card is authorized for ATM use (timer 97). Valid values are 1 through 204 seconds. Values greater than 120 are set to 120.

Note: Not all ATMs offer the vestibule door access feature.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

FORM INSERTION — The number of seconds allowed for the cardholder to insert a form into the validation printer after the terminal screen prompt (timer 98). If the cardholder does not respond within this time, the terminal enters the timeout state. Valid values are 1 through 204 seconds.

Note: This timer is used with NDC release 8.0 and previous releases.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

VANDAL GUARD CLOSE — The number of seconds the Vandal Guard remains unlocked without a card being inserted (timer 99). If a value of 0 is specified, the Vandal Guard remains unlocked at all times while the terminal is in service. Valid values are 0 through 204 seconds.

Note: This timer is used with NDC release 8.0 and previous releases.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

SOH CYCLICAL DISPLAY / FAULT DISPLAY — The number of seconds each state of health or terminal management (SOH/TM) or fault display entry is displayed without a next or previous entry from the cardholder at the terminal (timer 92). Valid values are as follows:

1–998 = State of health or terminal management (SOH/TM) cyclical display or fault display timer (in seconds).
999 = Do not download the state of health or terminal management (SOH/TM) cyclical display or fault display timer.

Note: This timer is used by NCR 56XX-series terminals only. It should not be downloaded to NCR 50XX-series ATMs.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 999

EJECT DOCUMENT — The number of seconds the document is presented to the cardholder to be taken before it is reaccepted (timer 94). Valid values are 1 through 204. Values under 10 are set to a value of 10, and values greater than 120 are set to a value of 120.

Note: This timer is used only with NCR 56XX-series terminals with a Document Processing Module (DPM) installed that are using the BASE24-atm self-service banking (SSB) add-on product.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 30

SOH NEXT/PREVIOUS — A multiplier for the value in the SOH CYCLICAL DISPLAY / FAULT DISPLAY field when the cardholder at the terminal makes a next or previous entry (timer 93). For example, a state of health or terminal management (SOH/TM) entry appears for 30 seconds if the SOH CYCLICAL DISPLAY / FAULT DISPLAY field contains a value of 5 and the SOH NEXT/PREVIOUS field contains a value of 6. Valid values are as follows:

- 0 = The next or previous selection is disabled.
- 1–998 = State of health or terminal management (SOH/TM) next or previous multiplier.
- 999 = Do not download the state of health or terminal management (SOH/TM) multiplier.

Note: Timer 93 is only used with NCR 56XX-series terminals running on NDC+ firmware for the state of health or terminal management (SOH/TM) entry. It should not be downloaded to NCR 50XX-series terminals. Also, this timer is not used with terminals running on Aprta Advance firmware and is ignored whenever timer 92 functions as the fault display.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 999

CAPTURE DOCUMENT — The number of seconds the DPM document captured screen is displayed to the cardholder after a document has been captured by the terminal (timer 87). Valid values are 1 through 255 and 999. A value of 999 indicates that this option is not used.

Note: This timer is used only with NCR 56XX-series terminals with a Document Processing Module (DPM) installed that are using the BASE24-atm self-service banking (SSB) add-on product.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 999

REAR SETTLEMENT — The number of seconds (times 10) that the rear keyboard is activated for operator input (timer 86). Valid values are 0 through 65, and 999. A value of 0 indicates an infinite wait for operator input. Values greater than 65 are set to 65. A value of 999 indicates that this option is not used.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 999

CAPTURE STATEMENT — The number of seconds a statement is presented to the cardholder. After this time interval, the statement is captured (timer 95). Valid values are 0 through 255 and 999. A value of 0 is set to 30. A value of 999 indicates that this option is not used.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 999

CAM DLY PRD — The number of tenths of seconds the camera should wait between recording digital images (timer 82). Valid values are 001 through 050 and 999. A value of 999 indicates that this option is not used.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 999

PICT FILE RETENT — The number of days the terminal should store digital images (timer 83). Valid values are 000 through 999. A value of 000 indicates that the images should not be deleted. A value of 999 indicates that this option is not used.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 999

CASH ACCEPTANCE — The number of seconds the customer has to enter notes into the Bunch Note Acceptor (timer 77). The keyboard beeps while this timer is active. Valid values are 010 through 204. A value of 999 indicates that this option is not used.

Note: This timer is used with the NCR Bunch Note Acceptor enhancement only.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 999

CASH REJECTION — The number of seconds the customer has to remove notes from the ATM before the transaction is completed (timer 78). Valid values are 010 through 204. A value of 999 indicates that this option is not used.

Note: This timer is used with the NCR Bunch Note Acceptor enhancement only.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 999

CARD REMOVAL FROM DASH CARD READER — The number of seconds the customer has to remove their ATM card from the DASH (Dip and Smart Hardware) card reader (timer 72). Valid values are 1 through 255. A value of 999 indicates that this option is not used.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 999

State Table Data Screens

State tables are a number of sequential instructions that provide the ATM with the information to perform financial and administrative transactions. The information in the state tables is arranged in a hierarchical manner using state types and groups. A state type is one action or condition of the ATM, such as waiting for the cardholder to first insert a card. A group is a set of state types and screens that provide a certain functionality, such as a fast cash withdrawal.

When a terminal is in service and a transaction is requested, the execution procedure accesses state tables beginning at state number 0 and either state type A or state type T. As each new state is entered, a corresponding screen is displayed for the cardholder at the ATM. The screen identifies the action required by the cardholder before the transaction continues. When the cardholder responds by pressing one or more keys, the transaction flow continues to the next state.

The BASE24-atm product provides a number of screens from which you can modify the NCR state table data. You can use these screens to access options in 41 states.

Note: Most of the values on the state table screens are 1–3 numeric characters. The leading zeros are optional.

Self-Service Banking States

The BASE24-atm self-service banking (SSB) add-on product is available for the NCR Device Handler process. Of the 41 state types in the NCR state table, the following can be used only with the SSB product installed and are documented in the ***BASE24-atm NCR 5XXX SSB Manual***.

- P for DPM CAV
- Q for DPM ACCEPT DOC (plus three Global Extension states)
- > for the Cash Accept state
- w for the Check Accept state
- r for Shared BNA
- q for Shared Check In

Several state table screens have individual fields or extension states that are applicable only to the SSB product. These fields and extension states are described in both this manual and the ***BASE24-atm NCR 5XXX SSB Manual***.

EMV States

The BASE24-atm EMV product is available for the NCR Device Handler process. The following state types can be used only with the EMV product installed and are documented in the ***BASE24-atm EMV Support Manual***:

- k for Smart FIT Check state
- + for Begin EMV Initialization state
- , for Complete EMV Initialization state
- ; for EMV Reinitialize state
- - for Automatic Language Selection state
- ? for Set EMV Transaction Data state
- . for Begin EMV Application Selection and Initialization state
- / for the Complete EMV Application Selection and Initialization state
- e for the Wincor EMV Chip Card state

State Table Data Menu

You access state screens from the State Table Data Menu. This screen is displayed when you place the cursor to the left of the STATE option and press the **F1** key on the NCR 5XXX Configuration File Menu. State screens are accessed by placing the appropriate screen number in the NEW PAGE field and pressing the **F7** key. You can return to this screen from any of the state screens by entering 1 in the NEW PAGE field and pressing the **F7** key.

The State Table Data Menu is shown below, followed by descriptions of its fields. The individual state screens are documented on the following pages. Refer to the *BASE24-atm NCR 5XXX SSB Manual* for descriptions of the Document Processing Module (DPM) Courtesy Amount Verification (CAV) State screen, DPM Accept Document State screen, Cash Accept State screen, and the Check Accept State screen. Refer to the *BASE24-atm EMV Support Manual* for descriptions of the Smart FIT Check State screen, Begin Initialization State screen, Complete Initialization State screen, EMV Reinitialize State screen, Automatic Language Select screen, Set EMV Transaction Data State screen, Begin EMV Application Selection and Initialization State screen, Complete EMV Application Selection and Initialization State screen, and the Wincor EMV Chip Card screen.

```

BASE24-DCT   STATE TABLE DATA           LLLL       YY/MM/DD  HH:MM  01 OF 42

      GROUP      0      STATE NUMBER      0

2:42 (A-CARD READ)           3:42 (B-PIN ENTRY)           4:42 (C-UNLOCK ENV DISP)
5:42 (D-PRESET OP-CODE)     6:42 (E-4 FDK SELECT)       7:42 (F-AMOUNT ENTRY)
8:42 (G-AMOUNT CHECK)       9:42 (H-4 FDK INFO)        10:42 (I-TRAN REQUEST)
11:42 (J-CLOSE)             12:42 (K-FIT SWITCH)        13:42 (L-CARD WRITE)
14:42 (M-ENH PIN ENTRY)     15:42 (N-CAMERA CONTROL)    16:42 (P-DPM CAV)
17:42 (Q-DPM ACCEPT DOC)    18:42 (R-ENH AMT ENTRY)     19:42 (S-LANG SWITCH)
20:42 (T-CRD READ PIN INIT) 21:42 (V-LANG FROM CARD)    22:42 (W-FDK SWITCH)
23:42 (X-8 FDK INFO)        24:42 (Y-8 FDK SELECT)     25:42 (Z-GLOBAL EXT)
26:42 (ENH FIT SWITCH)      27:42 (i-AUDIO CONTROL)    28:42 (k-SMART FIT CHECK)
29:42 (+-BEGIN EMV INIT)    30:42 (,-COMPL EMV INIT)    31:42 (;-EMV RE-INIT)
32:42 (--AUTO LANG SELCT)   33:42 (?-SET EMV TXN DATA) 34:42 (.-BEGIN EMV SELCT)
35:42 (/ -COMPL EMV SELCT)  36:42 (b-CUST SELCT PIN)    37:42 (>-CASH ACCEPT)
38:42 (w-CHECK ACCEPT)      39:42 (e-WINCOR EMV CARD)   40:42 (m-PIN & LANG SLCT)
41:42 (r-SHARED BNA)        42:42 (q-SHARED CHECK IN)

```

TYPE THE NUMBER OF THE DESIRED FUNCTION IN NEW PAGE FIELD AND KEY F7

```

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP

```

GROUP — The number used to identify a particular load group (i.e., a collection of states, screens, tables, and parameters) that is downloaded to the ATM. Each group provides certain functionality (e.g., group 0 is the basic group, group 1

provides Fast Cash support, etc.). The group number of each ATM is assigned in the TERMINAL GROUP field on ATD screen 7. Refer to the topic [“NCR 5XXX Configuration File Menu”](#) for a list of valid values for this field.

The value in this field was entered in the GROUP NUMBER field on the NCR 5XXX Configuration File Menu. You can modify this entry on this screen and subsequent state screens.

Field Length: 1–4 numeric characters
 Required Field: No
 Default Value: 0

STATE NUMBER — The number of the state you want to edit. You can modify this entry on subsequent state screens. The state number is part of the key to the record. State A or state T must be state number 000. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
 Required Field: No
 Default Value: 0

Menu Choices

Each state screen on the State Table Data Menu is identified by a code, which corresponds to the state type used in the NCR documentation. The following table lists the NCR state type codes, descriptions of the valid states, and the screen page numbers used to access the state tables.

State Type	Description	Page Number
A	Card Read State screen	2
B	PIN Entry State screen	3
C	Unlock Envelope Dispenser State screen	4
D	Preset Operations Code Buffer State screen	5
E	Four FDK Selection State screen	6
F	Amount Entry State screen	7

State Type	Description	Page Number
G	Amount Entry Check State screen	8
H	Four FDK Information Entry State screen	9
I	Transaction Request State screen	10
J	Close State screen	11
K	FIT Switch State screen	12
L	Card Write State screen	13
M	Enhanced PIN Entry State screen	14
N	Camera Control State screen	15
P	Document Processing Module (DPM) Courtesy Amount Verification (CAV) State screen	16
Q	DPM Accept Document State screen	17
R	Enhanced Amount Entry State screen	18
S	Language Code Switch State screen	19
T	Card Read PIN Entry Initiation State screen	20
V	Language From Card State screen	21
W	FDK Switch State screen	22
X	Eight FDK Information Entry State screen	23
Y	Eight FDK Selection State screen	24
Z	Global Extension screen	25
_	Enhanced FIT Switch State screen	26
i	Audio Control State screen	27
k	Smart FIT Check state screen	28
+	Begin EMV Initialization screen	29

State Type	Description	Page Number
,	Complete EMV Initialization screen	30
;	EMV Reinitialize screen	31
-	Automatic Language Selection screen	32
?	Set EMV Transaction Data screen	33
.	Begin EMV Application Selection and Initialization screen	34
/	Complete EMV Application Selection and Initialization screen	35
b	Customer Selectable PIN State screen	36
>	Cash Accept State screen	37
w	Check Accept State screen	38
e	Wincor EMV Chip Card State screen	39
m	PIN & Language Select State screen	40
r	Shared BNA	41
q	Shared Check In	42

Card Read State Screen

The ATM enters the Card Read state automatically when it is initialized into service. The Card Read state screen is accessed from the State Table Data Menu by entering a value of 2 in the NEW PAGE field and pressing the **F7** key.

The Card Read State screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT    STATE TABLE DATA    LLLL    YY/MM/DD  hh:mm  02 OF 42

      GROUP      0      STATE NUMBER  0

                        CARD READ STATE

STATE  SCREEN  GOOD-READ  ERROR  READ  READ  READ
TYPE  NUMBER  NEXT-STATE  SCREEN COND1 COND2 COND3
  A      0      0          0      0      0      0

      CARD-RETAIN  NO-FIT
      FLAG        NEXT-STATE
      0          0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP

```

STATE TYPE — The value A identifies the Card Read state.

Field Length: System protected

SCREEN NUMBER — The number of the display screen that prompts the cardholder to enter a card. This screen is displayed while the terminal is awaiting card entry. Valid values are 0 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

GOOD-READ NEXT-STATE — The number of the state to which the terminal proceeds following a good read of the card. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

ERROR SCREEN — The number of the screen displayed if the read conditions are not satisfied due to a bad card or mechanism error. The card is ejected. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

READ COND1 — A number sequence that specifies which track or tracks on the card are read. If the specified track is read, the state specified in the GOOD-READ NEXT-STATE field is performed. If the track could not be read, the action specified in the READ COND2 field is attempted. This field can be zero filled to the left. Valid values are 1 through 7.

Note: For specific information on what values to enter in the READ COND1 field, refer to the *NCR NDC Programmer's Reference Manual* for NCR 50XX-series terminals, and the *NCR NDC+ Programmer's Reference Manual* for NCR 56XX-series terminals.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

READ COND2 — A number sequence that specifies which track or tracks on the card are read. If the specified track is read, the state specified in the GOOD-READ NEXT-STATE field is performed. If the track could not be read, the action specified in the READ COND3 field is attempted. This field can be zero filled to the left. Valid values are 1 through 7.

Note: For specific information on what values to enter in the READ COND2 field, refer to the *NCR NDC Programmer's Reference Manual* for NCR 50XX-series terminals, and the *NCR NDC+ Programmer's Reference Manual* for NCR 56XX-series terminals.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

READ COND3 — A number sequence that specifies which track or tracks on the card are read. If the specified track is read, the state specified in the GOOD-READ NEXT-STATE field is performed. If the track could not be read, the card is ejected and the screen specified in the ERROR SCREEN field is displayed. This field can be zero filled to the left. Valid values are 1 through 7.

Note: For specific information on what values to enter in the READ COND3 field, refer to the *NCR NDC Programmer's Reference Manual* for NCR 50XX-series terminals, and the *NCR NDC+ Programmer's Reference Manual* for NCR 56XX-series terminals.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

CARD-RETAIN FLAG — Indicates when the ATM is to return the card. This field can be zero filled to the left. Valid values are as follows:

- 0 = Terminal returns the card immediately.
- 1 = Terminal returns the card as specified by a transaction reply command message.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NO-FIT NEXT-STATE — The number of the state to which the terminal proceeds if the financial institution number on a card does not match any FIT. If FITs are not used, this value must be 0. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

PIN Entry State Screen

The PIN Entry state displays a screen to prompt the cardholder to enter their PIN, and also enables the function display keys (FDK) along the side of the screen. The PIN Entry State screen is accessed from the State Table Data Menu by entering a value of 3 in the NEW PAGE field and pressing the **F7** key.

The PIN Entry State screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT    STATE TABLE DATA    LLLL    YY/MM/DD  hh:mm  03 OF 42

      GROUP      0      STATE NUMBER  0

                                PIN ENTRY STATE

STATE          SCREEN  TIMEOUT  CANCEL  GOOD PIN  BAD PIN
TYPE           NUMBER   STATE   STATE   CHECK    CHECK
  B             0         0         0         0         0

                LOCAL PIN    REMOTE PIN    LOCAL PIN
                ERROR        CHECK        MAX TRIES
                  0           0           9

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP

```

STATE TYPE — The value B identifies the PIN Entry state.

Field Length: System protected

SCREEN NUMBER — The number of the display screen that prompts the cardholder to enter the PIN at initial approach. This screen, displayed upon entry into the PIN Entry state, must position the CRT cursor at the start of the PIN entry display area. Valid values are 0 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

TIMEOUT STATE — The number of the state to which the terminal proceeds when the terminal times out before the cardholder enters a PIN. Valid values are 0 through 999. A value of 255 indicates that the timeout function is not required.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CANCEL STATE — The number of the state to which the terminal proceeds when the cardholder presses the **Cancel** key before entering the PIN correctly. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

GOOD PIN CHECK — The number of the state to which the terminal proceeds after verification of a correct PIN entry. A Preset Operation Code Buffer state, which clears the function key buffers before transaction selection, usually follows a correct PIN entry. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

BAD PIN CHECK — The number of the state to which the terminal proceeds when an incorrect PIN is entered after the maximum number of attempts set in the LOCAL PIN MAX TRIES field has been exceeded. The ATM normally enters a Close state (i.e., the session is terminated) if this occurs. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LOCAL PIN ERROR — The number of the display screen that prompts the cardholder to reenter a PIN. This screen is displayed after each incorrect PIN entry attempt; X is displayed for each key that is pressed. After the number of reentry

attempts specified in the LOCAL PIN MAX TRIES field of this screen is exceeded, the ATM enters the state set in the BAD PIN CHECK field. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

REMOTE PIN CHECK — The number of the state to which the terminal proceeds when local PIN verification is not performed. Note that the BASE24 product does not keep track of the number of tries when you choose this option. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LOCAL PIN MAX TRIES — The maximum number of incorrect PIN entries before the ATM enters the state set in the BAD PIN CHECK field. This field can be zero filled to the left. Valid values are 1 through 9.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 9

Unlock Envelope Dispenser State Screen

If the Unlock Envelope Dispenser state is used on a terminal with an envelope presenter, this state causes an envelope to be dispensed to the cardholder. If this state is used on a terminal without an envelope presenter, this state does nothing, and processing proceeds to the next state. The Unlock Envelope Dispenser State screen is accessed from the State Table Data Menu by entering a value of 4 in the NEW PAGE field and pressing the **F7** key.

The Unlock Envelope Dispenser State screen is shown below, followed by descriptions of its fields.

BASE24-DCT		STATE TABLE DATA		LLLL	YY/MM/DD	hh:mm	04 OF 42
GROUP	0	STATE NUMBER	0				
UNLOCK ENVELOPE DISPENSER STATE							
STATE TYPE	C	NEXT STATE	0				
***** BASE24 *****							
NEW PAGE:	FILE DESTINATION:		NEW LOGICAL NETWORK ID:				
	F12-HELP						

STATE TYPE — The value C identifies the Unlock Envelope Dispenser state.

Field Length: System protected

NEXT STATE — The number of the state to which the terminal proceeds immediately after unlocking the envelope dispenser. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

Preset Operations Code Buffer State Screen

The Preset Operations Code Buffer state is used to fill a buffer to signify the specific operation that the cardholder wants to perform. The Preset Operations Code Buffer State screen is accessed from the State Table Data Menu by entering a value of 5 in the NEW PAGE field and pressing the **F7** key.

The Preset Operations Code Buffer State screen is shown below, followed by descriptions of its fields.

BASE24-DCT		STATE TABLE DATA		LLLL	YY/MM/DD	hh:mm	05 OF 42
GROUP	0	STATE NUMBER	0				
PRESET OPCODE BUFFER							
STATE	NEXT	PRESET	NDC+ REL 4.0				
TYPE	STATE	MASKS	EXTENSION				
D	0	- - - - -	STATE (56XX)				
			0				

***** BASE24 *****		
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:
	F12-HELP	

STATE TYPE — The value D identifies the Preset Operations Code Buffer state.

Field Length: System protected

NEXT STATE — The number of the state to which the terminal proceeds after presetting the operation code buffer. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

PRESET MASKS — Defines the characters and buffer location to place in the operations code buffer. Valid values are A through D and F through I, and the following special characters:

b = Overwrites the current value with a space. (*b* designates a blank character.)
@ = Retains the current value.

Note: If you use values F through I, you must assign an extension state number in the following field.

Field Length: 1 alphanumeric character
Occurs: Up to eight times
Required Field: No
Default Value: No default value

NDC+ REL 4.0 EXTENSION STATE (56xx) — The state number of the extension state. You are required to assign an extension state to state D if you use operation codes F through I. Valid values are 0 through 254 and 256 through 999.

Note: This extension state is only applicable for NCR 56XX-series terminals that are running NDC+ release 4.0 or greater. The extended values for state D are found on this screen so you do not need to go to state Z to enter the additional values. However, internally these values are saved in an extension state, so you still need to assign an extension state number.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Four FDK Selection State Screen

The Four Function Display Key (FDK) Selection state permits the cardholder to select one function from four functions displayed on the screen, by pressing one of the four active display keys, A through D. Note that a local option exists to use the four display keys on the left side of the terminal (F through I) instead of the right keys. Refer to the *NCR NDC+ Supervisor's Reference Manual* for information on setting local options. The Four FDK Selection State screen is accessed from the State Table Data Menu by entering a value of 6 in the NEW PAGE field and pressing the **F7** key.

The Four FDK Selection State screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT   STATE TABLE DATA           LLLL           YY/MM/DD  hh:mm  06 OF 42

      GROUP      0      STATE NUMBER      0

                        4 FDK SELECTION STATE

STATE         SCREEN   TIMEOUT   CANCEL   KEY-A   KEY-B   KEY-C
TYPE          NUMBER   STATE    STATE    STATE   STATE   STATE
  E           0        0         0         0     0     0

                KEY-D   BUFFER
                STATE   LOCATION
                 0      0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

STATE TYPE — The value E identifies the Four FDK Selection state.

Field Length: System protected

SCREEN NUMBER — The number of the display screen that prompts the cardholder to make a choice between two or more alternatives. This screen, aligned with the active display keys (FDKs), is displayed when the ATM enters the select function state. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TIMEOUT STATE — The number of the state to which the terminal proceeds when the terminal times out before the cardholder enters a transaction type, account name, etc. Valid values are 0 through 999. A value of 255 indicates that the timeout function is not required.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CANCEL STATE — The number of the state to which the terminal proceeds when the cardholder presses the **Cancel** key instead of an operation key. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-A STATE — The number of the state to which the terminal proceeds when the cardholder presses operation key A. Valid values are 0 through 999. A value of 255 indicates that key A is deactivated.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-B STATE — The number of the state to which the terminal proceeds when the cardholder presses operation key B. Valid values are 0 through 999. A value of 255 indicates that key B is deactivated.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-C STATE — The number of the state to which the terminal proceeds when the cardholder presses operation key C. Valid values are 0 through 999. A value of 255 indicates that key C is deactivated.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-D STATE — The number of the state to which the terminal proceeds when the cardholder presses operation key D. Valid values are 0 through 999. A value of 255 indicates that key D is deactivated.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

BUFFER LOCATION — Specifies the operation key buffer byte in which the key code for the selected operation key is to be stored. This field can be zero filled to the left. Valid values are as follows:

0 = Byte number 1
1 = Byte number 2
2 = Byte number 3
3 = Byte number 4
4 = Byte number 5
5 = Byte number 6
6 = Byte number 7
7 = Byte number 8

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Amount Entry State Screen

The Amount Entry state procedure reads the amount entered by the cardholder, displays it on the ATM screen, and saves it in the amount buffer. The four active keys are A through D. Note that a local option exists to use the four display keys on the left side of the terminal (F through I) instead of the right keys. Refer to the *NCR NDC+ Supervisor's Reference Manual* for information on setting local options. The Amount Entry State screen is accessed from the State Table Data Menu by entering a value of 7 in the NEW PAGE field and pressing the **F7** key.

The Amount Entry State screen is shown below, followed by descriptions of its fields.

BASE24-DCT		STATE TABLE DATA		LLLL	YY/MM/DD	hh:mm	07 OF 42
GROUP	0	STATE NUMBER	0				
AMOUNT ENTRY STATE							
STATE TYPE	SCREEN NUMBER	TIMEOUT STATE	CANCEL STATE	KEY-A STATE	KEY-B STATE	KEY-C STATE	
F	0	0	0	0	0	0	
	KEY-D STATE	DISPLAY SCREEN					
	0	0					
***** BASE24 *****							
NEW PAGE:		FILE DESTINATION:		NEW LOGICAL NETWORK ID:			
		F12-HELP					

STATE TYPE — The value F identifies the Amount Entry state.

Field Length: System protected

SCREEN NUMBER — The number of the display screen that prompts for an amount entry. This screen is displayed when the ATM enters the Amount Entry state. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TIMEOUT STATE — The number of the state to which the terminal proceeds when an amount entry is not made within the permitted time interval. Valid values are 0 through 999. A value of 255 indicates that the timeout function is not required.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CANCEL STATE — The number of the state to which the terminal proceeds when the cardholder presses the **Cancel** key instead of an operation key. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-A STATE — The number of the state to which the terminal proceeds when the cardholder presses operation key A. Normally, one key is designated for correct amount entries and another for incorrect amount entries. Two are inactive. Valid values are 0 through 999. A value of 255 indicates that key A is deactivated.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-B STATE — The number of the state to which the terminal proceeds when the cardholder presses operation key B. Normally, one key is designated for correct amount entries and another for incorrect amount entries. Two are inactive. Valid values are 0 through 999. A value of 255 indicates that key B is deactivated.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-C STATE — The number of the state to which the terminal proceeds when the cardholder presses operation key C. Normally, one key is designated for correct amount entries and another for incorrect amount entries. Two are inactive. Valid values are 0 through 999. A value of 255 indicates that key C is deactivated.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-D STATE — The number of the state to which the terminal proceeds when the cardholder presses operation key D. Normally, one key is designated for correct amount entries and another for incorrect amount entries. Two are inactive. Valid values are 0 through 999. A value of 255 indicates that key D is deactivated.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

DISPLAY SCREEN — The number of the overlay screen used for setting the CRT cursor position. This screen optionally defines the initial display format before amount entry by the cardholder. The cursor should be set on the left-hand side of the echo field on the CRT, which is space filled up to the required start position for the currency character. The default echo field on the terminal screen is `bbbbbb$0.00` (where `b` is a blank character). Twelve-digit amounts can be used depending on the value you have set in the READY/SUPPLY field on the Enhanced Configuration Parameters screen. Valid values for the DISPLAY SCREEN field are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Amount Entry Check State Screen

The Amount Entry Check state accesses the amount buffer to verify that the amount entered is a whole amount, a dispensable amount, or a multiple of a predetermined value. This screen is accessed from the State Table Data Menu by entering a value of 8 in the NEW PAGE field and pressing the **F7** key.

The Amount Entry Check State screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT  STATE TABLE DATA          LLLL          YY/MM/DD  hh:mm  08 OF 42

      GROUP      0      STATE NUMBER      0

                        AMOUNT ENTRY CHECK STATE

STATE      AMOUNT CHECK      AMOUNT CHECK      BUFFER TO      INTEGER
TYPE      COND TRUE      COND FALSE      BE CHECKED      MULTIPLE VALUE
  G          0          0          0          0

      NUMBER OF      CURRENCY      AMOUNT CHECK
      DECIMAL PLACES      TYPE      CONDITION
          0          0          0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP
  
```

STATE TYPE — The value G identifies the Amount Check state.

Field Length: System protected

AMOUNT CHECK COND TRUE STATE — The state number to which the terminal goes when the condition indicated by the value of the AMOUNT CHECK CONDITION field is true.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

AMOUNT CHECK COND FALSE STATE — The state number to which the terminal goes when the condition indicated by the value of the AMOUNT CHECK CONDITION field is false.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

BUFFER TO BE CHECKED — A code identifying which buffer is to be checked for the transaction amount. This field can be zero filled to the left. Valid values are as follows:

0 = Amount buffer
1 = Buffer B
2 = Buffer C

Note: Values 1 and 2 are valid for NCR 5085 and 56XX-series ATMs only. This field should be set to 0 for all other ATM models.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INTEGER MULTIPLE VALUE — The integer multiple value used in checking the specified buffer. For example, to check that the specified buffer is a multiple of the minimum denomination of \$5 U.S., this field is set to a value of 005. This field is ignored if the AMOUNT CHECK CONDITION field is not set to a value of 001. Valid values are 000–999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NUMBER OF DECIMAL PLACES — The number of decimal places used in the specified buffer. Valid values are 000–004.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CURRENCY TYPE — The currency type of the amount in the specified buffer. Valid values are 000–255. This field is ignored if the AMOUNT CHECK CONDITION field is not set to a value of 002.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

AMOUNT CHECK CONDITION — A code indicating what the amount check condition is evaluating. Valid values are as follows:

- 000 = The amount check state evaluates whether the value to the right of the decimal position is zero.
- 001 = The amount check state evaluates whether the value is a whole number multiple of the value identified in the INTEGER MULTIPLE VALUE field.
- 002 = The amount check state evaluates whether the value can be dispensed by the ATM (taking into account the value of the CURRENCY TYPE field, currency type denominations available, dispenser status and cassette status).

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Four FDK Information Entry State Screen

The Four Function Display Key (FDK) Information Entry state displays a screen to prompt the cardholder for information. The four active keys are A through D. Note that a local option exists to use the four display keys on the left side of the terminal (F through I) instead of the right keys. Refer to the *NCR NDC+ Supervisor's Reference Manual* for information on setting local options. The Four FDK Information Entry State screen is accessed from the State Table Data Menu by entering a value of 9 in the NEW PAGE field and pressing the **F7** key.

The Four FDK Information Entry State screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT    STATE TABLE DATA    LLLL    YY/MM/DD  hh:mm  09 OF 42

      GROUP      0      STATE NUMBER  0

                                INFORMATION ENTRY STATE

STATE TYPE    SCREEN NUMBER  TIMEOUT STATE  CANCEL STATE  KEY-A STATE  KEY-B STATE  KEY-C STATE
H              0              0              0              0              0              0

                                KEY-D STATE  BUFFER PARAMS
                                0              0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                      F12-HELP

```

STATE TYPE — The value H identifies the Four FDK Information Entry state.

Field Length: System protected

SCREEN NUMBER — The number of the display screen that prompts the cardholder to enter data. This screen, which is displayed upon entry into the Four FDK Information Entry state, must return the cursor to the beginning of the display area. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TIMEOUT STATE — The number of the state to which the terminal proceeds when the cardholder does not respond to an instruction displayed on the CRT within the permitted time limit. Valid values are 0 through 999. A value of 255 indicates that the timeout function is not required.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CANCEL STATE — The number of the state to which the terminal proceeds when a cardholder presses the **Cancel** key instead of pressing an operation key. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-A STATE — The number of the state to which the terminal proceeds when the cardholder presses operation key A. Valid values are 0 through 999. A value of 255 indicates that operation key A is deactivated.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-B STATE — The number of the state to which the terminal proceeds when the cardholder presses operation key B. Valid values are 0 through 999. A value of 255 indicates that operation key B is deactivated.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-C STATE — The number of the state to which the terminal proceeds when the cardholder presses operation key C. Valid values are 0 through 999. A value of 255 indicates that operation key C is deactivated.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-D STATE — The number of the state to which the terminal proceeds when the cardholder presses operation key D. Valid values are 0 through 999. A value of 255 indicates that operation key D is deactivated.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

BUFFER PARAMS — Specifies the general purpose buffer and display parameters for storing the numeric data input. This field also specifies whether the actual data entered, or X, is to be displayed, beginning at the current cursor position. This field can be zero filled to the left. Valid values are as follows:

- 0 = Display X for each numeric key pressed. Store data in General Purpose Buffer C.
- 1 = Display data as keyed in. Store data in General Purpose Buffer C.
- 2 = Display X for each numeric key pressed. Store data in General Purpose Buffer B.
- 3 = Display data as keyed in. Store data in General Purpose Buffer B.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Transaction Request State Screen

The Transaction Request state transmits the Transaction Request message to the BASE24 system and executes Transaction Reply commands received from the BASE24 system. The Transaction Request State screen is accessed from the State Table Data Menu by entering a value of 10 in the NEW PAGE field and pressing the **F7** key.

Note: The Transaction Request State screen has an extension state associated with it. The extension state extends the functions of this state by allowing the optional data fields to be sent in the Transaction Request message. The Transaction Request state extension is accessed through the Global Extension screen (state Z), which is described later in this section.

The Transaction Request State screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT   STATE TABLE DATA           LLLL           YY/MM/DD  hh:mm  10 OF 42

      GROUP      0      STATE NUMBER      0

                                TRANSACTION REQUEST STATE

STATE  SCREEN  TIMEOUT  SEND      SEND      SEND      SEND
TYPE  NUMBER  STATE    TRACK2   TRACK1/3  OP-CODE   AMOUNT
  I         0         0         0         0         0         0

      SEND      SEND
      PIN BUF A  GEN BUF B/C
        0         0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

STATE TYPE — The value I identifies the Transaction Request state.

Field Length: System protected

SCREEN NUMBER — The number of the display screen presented on the CRT while the ATM sends a transaction request to the BASE24 system and awaits a reply. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
 Required Field: No
 Default Value: 0

TIMEOUT STATE — The number of the state to which the terminal proceeds when the BASE24 system fails to reply to the transaction request within the permitted time limit specified in the Communications Response timer. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
 Required Field: No
 Default Value: 0

SEND TRACK2 — A code that determines if Track 2 information is included in the transaction request message transmitted to the BASE24 system. This field can be zero filled to the left. Valid values are as follows:

0 = Do not send Track 2 data
 1 = Send Track 2 data

Field Length: 1–3 numeric characters
 Required Field: No
 Default Value: 0

SEND TRACK1/3 — A code that determines if Track 1 and 3 information is included in the transaction request message transmitted to the BASE24 system. This field can be zero filled to the left. Valid values are 0 through 3.

The following table shows how Track 1 and 3 information is selected depending on the value set in this field.

SEND TRACK 1/3 Value	Send Track 1	Send Track 3
0	No	No
1	No	Yes

SEND TRACK 1/3 Value	Send Track 1	Send Track 3
2	Yes	No
3	Yes	Yes

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

SEND OP-CODE — A code that determines if operation code information is included in the transaction request message transmitted to the BASE24 system. This field can be zero filled to the left. Valid values are as follows:

0 = Do not send data.

1 = Send data.

2 = Send Smart Card data packet.

3 = Send Smart Card data packet and operation code buffer.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

SEND AMOUNT — A code that determines if amount information is included in the transaction request message transmitted to the BASE24 system. This field can be zero filled to the left. Valid values are as follows:

0 = Do not send data

1 = Send data

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

SEND PIN BUF A — A code that determines if PIN buffer A information is included in the transaction request message transmitted to the BASE24 system. This field also determines what kind of data the next field, SEND GEN BUF B/C, contains. This field can be zero filled to the left. Valid values are as follows:

- 0 = Do not send buffer A. The next field, SEND GEN BUF B/C, contains a code to determine if buffers B and C are sent.
- 1 = Send buffer A. The next field, SEND GEN BUF B/C, contains a code to determine if buffers B and C are sent.
- 128 = Do not send buffer A. The next field, SEND GEN BUF B/C, contains the extension state number for additional information.
- 129 = Send buffer A. The next field, SEND GEN BUF B/C, contains the extension state number for additional information.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

SEND GEN BUF B/C — This field can have two meanings, depending on the value in the last field, SEND PIN BUF A.

If the SEND PIN BUF A field contains a value of 0 or 1, this field contains a code that determines if general purpose buffers B and C information is included in the transaction request message transmitted to the BASE24 system. This field can be zero filled to the left. Valid values in this case are as follows:

- 0 = Send no buffer
- 1 = Send buffer B
- 2 = Send buffer C
- 3 = Send buffers B and C

If the SEND BUF A field contains a value of 128 or 129, this field contains the state number of the extension state. Valid values in this case are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

Close State Screen

The Close state completes the transaction and terminates the cardholder's current terminal session. The Close State screen is accessed from the State Table Data Menu by entering a value of 11 in the NEW PAGE field and pressing the **F7** key.

Note: The Close State screen has an extension state associated with it. The extension state extends the functions of this state by allowing additional fields for the BASE24-atm self-service banking (SSB) product. The Close state extension is accessed through the Global Extension screen (state Z), which is described later in this section.

The Close State screen is shown below, followed by descriptions of its fields.

BASE24-DCT	STATE TABLE DATA	LLLL	YY/MM/DD	hh:mm	11 OF 42
GROUP	0	STATE NUMBER	0		
CLOSE STATE					
STATE TYPE	RECEIPT DEL SCR N	NEXT STATE	NO RECEIPT DEL SCR N	CARD RETAINED	STATEMENT DEL SCR N
J	0	0	0	0	0
DPM DOCUMENT RETAINED SCREEN		NDC+ REL 5.0 EXTENSION STATE (56xx)			
0		0			
***** BASE24 *****					
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:			
	F12-HELP				

STATE TYPE — The value J identifies the Close state.

Field Length: System protected

RECEIPT DEL SCRIN — The number of the display screen that is displayed upon entry into the Close state if the card is returned and the terminal has printed and delivered a receipt. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NEXT STATE — The number of the state to which the terminal proceeds after the Close state is completed. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NO RECEIPT DEL SCRIN — The number of the display screen that is displayed upon entry into the Close state if the card is returned and no receipt slip has been printed or when the receipt could not be delivered. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CARD RETAINED — The number of the display screen that informs a cardholder that the card has been retained. This screen is displayed upon entry into the Close state only if the cardholder's card has been retained on instruction from a transaction reply message. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

STATEMENT DEL SCRIN — The number of the display screen that is overlaid if the statement is burst and delivered in a Close state and the terminal has not been instructed by the last transaction reply message to capture the card. The screen is overlaid when the receipt is delivered or when there is no receipt delivery. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

DPM DOCUMENT RETAINED SCREEN — The number of the display screen that is displayed on entry to the Close state informing the cardholder that the document has been captured. Valid values are 0 through 999.

Note: This field is only applicable for NCR 56XX-series terminals that are running NDC+ release 5.0 or greater.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NDC+ REL 5.0 EXTENSION STATE (56XX) — The state number of the extension state containing additional information needed to close the terminal. Valid values are 0 through 254 and 256 through 999.

Note: This extension state is only applicable for NCR 56XX-series terminals that are running NDC+ release 5.0 or greater.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

FIT Switch State Screen

The FIT Switch state contains indexing data for selecting the next state according to a particular financial institution. The FIT Switch state can contain next states for up to eight financial institutions; if you have more financial institutions that need to be switched, use the Enhanced FIT Switch state, which is described later in this section. The FIT Switch State screen is accessed from the State Table Data Menu by entering a value of 12 in the NEW PAGE field and pressing the **F7** key.

The FIT Switch State screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT  STATE TABLE DATA          LLLL          YY/MM/DD  hh:mm  12 OF 42

      GROUP      0      STATE NUMBER      0

                                FIT SWITCH STATE

      STATE      NEXT      NEXT      NEXT      NEXT      NEXT      NEXT
      TYPE      STATE      STATE      STATE      STATE      STATE      STATE
      K          0          0          0          0          0          0

                                NEXT      NEXT
                                STATE      STATE
                                0          0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                                F12-HELP
  
```

STATE TYPE — The value K identifies the FIT Switch state.

Field Length: System protected

NEXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 0. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

NEXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 1. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NEXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 2. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NEXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 3. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NEXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 4. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NEXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 5. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NEXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 6. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NEXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 7. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Card Write State Screen

The Card Write state writes the current contents of the Track 3 data buffer onto the magnetic stripe of the card and displays the screen specified in the SCREEN NUMBER field. The Card Write State screen is accessed from the State Table Data Menu by entering a value of 13 in the NEW PAGE field and pressing the **F7** key.

The Card Write State screen is shown below, followed by descriptions of its fields.

```
BASE24-DCT   STATE TABLE DATA          LLLL          YY/MM/DD  hh:mm  13 OF 42

      GROUP      0      STATE NUMBER      0

                        CARD WRITE STATE

STATE         SCREEN      GOOD WRITE      BAD WRITE      NO WRITE
TYPE          NUMBER      STATE           STATE          STATE
  L             0           0             0             0
```

```
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP
```

STATE TYPE — The value L identifies the Card Write state.

Field Length: System protected

SCREEN NUMBER — The number of the screen displayed during a Card Write operation. Valid values are 0 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

GOOD WRITE STATE — The number of the state to which the terminal proceeds after a satisfactory Track 3 write operation. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

BAD WRITE STATE — The number of the state to which the terminal proceeds when a Track 3 write is unsuccessful. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NO WRITE STATE — The number of the state to which the terminal proceeds when no write information is present in the Track 3 buffer. This state is also used if a start or stop sentinel is missing. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Enhanced PIN Entry State Screen

The Enhanced PIN Entry state is similar to the PIN Entry state, but it also supports Track 3 retries if the FIT specifies local PIN verification and it also specifies that there is a Track 3 retry field on the card. The Enhanced PIN Entry State screen is accessed from the State Table Data Menu by entering a value of 14 in the NEW PAGE field and pressing the **F7** key.

The Enhanced PIN Entry State screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT   STATE TABLE DATA           LLLL           YY/MM/DD  hh:mm  14 OF 42

      GROUP      0      STATE NUMBER      0

                                ENHANCED PIN ENTRY

STATE         SCREEN      TIMEOUT      CANCEL      GOOD PIN      BAD PIN
TYPE          NUMBER      STATE        STATE        CHECK         CHECK
  M           0           0           0           0           0

      LOCAL PIN      REMOTE PIN      LOCAL PIN
      CHECK ERROR    CHECK           MAX TRIES
      0              0              9

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

STATE TYPE — The value M identifies the Enhanced PIN Entry state.

Field Length: System protected

SCREEN NUMBER — The number of the display screen that initially requests the cardholder to enter a PIN. This screen must return the cursor to the beginning of the PIN entry display area. Valid values are 0 through 999.

Field Length: 1–3 numeric characters

Required Field: Yes

Default Value: 0

TIMEOUT STATE — The number of the state to which the terminal proceeds when a cardholder does not enter the PIN within the permitted time interval. Valid values are 0 through 999. A value of 255 indicates that the timeout function is not required.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CANCEL STATE — The number of the state to which the terminal proceeds when a cardholder presses the **Cancel** key after an incorrect PIN entry has been made. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

GOOD PIN CHECK — The number of the state to which the terminal proceeds after verification of a correct PIN entry. A Preset Operation Code Buffer state, which clears the function key buffer before transaction selection, usually follows a correct PIN entry. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

BAD PIN CHECK — The number of the state to which the terminal proceeds when the correct PIN has not been entered after the number of attempts set in the LOCAL PIN MAX TRIES field. This could be a card write state to update the Track 3 PIN retry field. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LOCAL PIN CHECK ERROR — The number of the display screen that prompts the cardholder to reenter a PIN. This screen is displayed after an incorrect PIN entry attempt. After the number of reentry attempts specified in the LOCAL PIN MAX TRIES field, the ATM enters the state determined by the setting in the BAD PIN CHECK field. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

REMOTE PIN CHECK — The number of the state to which the terminal proceeds when local PIN verification is not performed. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LOCAL PIN MAX TRIES — The maximum number of incorrect PIN entries allowed before the ATM enters the state determined by the setting in the BAD PIN CHECK field. This field can be zero filled to the left. Valid values are 1 through 9.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 9

Camera Control State Screen

The Camera Control state controls the operation of the optional security camera and specifies the data to be overlaid on the photograph. The Camera Control State screen is accessed from the State Table Data Menu by entering a value of 15 in the NEW PAGE field and pressing the **F7** key.

The Camera Control State screen is shown below, followed by descriptions of its fields.

BASE24-DCT	STATE TABLE DATA	LLLL	YY/MM/DD	hh:mm	15 OF 42
GROUP	0	STATE NUMBER	0		
CAMERA CONTROL STATE					
STATE TYPE	PICTURE TYPE	NEXT STATE			
N	0	0			
***** BASE24 *****					
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:			
	F12-HELP				

STATE TYPE — The value N identifies the Camera Control state.

Field Length: System protected

PICTURE TYPE — Specifications for a camera record of the person performing an ATM transaction. This field can be zero filled to the left. Valid values are identified in the following table.

Camera Control State Picture Type Specification			
Field Entry	Picture Type	Camera Format	Picture Annotations Included
0	0	Fixed	Time (hhmm)
			Date (YYMMDD)
			Machine (MMMMMM)
		Variable	Time (hhmm)
			Date (MMDD) or (DDMM)
1	1	Fixed	Time (hhmm)
			Date (YYMMDD)
			Transaction Serial Number (TTTT)
			Machine (MMMMMM)
		Variable	Transaction Serial Number (TTTT)
			Date (YYMMDD)

Note: Refer to the *NCR NDC+ Programmer's Reference Manual* for a complete description of this field.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

NEXT STATE — The number of the state to which the terminal proceeds after capturing the image of the cardholder at the terminal. If the terminal does not have a camera, then this next state is entered immediately. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Enhanced Amount Entry State Screen

The Enhanced Amount Entry state reads the amount entered by the cardholder, displays it on the CRT screen, and saves it in the buffer specified by the state table. This state only applies if you have NDC release 8.0 or NDC+ release 5.0 or greater. The Enhanced Amount Entry State screen is accessed from the State Table Data Menu by entering a value of 18 in the NEW PAGE field and pressing the **F7** key.

Note: The Enhanced Amount Entry State screen has an extension state associated with it. The extension state extends the functions of this state by providing additional amount configurations. The Enhanced Amount Entry state extension is accessed through the Global Extension screen (state Z), which is described later in this section.

The Enhanced Amount Entry State screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT   STATE TABLE DATA      LLLL      YY/MM/DD  hh:mm  18 OF 42

      GROUP      0      STATE NUMBER      0

                        ENHANCED AMOUNT ENTRY STATE

STATE  SCREEN  TIMEOUT  CANCEL  KEY-A  KEY-B  KEY-C
TYPE  NUMBER  STATE   STATE   STATE  STATE  STATE
  R      0      0      0      0      0      0

      KEY-D  EXTENSION
      STATE  STATE
      0      0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP

```

STATE TYPE — The value R identifies the Enhanced Amount Entry state.

Field Length: System protected

SCREEN NUMBER — The number of the display screen that prompts for an amount entry. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TIMEOUT STATE — The state number to which the terminal proceeds if the cardholder fails to enter the amount of the transaction within the specified time limit. Enter 255 if the time out function is not required. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CANCEL STATE — The state number to which the terminal proceeds if the cardholder presses the **Cancel** key. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-A STATE — The state number to which the terminal proceeds if the cardholder presses the specified function display key (FDK). Normally one FDK is used to specify whether the amount entry is correct and another to specify whether the amount entry is incorrect. The other keys are inactive. Enter 255 to deactivate any FDK or touch area. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-B STATE — The state number to which the terminal proceeds if the cardholder presses the specified function display key (FDK). Normally one FDK is used to specify whether the amount entry is correct and another to specify whether the amount entry is incorrect. The other keys are inactive. Enter 255 to deactivate any FDK or touch area. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-C STATE — The state number to which the terminal proceeds if the cardholder presses the specified function display key (FDK). Normally one FDK is used to specify whether the amount entry is correct and another to specify whether the amount entry is incorrect. The other keys are inactive. Enter 255 to deactivate any FDK or touch area. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-D STATE — The state number to which the terminal proceeds if the cardholder presses the specified function display key (FDK). Normally one FDK is used to specify whether the amount entry is correct and another to specify whether the amount entry is incorrect. The other keys are inactive. Enter 255 to deactivate any FDK or touch area. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EXTENSION STATE — The state number of the extension state containing the additional values for the Enhanced Amount Entry state. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Language Code Switch State Screen

The Language Code Switch state switches the language to be used for a transaction. Which language to use can be automatically read from a code on the ATM card, or you can prompt the cardholder to manually select the choice of language. The Language Code Switch State screen is accessed from the State Table Data Menu by entering a value of 19 in the NEW PAGE field and pressing the **F7** key.

Note: The Language Code Switch State screen has an extension state associated with it. The extension state extends the functions of this state by providing additional next state entries for up to nine languages. The Language Code Switch State screen is accessed through the Global Extension screen (state Z), which is described later in this section.

The Language Code Switch State screen is shown below, followed by descriptions of its fields.

BASE24-DCT	STATE TABLE DATA	LLLL	YY/MM/DD	hh:mm	19 OF 42
GROUP	0	STATE NUMBER	0		
LANGUAGE SWITCH STATE					
STATE	NO LANG CODE	LANG CODE '0'	LANG CODE '1'	LANG CODE '2'	
TYPE	NEXT STATE	NEXT STATE	NEXT STATE	NEXT STATE	
S	0	0	0	0	
	LANG CODE '3'	LANG CODE '4'	LANG CODE '5'	EXTENSION	
	NEXT STATE	NEXT STATE	NEXT STATE	STATE	
	0	0	0	0	
***** BASE24 *****					
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:			
	F12-HELP				

STATE TYPE — The value S identifies the Language Code Switch state.

Field Length: System protected

NO LANG CODE NEXT STATE — The state number to which the terminal proceeds if the language code cannot be read or is invalid. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANG CODE '0' NEXT STATE — The state number to which the terminal proceeds if the language code is 0. Valid values are 0 through 999. A value of 255 indicates that language 0 is not required; however, if a card has a language code with this value, the terminal proceeds to the default Close state.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANG CODE '1' NEXT STATE — The state number to which the terminal proceeds if the language code is 1. Valid values are 0 through 999. A value of 255 indicates that language 1 is not required; however, if a card has a language code with this value, the terminal proceeds to the default Close state.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANG CODE '2' NEXT STATE — The state number to which the terminal proceeds if the language code is 2. Valid values are 0 through 999. A value of 255 indicates that language 2 is not required; however, if a card has a language code with this value, the terminal proceeds to the default Close state.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANG CODE '3' NEXT STATE — The state number to which the terminal proceeds if the language code is 3. Valid values are 0 through 999. A value of 255 indicates that language 3 is not required; however, if a card has a language code with this value, the terminal proceeds to the default Close state.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANG CODE '4' NEXT STATE — The state number to which the terminal proceeds if the language code is 4. Valid values are 0 through 999. A value of 255 indicates that language 4 is not required; however, if a card has a language code with this value, the terminal proceeds to the default Close state.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANG CODE '5' NEXT STATE — The state number to which the terminal proceeds if the language code is 5. Valid values are 0 through 999. A value of 255 indicates that language 5 is not required; however, if a card has a language code with this value, the terminal proceeds to the default Close state.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EXTENSION STATE — The state number of the extension state containing the next states for languages 6 through 9. Valid values are 0 through 999. A value of 255 indicates that no extension state exists so that a card with a language code value of 6 through 9 proceeds to the state specified in the NO LANG CODE NEXT STATE field described above.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

PIN Entry Initiation State Screen

The PIN Entry Initiation state can be used as a faster alternative to the Card Read state to initiate PIN entry by the cardholder at the same time the card is being read. The state number should be 000 if you use this state instead of the Card Read state. The PIN Entry Initiation State screen is accessed from the State Table Data Menu by entering a value of 20 in the NEW PAGE field and pressing the **F7** key.

Note: The PIN Entry Initiation State screen has an extension state associated with it. The extension state extends the functions of this state by allowing additional entries for PIN entry configurations. The PIN Entry Initiation state extension is accessed through the Global Extension screen (state Z), which is described later in this section.

The PIN Entry Initiation State screen is shown below, followed by descriptions of its fields.

```
BASE24-DCT  STATE TABLE DATA          LLLL          YY/MM/DD  hh:mm  20 OF 42

      GROUP      0      STATE NUMBER      0

      CARD READ - PIN ENTRY INIT STATE

STATE  SCREEN  GOOD READ  ERROR  READ  READ  READ
TYPE  NUMBER  STATE      SCREEN COND1 COND2 COND3
  T      0      0          0      0      0      0

EXTENSION
STATE
  0
```

```
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP
```

STATE TYPE — The value T identifies the PIN Entry Initiation state.

Field Length: System protected

SCREEN NUMBER — The number of the display screen that tells a cardholder to insert a card. While the ATM is awaiting a card entry, this screen is displayed on the terminal. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

GOOD READ STATE — The number of the state to which the terminal proceeds following the successful read of a card when the financial institution number on the card matches a financial institution number in a FIT. This field also specifies the number of the state to which the terminal proceeds following the successful read of a card if FITs are not being used. If FITs are being used, but the number on the card does not match a financial institution number in a FIT, the number of the state to which the terminal proceeds following a successful card read is specified in the No FIT Match Next State Number field of the Global Extension screen. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

ERROR SCREEN — The number of the display screen to display following an unsuccessful read of the card. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

READ COND1 — A number sequence that specifies which track or tracks on the card are read. If the specified track is read, the state specified in the GOOD READ STATE field is performed. If the track could not be read, the action specified in the READ COND2 field is attempted. This field can be zero filled to the left. Valid values are 1 through 7.

Note: For specific information on what values to enter in the READ COND1 field, refer to the *NCR NDC Programmer's Reference Manual* for NCR 50XX-series terminals, and the *NCR NDC+ Programmer's Reference Manual* for NCR 56XX-series terminals.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

READ COND2 — A number sequence that specifies which track or tracks on the card are read. If the specified track is read, the state specified in the GOOD READ STATE field is performed. If the track could not be read, the action specified in the READ COND3 field is attempted. This field can be zero filled to the left. Valid values are 1 through 7.

Note: For specific information on what values to enter in the READ COND2 field, refer to the *NCR NDC Programmer's Reference Manual* for NCR 50XX-series terminals, and the *NCR NDC+ Programmer's Reference Manual* for NCR 56XX-series terminals.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

READ COND3 — A number sequence that specifies which track or tracks on the card are read. If the specified track is read, the state specified in the GOOD READ STATE field is performed. If the track could not be read, the card is ejected and the screen specified in the ERROR SCREEN field is displayed. This field can be zero filled to the left. Valid values are 1 through 7.

Note: For specific information on what values to enter in the READ COND3 field, refer to the *NCR NDC Programmer's Reference Manual* for NCR 50XX-series terminals, and the *NCR NDC+ Programmer's Reference Manual* for NCR 56XX-series terminals.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

EXTENSION STATE — The number of the extension state containing information for the PIN Entry Initiation state. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Language From Card State Screen

The Language From Card state allows you to change the screen group from which screens are displayed. The language code stored on the card identifies the screen group that should be used. The Language From Card State screen is accessed from the State Table Data Menu by entering a value of 21 in the NEW PAGE field and pressing the **F7** key.

The Language From Card State screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT  STATE TABLE DATA          LLLL      YY/MM/DD  hh:mm  21 OF 42

      GROUP      0      STATE NUMBER      0

                LANGUAGE SELECTION FROM CARD STATE

STATE      NEXT STATE      LANG CODE      LANG CODE      LANG CODE
TYPE      TYPE      TYPE      GROUP 1      GROUP 2      GROUP 3
  V              0              0              0              0

                LANG CODE      LANG CODE      LANG CODE      SCREEN
                GROUP 4      GROUP 5      GROUP 6      GROUP SIZE
                  0              0              0              0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP
  
```

STATE TYPE — The value V identifies the Language From Card state.

Field Length: System protected

NEXT STATE — The state number to which the terminal proceeds after selecting the screen group. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

LANGUAGE CODE GROUP 1 — The language code number for screen group 1. This field can be zero filled to the left. Valid values are 0 through 9.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANGUAGE CODE GROUP 2 — The language code number for screen group 2. This field can be zero filled to the left. Valid values are 0 through 9.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANGUAGE CODE GROUP 3 — The language code number for screen group 3. This field can be zero filled to the left. Valid values are 0 through 9.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANGUAGE CODE GROUP 4 — The language code number for screen group 4. This field can be zero filled to the left. Valid values are 0 through 9.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANGUAGE CODE GROUP 5 — The language code number for screen group 5. This field can be zero filled to the left. Valid values are 0 through 9.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANGUAGE CODE GROUP 6 — The language code number for screen group 6. This field can be zero filled to the left. Valid values are 0 through 9.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

SCREEN GROUP SIZE — The number of screens in each language screen group. Valid values are 0 through 400.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Function Display Key Switch State Screen

The Function Display Key (FDK) Switch state follows an Eight FDK Select Function state or Four FDK Information Entry state. It uses the contents of the FDK buffer set up by the previous state to identify the next state to be entered for subsequent transaction processing. The Function Display Key Switch State screen is accessed from the State Table Data Menu by entering a value of 22 in the NEW PAGE field and pressing the **F7** key.

The Function Display Key Switch State screen is shown below, followed by descriptions of its fields.

BASE24-DCT		STATE TABLE DATA		LLLL		YY/MM/DD hh:mm		22 OF 42	
GROUP		0		STATE NUMBER		0			
FUNCTION DISPLAY KEY SWITCH									
STATE TYPE	KEY-A STATE	KEY-B STATE	KEY-C STATE	KEY-D STATE	KEY-F STATE	KEY-G STATE			
W	0	0	0	0	0	0			
	KEY-H STATE	KEY-I STATE							
	0	0							
***** BASE24 *****									
NEW PAGE:		FILE DESTINATION:		NEW LOGICAL NETWORK ID:					
		F12-HELP							

STATE TYPE — The value W identifies the Function Display Key Switch state.

Field Length: System protected

KEY-A STATE — The number of the state to which the terminal proceeds when operation key A is pressed. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

KEY-B STATE — The number of the state to which the terminal proceeds when operation key B is pressed. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-C STATE — The number of the state to which the terminal proceeds when operation key C is pressed. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-D STATE — The number of the state to which the terminal proceeds when operation key D is pressed. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-F STATE — The number of the state to which the terminal proceeds when operation key F is pressed. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-G STATE — The number of the state to which the terminal proceeds when operation key G is pressed. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY-H STATE — The number of the state to which the terminal proceeds when operation key H is pressed. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

KEY-I STATE — The number of the state to which the terminal proceeds when operation key I is pressed. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

Eight FDK Information Entry State Screen

The Eight Function Display Key (FDK) Information Entry state edits the contents of one of the general purpose buffers or the amount buffer with the value contained in the specified extension state for the FDK selected by the cardholder. The Eight FDK Information Entry State screen is accessed from the State Table Data Menu by entering a value of 23 in the NEW PAGE field and pressing the **F7** key.

Note: The Eight FDK Information Entry State screen has an extension state associated with it. The extension state extends the functions of this state by adding eight values to be stored in buffers. The Eight FDK Information Entry state extension is accessed through the Global Extension screen (state Z), which is described later in this section.

The Eight FDK Information Entry State screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT  STATE TABLE DATA          LLLL          YY/MM/DD  hh:mm  23 OF 42

      GROUP      0      STATE NUMBER      0

                        8 FDK INFORMATION ENTRY STATE

STATE  SCREEN  TIMEOUT  CANCEL  KEY  EXTENSION
TYPE  NUMBER  STATE   STATE   STATE STATE
  X      0      0      0      0      0

      BUFFER  KEYS ACTIVE
      IDENTITY MASK
        0      0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

STATE TYPE — The value X identifies the Eight FDK Information Entry state.

Field Length: System protected

SCREEN NUMBER — The number of the display screen that prompts function key selection by cardholder. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TIMEOUT STATE — The number of the state to which the terminal proceeds when a cardholder fails to respond to a CRT message within the applicable time limit. Valid values are 0 through 999. A value of 255 indicates that the timeout function is not required.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CANCEL STATE — The number of the state to which the terminal proceeds when the **Cancel** key is pressed. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY STATE — The number of the state to which the terminal proceeds when an operations key is pressed. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EXTENSION STATE — The state number of the extension state containing assigned values for the eight operations keys. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

BUFFER IDENTITY — Identifies which buffer is used to store information and indicates the number of zeros to append to the values specified in the extension state.

The first two positions in the field indicate the buffer. Valid values for the first two positions of the field are as follows:

- 01 = General Purpose Buffer B.
- 02 = General Purpose Buffer C.
- 03 = Amount Buffer comprised of eight or twelve digit amounts. The number of digits is specified in the READY/SUPPLY field on the Enhanced Configuration Parameters screen.

The last position of the field indicates the number of zeros to append to the values specified in the extension state. The range of valid values for the last position of the field is 0 through 9.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 0

KEYS ACTIVE MASK — The decimal representation for an eight-bit mask, which indicates if operation keys A through I are active. To calculate the value for this field, add the values in the following table for the operation keys that are active. Valid values for the decimal total are 0 through 255.

Operation Key	Value
A	1
B	2
C	4
D	8
F	16
G	32
H	64
I	128

Field Length:	1–3 numeric characters
Required Field:	No
Default Value:	0

Eight FDK Selection State Screen

The Eight Function Display Key (FDK) Selection state displays a menu and enables the FDKs to allow the cardholder to make a choice. The key code of the selected FDK is stored in a buffer that can be acted on later by a subsequent FDK Switch state. The Eight FDK Selection State screen is accessed from the State Table Data Menu by entering a value of 24 in the NEW PAGE field and pressing the **F7** key.

Note: The Eight FDK Selection State screen has two extension states associated with it. These extension states extend the functions of this state by adding eight operation codes that can be stored in buffers, and seven base screens for other languages. The Eight FDK Selection state extensions are accessed through the Global Extension screen (state Z), which is described later in this section.

The Eight FDK Selection State screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT    STATE TABLE DATA    LLLL    YY/MM/DD  hh:mm  24 OF 42

      GROUP      0      STATE NUMBER  0

                        8 FDK SELECTION STATE

STATE  SCREEN  TIMEOUT  CANCEL  KEY NEXT  EXTENSION
TYPE  NUMBER  STATE   STATE   STATE     STATE
  Y      0      0      0      0         0

      BUFFER  KEYS ACTIVE  MULTI-LANGUAGE
      POSITION MASK        SCREEN SELECT
        0      0          0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:  NEW LOGICAL NETWORK ID:
                   F12-HELP

```

STATE TYPE — The value Y identifies the Eight FDK Selection state.

Field Length: System protected

SCREEN NUMBER — The number of the display screen that prompts function key selection by cardholder. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TIMEOUT STATE — The number of the state to which the terminal proceeds when the terminal times out before the cardholder reacts to the prompt within a permitted time interval. Valid values are 0 through 999. A value of 255 indicates that the timeout function is not required.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CANCEL STATE — The number of the state to which the terminal proceeds when the **Cancel** key is pressed. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

KEY NEXT STATE — The number of the state to which the terminal proceeds when the cardholder presses an operations key. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EXTENSION STATE — The state number of the first extension state containing the operation code for the eight operations keys. Valid values are 0 through 999. A value of 255 indicates that the extension state is not required.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

BUFFER POSITION — The buffer position to be edited. The meaning and values entered in this field differ depending on whether or not the first extension state is used with this state, as defined in the previous field.

When the first extension state is not used, this field defines the single buffer position to be edited if any key is pressed. This field can be zero filled to the left. Valid values are 0 through 7.

If there is a first extension state to the buffer position, this field specifies three bytes of the operation code buffer to be set to the values specified in the extension state. Three numeric values each in the range 0 through 7 are specified to define three positions in the eight-byte operation code buffer to be edited. The values must correspond with the edit characters in the extension state. Valid values are 0 through 777.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

KEYS ACTIVE MASK — The decimal representation for an eight-bit mask, which indicates if operation keys A through I are active. To calculate the value for this field, add the values in the following table for the operation keys that are active. Valid values for the decimal total are 0 through 255.

Operation Key	Value
A	1
B	2
C	4
D	8
F	16
G	32
H	64
I	128

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

MULTI-LANGUAGE SCREEN SELECT — The state number of the second extension state specifying which of the multiple languages is being used on a terminal screen. Valid values are 0 through 999. A value of 0 or 255 indicates that no extension state exists for this option, and the multi-language display is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Global Extension Screen

The Global Extension screen is a generic screen that is used to access extension state table data for all states that have an extension state. An extension state is used to extend, or expand, the functions of a particular state. The state tables in general only allow eight values to be set. For those states requiring more than eight configuration values, the global extension table is used to hold the excess values.

The extension states are as follows:

- Transaction Request state extension (state I)
- Close state extension (state J)
- DPM Accept Document (three extensions for state Q; described in the *BASE24-atm NCR 5XXX SSB Manual*)
- Enhanced Amount Entry state extension (state R)
- Language Code Switch state extension (state S)
- PIN Entry Initiation state extension (state T)
- Eight FDK Information Entry state extension (state X)
- Eight FDK Selection state extensions (two extensions for state Y)
- Begin EMV Application Selection and Initialization state extensions (three extensions for state . (period); described in the *BASE24-atm EMV Support Manual*)
- Complete EMV Application Selection and Initialization state extension (state / (slash); described in the *BASE24-atm EMV Support Manual*)
- Cash Accept (three extensions for state >; described in the *BASE24-atm NCR 5XXX SSB Manual*)
- Check Accept (two extensions for state w; described in the *BASE24-atm NCR 5XXX SSB Manual*)
- PIN & Language Select (three extensions for state m)

The Global Extension screen is used to input extension state table data for all of the extension states mentioned above. The additional configuration values on the Global Extension screen have generic field names, and the meaning of a specific field changes depending on the original state screen that referenced this screen. The table entry fields on the Global Extension screen correspond to the extension state table entries described in the *NCR NDC Programmer's Reference Manual* for NCR 50XX-series terminals, and the *NCR NDC+ Programmer's Reference Manual* for NCR 56XX-series terminals.

The contents of the table entry fields for each of the extension states are described on the pages following the Global Extension screen. The GROUP field was described previously for the State Table Data Menu.

The Global Extension screen is accessed from the State Table Data Menu by entering a value of 25 in the NEW PAGE field and pressing the **F7** key. The Global Extension screen is shown below. Field descriptions are provided for each extension state on the following pages.

```
BASE24-DCT    STATE TABLE DATA    LLLL    YY/MM/DD  hh:mm  25 OF 42
```

```
GROUP      0    STATE NUMBER    0
```

```
GLOBAL EXTENSION
```

STATE TYPE	TABLE ENTRY-2	TABLE ENTRY-3	TABLE ENTRY-4	TABLE ENTRY-5	TABLE ENTRY-6	TABLE ENTRY-7
Z	0	0	0	0	0	0

TABLE ENTRY-8	TABLE ENTRY-9
0	0

```
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                   F12-HELP
```

Transaction Request State Extension

The following fields are used for the Transaction Request state extension (state I). Each field is input as a table entry on the Global Extension screen. For example, the Send General Purpose Buffers B or C data (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen. The fields TABLE ENTRY-6 through TABLE ENTRY-9 are not used with the Transaction Request state extension.

STATE NUMBER — The number of the extension state as specified on the Transaction Request State screen. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

STATE TYPE — The value Z identifies the Global Extension state.

Field Length: System protected

TABLE ENTRY-2 (Send General Purpose Buffers B or C) — Determines whether or not buffers B or C are sent. Valid values are as follows:

0 = Send no buffer
1 = Send buffer B
2 = Send buffer C
3 = Send buffers B and C

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (Send Optional Data Fields A through H) — The decimal representation for an eight-bit mask that indicates which individual optional data fields in the Transaction Request message are sent. To convert the

binary mask to a three-character decimal value, add the values in the following table for the optional data fields that are active. Reserved fields cannot be sent. Valid values for the decimal total are 0 through 255.

Optional Data Field	Value
A (Reserved)	1
B (Reserved)	2
C	4
D	8
E	16
F	32
G	64
H	128

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

TABLE ENTRY-4 (Send Optional Data Fields I through L) — The decimal representation for an eight-bit mask that indicates which individual optional data fields in the Transaction Request message are sent. To calculate the value for this field, add the values in the following table for the optional data fields that are active. This field can be zero filled to the left. Valid values for the decimal total are 0 through 15.

Optional Data Field	Value
I	1
J	2
K	4
L	8

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (Send Optional Data Fields Q through S) — The decimal representation for an eight-bit mask that indicates which individual optional data fields in the Transaction Request message are sent. To calculate the value for this field, add the values in the following table for the optional data fields that are active. This field can be zero filled to the left. Valid values for the decimal total are 0 through 07.

Optional Data Field	Value
Q	1
R	2
S	4

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Close State Extension

The following fields are used for the Close state extension (state J). This extension is used only for terminals that have a Document Processing Module (DPM) and use the BASE24-atm self-service banking (SSB) add-on product. Each field is input as a table entry on the Global Extension screen. For example, the DPM Take Document Screen Number data (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen. The fields TABLE ENTRY-6 through TABLE ENTRY-9 are not used with the Close state extension.

STATE NUMBER — The number of the extension state as specified on the Close State screen. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

STATE TYPE — The value Z identifies the Global Extension state.

Field Length: System protected

TABLE ENTRY-2 (DPM Take Document Screen Number) — The number of the screen that is displayed during error recovery actions, which can take place in this state if a Document Processing Module (DPM) is present in the terminal. Valid values are 0 to 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (DPM Retain/Return Flag) — Indicates the error recovery actions that can take place in this state if a Document Processing Module (DPM) is present in the terminal. Valid values are as follows:

000 = Retain the document
001 = Return the document

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Next State if Card Capture) — The number of the next state to which the terminal proceeds if the card is captured. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (Special Close Flag) — A flag identifying whether the terminal is to close normally. Valid values are as follows:

000 = Normal close.
001 = Eject only; bypass receipt or statement cut, eject, and so on.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Enhanced Amount Entry State Extension

The following fields are used for the Enhanced Amount Entry state extension (state R). Each field is input as a table entry on the Global Extension screen. For example, the Document Amount Buffer data (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen. The fields TABLE ENTRY-6 through TABLE ENTRY-9 are not used with the Enhanced Amount Entry state extension.

Note: This extension only applies if you have NDC release 8.0 or NDC+ release 5.0.

STATE NUMBER — The number of the extension state as specified on the Enhanced Amount Entry State screen. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

STATE TYPE — The value Z identifies the Global Extension state.

Field Length: System protected

TABLE ENTRY-2 (Document Amount Buffer) — Determines which buffer the amount entered is stored in. Valid values are as follows:

000 = Amount buffer
001 = Buffer B
002 = Buffer C

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (Amount Display Screen Number) — The number of the overlay screen that sets the CRT cursor position. This screen optionally defines the initial display format before amount entry by the cardholder. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Start CAV Command) — Specifies whether or not courtesy amount verification (CAV) is started. This field can be zero filled to the left. Valid values are as follows:

0 = Do not start CAV.
1 = Start CAV processing if a document is available and the Document Processing Module (DPM) is properly configured.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Language Code Switch State Extension

The following fields are used for the Language Code Switch state extension (state S). Each field is input as a table entry on the Global Extension screen. For example, the Lang Code '6' Next State data (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen. The fields TABLE ENTRY-6 through TABLE ENTRY-9 are not used with the Language Code Switch state extension.

STATE NUMBER — The number of the extension state as specified on the Language Code Switch State screen. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

STATE TYPE — The value Z identifies the Global Extension state.

Field Length: System protected

TABLE ENTRY-2 (Lang Code '6' Next State) — The state number to which the terminal proceeds if the language code is 6. Valid values are 0 through 999. A value of 255 indicates that language 6 is not required; however, if a card has a language code with this value, the terminal proceeds to the default Close state.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (Lang Code '7' Next State) — The state number to which the terminal proceeds if the language code is 7. Valid values are 0 through 999. A value of 255 indicates that language 7 is not required; however, if a card has a language code with this value, the terminal proceeds to the default Close state.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Lang Code '8' Next State) — The state number to which the terminal proceeds if the language code is 8. Valid values are 0 through 999. A value of 255 indicates that language 8 is not required; however, if a card has a language code with this value, the terminal proceeds to the default Close state.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (Lang Code '9' Next State) — The state number to which the terminal proceeds if the language code is 9. Valid values are 0 through 999. A value of 255 indicates that language 9 is not required; however, if a card has a language code with this value, the terminal proceeds to the default Close state.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

PIN Entry Initiation State Extension

The following fields are used for the PIN Entry Initiation state extension (state T). Each field is input as a table entry on the Global Extension screen. For example, the Screen Number data (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen. The fields TABLE ENTRY-6 through TABLE ENTRY-9 are not used with the PIN Entry Initiation state extension.

STATE NUMBER — The number of the extension state as specified on the PIN Entry Initiation State screen. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

STATE TYPE — The value Z identifies the Global Extension state.

Field Length: System protected

TABLE ENTRY-2 (Screen Number) — The number of the display screen that prompts the cardholder to enter a PIN. This screen is displayed upon card entry. The identified screen must leave the CRT cursor positioned at the beginning of the PIN entry display area; X is displayed for each key pressed. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (No FIT Match Next State Number) — The number of the state to which the terminal proceeds when the financial institution number on the card does not match any FIT. If FITs are not used, this entry must be 0. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Clear Key Mask) — The decimal representation for an eight-bit mask that indicates if operation keys A through I are active as clear keys during PIN entry. To calculate the value for this field, add the values in the following table for the operation keys that are active. Valid values for the decimal total are 0 through 255.

Operation Key	Value
A	1
B	2
C	4
D	8
F	16
G	32
H	64
I	128

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

TABLE ENTRY-5 (Accept Key Mask) — The decimal representation for an eight-bit mask that indicates if operation keys A through I are active as accept keys during PIN entry. To calculate the value for this field, add the values in the following table for the operation keys that are active. Valid values for the decimal total are 0 through 255.

Operation Key	Value
A	1
B	2
C	4
D	8
F	16

Operation Key	Value
G	32
H	64
I	128

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

Eight FDK Information Entry State Extension

The following fields are used for the Eight Function Display Key (FDK) Information Entry state extension (state X). Each field is input as a table entry on the Global Extension screen. For example, the Key-A data (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen.

STATE NUMBER — The number of the extension state as specified on the Eight FDK Information Entry State screen. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

STATE TYPE — The value Z identifies the Extension state.

Field Length: System protected

TABLE ENTRY-2 (Key-A) — The value stored in the buffer identified in the BUFFER IDENTITY field of the Eight FDK Information Entry State screen if operation key A is pressed. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (Key-B) — The value stored in the buffer identified in the BUFFER IDENTITY field of the Eight FDK Information Entry State screen if operation key B is pressed. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Key-C) — The value stored in the buffer identified in the BUFFER IDENTITY field of the Eight FDK Information Entry State screen if operation key C is pressed. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (Key-D) — The value stored in the buffer identified in the BUFFER IDENTITY field of the Eight FDK Information Entry State screen if operation key D is pressed. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-6 (Key-F) — The value stored in the buffer identified in the BUFFER IDENTITY field of the Eight FDK Information Entry State screen if operation key F is pressed. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-7 (Key-G) — The value stored in the buffer identified in the BUFFER IDENTITY field of the Eight FDK Information Entry State screen if operation key G is pressed. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-8 (Key-H) — The value stored in the buffer identified in the BUFFER IDENTITY field of the Eight FDK Information Entry State screen if operation key H is pressed. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-9 (Key-I) — The value stored in the buffer identified in the BUFFER IDENTITY field of the Eight FDK Information Entry State screen if operation key I is pressed. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Eight FDK Selection State Extensions

The Eight FDK Selection state has two extensions that provide additional values for state Y.

Extension 1 to State Y

The following fields are used for the first extension to the Eight FDK Selection state (state Y). Each field is input as a table entry on the Global Extension screen. For example, the Key-A data (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen.

Note: Because the Global Extension screen allows only numeric entries, the Key-A through Key-I fields use a three-digit code to represent up to three alphabetic entries. Each of the three characters has one digit of the three-digit number, and is mapped as follows:

0 = no change
1 = A
2 = B
3 = C
4 = D
5 = E
6 = F
7 = G
8 = H
9 = I

STATE NUMBER — The number of the extension state as specified in the EXTENSION STATE field on the Eight FDK Selection State screen. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

STATE TYPE — The value Z identifies the Global Extension state.

Field Length: System protected

TABLE ENTRY-2 (Key-A) — The three operation code characters written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key A is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (Key-B) — The three operation code characters written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key B is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Key-C) — The three operation code characters written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key C is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (Key-D) — The three operation code characters written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key D is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-6 (Key-F) — The three operation code characters written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key F is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-7 (Key-G) — The three operation code characters written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key G is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-8 (Key-H) — The three operation code characters written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key H is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-9 (Key-I) — The three operation code characters are written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key I is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Extension 2 to State Y

The following fields are used for the second extension to the Eight FDK Selection state (state Y). Each field is input as a table entry on the Global Extension screen. For example, the Screen Base A data (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen.

STATE NUMBER — The number of the extension state as specified in the MULTI-LANGUAGE SCREEN SELECT field on the Eight FDK Selection State screen. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

STATE TYPE — The value Z identifies the Global Extension state.

Field Length: System protected

TABLE ENTRY-2 (Screen Base A) — The screen number of the first screen in language 0 if operation key A is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (Screen Base B) — The screen number of the first screen in language 1 if operation key B is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Screen Base C) — The screen number of the first screen in language 2 if operation key C is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (Screen Base D) — The screen number of the first screen in language 3 if operation key D is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-6 (Screen Base F) — The screen number of the first screen in language 4 if operation key F is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-7 (Screen Base G) — The screen number of the first screen in language 5 if operation key G is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-8 (Screen Base H) — The screen number of the first screen in language 6 if operation key H is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-9 (Screen Base I) — The screen number of the first screen in language 7 if operation key I is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

PIN & Language Select State Extensions

The PIN & Language Select state has three extensions that provide additional values for state m.

Extension 1 to State m

The following fields are used for the first extension to the PIN & Language Select state (state m). Each field is input as a table entry on the Global Extension screen. For example, the No PIN Entered Next State data (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen.

STATE NUMBER — The number of the extension state as specified on the PIN & Language Select state screen. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

STATE TYPE — The value Z identifies the Global Extension state.

Field Length: System protected

TABLE ENTRY-2 (No Pin Entered Next State Number) — The number of the state to which the terminal proceeds if the cardholder does not enter a PIN, but selects a language from an active FDK. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

TABLE ENTRY-3 (Local PIN Check - Good PIN Next State

Number) — The number of the state to which the terminal proceeds after verification of a correct PIN entry. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: Yes

Default Value: 0

TABLE ENTRY-4 (Local PIN Check - Maximum Bad PINs Next State

Number) — The number of the state to which the terminal proceeds if the incorrect PIN has been entered after the number of attempts allowed by TABLE ENTRY-7. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: Yes

Default Value: 0

TABLE ENTRY-5 (Local PIN Check Error Next State Number) — The number of the state to which the terminal proceeds if an incorrect PIN is entered. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: Yes

Default Value: 0

TABLE ENTRY-6 (Remote PIN Check Next State Number) — The number of the state to which the terminal proceeds when the local PIN verification is not performed. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: Yes

Default Value: 0

TABLE ENTRY-7 (Local PIN Check Maximum PIN Retries) — The maximum number of incorrect PIN entries allowed before the terminal enters the state determined by TABLE ENTRY-4. Valid values are 0 through 9.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

TABLE ENTRY-8 (State ‘B’ or ‘M’ Option) — Specifies whether to use Track 3 retries on the card or from the state table. Valid values are as follows:

000 = PIN Entry State B functionality is needed.
001 = Enhanced PIN Entry State M functionality is needed.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

TABLE ENTRY-9 (FDK Enter Key Default Mask) — Specifies the FDK that is the default language if the Enter key on the key pad is pressed instead of an FDK. Each bit relates to an FDK. Bit 0 relates to FDK A; bit 7 relates to FDK I.

Set the bit to 1 to make that FDK the default Enter key value. The mapped FDK will be set even if the FDK is inactive. Only one bit should be set to 1; all other bits should be 0. If all bits are set to 0, the Enter key will not be active. Valid values are 0 through 128.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

Extension 2 to State m

The following fields are used for the second extension to the PIN & Language Select state (state m). Each field is input as a table entry on the Global Extension screen. For example, the Key-A data (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen.

Note: Because the Global Extension screen allows only numeric entries, the Key-A through Key-I fields use a three-digit code to represent up to three alphabetic entries. Each of the three characters has one digit of the three-digit number, and is mapped as follows:

0 = no change
1 = A
2 = B
3 = C
4 = D
5 = E
6 = F
7 = G
8 = H
9 = I

STATE NUMBER — The number of the extension state as specified on the PIN & Language Select state screen. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

STATE TYPE — The value Z identifies the Global Extension state.

Field Length: System protected

TABLE ENTRY-2 (Key-A) — The three operation code characters written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key A is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (Key-B) — The three operation code characters written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key B is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Key-C) — The three operation code characters written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key C is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (Key-D) — The three operation code characters written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key D is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-6 (Key-F) — The three operation code characters written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key F is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-7 (Key-G) — The three operation code characters written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key G is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-8 (Key-H) — The three operation code characters written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key H is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-9 (Key-I) — The three operation code characters are written in order to the buffer position in the associated Eight FDK Selection state table BUFFER POSITION field if operation key I is pressed. Refer to the note at the beginning of the topic for how these digits are mapped to characters.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Extension 3 to State m

The following fields are used for the third extension to the PIN & Language Select state (state m). Each field is input as a table entry on the Global Extension screen. For example, the Screen Base A data (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen.

STATE NUMBER — The number of the extension state as specified in the MULTI-LANGUAGE SCREENS EXTENSION STATE field on the PIN & Language Select state screen. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

STATE TYPE — The value Z identifies the Global Extension state.

Field Length: System protected

TABLE ENTRY-2 (Screen Base A) — The screen number of the first screen in language 0 if operation key A is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (Screen Base B) — The screen number of the first screen in language 1 if operation key B is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Screen Base C) — The screen number of the first screen in language 2 if operation key C is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (Screen Base D) — The screen number of the first screen in language 3 if operation key D is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-6 (Screen Base F) — The screen number of the first screen in language 4 if operation key F is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-7 (Screen Base G) — The screen number of the first screen in language 5 if operation key G is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-8 (Screen Base H) — The screen number of the first screen in language 6 if operation key H is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-9 (Screen Base I) — The screen number of the first screen in language 7 if operation key I is pressed. Valid values are 0 through 999. A value of 0 indicates that the operation key is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Enhanced FIT Switch State Screen

The Enhanced FIT Switch state contains indexing data for selecting the next state according to a particular financial institution. This state is similar to the FIT Switch state (state K), but this state provides additional switches using an extension state. The Enhance FIT Switch State screen is accessed from the State Table Data Menu by entering a value of 26 in the NEW PAGE field and pressing the **F7** key.

The Enhanced FIT Switch State screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT      STATE TABLE DATA      LLLL      YY/MM/DD  hh:mm  26 OF 42

      GROUP      0      STATE NUMBER      0

      ENHANCED FIT SWITCH STATE (56XX NDC+ REL 4.02)

STATE      INDEX      INDEX      INDEX      INDEX      INDEX      INDEX      INDEX      EXT
TYPE      0 NXT      1 NXT      2 NXT      3 NXT      4 NXT      5 NXT      6 NXT      STATE
      STATE      STATE      STATE      STATE      STATE      STATE      STATE      NUM
      —          0          0          0          0          0          0          0          0

      INDEX      INDEX      INDEX      INDEX      INDEX      INDEX      INDEX      INDEX
      7 NXT      8 NXT      9 NXT      10 NXT     11 NXT     12 NXT     13 NXT     14 NXT
      STATE      STATE      STATE      STATE      STATE      STATE      STATE      STATE
      0          0          0          0          0          0          0          0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP
  
```

STATE TYPE — The value of an underbar () identifies the Enhanced FIT Switch state.

Field Length: System protected

INDEX 0 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 0. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INDEX 1 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 1. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INDEX 2 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 2. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INDEX 3 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 3. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INDEX 4 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 4. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INDEX 5 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 5. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INDEX 6 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 6. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EXT STATE NUM — The state number of the extension state containing additional state switches. Valid values are 0 through 254 and 256 through 999.

Note: The extended values for state _ are found on this screen so you do not need to go to state Z to enter the additional values. However, internally these values are saved in an extension state, so you still need to assign an extension state number.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INDEX 7 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 7. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INDEX 8 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 8. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INDEX 9 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 9. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INDEX 10 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 10. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INDEX 11 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 11. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INDEX 12 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 12. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INDEX 13 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 13. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

INDEX 14 NXT STATE — The number of the state to which the terminal proceeds if the PSTDX field in the FIT contains a value of 14. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Audio Control State Screen

The Audio Control State screen is used to select the audio channel and volume controls used with the NDC+ Digital Audio Service (DAS). This screen is accessed from the State Table Data Menu by entering a value of 27 in the NEW PAGE field and pressing the **F7** key.

The Audio Control State screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT   STATE TABLE DATA           LLLL           YY/MM/DD  hh:mm  27 OF 42

      GROUP      0      STATE NUMBER      0

                                AUDIO CONTROL STATE

STATE  SCREEN  TIMEOUT  COMPL/CNCL  INCR-VOL  DECR-VOL
TYPE   NUMBER  STATE   STATE      KEY MASK  KEY MASK
  i      0      0      0          0          0

      AUD-CNTL  AUD-MSG
      BIT-MAP  NUMBER
        0      0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP
  
```

STATE TYPE — The value i identifies the Audio Control State.

Field Length: System protected

SCREEN NUMBER — The number of the screen that prompts the cardholder to adjust the volume of the terminal audio system. This screen is displayed only when the audio control bit map entry indicates the volume is to be adjusted. Valid values are 000 through 999.

Field Length: 1–3 numeric characters
 Required Field: Yes
 Default Value: 0

TIMEOUT STATE — The number of the state to which the terminal proceeds if the cardholder has not responded within the time specified by timer 0. Valid values are 000 through 999. A value of 255 disables the timeout.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

COMPL/CNCL STATE — The number of the state to which the terminal proceeds when one of the following events occurs:

- The cardholder presses the **Cancel** or **Enter** key.
- The volume adjust feature is disabled and the audio channel has been selected.
- No audio file is present on the terminal.

Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

INCR-VOL KEY MASK — The decimal representation for a bit mask that indicates the bit number of the function key that increases the volume when the terminal is in the Audio Control state. Valid values are as follows:

0 = Increase volume key mask disabled
1 = FDK A
2 = FDK B
4 = FDK C
8 = FDK D
16 = FDK F
32 = FDK G
64 = FDK H
128 = FDK I

The FDKs are not enabled unless the volume adjustment is on (i.e., the AUD-CNTL BIT-MAP field is set to a value of 1, 5, 7, 9, 11, 13, or 15).

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

DECR-VOL KEY MASK — The decimal representation for a bit mask that indicates the bit number of the function display keys that decrease the volume when the terminal is in the Audio Control state. Valid values are as follows:

0 = Decrease volume key mask disabled
1 = FDK A
2 = FDK B
4 = FDK C
8 = FDK D
16 = FDK F
32 = FDK G
64 = FDK H
128 = FDK I

The FDKs are not enabled unless the volume adjustment is on (i.e., the AUD-CNTL BIT-MAP field is set to a value of 1, 5, 7, 9, 11, 13, or 15).

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

AUD-CNTL BIT-MAP — The bit setting used to adjust the volume or select the audio channel. Bit assignments are as follows:

- Bit 0 turns the volume adjustment on (1) or off (0)
- Bit 1 enables (1) or disables (0) the effect of bits 2 and 3
- Bit 2 enables (1) or disables (0) the terminal loudspeaker
- Bit 3 enables (1) or disables (0) the audio jack

This field can be zero filled to the left. Valid values are as follows:

Value	Audio Jack	Terminal Loudspeaker	Enable Bits 2 and 3	Volume Adjustment
000	disabled	disabled	disabled	disabled
001	disabled	disabled	disabled	enabled
002	disabled	disabled	enabled	disabled
004	disabled	enabled	disabled	disabled
005	disabled	enabled	disabled	enabled
006	disabled	enabled	enabled	disabled
007	disabled	enabled	enabled	enabled
008	enabled	disabled	disabled	disabled
009	enabled	disabled	disabled	enabled
010	enabled	disabled	enabled	disabled
011	enabled	disabled	enabled	enabled
012	enabled	enabled	disabled	disabled
013	enabled	enabled	disabled	enabled
014	enabled	enabled	enabled	disabled
015	enabled	enabled	enabled	enabled

Field Length: 1–3 numeric characters

Required Field: Yes

Default Value: 0

AUD-MSG NUMBER — The number of the audio message to be played for volume adjustment. Valid values are 000 through 999.

If the volume adjustment is on (i.e., the AUD-CNTL BIT-MAP field is set to a value of 1, 5, 7, 9, 11, 13, or 15), the audio message is played repeatedly until the state is exited. If no audio file is present on the terminal, the terminal executes the Function Complete Next State exit.

Field Length: 1–3 numeric characters

Required Field: Yes

Default Value: 0

Customer Selectable PIN State Screen

The Customer Selectable PIN State enables the customer to select a new PIN. It differs from the PIN Entry state in the number of retries. The Customer Selectable PIN State allows two retries and proceeds to the accepted state if both retries are the same and the terminal checking feature is enabled. The Customer Selectable PIN State screen is accessed from the State Table Data Menu by entering a value of 36 in the NEW PAGE field and pressing the **F7** key.

The Customer Selectable PIN State screen is shown below, followed by descriptions of its fields.

```
BASE24-DCT    STATE TABLE DATA    LLLL    YY/MM/DD  hh:mm  36 OF 42
```

```
GROUP    0    STATE NUMBER    0
```

CUSTOMER SELECTABLE PIN

```
STATE      FIRST PIN    TIMEOUT    CANCEL    GOOD PIN
TYPE       SCREEN      STATE      STATE     ENTRY STATE
b          0           0         0         0
```

```
BAD PIN    SECOND PIN    PIN MIS-    EXTENSION
ENTRY STATE SCREEN      MATCH SCREEN STATE
0          0           0         0
```

```
***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP
```

STATE TYPE — The value *i* identifies the Customer Selectable PIN State.

Field Length: System protected

FIRST PIN SCREEN — The number of the display screen that prompts the cardholder to enter the PIN the first time.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

TIMEOUT STATE — The number of the state to which the terminal proceeds if the cardholder does not select a PIN within the permitted time interval. Valid values are 000 through 999. A value of 255 indicates the timeout function is not required.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CANCEL STATE — The number of the state to which the terminal proceeds when the cardholder presses the **Cancel** key instead of an operation key. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

GOOD PIN ENTRY STATE — The number of the state to which the terminal proceeds when the cardholder enters a matching pair of new PIN values. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

BAD PIN ENTRY STATE — The number of the state to which the terminal proceeds when the cardholder fails to enter a matching pair of new PIN digits or when the maximum number of PIN retries for local PIN checking is reached. This field also specifies the number of the state to which the terminal proceeds if the financial institution number on a card does not match any FIT, cardholder tampering is suspected, or the PIN block generation fails. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

SECOND PIN SCREEN — The number of the display screen that prompts for the confirmation of the customer-selected PIN. Valid values are 000 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

PIN MIS-MATCH SCREEN — The number of the display screen that replaces the screen for table entry 2. This screen is displayed on each attempt after a customer-selected PIN pair attempt did not match (or an error has occurred) and there are still further entries allowed. Valid values are 000 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EXTENSION STATE — The number of the extension state containing information for the Customer Selectable PIN state. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

PIN & Language Select State Screen

The PIN & Language Select state displays a screen to prompt the cardholder to enter their PIN, and also enables the function display keys (FDK) along the side of the screen. The cardholder can then select the language to be used. The PIN & Language Select state screen is accessed from the State Table Data Menu by entering a value of 40 in the NEW PAGE field and pressing the **F7** key.

Note: The PIN & Language Select screen has three extension states associated with it. These extension states extend the functions of this state by providing additional PIN and language options. The PIN & Language Select state extensions are accessed through the Global Extension screen (state Z), which is described earlier in this section.

The PIN & Language Select state screen is shown below, followed by descriptions of its fields.

BASE24-DCT	STATE TABLE DATA	LLLL	YY/MM/DD	hh:mm	40 OF 42
GROUP	0	STATE NUMBER	0		
PIN & LANGUAGE SELECT					
STATE TYPE	SCREEN NUMBER	TIMEOUT STATE	CANCEL STATE	OPTIONS EXTENSION STATE	
m	0	0	0	0	
OPERATIONS CODES EXTENSION STATE		BUFFER POSITIONS	FDKS ACTIVE MASK	MULTI-LANGUAGE SCREENS EXTENSION STATE	
0		0	0	0	
***** BASE24 *****					
NEW PAGE:		FILE DESTINATION:	NEW LOGICAL NETWORK ID:		
		F12-HELP			

STATE TYPE — The value m identifies the PIN & Language Select state.

Field Length: System protected

SCREEN NUMBER — The number of the display screen that initially requests the cardholder to enter a PIN. This screen must return the cursor to the beginning of the PIN entry display area. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TIMEOUT STATE — The number of the state to which the terminal proceeds when a cardholder does not enter the PIN within the permitted time limit. Valid values are 0 through 999. A value of 255 indicates that the timeout function is not required.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CANCEL STATE — The number of the state to which the terminal proceeds when a cardholder presses the **Cancel** key before the maximum number of digits specified for the PIN have been entered. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

OPTIONS EXTENSION STATE — The number of the extension state containing the PIN options. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

OPERATIONS CODES EXTENSION STATE — The number of the extension state containing assigned operation codes for the eight FDKs. Valid values are 0 through 999. A value of 255 indicates that no extension state exists for this option.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

BUFFER POSITIONS — The buffer position to be edited. The meaning and values entered in this field differ depending on whether or not the first extension state is used with this state, as defined in the previous field.

When the first extension state is not used, this field defines the single buffer position to be edited if any key is pressed. This field can be zero filled to the left. Valid values are 0 through 7.

If there is a first extension state to the buffer position, this field specifies three bytes of the operation code buffer to be set to the values specified in the extension state. Three numeric values each in the range 0 through 7 are specified to define three positions in the eight-byte operation code buffer to be edited. The values must correspond with the edit characters in the extension state. Valid values are 0 through 777.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

FDKS ACTIVE MASK — The decimal representation for an eight-bit mask, which indicates if operation keys A through I are active. To calculate the value for this field, add the values in the following table for the operation keys that are active. Valid values for the decimal total are 0 through 255.

Operation Key	Value
A	1
B	2
C	4
D	8
F	16
G	32
H	64
I	128

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

MULTI-LANGUAGE SCREENS EXTENSION STATE — The state number of the third extension state specifying which of the multiple languages is being used on a terminal screen. Valid values are 0 through 999. A value of 0 or 255 indicates that no extension state exists for this option, and the multi-language display is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Screen Data Format Screen

The Screen Data Format screen enables customization of screen display information by controlling the appearance of the display screens, overlay screens that can be displayed on top of existing screens, character size and placement, special graphics, control characters, and character blinking. For more detailed information on customization of screen displays, refer to the *NCR NDC Programmer's Reference Manual* for NCR 50XX-series terminals, and the *NCR NDC+ Programmer's Reference Manual* for NCR 56XX-series terminals.

This screen is divided into two screen layouts. The left-side screen layout is used to enter the screen information that the cardholder sees. The right-side screen layout contains special characters and control data. Horizontal and vertical rows of characters outline the layouts. These characters can serve as coordinates to help you move from one layout to the other. ATM keys A, B, C, D, F, G, H, and I are noted with an angle bracket character (< or >).

Information entered in the right-side section of the screen layout instructs the editor to insert control codes. These control codes determine which character set to display, when to overlay a screen with another screen number, where to position the cursor in a specific place, when to blink a character on the screen, or where to display the cardholder's name from Track 1. For example, the value C indicates where the cursor is to be placed.

To move the cursor from the left-side screen layout to the same position on the right-side screen layout, press the **F7** key. The cursor jumps to the right-side screen layout.

When the cursor is in the correct position on the right-side screen layout, you can enter special characters or control data. To return to the screen layout on the left, press the **F7** key. The cursor jumps back to the same position, but on the left-side screen.

This screen is displayed when you place the cursor to the left of the Screen option and press the **F1** key on the NCR 5XXX Configuration File Menu. All display screen maintenance functions are performed on the Screen Data Format screen.

The Screen Data Format screen is shown below, followed by field descriptions, function key definitions, and examples for using the screen.

BASE24-DCT SCREEN DATA FORMAT LLLL YY/MM/DD hh:mm 01 OF 01

GROUP 0 SCREEN 0 00 LANG 0 DEF CHAR DESIG 1 CLR SCR N PRIOR N VOICE _ _

@ABCDEFGHIJKLMNO0123456789:;<=>? @ABCDEFGHIJKLMNO0123456789:;<=>?

@ @ @

A A A

B B B

C C C

D D D

E E E

I>F F<A F

G G G

H H H

H>I I<B I

J J J

K K K

G>L L<C L

M M M

N N N

F>O O<D O

@ABCDEFGHIJKLMNO0123456789:;<=>? @ABCDEFGHIJKLMNO0123456789:;<=>?

***** BASE24 *****

NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:

F12-HELP

GROUP — The number used to identify a particular load group (i.e., a collection of states, screens, tables, and parameters) that is downloaded to the ATM. Each group provides certain functionality (e.g., group 0 is the basic group, group 1 provides Fast Cash support, etc.). The group number of each ATM is assigned in the TERMINAL GROUP field on ATD screen 7. Refer to the topic “[NCR 5XXX Configuration File Menu](#)” for a list of valid values for this field.

Field Length: 1–4 numeric characters
Required Field: No
Default Value: 0

SCREEN — The screen number of the screen on which an action is being performed. Valid values are 0 through 999, C00 through C06, R00, or R01.

Note: R00 and R01 are reserved for preprinted receipt messages. Type C screens are reserved for terminal use and, if used, provide the following set of screens to be displayed at specified times.

- C00 = Displayed when a cardholder fails to respond within a specified interval to a keyboard input, envelope, or form insertion request.
- C01 = Displayed when the terminal is offline.
- C02 = Displayed when the terminal is out-of-service (e.g., when communication lines are down).
- C03 = Displayed when the terminal is in Supervisor mode.
- C04 = Displayed when the terminal captures a card due to a card reader failure.
- C05 = Displayed when cardholder tampering is suspected and the terminal enters Suspend mode. When the terminal enters Suspend mode, a timer is started and the Suspend screen is displayed. The terminal returns to service when the timer expires.
- C06 = Displayed when the terminal is temporarily out-of-service (e.g., when the terminal is changing modes).

Field Length: 3 alphanumeric characters
Required Field: Yes
Default Value: 000

LANG — A code identifying the language used on the screen. A value of 0 indicates the primary language for the terminal. Valid values are 0 through 9.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0

DEF CHAR DESIG — A code identifying the default character set or font. The default can be changed for specific characters on the screen by entering control data on the right-hand section of the Screen Data Format screen. For more information on the available character sets, refer to the *NCR NDC Programmer's Reference Manual* for NCR 50XX-series terminals, and the *NCR NDC+ Programmer's Reference Manual* for NCR 56XX-series terminals.

The ATM can be configured to display any character from the following character sets. The values in the Character Set Designator column are valid for this field.

Character Set (Font)	Character Set Designator	Number of Characters	ASCII Code (Hexadecimal)
Single-size Alphanumeric 1	1	96	31
Single-size Alphanumeric 2	2	96	32
Standard Graphics 1	3	64	33
Standard Graphics 2	4	64	34
Standard Graphics 3	5	64	35
Customer Graphics 1	6	64	36
Customer Graphics 2 (Arabic)	7	64	37
Customer Graphics 3	8	64	38
Customer Graphics 4 (pixels)	9	64	39
Double-size Alphanumeric 1	>	96	3E
Double-size Alphanumeric 2	?	96	3F
Chinese 1	:	64	3A
Chinese 2	;	64	3B
Chinese 3	<	64	3C
Chinese 4	'	64	3D
Chinese 7	@	64	40
Chinese 8	A	64	41
Double-size Arabic 3	B	64	42

Field Length: 1 alphanumeric character
 Required Field: No
 Default Value: 1

CLR SCR N PRIOR — A flag identifying whether the screen is to be cleared prior to the display of this screen. Usually, the screen should be cleared. The screen should not be cleared, however, if an overlay screen is being formatted. Valid values are as follows:

Y = Yes, clear screen.

N = No, do not clear screen.

Field Length: 1 alphabetic character

Required Field: Yes

Default Value: N

VOICE — Indicates if the voice responder is in operation. A voice response system is limited to the following operations:

- Card Entry
- Transaction Requests
- Card Removal

The first 3-digit positions of this field indicate the voice message number. The last 1-digit position indicates the number of messages that can be concatenated. The following table describes the valid values for the voice response system.

Positions 1–3 Voice Message Number Range	Position 4 Number of Messages Concatenated
000 through 999	0 through 3
blank	blank

If positions 1 through 3 are set to 000 through 999, position 4 must be set to a value in the range of 0 through 3. If positions 1 through 3 are blank, position 4 must be blank.

Field Length: 4 alphanumeric characters

Required Field: No

Default Value: No default value

Function Keys

The use of the following keys varies from the standard function keys defined earlier in this section for the NCR 5XXX Configuration Menu. The first column of the table shows the BASE24 key. The second column describes the functions that can be accomplished with these function keys.

Key	Description
Shift-F2	Read With Overlays — Displays the screen with all of its associated overlays.
Shift-F6	Read Next With Overlays — Displays the next screen with all of its associated overlays.
F7	Jump Screen — Moves the cursor between the left-side and right-side screen layouts, maintaining the same cursor position in the original layout.
Shift-F7	Next Overlay — Pages through overlays that are associated with one base screen. This occurs only if an overlay on a screen is configured to clear the screen first.
F13	Change Logical Network — Moves from one logical network to another.

Special Characters

The following special characters are entered on the right-side screen data layout. These characters are not displayed on the cardholder screen, but have a special meaning to the configuration editor.

The uppercase character C tells the editor where to position the cursor on a cardholder screen. The uppercase character C must be entered on a line by itself. Refer to the topic “Screen Data Format Screen Examples” for a sample screen using this special character for cursor position.

The uppercase character N instructs the editor to display the cardholder’s name from Track 1.

Control Characters

The following control characters are entered on the right-side screen data layout. These characters are not displayed to the cardholder but have a meaning to the configuration editor.

Overlay Screen and Screen Idle

Lowercase *s* represents an overlay screen. The screen called by the next character set, such as *s100*, overlays the existing screen with screen 100 that is stored in the ATM. Refer to the topic “Screen Data Format Screen Examples” for sample screens using the screen overlay control characters.

Lowercase *i* represents a changing screen display when the ATM is idle, such as waiting for the next card to be inserted. If a delay control sequence is encountered at the end of the idle screen, the ATM cycles through a specified display sequence displaying and delaying as required until the next card is inserted.

Overlay screens can be invoked from the idle screen to increase the amount of data that can be displayed during the idle sequence.

To set up a display sequence, such as advertising screens that include a delay control function, you must store the required Screen Data Format screen in the NCR 5XXX Configuration File. The cursor must then be placed in the right-side section of the Screen Data Format screen by pressing the **F7** key.

You then enter a string in the format *snnnit* on the right-side section of the Screen Data Format screen to set up a display format, where *nnn* is the 3-digit overlay screen number and *t* is the 3-digit length of time (in 100 millisecond intervals or 1/10 of a second) that the screen is displayed. You can repeat this string up to four times on one line, and have up to sixteen lines as needed to display a series of screens. For example, the following string displays screen 011 for 500 milliseconds (5/10 seconds), then displays screen 012 for 100 milliseconds (1/10 seconds), then displays screen 025 for 1000 milliseconds (1 second), and finally displays screen 024 for 30,000 milliseconds (30 seconds).

```
s011i005s012i001s025i010s024i300
```

Picture Control

The right brace character (}) is the picture control command identifier. This identifier is entered in the right-side layout at the position of the upper left corner of the graphic. The format is }*nnn**b*, where *nnn* is a picture code stored on the system disk and *b* is a space.

If your institution has more than 999 images in its files, you can enter the following sequence in the **left-side** layout at the position of the upper left corner of the graphic: lc:\s4pict\filename, where *filename* is the name of a custom picture file.

Refer to the topic “Screen Data Format Screen Examples” for sample screens using the picture control characters.

Logo Control

The number sign character (#) is the logo control command identifier. The logo (picture) is displayed from the current cursor position. The format is #*nn*, where *nn* is a logo stored on the system disk.

Character Blink

You can also use control characters to display blinking characters. For example, you could highlight special instructions by making the characters blink. To make a character blink, add hex value 20 to the appropriate ASCII code of the character designator. (Refer to the description of the DEF CHAR DESIG field on the Screen Data Format screen for a table describing character sets and character set designators.) Then, enter the text that is to blink on the left side of the Screen Data Format screen and the corresponding blink code on the right side of the screen.

The following table is used to determine appropriate blink codes.

Character Designator	ASCII Code (Hexadecimal)	ASCII Code Plus 20	Blink Code
1	31	51	Q
2	32	52	R
3	33	53	S

Character Designator	ASCII Code (Hexadecimal)	ASCII Code Plus 20	Blink Code
4	34	54	T
5	35	55	U
6	36	56	V
7	37	57	W
8	38	58	X
9	39	59	Y
>	3E	5E	^
?	3F	5F	_
:	3A	5A	Z
;	3B	5B	[
<	3C	5C	\
=	3D	5D	]
@	40	60	'
A	41	61	a
B	42	62	b

Refer to the topic “Screen Data Format Screen Examples” for a sample screen using the character blink control characters.

Screen Data Format Screen Examples

The following screen examples illustrate how you can use the Screen Data Format screen to customize ATM screen layouts. The following features are explained in the paragraphs that follow:

- Cursor position
- Screen overlay
- Character blink
- Self-service banking
- Picture control

Cursor Position

An example of the Screen Data Format screen showing correct use of the special character function for cursor position is shown below.

```
BASE24-DCT    SCREEN DATA FORMAT    LLLL    YY/MM/DD  hh:mm  01 OF 01
GROUP      0 SCREEN 0 00 LANG 0 DEF CHAR DESIG 1 CLR SCRN PRIOR N VOICE ____ _
@ABCDEFGHIJKLMNO0123456789;,<=>? @ABCDEFGHIJKLMNO0123456789;,<=>? @
A  YOU DID NOT ENTER YOUR CORRECT A  A  A
B  B  B  B
C  PERSONAL IDENTIFICATION NUMBER C  C  C
D  D  D  D
E  E  E  E
I>F AFTER PIN ENTERED-PRESS HERE--> F<A  F  F
G  G  G  G
H  H  H  H
H>I I<B  I  C  I
J  J  J  J
K  K  K  K
G>L L<C  L  L
M  M  M  M
N  TO OBTAIN CARD PRESS CANCEL N  N  N
F>O O<D  O  O
@ABCDEFGHIJKLMNO0123456789;,<=>? @ABCDEFGHIJKLMNO0123456789;,<=>?
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                      F12-HELP
```

In this example, the text in the left-side layout is what cardholders see when they enter an incorrect PIN. The character in the right-side layout indicates the cursor position.

Screen Overlay

An example of the Screen Data Format screen showing the correct use of overlay control character functions is shown below.

```

BASE24-DCT  SCREEN DATA FORMAT  LLLL  YY/MM/DD hh:mm 01 OF 01
GROUP 0 SCREEN 0 00 LANG 0 DEF CHAR DESIG 1 CLR SCRN PRIOR N VOICE ____ _
@ABCDEFGH IJKLMNO0123456789:;<=>? @ABCDEFGH IJKLMNO0123456789:;<=>?
@ @ @
A A A sG05 A
B B B
C C C
D D D
E E E
I>F F<A F
G G G
H H H
H>I I<B I
J J J
K K K
G>L L<C L
M PLEASE TAKE YOUR MONEY M M
N N N
F>O O<D O
@ABCDEFGH IJKLMNO0123456789:;<=>? @ABCDEFGH IJKLMNO0123456789:;<=>?
***** BASE24 *****
NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

```

Lowercase s represents an overlay screen. The screen called by the next character set, such as s100, overlays the existing screen with screen 100. The screen G05 used in the example is a predefined graphic that is stored in the ATM.

Character Blink

An example of the Screen Data Format screen using blink characters is shown below. In this example, each letter Q causes the corresponding text line in the single-size Alphanumeric 1 character set to blink.

```
BASE24-DCT    SCREEN DATA FORMAT    LLLL    YY/MM/DD  hh:mm  01 OF 01
GROUP      0 SCREEN 0 00 LANG 0 DEF CHAR DESIG 1 CLR SCRN PRIOR N VOICE ____ _
@ABCDEFGHIJKLMNO0123456789:;<=>?    @ABCDEFGHIJKLMNO0123456789:;<=>?
@
A
B
C      PLEASE
D
E      REMOVE
I>F
G      CARD
H
H>I
J
K
G>L
M
N
F>O
@ABCDEFGHIJKLMNO0123456789:;<=>?
*****
NEW PAGE:      FILE DESTINATION:      BASE24 *****
                F12-HELP                NEW LOGICAL NETWORK ID:
```

Self-Service Banking

The BASE24-atm self-service banking (SSB) product provides the means for displaying a scanned image of the document that the cardholder inserted into the terminal. SSB-specific screens are provided in the sample CNFG5070 that is provided with the SSB Device Handler extension.

The names of the files containing the scanned images are displayed on the left-hand side of the Screen Data Format screen for these SSB-specific screens. Refer to the *BASE24-atm NCR 5XXX SSB Manual* for additional information on these screens.

Picture Control

The NCR ATM screen can also display a customized graphic, such as a logo. The right brace character (}) is the picture control command identifier. This identifier is entered in the right-side layout at the position of the upper left corner of the graphic. The picture number contains up to three digits followed by at least one space. This method can be used only when the institution has 999 or fewer graphics files.

An example of the Screen Data Format screen showing the correct use of picture control functions is displayed below. In the example, the character *b* represents a space.

```

BASE24-DCT   SCREEN DATA FORMAT           LLLL           YY/MM/DD  hh:mm  01 OF 01
GROUP      0 SCREEN 0 00 LANG 0 DEF CHAR DESIG 1 CLR SCRN PRIOR N VOICE ____ _
@ABCDEFGHIJKLMNO0123456789:;<=>?           @ABCDEFGHIJKLMNO0123456789:;<=>?
@
A
B
C
D
E
I>F
G
H
H>I
J
K
G>L
M
N
F>O
@ABCDEFGHIJKLMNO0123456789:;<=>?
*****
NEW PAGE:
FILE DESTINATION:
F12-HELP
@
A
B
C
D
E
F<A
G
H
I<B
J
K
L<C
M
N
O<D
O
@
A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
@ABCDEFGHIJKLMNO0123456789:;<=>?
*****
BASE24 *****
NEW LOGICAL NETWORK ID:

```

In this example, picture number 125 is displayed on the ATM above the text PLEASE WAIT. The upper left corner of the graphic is in position AA.

Institutions also can use the `lc:\s4pict\filename` sequence, where *filename* is the name of a graphics file, on the left-side layout to specify a graphics file. This sequence can be used with any number of graphics files (i.e., you are not limited to 999 graphics files). The following example of the Screen Data Format screen shows the correct use of the `lc:\s4pict` sequence.

```

BASE24-DCT  SCREEN DATA FORMAT      LLLL      YY/MM/DD  hh:mm  01 OF 01
GROUP      0 SCREEN 0 00 LANG 0 DEF CHAR DESIG 1 CLR SCR N PRIOR N VOICE ____ _
@ABCDEFGHIJKLMNO0123456789:;<=>? @ABCDEFGHIJKLMNO0123456789:;<=>?
@ |c:\s4pict\pic1001.pcx @ @ @
A A A A
B B B B
C C C C
D D D D
E E E E
I>F F<A F F
G PLEASE ENTER G G G
H H H H
H>I YOUR PERSONAL I<B I I
J J J J
K IDENTIFICATION K K K
G>L L<C L C L
M NUMBER, THEN M M M
N N N N
F>O PRESS ENTER O<D O O
@ABCDEFGHIJKLMNO0123456789:;<=>? @ABCDEFGHIJKLMNO0123456789:;<=>?
***** BASE24 *****
NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

```

In this example the graphic called `pic1001.pcx` is displayed above the text `PLEASE ENTER YOUR PERSONAL IDENTIFICATION NUMBER, THEN PRESS ENTER`.

Color Support

The BASE24-atm product supports the use of color for NCR terminals using NDC Color Support. This level of color support enables you to control the color display at the terminal by providing the ability to perform the following functions:

- Set the default foreground and background colors
- Set a transparent background color for screen characters

These features enable you to highlight important information on the terminal screen for the cardholder.

In addition, with color support, the Reset Attributes command is used to turn character blinking off and to set default colors back to those in the configuration file (CONFIG.CON) stored on the NDC system disk.

Set Default Foreground and Background Colors

This feature enables you to set the default foreground and background colors for the ATM screen. The defaults that are set remain in effect until another Set Default Color Control command or a Reset Attributes command is sent.

Set Transparent Background Colors

This feature enables you to set a transparent background for screen characters. The transparent background feature allows characters to be displayed directly on the current screen background. For example, this feature could be used to overlay text on a picture background.

To turn the transparent background off, another control string setting the background color to its previous color is sent. This can be done by setting the default background color or by specifically setting the background color.

Reset Attributes

The Reset Attributes command sets the default colors for ATM screens back to those in the configuration file (CONFIG.CON) on the NDC system disk. It is also used to turn the character blinking feature off.

Supported Color Parameters

Each feature of color support is specified by entering a special character followed by a control string with a separate and unique parameter sequence on the right side of the Screen Data Format screen opposite the corresponding text on the left side of the screen.

The table below shows the control strings for color parameters supported by the BASE24-atm product.

Color Feature	Entry String	Description
Set Default Foreground/ Background	<i>%f1;b1</i>	The percent symbol (%) sets default background/ foreground colors. Following this character can be one or two sets of two-digit numbers, separated by a semicolon (;). The format for setting the default background and foreground colors is <i>%f1;b1</i> where <i>f1</i> is the new default foreground color and <i>b1</i> is the new default background color. The <i>f1</i> code must be in the range of 60 through 67, or F0 through F7. The <i>b1</i> code must be in the range of 70 through 77, or G0 through G7. The <i>f1</i> code does not have to precede <i>b1</i> ; there is no order of precedence. Values for the default foreground and background colors are described in the topic “Default Colors,” following this table.
Set Transparent Background	=	The character used for setting a transparent background.
Reset Attributes	{00	The character string for the Reset Attributes command.
Set Color Control	{f2;b2	The special character { is used for changing colors. Following this character can be one or two sets of two-digit numbers, separated by a semicolon (;). The format for setting colors is {f2;b2 where <i>f2</i> is the new foreground color and <i>b2</i> is the new background color. The <i>f2</i> code must be in the range of 20 through 27, or B0 through B7. The <i>b2</i> code must be in the range of 30 through 37, 80, or C0 through C7. The <i>f2</i> code does not have to precede <i>b2</i> ; there is no order of precedence. Values for changing colors are described in the topic “Character Colors” following this table.

Color Feature	Entry String	Description
Set Blinking Off/Color Control	{00; <i>f2</i> ; <i>b2</i>	The entry string used to turn off the blinking feature and set the default colors to those in the configuration file (CONFIG.CON). See the previous table entry for an explanation of the <i>f2</i> ; <i>b2</i> variables.

Default Colors

The following table contains the characters used to define default foreground and background colors using the Screen Data Format screen.

Default Color Controls	Intensity	
	Low	High
Black foreground	60	F0
Red foreground	61	F1
Green foreground	F2	62
Yellow foreground	F3	63
Blue foreground	64	F4
Magenta foreground	65	F5
Cyan foreground	66	F6
White foreground	F7	67
Black background	70	G0
Red background	71	G1
Green background	G2	72
Yellow background	G3	73
Blue background	74	G4
Magenta background	75	G5

Default Color Controls	Intensity	
	Low	High
Cyan background	76	G6
White background	G7	77

Character Colors

The following table contains the characters used to define the character background and foreground colors that override the default colors.

Character Color Controls	Intensity	
	Low	High
Black foreground	20	B0
Red foreground	21	B1
Green foreground	B2	22
Yellow foreground	B3	23
Blue foreground	24	B4
Magenta foreground	25	B5
Cyan foreground	26	B6
White foreground	B7	27
Black background	30	C0
Red background	31	C1
Green background	C2	32
Yellow background	C3	33
Blue background	34	C4
Magenta background	35	C5
Cyan background	36	C6

Character Color Controls	Intensity	
	Low	High
White background	C7	37
Transparent background	Not applicable	80

Example for Setting Default Colors

The following example of the Screen Data Format screen shows a control string sent to the ATM to set the default colors.

```

BASE24-DCT  SCREEN DATA FORMAT  LLLL  YY/MM/DD  hh:mm  01 OF 01
GROUP      0 SCREEN 0 00 LANG 0 DEF CHAR DESIG 1 CLR SCRN PRIOR N VOICE ____ _
@ABCDEFGHIJKLMNO0123456789:;<=>? @ABCDEFGHIJKLMNO0123456789:;<=>?
@          @          @
A          A          A
B          B          B
C          PLEASE     C          C          %63;74
D          D          D
E          REMOVE     E          E
I>F        F<A       F
G          CARD       G          G
H          H          H
H>I        I<B       I
J          J          J
K          K          K
G>L        L<C       L
M          M          M
N          N          N
F>O        O<D       O
@ABCDEFGHIJKLMNO0123456789:;<=>? @ABCDEFGHIJKLMNO0123456789:;<=>?
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:  NEW LOGICAL NETWORK ID:
                      F12-HELP

```

In this example, the default foreground color is set to yellow and the default background color is set to blue.

Example for Setting Transparent Background

The following example shows the control string that is sent to the ATM to set a transparent background. In this example, a transparent background is set behind the text “PLEASE REMOVE CARD.”

```

BASE24-DCT    SCREEN DATA FORMAT    LLLL    YY/MM/DD hh:mm 01 OF 01
GROUP    0 SCREEN 0 00 LANG 0 DEF CHAR DESIG 1 CLR SCRN PRIOR N VOICE ____ _
@ABCDEFGHJKLMNO0123456789:;<=>?    @ABCDEFGHJKLMNO0123456789:;<=>?
@
A
B
C
D
E
I>F
G
H
H>I
J
K
G>L
M
N
F>O
@ABCDEFGHJKLMNO0123456789:;<=>?
*****
NEW PAGE:
FILE DESTINATION:
F12-HELP
@
A
B
C
D
E
F<A
G
H
I<B
J
K
L<C
M
N
O<D
O
@
A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
@
A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
=
*****
NEW LOGICAL NETWORK ID:

```

Example for Turning Off Transparent Background Feature

To turn off the transparent background feature, another control string setting the background color to its previous color is sent. This can be accomplished by setting the default background color or specifically setting the background color. The following example shows how the transparent background is turned off by setting the background to the previous color, which in this example is blue.

```
BASE24-DCT    SCREEN DATA FORMAT    LLLL    YY/MM/DD hh:mm 01 OF 01
GROUP      0 SCREEN 0 00 LANG 0 DEF CHAR DESIG 1 CLR SCRN PRIOR N VOICE ____ _
@ABCDEFGHIJKLMNO0123456789:;<=>?    @ABCDEFGHIJKLMNO0123456789:;<=>?
@          @          @          @
A          A  A  =          A
B          B  B          B
C          PLEASE          C  C          %74
D          D  D          D
E          REMOVE          E  E          E
I>F          F<A  F          F
G          CARD          G  G          G
H          H  H          H
H>I          I<B  I          I
J          J  J          J
K          K  K          K
G>L          L<C  L          L
M          M  M          M
N          N  N          N
F>O          O<D  O          O
@ABCDEFGHIJKLMNO0123456789:;<=>?    @ABCDEFGHIJKLMNO0123456789:;<=>?
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                      F12-HELP
```

Note: The transparent control string and the control string for setting the background color should be entered on the same screen.

Financial Institution Table Screen

The Financial Institution Table screen is used only when NCR ATMs are set up to perform DES PIN encryption. The financial institution table (FIT) is a set of data and instructions used for PIN encryption and verification. Hexadecimal numbers are used for identifying fields on the Financial Institution Table screen.

The terminal can be downloaded with up to 255 FITs. The format of a FIT is described in the FIT data of the *NCR NDC Programmer's Reference Manual* for NCR 50XX-series terminals, and the *NCR NDC+ Programmer's Reference Manual* for NCR 56XX-series terminals. When a card is read, the FITs are searched to find the FIT that matches the information on the card. Parameters in this FIT and following linked FITs, are then used for all subsequent PIN processing and also, if required, to direct state table sequencing.

This screen is displayed when you place the cursor to the left of the FINANCIAL INSTITUTION TABLES option and press the **F1** key on the NCR 5XXX Configuration File Menu. The Financial Institution Table screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT   FINANCIAL INST TABLE   LLLL           YY/MM/DD  hh:mm  01 OF 01

GROUP:      0   FIT NUMBER:      0
PIDDx:  ____ ( INSTITUTION ID INDEX )
PFIID:  _____ ( INSTITUTION ID NUMBER )
PSTDx:  ____ ( INDIRECT NEXT STATE INDEX/LOGO NUMBER )
PAGDx:  ____ ( ALGORITHM/BANK ID INDEX )
PMXPN:  ____ ( MAXIMUM PIN DIGITS )
PCKLN:  ____ ( PIN CHECK LENGTH )
PINPD:  ____ ( PIN PAD )
PANDx:  ____ ( PRIMARY ACCOUNT NUMBER INDEX )
PANLN:  ____ ( PRIMARY ACCOUNT NUMBER LENGTH )
PANPD:  ____ ( PIN PAD )
PRCNT:  ____ ( PIN RETRY COUNT INDEX )
POFDx:  ____ ( PIN OFFSET INDEX )
PDCTB:  _____ ( DECIMALISATION TABLE )
PEKEY:  _____ ( ENCRYPTED PIN KEY )
PINDx:  ____ ( INDEX REFERENCE POINTS )
PLNDx:  ____ ( LANGUAGE CODE INDEX )

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP

```

GROUP — The group number associated with the FIT group indicated in the FIT NUMBER field.

Field Length: 1–4 numeric characters
Required Field: No
Default Value: 0

FIT NUMBER — The number associated with the financial institution table (FIT) group to be impacted. Valid values are 0 to 254. The FIT numbers within a FIT group must be consecutive; when adding a record you cannot skip a value.

Field Length: 1–3 numeric characters
Required Field: Yes
Default Value: 0

PIDDX (INSTITUTION ID INDEX) — The index that specifies the location of the financial institution identification (FIID) number on the card.

Field Length: 2 hexadecimal characters
Required Field: Yes
Default Value: No default value

PFIID (INSTITUTION ID NUMBER) — The financial institution identification (FIID) number on the card.

Field Length: 10 hexadecimal characters
Required Field: Yes
Default Value: No default value

PSTDY (INDIRECT NEXT STATE INDEX/LOG NUMBER) — The index for entries in the indirect next state table. The indirect next state is either the FIT Switch state or the Enhanced FIT Switch state.

Field Length: 2 hexadecimal characters
Required Field: Yes
Default Value: No default value

PAGDX (ALGORITHM/BANK ID INDEX) — A two-digit hexadecimal number that specifies the location of the Diebold algorithm number on the card. Valid values are 00 through 7F. A value of 00 indicates that the DES verification method is used; any other valid value indicates that the Diebold verification method is used.

Field Length: 2 hexadecimal characters
Required Field: Yes
Default Value: No default value

PMXPN (MAXIMUM PIN DIGITS) — The maximum number of PIN digits that a cardholder is allowed to enter as a PIN.

Field Length: 2 hexadecimal characters
Required Field: Yes
Default Value: No default value

PCKLN (PIN CHECK LENGTH) — The maximum number of PIN digits used for local PIN verification. Valid values are 00–44.

Field Length: 2 hexadecimal characters
Required Field: Yes
Default Value: No default value

PINPD (PIN PAD) — The character used to pad PINs for transmission to the BASE24 system and to designate the encryption method used.

Field Length: 2 hexadecimal characters
Required Field: Yes
Default Value: No default value

PANDX (PRIMARY ACCOUNT NUMBER INDEX) — The index that specifies the location of the primary account number (PAN) on the cardholder's card.

Field Length: 2 hexadecimal characters
Required Field: Yes
Default Value: No default value

PANLN (PRIMARY ACCOUNT NUMBER LENGTH) — The length of the primary account number (PAN).

Field Length: 2 hexadecimal characters
Required Field: Yes
Default Value: No default value

PANPD (PIN PAD) — The character used to pad the Primary Account Number (PAN) field for encryption.

Field Length: 2 hexadecimal characters
Required Field: Yes
Default Value: No default value

PRCNT (PIN RETRY COUNT INDEX) — The index that specifies the location of the PIN Retry Count field on the card.

Field Length: 2 hexadecimal characters
Required Field: Yes
Default Value: No default value

POFDX (PIN OFFSET INDEX) — The index that specifies the location of the PIN Offset field on the card.

Field Length: 2 hexadecimal characters
Required Field: Yes
Default Value: No default value

PDCTB (DECIMALISATION TABLE) — The decimalization table used in the encryption process.

Field Length: 16 hexadecimal characters
Required Field: Yes
Default Value: No default value

PEKEY (ENCRYPTED PIN KEY) — The encrypted PIN verification key, using the DES verification method. This method is used in the BASE24-atm system only if local PIN verification is used.

Field Length: 16 hexadecimal characters
Required Field: Yes
Default Value: No default value

PINDX (INDEX REFERENCE POINTS) — The track and index reference point for all card-related entries in the FIT.

Field Length: 6 hexadecimal characters
Required Field: Yes
Default Value: No default value

PLNDX (PIN LANGUAGE INDEX) — A two-digit hexadecimal number that specifies the location of the language code on the card as the offset from a delimiter specified in the PINDX field (digit 5). The terminal uses the language code in conjunction with the Language Code Switch state to display multilanguage screens during a transaction. Valid values are 00 through 7F. A value of 00 is not allowed if the language code is used.

Field Length: 2 hexadecimal characters
Required Field: Yes
Default Value: No default value

Supervisor Card Data Screen

The Supervisor Card Data screen allows the configuration of predefined supervisor transaction data to support the Rear Supervisor/Settlement transaction. The Rear Supervisor/Settlement transaction enables personnel to initiate administrative transactions from the rear of the ATM without the use of an administrative card.

The Supervisor Card Data screen incorporates the SPVR TRANS SCREEN facility and, when downloaded to the ATM, overrides the STDATA File on the NDC system disk. After SPVR TRANS SCREEN data has been configured and downloaded to the ATM and the 15 SPVR TRAN option is selected from the replenishment menu, administrative personnel are prompted to enter additional data that further defines the administrative transaction. Up to ten SPVR TRANS SCREENs can be defined, using either screen C09, or if the terminal is running NDC+ release 4.0 or greater, using screens C10 through C19. Each of these screens can be defined with different supervisor transactions.

For further information on the SPVR TRANS SCREEN facility and the Rear Supervisor/Settlement transaction, refer to the *NCR NDC+ Supervisor Reference Manual*.

The Supervisor Card Data screen is displayed when you place the cursor to the left of the Supervisor Card Data option and press the **F1** key on the NCR 5XXX Configuration File Menu. The Supervisor Card Data screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT   SUPERVISOR CARD DATA   LLLL   YY/MM/DD  hh:mm  01 OF 01

GROUP:      0  SCREEN:                TRAN REQUEST STATE NUM:    0      PIN COLLECT: Y
TRACK 1 DATA:

TRACK 2 DATA:
TRACK 3 DATA:

      ATM FUNC-   FUNCTION      CASSETTE      NUMBER OF BILLS
          0
          1
          2
          3
          4
          5
          6
          7
          8
          9

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP

```

GROUP — The group number associated with this screen. Refer to the topic [“NCR 5XXX Configuration File Menu”](#) for a list of valid values for this field.

Field Length: 1–4 numeric characters
 Required Field: No
 Default Value: 0

TRAN REQUEST STATE NUM — The transaction request state number for this administrative request. The number in this field must match the state number that is entered for the Transaction Request state (state I), which is added to the configuration file for the screen C09 administrative functions. The default state number for the Transaction Request state (state I) is 109. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
 Required Field: No
 Default Value: 0

PIN COLLECT — A flag that indicates if the PIN should be entered for the transaction. The only valid value is Y.

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: Y

TRACK 1 DATA — The Track 1 data as it would appear on the administrative card, using valid track values including start (;) and end (?) sentinels. If no Track 1 data is used, enter the end sentinel only.

Field Length: 1–78 alphanumeric characters
Required Field: Yes
Default Value: No default value

TRACK 2 DATA — The Track 2 data as it would appear on the administrative card, using valid track values that include start (;) and end (?) sentinels. If no Track 2 data is used, enter the field separators only.

Field Length: 1–39 alphanumeric characters
Required Field: Yes
Default Value: No default value

TRACK 3 DATA — The Track 3 data as it would appear on the administrative card, using valid track values that include start (;) and end (?) sentinels. If no Track 3 data is used, enter the end sentinel only.

Field Length: 1–106 alphanumeric characters
Required Field: Yes
Default Value: No default value

ATM FUNC — The function number. Up to 10 functions can be defined for each screen.

Field Length: System protected
Occurs: 10 times

FUNCTION — A code indicating the administrative function to be performed. Valid values are as follows:

- 1 = Balance, current cash
- 2 = Balance, standard cash
- 3 = Cash increase
- 4 = Cash decrease
- 5 = Subtotals

Field Length: 1 alphanumeric character
Occurs: 1–10 times
Required Field: No
Default Value: No default value

CASSETTE — The number of the cassette for which the cash increase or decrease function is being performed. Valid values are 1 through 4 and (?). A value of (?) indicates that this field is entered at the ATM during servicing.

Field Length: 1 alphanumeric character
Occurs: 1–10 times
Required Field: Yes, if the value in the FUNCTION field indicates a cash increase or cash decrease administrative function.
Default Value: No default value

NUMBER OF BILLS — The number of bills to increase or decrease in the specified cassette. Valid values are any whole amount or (?). A value of (?) indicates that this field is entered at the ATM during servicing.

Field Length: 1–8 numeric characters
Occurs: 1–10 times
Required Field: Yes, if the value in the FUNCTION field indicates a cash increase or cash decrease administrative function.
Default Value: No default value

Kxx Screen Data Format Screen

The Kxx Screen Data Format screen allows you to customize the printing of receipts. Up to nine blank lines allow the entry of as many control codes as you want to set. This screen is displayed when you place the cursor to the left of the Printer Control Screen (Kxx Screens) option and press the **F1** key on the NCR 5XXX Configuration File Menu.

Note: If you press the **F12** key from this screen, the typical function key help screen is displayed. However, this help screen has additional help screens available that list information on the printer control codes. Press the **Shift-F2** key while the original help screen is displayed to display the first additional help screen. Press the **Shift-F2** key while the first additional help screen is displayed to display the second additional help screen.

The Printer Control (Kxx Screens) screen is shown below, followed by descriptions of its fields.

```
BASE24-DCT   KXX SCREEN DATA FRMT   LLLL           YY/MM/DD  hh:mm  01 OF 01
GROUP       0  SCREEN  K    1  (RECEIPT-INITIAL CONTROLS)_____
```

<print control codes>

VALID CONTROL CODES:

ESC-TM, <nnn>	Set Top Margin	ESC-DC, <fname>	Set Char Set File Name
ESC-LF, <n>	Set Line Feed	ESC-BI, <n>, <fname>	Set Bit Image File Name
ESC-G0, <x>	Set G0 Char Set	ESC-HRI, <n>	Set HRI Print Position
ESC-G1, <x>	Set G1 Char Set	ESC-BCW, <n>	Set Bar Code Width
ESC-LM, <nnn>	Set Left Margin	ESC-BCH, <nnn>	Set Bar Code Height
ESC-RM, <nnn>	Set Right Margin	ESC-PTH, <path>	Set Printer Path (K07)
SI	Select G0 Char Set		

***** BASE24 *****

```
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP
```

GROUP — The group number associated with this screen. Refer to the topic [“NCR 5XXX Configuration File Menu”](#) for a list of valid values for this field.

Field Length: 1–4 numeric characters

Required Field: Yes

Default Value: 0

SCREEN K — The screen number that contains the printer control codes. Valid values are 1 through 3, 6 through 10, and 14. For more information about screen numbers, refer to the *NCR NDC Programmer’s Reference Manual* for NCR 50XX-series terminals, and the *NCR NDC+ Programmer’s Reference Manual* for NCR 56XX-series terminals.

Field Length: 1–2 numeric characters followed by a system-protected text description

Required Field: Yes

Default Value: 1

<print control codes> — A code or codes indicating how the text on the receipt looks. You can enter as many codes as necessary to format the printing on the receipts. If you enter more than one code, use a comma as a separator. For more information on printer control codes, refer to the *NCR NDC Programmer’s Reference Manual* for NCR 50XX-series terminals, and the *NCR NDC+ Programmer’s Reference Manual* for NCR 56XX-series terminals. Valid codes are as follows:

Code	Description
ESC-TM, <i>nnn</i>	Indicates where the printing starts on the receipt, where <i>nnn</i> is a value divided by 144 to obtain the number of inches from the bottom of the receipt. Valid values are 18 through 288.
ESC-LF, <i>n</i>	Specifies how many lines per inch (LPI) are printed on the receipt, where <i>n</i> is the number of lines. Valid values are as follows: 0–1 = 7.5 LPI for journal and consumer receipts; 8.0 LPI for statement print receipts 2–3 = 6.0 LPI for all receipts 4–7 = 5.0 LPI for journal and consumer receipts; ignored for statement print receipts

Code	Description
ESC-G0, <i>x</i>	Specifies which G0 character set prints on the receipt, where <i>x</i> is a G0 character set. Valid values are 1 through P (hexadecimal 31 through 50).
ESC-G1, <i>x</i>	Specifies which G1 character set prints on the receipt, where <i>x</i> is a G1 character set. Valid values are 1 through P (hexadecimal 31 through 50).
ESC-LM, <i>nnn</i>	Specifies how far from the left edge of the receipt the printing starts, where <i>nnn</i> is the number of characters from the left edge of the receipt paper. Valid values are 0 through 133 and are determined by receipt type. Refer to the appropriate NCR programmer's manual for more information about valid values for each receipt type.
ESC-RM, <i>nnn</i>	Specifies how far from the right edge of the receipt where the printing stops, where <i>nnn</i> is the number of characters from the left edge of the receipt paper. Valid values are 2 through 147 and are determined by receipt type. Refer to the appropriate NCR programmer's manual for more information about valid values for each receipt type.
SI	Specifies that the G0 character set is used.
ESC-DC, <i>fname</i>	Specifies the file name of the file where a user-defined character set is stored, where <i>fname</i> is the valid OS/2 file name.
ESC-BI, <i>n, fname</i>	Specifies which bit image is to be displayed and the file name where the bit images are stored, where <i>n</i> is the bit image (valid values are 1 or 2) and <i>fname</i> is the valid OS/2 file name. For K14 screens only, valid values for bit image <i>n</i> are 1 through 9.

Code	Description
ESC-HRI, <i>n</i>	Specifies whether or not the value associated with the bar code (also known as a human-readable interface code) is printed with the bar code, and if so, where it is printed in relationship to the bar code, where <i>n</i> is the position of the value. Valid values are as follows: 1 = The value associated with the bar code is not printed. 2 = Print the value associated with the bar code above the bar code. 3 = Print the value associated with the bar code below the bar code. 4 = Print the value associated with the bar code above and below the bar code.
ESC-BCW, <i>n</i>	Specifies how wide the printed bar code is, where <i>n</i> is the width of the bar code in dots. Valid values are 2 through 4.
ESC-BCH, <i>nnn</i>	Specifies how high the printed bar code is, where <i>nnn</i> is the height of the bar code in dots. Valid values are 1 through 255.
ESC-PTH, <i>path</i>	Specifies the location of the printer data files, where <i>path</i> is a valid OS/2 path that defines the location of the printer files. This code is allowed for screen K07 only.

Configuration Identity Number Screen

The Configuration Identity Number screen is used to establish the current configuration version present at the terminal. You are then able to identify which configuration data ATMs are using.

This screen is displayed when you place the cursor to the left of the Configuration Identity Number option and press the **F1** key on the NCR 5XXX Configuration File Menu. The Configuration Identity Number screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT   CONFIGURATION ID NUM   LLLL       YY/MM/DD  hh:mm  01 OF 01

GROUP       0

CONFIGURATION ID NUMBER      0

```

```

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP

```

GROUP — The number used to identify a particular load group (i.e., a collection of states, screens, tables, and parameters) that is downloaded to the ATM. Each group provides certain functionality (e.g., group 0 is the basic group, group 1 provides Fast Cash support, etc.). The group number of each ATM is assigned in the TERMINAL GROUP field on ATD screen 7. Refer to the topic [“NCR 5XXX Configuration File Menu”](#) for a list of valid values for this field.

Field Length: 1–4 numeric characters
 Required Field: No
 Default Value: 0

CONFIGURATION ID NUMBER — The configuration ID number identifies the current configuration version present at the terminal. At the time of installation, the configuration ID number is set to 1. When configuration data is updated, the configuration ID number must be set to the new configuration version, and the Configuration ID Number Load message must be the last item of data downloaded to the terminal. Valid values are 1 through 9999.

Field Length: 1–4 numeric characters
Required Field: Yes
Default Value: 0

Hopper Mapping Table Screen

The Hopper Mapping Table screen enables you to determine the currency denominations and currency types present within each hopper.

This screen is displayed when you place the cursor to the left of the Hopper Mapping Table option and press the **F1** key on the NCR 5XXX Configuration File Menu. The Hopper Mapping Table screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT  CONFIGURATION ID NUM  LLLL  YY/MM/DD  hh:mm  01 OF 01

      GROUP      0

      HOPPER      HOPPER      CURRENCY
      NUMBER      VALUE      TYPE
      1           0           0
      2           0           0
      3           0           0
      4           0           0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP
  
```

GROUP — The number used to identify a particular load group (i.e., a collection of states, screens, tables, and parameters) that is downloaded to the ATM. Each group provides certain functionality (e.g., group 0 is the basic group, group 1 provides Fast Cash support, etc.). The group number of each ATM is assigned in the **TERMINAL GROUP** field on ATD screen 7. Refer to the topic [“NCR 5XXX Configuration File Menu”](#) for a list of valid values for this field.

Field Length: 1–4 numeric characters
 Required Field: No
 Default Value: 0

HOPPER NUMBER — The number of the hopper being edited. You do not have to use all of the hopper numbers or use consecutive hopper numbers.

Field Length: System protected
Occurs: 4 times

HOPPER VALUE — The denomination of the currency in the hopper.

Field Length: 1–5 numeric characters
Occurs: 4 times
Required Field: No
Default Value: 0

CURRENCY TYPE — The type of currency in the hopper. The only valid value is 01.

Field Length: 1–2 numeric characters
Occurs: 4 times
Required Field: Yes, if the HOPPER VALUE field contains a value.
Default Value: 0

Print Menu

The Print Menu allows you to print part or all of the configuration file information. You enter the configuration file name that you want to print on the NCR 5XXX Configuration File Menu before selecting the Print option.

The information to be printed is sent to the printer location entered on the Logon screen. If you do not designate a printer at logon, the system defaults to \$S.#HOLD, causing the print job to be placed in a print queue but not printed. The print location must then be changed to a valid print destination before it is printed.

Additional commands cannot be executed while a PRINT request is in progress. Once the print job is completed, the screen displays a text line indicating "PRINT REQUEST COMPLETE", and the cursor is displayed at the particular print option selected.

To use these fields, place the cursor to the left of the field name and press the **F1** key.

This menu is displayed when you place the cursor to the left of the Print option and press the **F1** key on the NCR 5XXX Configuration File Menu. The Print Menu is shown below, followed by descriptions of its fields and function keys.

```
BASE24-DCT   NCR 50 PRINT MENU           LLLL           YY/MM/DD  hh:mm  01 OF 01
```

```
_  PRINT ALL LOAD INFORMATION
```

```
GROUP NUMBER:  0000
```

```
_  SCREENS
```

```
_  STATES
```

```
_  FINANCIAL INSTITUTION TABLES  ( FIT )
```

```
_  XREF  ( STATES CALLING SCREENS )
```

```
_  ALL IN GROUP ( INCLUDES CONFIG IDENTITY AND
                     ENHANCED CONFIG PARAMETERS )
```

```
PLACE CURSOR IN FRONT OF DESIRED FUNCTION AND DEPRESS F1
```

```
***** BASE24 *****
```

```
NEW PAGE:           FILE DESTINATION:           NEW LOGICAL NETWORK ID:
F1-SELECT           F10-PRINT           F16-EXIT           SF16-LOGOFF
```

PRINT ALL LOAD INFORMATION — Prints all load information for all group numbers in the ATD, such as screens, states, FITs, state/screen cross reference tables, identity numbers, and configuration parameters.

Field Length: 1 alphanumeric character
Required Field: No
Default Value: No default value

GROUP NUMBER — Specifies a group number for printing load information for a specific group. Refer to the topic [“NCR 5XXX Configuration File Menu”](#) for a list of valid values for this field.

Field Length: 4 numeric characters
Required Field: No
Default Value: 0000

SCREENS — Prints the base screen, then prints the base screen including overlays, then prints the next base screen, for the group number specified in the GROUP NUMBER field.

Field Length: Cursor placement only

STATES — Prints all states for the group number specified in the GROUP NUMBER field.

Field Length: Cursor placement only

FINANCIAL INSTITUTION TABLES (FIT) — Prints all FITs for the group number specified in the GROUP NUMBER field.

Field Length: Cursor placement only

XREF (STATES CALLING SCREENS) — Prints the state/screen cross reference tables for the group number specified in the GROUP NUMBER field. These tables provide a summary of the states that access each screen number.

Field Length: Cursor placement only

ALL IN GROUP (INCLUDES CONFIG IDENTITY AND ENHANCED CONFIG PARAMETERS) — Prints all load information for the group number specified in the GROUP NUMBER field, including screens, states, FITs, state/screen cross reference tables, identity numbers, and configuration parameters.

Field Length: Cursor placement only

Function Keys

The use of one function key on the Print Menu varies from the standard function keys explained earlier in this section for the NCR 5XXX Configuration Menu. The use of this function key is explained below.

The first column of information below shows the BASE24 key. The second column describes the function that can be accomplished with this function key.

Key	Description
F1	Select — Checks the selected field for errors and, if the selection is valid, prints the selection at the specified print location.

PIN Processing Requirements

The BASE24-atm product supports local PIN verification and ANSI (PIN/PAN) PIN block formats for NCR terminals. Specific configuration requirements exist for each of these features, as described below.

Local PIN Verification

The BASE24-atm product supports local PIN verification, in which NCR terminals that have PIN verification capability perform PIN verification at the device for financial and administrative transactions without the involvement of the BASE24 network.

Local PIN verification requires changes to the PIN Entry state, the Transaction Request state, and the Financial Institution Table (FIT) in the NCR 5XXX Configuration File (CNFG5070). The following changes must be made to support local PIN verification.

PIN Entry State

The GOOD PIN CHECK field on the PIN Entry State screen must be configured to specify the next state to which the terminal proceeds upon successful local PIN verification. The REMOTE PIN CHECK field on the PIN Entry State screen must contain the value 255 (not used).

Transaction Request State

The SEND PIN BUF A field on the Transaction Request State screen must be set to a value of 0 (do not send PIN buffer A data).

Financial Institution Table

The financial institution table (FIT) data specifies parameters to be used depending on the terminal PIN verification method to be used. The following field values are important to support local PIN verification. These fields are described in the topic “Financial Institution Table Screen.”

Field	Value	
PINPD	This field must be set to 00.	
PRCNT	The value is FF indicating that the PIN-retry count is not on the ATM card, but in the PIN Entry state (B) table.	
PDCTB	You are responsible for the values for this field.	
PEKEY	You are responsible for the values for this field.	
PINDX	The fifth PINDX digit is used for the PIN Language Code Index.	
PCKLN	The valid values for the entire PCKLN byte, excluding Identkey, are shown below.	
	Terminal Verification	PCKLN Value (Hexadecimal)
	None *	00
	DES	04–10
	Diebold	40–44
	VISA	20–24
PAGDX	The valid values for the PAGDX byte are shown below.	
	Terminal Verification	PAGDX Value (Hexadecimal)
	DES	00
	Diebold	01–7F

* The terminal does not perform local PIN verification.

ANSI PIN Block Format

To transmit the ANSI PIN block format, the FIT must be modified so that the ATM device transmits the ANSI format. The following field values must be set in the FIT so that devices can transmit this PIN block format. These fields are described in the topic “Financial Institution Screen.”

FIT Field	Setting
PMXPN	90
PCKLN	00
PINPD	0F
PANLN	1F

Multiple Language Support Requirements

The BASE24-atm product supports the Multiple Language Support feature, in which NCR 5XXX terminals can provide cardholder screens and receipts in up to three languages.

The BASE24-atm NCR 5XXX Device Handler process supports the Multiple Language Support feature in the Language Code Switch state (state S). The Language Code Switch State screen enables you to configure the next state for each language. The Language Code Switch State screen is described earlier in this section.

The following states require changes when using the Multiple Language Support feature:

- PIN Entry state 10
- 8 FDK Selection state 11
- Global Extension state 12

In addition, the Language Selection screen (screen 12) must be modified when using the Multiple Language Support feature.

Section 5 contains information about formatting receipts for the Multiple Language Support feature.

PIN Entry State 10

The REMOTE PIN CHECK field on the PIN Entry State screen for state 10 must be set to a value of 11 to support the Multiple Language Support feature. Currently this state is configured to bypass any language selection. The following screen illustration shows the PIN Entry State screen when it is configured for the Multiple Language Support feature.

```

BASE24-DCT  STATE TABLE DATA          LLLL          YY/MM/DD  hh:mm  03 OF 40

      GROUP      0      STATE NUMBER  10

                                PIN ENTRY STATE

      STATE      SCREEN      TIMEOUT      CANCEL      GOOD PIN      BAD PIN
      TYPE      NUMBER      STATE      STATE      CHECK      CHECK
      B          11          141          137          0          0

                                LOCAL PIN      REMOTE PIN      LOCAL PIN
                                ERROR      CHECK      MAX TRIES
                                255          11          9

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                      F12-HELP

```

8 FDK Selection State 11

State 11 is configured to support three languages. If you are using two languages you must update the KEYS ACTIVE MASK field on the 8 FDK Selection State screen for state 11 to a value of 3. This activates keys A and B on terminal screen 012 (the language selection screen). See the topic, “Language Selection Screens,” below, for an illustration of screen 012 for a terminal that supports English and Spanish.

State 12 Extension Screen

State 12 can be configured for up to three languages. The Global extension screen for this state enables you to define the number of languages in use for the ATM. The TABLE ENTRY-2 field is reserved for the primary language of the ATM. This field is set to a default value of 0. TABLE ENTRY fields 3 and 4 are for additional languages. The TABLE ENTRY-3 field is for the first additional language. If your ATM uses a second language, you must change the value in this field to 1. The TABLE ENTRY-4 field is for the second additional language and must contain a value of 2 if a second additional language is in use. The TABLE ENTRY-4 through TABLE ENTRY-9 fields should contain a value of 0. A sample Global Extension screen for state 12 is shown below. In this sample, the ATM is configured to support one additional language.

BASE24-DCT STATE TABLE DATA LLLL YY/MM/DD hh:mm 25 OF 40

GROUP 0 STATE NUMBER 12

GLOBAL EXTENSION

STATE	TABLE	TABLE	TABLE	TABLE	TABLE	TABLE
TYPE	ENTRY-2	ENTRY-3	ENTRY-4	ENTRY-5	ENTRY-6	ENTRY-7
Z	0	1	0	0	0	0

TABLE TABLE

ENTRY-8 ENTRY-9

0 0

***** BASE24 *****

NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:

F12-HELP

Language Selection Screens

Cardholder screens for each language supported by the ATM can be formatted using the Screen Data Format screen. A language selection screen (screen 012) is the language selection screen and enables cardholders to select the language to be used. The following screen layout example shows screen 012 when the Multiple Language Support feature is configured in English as the primary language and Spanish as an additional language.

```

BASE24-DCT    SCREEN DATA FORMAT          LLLL          YY/MM/DD  hh:mm  01 OF 01
GROUP      0 SCREEN 0 12 LANG 0 DEF CHAR DESIG 1 CLR SCR N PRIOR N VOICE ____ _
@ABCDEFGHIJKLMNO0123456789:;<=>?          @ABCDEFGHIJKLMNO0123456789:;<=>?
@
A          PLEASE INDICATE YOUR              A    A
B          LANGUAGE PREFERENCE                B    B
C
D
E
I>F          ENGLISH-----> F<A    F
G
H
H>I          ESPANOL-----> I<B    I
J
K
G>L
M
N
F>O          O<D    O
@ABCDEFGHIJKLMNO0123456789:;<=>?          @ABCDEFGHIJKLMNO0123456789:;<=>?
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

You also must translate all other screens into the second language. These screens must be added to the configuration file. For each of these screens, the LANG field is set to a value of 1. If you are using a third language, the LANG field is set to a value of 2.

The following is an example of the Screen Data Format screen used to format the Transaction Selection screen (screen 014) in English. Note that the SCREEN field is set to a value of 014 and the LANG field is set to a value of 0.

```
BASE24-DCT    SCREEN DATA FORMAT    LLLL    YY/MM/DD  hh:mm  01 OF 01
GROUP      0  SCREEN 0 14 LANG 0 DEF CHAR DESIG 1 CLR SCRN PRIOR N VOICE  _ _

      @ABCDEFHIJKLMNO0123456789:;<=>?      @ABCDEFHIJKLMNO0123456789:;<=>?
      @                                     @ @                                     @
      A  PLEASE SELECT YOUR TRANSACTION      A  A                                     A
      B      BY PRESSING THE KEY              B  B                                     B
      C                                         C  C                                     C
      D                                         D  D                                     D
      E                                         E  E                                     E
      I>F <-DEPOSIT          WITHDRAWAL-----> F<A  F                                     F
      G                                         G  G                                     G
      H                                         H  H                                     H
      H>I <-TRANSFER          BALANCE INQUIRY-> I<B  I                                     I
      J                                         J  J                                     J
      K                                         K  K                                     K
      G>L <-PAYMENT          PIN CHANGE-----> L<C  L                                     L
      M                                         M  M                                     M
      N                                         N  N                                     N
      F>O          OTHER-----> O<D  O                                     O
      @ABCDEFHIJKLMNO0123456789:;<=>?      @ABCDEFHIJKLMNO0123456789:;<=>?
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                      F12-HELP
```

The following screen is screen 014 translated into the second language of the ATM, Spanish. Note that the SCREEN field is defined with a value of 014 and the LANG field is defined with a value of 1.

```

BASE24-DCT  SCREEN DATA FORMAT  LLLL  YY/MM/DD  hh:mm  01 OF 01
GROUP      0  SCREEN 0 14 LANG 1 DEF CHAR DESIG 1 CLR SCRN PRIOR N VOICE  _ _

      @ABCDEFGHIJKLMNO0123456789:;<=>?      @ABCDEFGHIJKLMNO0123456789:;<=>?
      @      @      @      @
      A      POR FAVOR SELECCIONE      A      A      A
      B      SU TRANSACCION      B      B      B
      C      OPRIMENDO EL BOTON      C      C      C
      D      D      D      D
      E      E      E      E
      I>F <-DEPOSITO      RETIRO-----> F<A      F      F
      G      G      G      G
      H      INQUISION      H      H      H
      H>I <-TRANSFERENCIA DEL SALDO-----> I<B      I      I
      J      J      J      J
      K      K      K      K
      G>L <-PAGO      CAMBIO DEL PIN-> L<C      L      L
      M      M      M      M
      N      OTROS-----> N      N      N
      F>O      O<D      O      O

      @ABCDEFGHIJKLMNO0123456789:;<=>?      @ABCDEFGHIJKLMNO0123456789:;<=>?
      ***** BASE24 *****
      NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                        F12-HELP

```

To provide this screen for a third language, change the value in the SCREEN field to 014 and the value in the LANG field to 2.

5: ATM Printer Support

Standard default receipt formats based on NCR guidelines are provided with the BASE24 product. You can customize these default receipt formats. This section provides examples of default financial and administrative receipt formats as well as instructions on how to configure customized receipts. Refer to NCR vendor documentation for information on specifications for the physical properties and standard print formats supported by NCR.

The BASE24-atm self-service banking (SSB) add-on product modifies some of the receipts shown here, and provides additional receipts. Refer to the ***BASE24-atm NCR 5XXX SSB Manual*** for information on these modifications.

BASE24-atm Receipts

The BASE24-atm product can produce a printed receipt for the cardholder, which provides a tangible record of the electronic transaction that just took place. Receipts are also printed on the journal printer, so that the ATM owner has a record of the transaction.

Types of Printed Receipts

The BASE24-atm product can print receipts for the following types of transactions:

- Financial transactions
- Statement print transactions
- Administrative transactions

Financial Transactions

Each financial transaction has its own type of receipt, so that the information provided to the cardholder is tailored to the requirements of the transaction. A denied transaction also has its own type of receipt. Whether or not a receipt is printed can be determined by you at the terminal level, or you can allow cardholders to decide if they want receipts to be printed.

Statement Print Transactions

Statement print transactions also produce receipts given to the cardholder. However, this manual does not group these transactions with financial transactions because the BASE24-atm product does not control the formatting of the receipt; the host computer is responsible for statement print transaction receipt formatting. Statement print transaction receipts must be printed; cardholders do not have the option of not printing them. The BASE24-atm product does provide methods for print compression of statement print transactions, which reduces the message traffic required. Print compression is described in [appendix A](#).

Administrative Transactions

Administrative transactions are performed by institution personnel to perform such tasks as replenishing supplies and correcting problems. All administrative transactions must be printed for security reasons.

Configurable Receipts

Configurable receipts provide you with the ability to establish ATM receipt formats that are unique to the institution. In addition, this customization option supports the display of an informational or marketing message on the cardholder receipt, the use of flexible monetary symbols, multiple languages, and the option to print the account description and account type on the receipt. The configurable receipt option allows for configuring approved and denied financial transaction receipts as well as administrative receipts.

Terminal Receipt File (TRF)

The Terminal Receipt File (TRF) is used to define the receipt requirements for a particular transaction for both approved and denied transactions. The transactions are defined using *ISO processing codes*. These codes consist of a two-character transaction code, a two-character *from* account type, and a two-character *to* account type.

The key to the TRF is the terminal receipt profile, message category (the second digit of the message type code), and the ISO processing code. The terminal receipt profile is specified in the BASE24-atm Terminal Data files (ATD) and can be assigned to a single terminal or a group of terminals. The Device Handler process calls a utility to map the internal BASE24 transaction code to the corresponding ISO processing code before reading the TRF extended memory table.

The Device Handler process attempts to read the TRF extended memory table using the terminal receipt profile value, the message category code from the transaction, and the ISO processing code for the transaction. The Device Handler process calls a utility to map the internal BASE24 transaction code for the transaction to the corresponding ISO processing code before reading the TRF extended memory table.

Using the terminal receipt profile, message category, and ISO processing code, the Device Handler process steps through the records in the TRF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection:

Preference	Terminal Receipt Profile	Message Category	Processing Code
1	Exact Match	Exact Match	Exact Match
2	Exact Match	*	Exact Match
2	*****	Exact Match	Exact Match
4	*****	*	Exact Match

If none of the searches return a match, the Device Handler process uses the default value of 0 (no printed receipt, no audit receipt) for a declined transaction or the value of 1 (print receipt only, mandatory) for an approved transaction.

Note: Every time the TRF is modified, the TRF extended memory table must be rebuilt using the Base Extended Memory Table Build utility. After the TRF extended memory table is rebuilt, the Device Handler process must reallocate the table. You can reallocate the TRF extended memory table by using the ALTERATTRIBUTE command from the EMT Control Command screen (screen 5) in the Device Control Terminal (DCT). You can also enter the ALTERATTRIBUTE command manually from a network control facility to warmboot an extended memory table for an individual process. Refer to the *BASE24-atm Transaction Processing Manual* for information about the Base Extended Memory Table Build utility. Refer to the *BASE24 Text Command Reference Manual* for more information about the ALTERATTRIBUTE command and the *BASE24 Device Control Manual* for more information on the EMT Control Command screen.

Receipt File (RCPT)

The Receipt File (RCPT) is the primary file used to configure receipts. It contains receipt templates that define the actual text and format to use for the various types of receipts.

The Device Handler process retrieves receipt templates in the RCPT from the RCPT extended memory table and then uses that information for creating receipts.

Initially, the RCPT contains default templates for each of the receipt types. You can use these default templates as they are, or you can modify the templates to print the information you want on the receipt.

Note: Every time the RCPT is modified, the RCPT extended memory table must be rebuilt using the Base Extended Memory Table Build utility. After the RCPT extended memory table is rebuilt, the Device Handler process must reallocate the table. You can reallocate the RCPT extended memory table by using the ALTERATTRIBUTE command from the EMT Control Command screen (screen 5) in the Device Control Terminal (DCT). You can also enter the ALTERATTRIBUTE command manually from a network control facility to warmboot an extended memory table for an individual process. Refer to the ***BASE24-atm Transaction Processing Manual*** for information about the Base Extended Memory Table Build utility. Refer to the ***BASE24 Text Command Reference Manual*** for more information about the ALTERATTRIBUTE command and the ***BASE24 Device Control Manual*** for more information on the EMT Control Command screen.

Receipt Templates

Receipt templates are made up of a series of field markers and field type codes that define the information to be included on the receipt and the format of the receipt. The field markers and field type codes that can be used in the receipt templates are defined later in this section.

Individual receipt templates can be created for the various types of approved and denied financial transactions and approved and denied administrative transactions. These same receipt templates can also be created in different languages.

To determine the text and format to use for a receipt, the Device Handler process scans a template to identify the field markers and field type codes and their positions in the template. Then, based on the field type codes, the Device Handler process retrieves the required information and formats the receipt information based on the field markers used and their positions in the template. Examples of the default templates provided in the RCPT extended memory table for various transactions and examples of actual receipts created from these default templates are included later in this section.

Receipt templates are divided into the following three general categories: receipt header templates, receipt body templates, and receipt trailer templates.

Receipt header templates control the information printed at the top of the receipt for all types of transactions.

Receipt body templates control the information printed on the remainder of the receipt; body templates are specific to the type of transaction (e.g., withdrawal or deposit, approved or denied, financial or administrative).

Receipt trailer templates allow additional marketing information to be placed at the end of each financial transaction. Trailers can be multiple lines, however, the length of potential receipts with header, body, and trailer should be taken into consideration so that the number of lines does not exceed the maximum. Trailers are only allowed on financial transactions.

In order to correctly format receipt information for a transaction, a Device Handler process must be able to retrieve the appropriate templates from its RCPT information.

Receipt Template Keys

Each receipt template is one record in the RCPT, and the key fields for RCPT records are as follows: DEVICE TYPE, RECEIPT TYPE, RECEIPT GROUP, LANG CODE/MSG ID, FIID, FROM ACCT, TO ACCT, and PRNTR TYPE. This group of key fields, described below, provides you with a great deal of flexibility in defining templates. For a comprehensive list of values that can be used in each of these key fields, refer to the ***BASE24-atm Files Maintenance Manual***.

Key Field	Description
DEVICE TYPE	Groups the RCPT records by type of device, allowing the Device Handler process to locate its records at initialization.
RECEIPT TYPE	Identifies the type of receipt template. Valid type codes have been defined by the BASE24 product.
RECEIPT GROUP	Allows the use of a single receipt template for a group of ATM terminals. Receipt groups are specified for each terminal on BASE24-atm Terminal Data files (ATD) screen 8. When selecting template data, the Device Handler process determines the group number assigned to an ATM in the ATD, and then searches for templates with the corresponding group. If an exact match cannot be found for the ATM group, the Device Handler process uses template records with asterisks in their group fields.
LANG CODE/ MSG ID	Allows you to define the same receipt templates for more than one language. That is, a separate receipt template can be created for each of the languages supported by the ATM. A language indicator is sent to the BASE24 system from the ATM after the cardholder selects a preferred language for the transaction. The Device Handler process then uses this language indicator to select the templates to use. For example, you could configure one receipt template in English and one in Spanish in the RCPT. Each would be assigned a corresponding language indicator, which would allow the Device Handler process to select the appropriate template based on the language indicator sent from the ATM.

Key Field	Description
FIID	Allows a receipt to be formatted differently depending on who issued the card. For instance, you may want different information printed for an on-us transaction than a not-on-us transaction.
FROM ACCT	Allows a receipt to be formatted differently depending on the type of <i>from</i> account. For example, you can truncate savings account numbers differently than credit card numbers.
TO ACCT	Allows a receipt to be formatted differently depending on the type of <i>to</i> account. For example, you can truncate savings account numbers differently than credit card numbers.
PRNTR TYPE	Allows a receipt to be formatted differently depending on the type of printer. For instance, you can truncate the account numbers on a financial receipt to help prevent fraud if the receipt falls into the wrong hands. However, you want the entire account number to print on the journal printer so that you have a complete record of the transaction. This field also is used to specify a Digital Audio Service (DAS) record.

Default Receipt Templates

Default receipt templates are provided in the RCPT for each type of transaction. These default receipt templates can be edited by you to customize specific types of information printed on the ATM receipts. Examples of the various default templates are described later in this section.

Multiple Languages

Receipt templates can be configured in more than one language. That is, a separate receipt template can be created for every type of transaction in each of the languages supported by the ATM. The language indicator is sent to the BASE24 system from the ATM after the cardholder selects a preferred language for the transaction. For example, by configuring one receipt template in English and one in Spanish and matching each receipt template to the appropriate language indicator in the RCPT, cardholder receipts can be printed in English or in Spanish.

The Device Handler process uses the eighth byte in the operations code buffer to determine the language to use in screen display and receipt printing.

The Device Handler process uses the language indicator in the RCPT extended memory table, along with the terminal receipt group identifier, receipt type, and device type, to determine the correct receipt template to use.

Components of the Receipt Template

A number of components make up a receipt template. They are as follows:

- Field markers
- Field type codes
- Static text
- Directives

Field Markers

Field markers are special characters that identify the type of data to be used on a receipt or specify an action such as a line feed. The field markers that can be used with financial and administrative configurable receipt templates in the RCPT are described below and on the following pages.

The following table describes the standard field markers that can be used in receipt templates to define receipt text and format.

Code	Description
#	Indicates that a variable data field is to be inserted. The number symbol (#) is immediately followed by a field type code that indicates the type of data to be inserted. The possible field type codes are discussed below in the topic “Field Type Codes.”
>	Indicates that a line feed character (hexadecimal 0A) is to be inserted.
@	Indicates that a literal data string follows.
?	Indicates that a directive statement follows. A directive specifies a command other than printing. The possible directives are discussed in the topic “Directives,” below.

Field Type Codes

Field type codes correspond to predetermined pieces of fixed-length data that the Device Handler process retrieves from the BASE24-atm Standard Internal Message (STM), the ATD, or the N50NAMS. The field type codes that can be used in configurable receipt templates in the RCPT are described on the following pages, and are divided into codes used by financial transactions and codes used by administrative transactions. The field type codes are divided as follows:

- Standard field type codes. These are used with all receipt types.
- Cartridge field type codes. These are used with administrative receipts only.
- Terminal totals field type codes. These are used with administrative receipts only.

Data names are provided throughout the field type code documentation, as applicable, to indicate the specific data fields from which the information is taken. For further information on the STM data fields, refer to the ***BASE24-atm Transaction Processing Manual***. For further information on the ATD data names, refer to the ***BASE24-atm Files Maintenance Manual*** (using the data name index) or refer to the DDLs.

Static Text

You can enter any text in the receipt template that you want to print on the receipt. Static text must be preceded by the at symbol (@) in the template. You must make sure that the length of the static text plus the length of any field type codes used does not exceed the maximum number of characters that can be printed on the receipt.

Special characters that are reserved for use with field markers (#, >, @, ~, and ^) can be printed in the text of the receipt by preceding them with the number symbol (#). For instance, to print the greater than character (>), you would enter #> in the template.

Directives

A directive indicates a command other than printing. The IF and ENDIF directives allow one template to print different text or data fields, depending upon a condition that is determined outside of the RCPT. The IF directive marks the beginning of a block of the receipt template that is printed if the condition is true; on the same line as the IF directive is the variable name of the condition to be tested. The ENDIF

directive marks the end of the block. If the condition is not true, the block is skipped, and the receipt continues printing with the receipt line following the ENDIF directive. Currently, the only valid condition values are as follows:

- FEES and NOFEES
- ISSFEES and NOISSFEES
- NCD and NONCD
- CASH and ENVLP

The FEES and NOFEES condition values can be used only in financial receipt templates. When the FEES condition is on the same line as the IF directive, the block that follows is printed if there is a surcharge fee assessed on the transaction. When the NOFEES condition is on the same line as the IF directive, the following block is printed if there is no acquirer fee for the transaction (i.e., surcharging is not configured, or surcharging is configured, but not applicable for this particular transaction).

The ISSFEES and NOISSFEES condition values can be used only in financial receipt templates. When the ISSFEES condition is on the same line as the IF directive, the block that follows is printed if there is an issuer fee or rebate for the transaction. When the NOISSFEES condition is on the same line as the IF directive, the following block is printed if there is no issuer fee or rebate for the transaction

The NCD and NONCD condition values enable noncurrency data to be displayed on withdrawal transaction receipts. When the NCD condition is on the same line as the IF directive, the block that follows is printed if the transaction is a noncurrency transaction (i.e., if the Non-Currency Dispense token is present). When the NONCD condition is on the same line as the IF directive, the following block is printed if there are no noncurrency items in the transaction (i.e., the Non-Currency Dispense add-on product is not configured, or the Non-Currency Dispense add-on product is configured but not applicable for this particular transaction).

The CASH and ENVLP condition values enable values associated with the NCR Bunch Note Acceptor and Check Processing Module to be printed on receipts. When the CASH condition is on the same line as the IF directive, the block that follows is printed if there was a cash deposit transaction made using the NCR Bunch Note Acceptor. When the ENVLP condition is on the same line as the IF directive, the following block is printed if there is an envelope deposit.

An example of directives is found in the topics “[Withdrawal from Checking, No Surcharge](#)” and “Withdrawal from Checking, with Surcharge,” found later in this section.

Financial Transaction Receipts

The BASE24-atm configurable receipt option supports receipts for both approved and denied transactions. The following pages describe the default receipt templates that are used to configure approved financial transaction receipts as well as denied financial transaction receipts.

Default receipt templates are provided in English only regardless of the language indicator associated with the template. You must then modify the appropriate template if you want another language. For example, default receipt templates for language indicators 0, 1, or 2 are all configured in English. You must modify the templates to correspond to the languages supported. This section shows one example for each default receipt template in the RCPT. However, if more than one language is supported by the device, a receipt template must be established for each language.

Printer data for the NCR receipt and journal printers is printed in columns 1 through 40 unless the ROLL WIDTH or LEFT PRINT COL fields are changed on the Enhanced Configuration Parameters screen in the CNFG5070. The number of characters per line depends on the character set. Printing for chained transactions on one receipt forces an automatic cut if an attempt is made to print on the twenty-fifth line. The automatic cut is not performed on the first transaction printed on the receipt. These restrictions must be considered when modifying receipt templates.

The Device Handler process logs event messages if receipt formatting errors occur when BASE24-atm transaction processing takes place. The Device Handler process also logs event messages if the maximum number of lines per receipt or the print buffer is exceeded when you add a trailer receipt type.

Balance inquiries requested by the cardholder can be printed, displayed on the ATM screen, or both, depending on how the ATD and IDF are configured.

Note: If the BASE24-atm Multiple Currency Dispense add-on product has been installed, the balance amount that appears on the receipt and on the ATM screen is in the Account currency. Because of the multiple currency conversions performed on the account balances, the balance on the receipt and on the screen may not be exactly the same as the balance amount as held by the account issuer. For more information about the BASE24-atm Multiple Currency Dispense add-on product., refer to appendix F.

Optional Receipts

You can determine whether or not receipts are printed for each transaction by several methods. The first method allows you to control receipt and audit printing by transaction type, and by whether the transaction was approved or denied. You enter this information using the Terminal Receipt File (TRF) screens. See the *BASE24-atm Files Maintenance Manual* for information on this file.

The second method allows the cardholder to make the final decision as to whether or not a receipt should be printed. If the cardholder chooses not to have a receipt printed, the journal receipt is still printed if the settings indicate that it should print. Note that the cardholder's choice with the second method overrides the settings in the TRF of the first method.

Configuring Optional Receipts

The optional receipts function is determined by settings on ATD screen 7, which is also a device-specific screen. The OPTIONAL RECEIPTS field on ATD screen 7 has four values, as follows:

- 0 = Optional receipts are not supported.
- 1 = The cardholder selects optional receipts if the printer fails. If the printer is working, the settings from the TRF extended memory table are used.
- 2 = The cardholder selects optional receipts on all approved transactions. If the transaction is denied, the settings from the TRF extended memory table are used. If the transaction is approved, but it cannot be completed because of a hardware fault, the Device Handler process uses the deny transaction template.
- 3 = The cardholder selects optional receipts on all approved and denied transactions.

One exception to the optional receipt selection, no matter which setting you choose, is that statement print transactions must have receipts printed. If the printer fails, the cardholder cannot complete a statement print transaction.

Two additional fields on ATD screen 7, PRINTER OPERABLE GROUP and PRINTER INOPERABLE GROUP, determine how often and where in the flow the cardholder is asked about optional receipts. The inoperable group contains the series of screens that are downloaded when the terminal sends an unsolicited status indicating that the printer is not working. The operable group contains the series of screens that are downloaded initially with group 0 and group 300, and then downloaded again when the printer is once again working. One of the following options is used, depending on the values that you enter in the two group fields.

- The cardholder selects whether or not the receipt is printed once at the beginning of the session (decision applies to all chained transactions).
- The cardholder selects whether or not the receipt is printed during each transaction (decision is for that transaction only).

The groups that ACI provides in a sample configuration file for the optional receipts function are as follows (the options are set in the **OPTIONAL RECEIPTS** field as described above):

- Group 300 = The basic group for the optional receipts function. This group is merged with group 0 and the operable group selected from the rest of this list.
- Group 301 = The printer currently is operational. The cardholder selects optional receipts once for all chained transactions, on all approved (option 2) or approved and denied (option 3) transactions.
- Group 302 = The printer currently is inoperable. The cardholder selects optional receipts once for all chained transactions (option 1, 2, or 3).
- Group 303 = The printer currently is operational. The cardholder selects optional receipts once for all chained transactions, only if the printer becomes inoperable (option 1).
- Group 311 = The printer currently is operational. The cardholder selects optional receipts before each transaction, on all approved (option 2) or approved and denied (option 3) transactions.
- Group 312 = The printer currently is inoperable. The cardholder selects optional receipts before each transaction (option 1, 2, or 3).
- Group 313 = The printer currently is operational. The cardholder selects optional receipts before each transaction, only if the printer becomes inoperable (option 1).

You can use the states, FITs, and screens to create subsets of cardholders that use various settings of the optional receipts functionality. Section 4 describes the screens used to create cardholder subsets. However, the sample configuration file provided by ACI shows only settings for all cardholders.

Transaction Log File

The Transaction Log File (TLF) contains a flag indicating whether or not a receipt was printed. Valid values for the receipt option flag are as follows:

- 0 = The cardholder was not asked whether a receipt was required

- 1 = A receipt was requested
- 2 = A receipt was not printed because the printer has faulted
- 3 = A receipt was not printed because the cardholder requested that one not be printed

You can view this flag in the TLF. Refer to the *BASE24-atm Files Maintenance Manual* for information on how to peruse the TLF. However, this flag is not displayed on any of the printed BASE24 reports.

Samples of Optional Receipt Settings

The following topics show how the different settings of the ATD work together to provide six different scenarios for optional receipts. These settings determine when the optional receipts function is used and how often the cardholders are prompted about whether or not they want to complete a transaction without a printed receipt.

Scenario 1—Prompt Once per Session Only if Printer Fails

To prompt the cardholder only once per session as to whether or not a receipt is printed if the printer fails, enter the following values in the ATD screen 7 fields:

Field	Value
OPTIONAL RECEIPTS	1
PRINTER OPERABLE GROUP	0303
PRINTER INOPERABLE GROUP	0302

If the printer is working, the settings in the TRF extended memory table are used. If the printer faults in the middle of a series of chained transactions, the cardholder is prompted with the message “Continue without a receipt?” during each transaction. If the cardholder selects Yes, the transaction is allowed without a receipt. If the cardholder selects No, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is prompted at the beginning of the session with the message “Continue without a receipt?” If the cardholder selects Yes, then all chained transactions are allowed without a receipt. If the cardholder selects No, the card is returned and the session is ended.

After the printer is serviced and the terminal exits maintenance mode, the printer operable group is downloaded.

Scenario 2—Prompt on Each Transaction Only if Printer Fails

To prompt the cardholder during each transaction as to whether or not a receipt is printed if the printer fails, enter the following values in the ATD fields:

Field	Value
OPTIONAL RECEIPTS	1
PRINTER OPERABLE GROUP	0313
PRINTER INOPERABLE GROUP	0312

If the printer is working, the settings in the TRF extended memory table are used. If the printer faults in the middle of a series of chained transactions, the cardholder is prompted with the message “Continue without a receipt?” during each transaction. If the cardholder selects Yes, the transaction is allowed without a receipt. If the cardholder selects No, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is prompted during each transaction with the message “Continue without a receipt?” If the cardholder selects Yes, the current transaction is allowed without a receipt. If the cardholder selects No, the Main Menu is displayed, and the cardholder can select another transaction.

Once the printer is serviced and the terminal exits maintenance mode, the printer operable group is downloaded.

Scenario 3—Prompt Once per Session on Approved Transactions

To prompt the cardholder once per session as to whether or not a receipt is printed on all approved transactions, enter the following values in the ATD fields:

Field	Value
OPTIONAL RECEIPTS	2
PRINTER OPERABLE GROUP	0301
PRINTER INOPERABLE GROUP	0302

When the printer is working, the cardholder is prompted once at the beginning of the session with the message “Do you want a receipt?” If the cardholder selects Yes, the receipts for all approved chained transactions are printed. If the cardholder selects No, receipts are not printed for all chained transactions, unless one of the following exceptions occurs:

- If the transaction can only complete a partial dispense, a receipt is always printed.
- If the transaction is denied, the TRF extended memory table settings are used to determine whether a receipt is printed.

If the printer faults in the middle of a series of chained transactions, the cardholder is prompted with the message “Continue without a receipt?” during each transaction. If the cardholder selects Yes, the transaction is allowed without a receipt. If the cardholder selects No, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is prompted at the beginning of the session with the message “Continue without a receipt?” If the cardholder selects Yes, then all approved chained transactions are allowed without a receipt. If the cardholder selects No, the card is returned and the session is ended.

Once the printer is serviced and the terminal exits maintenance mode, the printer operable group is downloaded.

Scenario 4—Prompt Once per Session on Approved and Denied Transactions

To prompt the cardholder once per session as to whether or not a receipt is printed on all approved and denied transactions, enter the following values in the ATD fields:

Field	Value
OPTIONAL RECEIPTS	3
PRINTER OPERABLE GROUP	0301
PRINTER INOPERABLE GROUP	0302

When the printer is working, the cardholder is prompted once at the beginning of the session with the message “Do you want a receipt?” If the cardholder selects Yes, the receipts for all approved and denied chained transactions are printed. If the cardholder selects No, receipts are not printed for all approved or denied chained transactions unless the transaction can only complete a partial dispense. In this case, a receipt is always printed.

If the printer faults in the middle of a series of chained transactions, the cardholder is prompted with the message “Continue without a receipt?” during each transaction. If the cardholder selects Yes, the transaction is allowed without a receipt. If the cardholder selects No, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is prompted at the beginning of the session with the message “Continue without a receipt?” If the cardholder selects Yes, then all approved and denied chained transactions are allowed without a receipt. If the cardholder selects No, the card is returned and the session is ended.

Once the printer is serviced and the terminal exits maintenance mode, the printer operable group is downloaded.

Scenario 5—Prompt on Each Transaction on Approved Transactions

To prompt the cardholder during each transaction as to whether or not a receipt is printed on all approved transactions, enter the following values in the ATD fields:

Field	Value
OPTIONAL RECEIPT	2
PRINTER OPERABLE GROUP	0311
PRINTER INOPERABLE GROUP	0312

When the printer is working, the cardholder is prompted during each transaction with the message “Do you want a receipt?” If the cardholder selects Yes, the receipt for the approved transaction is printed. If the cardholder selects No, a receipt is not printed for the approved transaction, unless one of the following exceptions occurs:

- If the transaction can only complete a partial dispense, a receipt is always printed.
- If the transaction is denied, the TRF extended memory table settings are used to determine whether a receipt is printed.

If the printer faults in the middle of a series of chained transactions, the cardholder is prompted with the message “Continue without a receipt?” during each transaction. If the cardholder selects Yes, the transaction is allowed without a receipt. If the cardholder selects No, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is prompted during each transaction with the message “Continue without a receipt?” If the cardholder selects Yes, the approved transaction is allowed without a receipt. If the cardholder selects No, the Main Menu is displayed, and the cardholder can select another transaction.

Once the printer is serviced and the terminal exits maintenance mode, the printer operable group is downloaded.

Scenario 6—Prompt on Each Transaction on Approved and Denied Transactions

To prompt the cardholder during each transaction as to whether or not a receipt is printed on all approved and denied transactions, enter the following values in the ATD fields:

Field	Value
OPTIONAL RECEIPT	3
PRINTER OPERABLE GROUP	0311
PRINTER INOPERABLE GROUP	0312

When the printer is working, the cardholder is prompted during each transaction with the message “Do you want a receipt?” If the cardholder selects Yes, the receipt for the approved or denied transaction is printed. If the cardholder selects No, a receipt is not printed for the approved or denied transaction, unless the transaction can only complete a partial dispense. In this case, a receipt is always printed.

If the printer faults in the middle of a series of chained transactions, the cardholder is prompted with the message “Continue without a receipt?” during each transaction. If the cardholder selects Yes, the transaction is allowed without a receipt. If the cardholder selects No, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is prompted during each transaction with the message “Continue without a receipt?” If the cardholder selects Yes, the approved transaction is allowed without a receipt. If the cardholder selects No, the Main Menu is displayed, and the cardholder can select another transaction.

Once the printer is serviced and the terminal exits maintenance mode, the printer operable group is downloaded.

Sample Financial Transaction Receipt

The Device Handler process sends all of the information needed to print a financial transaction receipt. NCR allows 24 print lines on a financial transaction receipt with 40 character positions per line.

Financial Receipt Descriptions

The table below describes the standard lines used on NCR 5XXX-series ATM default financial receipts. Because receipts are configurable, however, the information on each receipt line could be different.

Line	Description
1–2	A space reserved for the name and address of the institution along with space for an institution's free-format marketing message. The NCR ATM printer can utilize preprinted receipt paper or can print these lines.
3	DATE TIME TERM
4	The date of the transaction, the time the transaction occurred, and the first 8 digits of the 16-digit terminal ID of the NCR ATM.
5	Blank
6	The card number that was used to initiate the transaction.
7	Blank
8	The ATM location; obtained from the ATD.
9	Blank
10	A label of RECORD NO. and the terminal sequence number assigned to this transaction.
11	The transaction type that has been performed and the amount of the transaction.
12	The account number of the affected account. If funds are being withdrawn <i>from</i> the affected account the account number is preceded by FR. If funds are being deposited <i>to</i> the affected account the account number is preceded by TO. If a two-sided transaction such as transfer, inquiry, or payment has been performed, the affected <i>from</i> account number is printed on this line, preceded by FR.
13	A space reserved for the LEDGER BALANCE field, if needed. The NCR ATM prints the ledger balance of the affected account immediately following the LEDGER BALANCE field for deposit, withdrawal, or inquiry transactions. If a two-sided transaction such as transfer or payment has been performed, the affected <i>to</i> account number is printed on this line, preceded by TO.

Line	Description
14	A space reserved for the AVAILABLE BALANCE field, if needed. The NCR ATM prints the available balance of the affected account immediately following the AVAILABLE BALANCE field for withdrawal, deposit, or inquiry transactions.

The following is a sample receipt that demonstrates the standard lines for an NCR receipt.

```

1          A C I   D E M O
2
3      DATE          TIME   TERM
4      MM/DD/YY   HH:MM S1AN5070
5
6      1234567000000003-003
7
8      330 SOUTH 108TH AVE
9
10     RECORD NO.          570
11     WITHDRAWAL          $20.00
12     FR      013333
13     LEDGER BAL          $3412.00
14     AVAIL BAL          $1760.00

```

1 2

123456789012345678901234567

Field Type Codes

The following table describes the standard field type codes that are used in receipt templates. Each field type code corresponds to a predetermined piece of fixed-length data that the Device Handler process retrieves from the BASE24-atm Standard Internal Message (STM), the ATD, or the N50NAMS. Data names are

included in parentheses after each field type code description to indicate the specific data field from which the information is taken. This list includes field type codes that are not used in the default receipt templates, but which you can use to customize the receipt templates.

Note: For statement print transactions, the RQST fields in the STM are redefined with the STMT-INFO fields.

For further information on the STM data fields, refer to the ***BASE24-atm Transaction Processing Manual***. For further information on the ATD data names, refer to the ***BASE24-atm Files Maintenance Manual*** (using the data name index) or refer to the DDLs.

Code	Description
A1	The transaction amount (STM.RQST.AMT-1), for instance, the withdrawal amount or deposit amount. In the case of a withdrawal with a surcharge fee, A1 contains the withdrawal amount plus any surcharge or issuer fees or rebates, which is the amount that is deducted from or added to (for negative surcharges) the cardholder's account.
A2	The transaction amount (STM.RQST.AMT-2).
A3	The transaction amount (STM.RQST.AMT-3).
AB	The available balance (STM.RQST.AMT-3). The available balance and its corresponding text from the ledg_avail_tbl_g table in the N50NAMS is only printed if response code 0 is returned from the BASE24-atm Authorization process and the CUSTOMER BALANCE INFO fields in the Institution Definition File (IDF) and the ATD indicate that Amount 3 (STM.RQST.AMT-3) is to be displayed. This code does not print the text, "AVAIL BAL" (Available Balance), on the receipt. Use code XA below to print the text on the receipt.
AC	Use the Arabic character set.
AI	The application identifier (DF-NAME field of the EMV Discretionary Data token). This field type code is used with the BASE24-atm EMV add-on product. Refer to the <i>BASE24-atm EMV Support Manual</i> for more information about the EMV Discretionary Data token.

Code	Description
AL	The Application Label that the ATM sent in the card authentication method (CAM) data on the request. This field type code is used with the BASE24-atm EMV add-on product. Refer to the <i>BASE24-atm EMV Support Manual</i> for more information about the EMV Request token.
AS	The application PAN sequence number (APPL-PAN-SEQ-NUM field of the EMV Status token). This field type code is used with the BASE24-atm EMV add-on product. Refer to the <i>BASE24-atm EMV Support Manual</i> for more information about the EMV Status token.
BD x ,@ t [^]	Print a bar code. Requires two parameters. Valid values for x are 0 through 6. Valid values for t are 1 to 12 alphanumeric characters. Only allowed on thermal printers.
BI x , y	Print a bit image. Requires two parameters. Valid values for x (the bit image number) are 1 and 2. Valid values for y (the width and height indicators) are 0 through 3. Only allowed on thermal printers.
CA	Indicates the amount for the particular currency deposited after it has been converted to the terminal currency (AMT-INFO.CONV-AMT of the BNA Multiple Currency token).
CN	The card number from the ATM card (STM.RQST.TRACK2) and member number (STM.RQST.MBR-NUM). To print the card number only, see CX below.
CV	The equivalent converted amount for the transaction. The Device Handler process retrieves this amount from the Multiple Currency token. The Device Handler process uses the amount specified in the currency of the account, if available. If it is not available, the Device Handler process uses the amount in the currency of the terminal. This code is used with the Multiple Currency Dispense add-on feature only.

Code	Description
CX <i>l,r,c</i>	The card number from the ATM card (STM.RQST.TRACK2). The printer does not print the member number. The three variables (<i>l</i> for characters starting from the left, <i>r</i> for characters starting from the right, and <i>c</i> for the character code) are used to control the truncation of the card number. If the variables are not present, the card number is printed in its entirety. Refer to the topic “Account Number Truncation” later in this section for additional information about these variables.
DA	The transaction date (STM.TRAN-DAT—MM/DD/YY or DD/MM/YY). The date displayed is dependent on the value specified in the DATE-DISPLAY-TYPE param in the Logical Network Configuration File (LCONF).
DC	The text CONVERTED AMOUNT” that is printed on the receipt. This text is configured in the mult_crncy_tbl_g table in the C100NAMS File. This code does not print the converted amount. Use code CA (above) to print the converted amount.
DE	The text EXCHANGE RATE” that is printed on the receipt. This text is configured in the mult_crncy_tbl_g table in the C100NAMS File. This code does not print the value of the exchange rate. Use code EA (below) to print the value of the exchange rate.
DO	The text ORIGINAL AMOUNT” that is printed on the receipt. This text is configured in the mult_crncy_tbl_g table in the C100NAMS File. This code does not print the original . Use code OA (below) to print the original amount.
EA	The rate used to convert the original amount into the currency used by the terminal (ATM-INFO.CONV-RATE of the BNA Multiple Currency token).
EG@ <i>f</i> [^]	Print Epson graphics. The graphic file name is specified in the variable <i>f</i> , which is a valid DOS file name with an extension, but no directory path. Only allowed on thermal printers.
E [^] <i>x</i>	Space compression code. This code places spaces on the receipt. The value <i>x</i> is the number of spaces in hexadecimal notation, and must be in the range of 1–F.

Code	Description
E(<i>x</i>)	Change character set. The value of <i>x</i> is one of the following: 1 = Single-size alphanumeric character set 1 2 = Single-size alphanumeric character set 2 3 = Single-size alphanumeric character set 3 4 = Single-size alphanumeric character set 4 5 = Single-size alphanumeric character set 5 #> = Double-size alphanumeric character set 1 ? = Double-size alphanumeric character set 2 - B = Double-size alphanumeric character set 3 - K = Double-size alphanumeric character set 4 - L = Double-size alphanumeric character set 5 Single-size character sets print 10 characters per inch, while the double-size alphanumeric character sets print 5 characters per inch. This field type code allows for the configuration of type size on the receipt. The receipt and journal printers support the first 64 characters from the single- and double-size alphanumeric character sets. For receipt and journal printers, the character set defaults to single-size alphanumeric 1 at the end of the printer data field until another character set control sequence appears. Note: The value for double-size alphanumeric character set 1 requires two characters in place of <i>x</i>. The values for other character sets require only one character in place of <i>x</i>.
EV	The exchange rate value. The Device Handler process retrieves the exchange rate from the Multiple Currency token. The Device Handler process uses the exchange rate for the account currency, if available. If it is not available, the Device Handler process uses the terminal currency rate. This code is used with the Multiple Currency Dispense add-on feature only.
FA	The <i>from</i> account type (located in N50NAMS acct_typ_tbl_g). The account type printed on the receipt depends on the language indicator and the value of STM.FROM-ACCT-TYP. The Device Handler process matches account type numbers to a two-character account type description in the acct_typ_tbl_g table in the N50NAMS.

Code	Description
FD	The <i>from</i> account description (located in N50NAMS appl_tbl_g). The account description printed on the receipt depends on the language indicator and the value of STM.FROM-ACCT-TYP. The Device Handler process retrieves the account description from the appl_tbl_g table in the N50NAMS.
FRL, <i>r,c</i>	The <i>from</i> account number (STM.RQST.FROM-ACCT). If the STM.RQST.FROM-ACCT is all blanks or zeros, the Device Handler process uses the account description from the appl_tbl_g table in the N50NAMS. The value of STM.RQST.FROM-ACCT-TYP and the language indicator determine which table entry is used.
FRL, <i>r,c</i> <i>continued</i>	The three variables (<i>l</i> for characters starting from the left, <i>r</i> for characters starting from the right, and <i>c</i> for the character code) are used to control the truncation of the card number. If the variables are not present, the card number is printed in its entirety. Refer to the topic “Account Number Truncation” later in this section for additional information about these variables.
IC	Use the International character set.
LB	The ledger balance (STM.RQST.AMT-2). The ledger balance and its corresponding text from the ledg_avail_tbl_g table in the N50NAMS is only printed if a response code of 0 is returned from the BASE24-atm Authorization process and the CUSTOMER BALANCE INFO fields in the IDF and the ATD indicate that Amount 2 (STM.RQST.AMT-2) is to be displayed. This code does not print the text, “LEDGER BAL”, on the receipt. Use code XL (below) to print the text on the receipt.
LGnn, <i>xx</i>	Print the specified image file that has already been downloaded into the ATM printer memory buffer. The value <i>nn</i> indicates the logo number (valid values are 00 through 09). The value <i>xx</i> indicates the number of lines the logo file requires when it is printed (valid values are 00 through 99). This code is available only when the value of the RECEIPT TYPE field in the RCPT is set to STHD and the value in the PRINTER TYPE field is set to a value of C.
MB	Cardholder member number (STM.RQST.MBR.NUM).
NC	Non-currency item description text (located in N50NAMS).

Code	Description
OA	The amount of the deposit in the original currency (AMT-INFO. ORIG-AMT of the BNA Multiple Currency token).
OL	The number of the log only transaction selected (e.g., Log Only #1). Four Log Only transactions are supported.
OS _{xxx}	Select OS/2 code page. Valid values for <i>xxx</i> are as follows: 000, 437, 850, 851, 852, 855, 857, 860, 861, 862, 863, 864, and 866. This field type code is only allowed on thermal printers.
RC	The response code (STM.RQST.RESP).
RT	The response text (located in N50NAMS response_text_g). This code is used with denied transaction receipts. The printed receipt text depends on the value of ATM.N50.TRAN-RESP-CDE. The Device Handler process searches the response_text_g table in the N50NAMS to find a match on the response code. The corresponding text is then moved to the print buffer. The response text is configurable and is described in the topic “Response Code Text Table for Financial Receipts,” later in this section.
S1	In the case of a withdrawal with a surcharge fee, this code is the amount to be dispensed, without including the surcharge amount (STM.RQST.AMT-1 minus the value of the TRAN-FEE field from the Surcharge Data token).
S2	In the case of a withdrawal with a surcharge fee, this code is the amount of the surcharge (TRAN-FEE, from the Surcharge Data token).
S3	In the case of a withdrawal with an issuer fee or rebate, this code is the amount of the issuer fee or rebate (TRAN-FEE, from the Issuer Fee/Rebate token).
SQ	The four-digit BASE24 sequence number assigned to the transaction (ATDD1.ATDD1-CORE.TRAN-SEQ-NUM).
T1	The time (hh:mm) the transaction took place—in 24-hour format (STM.TRAN-TIM).

Code	Description
T2xx/yy	The time the transaction took place in a 12-hour format as opposed to the 24-hour format (STM.TRAN-TIM). If the time is less than 1200, the time appears as <i>hh:mmxx</i> . If the time is greater than or equal to 1200, the time appears as <i>hh:mmyy</i> . In these formats <i>hh</i> has valid values of 1 through 12. <i>xx</i> and <i>yy</i> are the configurable variable length symbols for a.m. and p.m., respectively. The format of this field type code is T2xx/yy. You can optionally insert spaces before <i>xx</i> and <i>yy</i> if you do not want the text to directly follow the time. <i>yy</i> is terminated by either the next field marker code (#, >, @, or ?), or by the end of the template.
TA	The <i>to</i> account type (located in N50NAMS acct_typ_tbl_g). The account type printed on the receipt depends on the language indicator and the value of STM.TO-ACCT-TYP. The Device Handler process matches account type numbers to a two-character account type description in the acct_typ_tbl_g table in the N50NAMS.
TC	The city where the terminal is located (STM.RQST.TERM-CITY).
TD	The <i>to</i> account description (located in N50NAMS appl_tbl_g). The account description printed on the receipt depends on the language indicator and the value of STM.TO-ACCT-TYP. The Device Handler process retrieves the account description in the appl_tbl_g table in the N50NAMS.
TE	All 16 bytes of the terminal ID (STM.TERM-ID).
TL	The terminal location (STM.RQST.TERM-NAME-LOC).
TM	The first eight bytes of the terminal ID (STM.TERM-ID).
TN	The name of the terminal owner (STM.RQST.TERM-OWNER-NAME).
TO _{l,r,c}	The <i>to</i> account number (STM.RQST.TO-ACCT). If the STM.RQST.TO-ACCT is all blanks or zeros, the Device Handler process uses the account description from the appl_tbl_g table in the N50NAMS. The value of STM.RQST.TO-ACCT-TYP and the language indicator determine which table entry is used.

Code	Description
TOL, <i>r,c</i> <i>continued</i>	The three variables (<i>l</i> for characters starting from the left, <i>r</i> for characters starting from the right, and <i>c</i> for the character code) are used to control the truncation of the card number. If the variables are not present, the card number is printed in its entirety. Refer to the topic “Account Number Truncation” later in this section for additional information about these variables.
TS	The state where the terminal is located (STM.RQST.TERM-ST-X).
TT	The transaction type (located in N50NAMS tran_tbl_g or xN50NAMS where <i>x</i> is the identifier for a bindable module). The transaction type printed on the receipt depends on the value of STM.TRAN-CDE. The Device Handler process searches bindable modules for a match on the transaction code. If a match is found, the Device Handler process uses the transaction type from the bindable module. If a match is not found, the Device Handler process searches the tran_tbl_g table in the N50NAMS to find a match on the transaction code. The corresponding text is then moved to the print buffer.
V0	The activation telephone number used to activate a mobile top-up voucher (Pre-Pay Top-Up token). If the token contains blanks, blank spaces are printed on the receipt. If the token cannot be retrieved, no activation information is printed. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
V1	The stock number (Inventory Voucher token) of a voucher. If the Inventory Voucher token is present, the Device Handler process uses the stock number in the token. If the token is not present, no stock number information is printed on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
V2	The expiration date (Pre-Pay Voucher token or Inventory Voucher token) of an inventory voucher. If the Pre-Pay Voucher token is present, the Device Handler process uses the shelf date in the token. If the Pre-Pay Voucher token is not present, the Device Handler process uses the value in the stock expiration date field in the Inventory Voucher token. If the Inventory Voucher token is not present, no expiration date is included on the receipt.

Code	Description
V2 <i>continued</i>	The Device Handler process uses the date format specified in the DATE-DISPLAY TYPE param in the LCONF. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
V3	The mobile telephone number that has been topped up (Pre-pay Top-Up token). If the Pre-Pay Top-Up token is present, the Device Handler process uses the telephone number in the token. If the Pre-Pay Top-Up token is not present, no information is included in the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
V4	The unique transaction number (Pre-Pay Top-Up token). If the Pre-Pay Top-Up token is present, the Device Handler process uses the reference number in the token. If the field contains a value, that value is printed on the receipt. If the field contains blanks or the token is not present, the unique transaction number is not printed on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
V5	A telecommunications company marketing message (Pre-Pay Receipt token). If the Pre-Pay Receipt token is present, the Device Handler process retrieves the operator message and the corresponding length field. If it cannot retrieve the token, no marketing message is printed on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
V6	The telecommunications company name (Pre-Pay Response token). If the token is not present, no telecommunications company information is included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
V7	The carrier telephone number (Pre-Pay Response token). The Device Handler process uses the telephone number from the Pre-Pay Response token. If the token is not present, the carrier telephone number is not included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.

Code	Description
VA	The new minutes balance on the cardholder's mobile telephone (Pre-Pay Top-Up token). The Device Handler process uses the service balance from the Pre-Pay Top-Up token. If the token is not present, no new minutes balance information is included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
VB	The batch ID (Pre-Pay Voucher Receipt token). The Device Handler process uses the batch ID from the Pre-Pay Voucher Receipt token. If the token is not present, no batch ID information is included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
VE	The number of expiration days (Pre-Pay Voucher Receipt token). The Device Handler process uses the expiration days from the Pre-Pay Voucher Receipt token. If the token is not present, no expiration days information is included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
VI	The control ID number (Pre-Pay Voucher Receipt token). The Device Handler process uses the control number from the Pre-Pay Voucher Receipt token. If the token is not present, control ID information is not included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
VN	The voucher number (Pre-Pay Voucher Receipt token). The Device Handler process uses the value of the NUM field in the Pre-Pay Voucher Receipt token. If the token is not present, no voucher number is included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
VS	The expiration date of the minutes purchased (Pre-Pay Online Receipt token). The Device Handler process uses the expiration date from the Pre-Pay Online Receipt token. If the token is not present no expiration date is included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.

Code	Description
VT	The tax amount (Pre-Pay Response token). The Device Handler process uses the tax amount from the Pre-Pay Response token. If the token is not present, no tax information is included on the receipt. This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
VX	The amount of the mobile top-up (STM.RQST.AMT-1 minus the amount value of the tax in the Pre-Pay Response token). This code is used with the BASE24-atm Mobile Top-Up Device Handler extension only.
XA	The text, “AVAIL BAL”, (Available Balance) is printed on the receipt if a response code 0 is returned from the BASE24-atm Authorization process and the CUSTOMER BALANCE INFO fields in the IDF and the ATD indicate that Amount 3 (STM.RQST.AMT-3) is to be displayed. This code does not print the <i>amount</i> of the available balance. Use code AB (above) to print the available balance amount.
XC	The converted currency description text. The Device Handler process retrieves the converted amount description text from the ncr_mult_crncy_tbl table in the N50NAMS File. This code is used with the Multiple Currency Dispense add-on feature only.
XE	The exchange rate description text. The Device Handler process retrieves the exchange rate text from the ncr_mult_crncy_tbl table in the N50NAMS File. This code is used with the Multiple Currency Dispense add-on feature only.
XL	The text, “LEDGER BAL”, is printed on the receipt if a response code of 0 is returned from the BASE24-atm Authorization process and the CUSTOMER BALANCE INFO fields in the IDF and the ATD indicate that Amount 2 (STM.RQST.AMT-2) is to be displayed. This code does not print the <i>amount</i> of the ledger balance. Use code LB (above) to print the ledger balance amount.
XS	The selected currency description text. The Device Handler process retrieves the selected amount description text from the ncr_mult_crncy_tbl table in the N50NAMS File. This field is used with the Multiple Currency Dispense add-on feature only.

Account Number Truncation

Three of the field type codes listed above have additional variables that allow the account number or card number to be truncated. This feature can help prevent fraud by making it more difficult for an unscrupulous person to use a carelessly discarded receipt to gain access to the account.

The three field type codes are CX (card number without member number), FR (*from* account number), and TO (*to* account number). The three variables that can be appended to these field type codes are *l*, a two-character numeric code indicating the number of digits starting from the left that are printed; *r*, a two-character numeric code indicating the number of digits starting from the right that are printed; and *c*, a one-alphanumeric character code indicating the pad character, or the character used to fill the remaining digits. If these three variables are not present, the card or account number is printed in its entirety.

Note: You can use a blank space as a pad character. To use a reserved special character, (#, @, ?, >, and ^), precede it with the number (#) symbol.

You must have another receipt template with a different printer type if you want the card or account number to print in its entirety on the journal printer. Also note that if you do truncate one or more of the numbers, this does not affect how the number is stored internally; the complete number is still shown on the TLF and reports.

The following examples use a 19-digit primary account number (PAN) value of 1123581321345589144.

Field Type Code in Receipt Template	PAN as Printed on Receipt
#CX00,04,%	%%%%%%%%%%%%%%9144
#CX06,04,#@	112358@@@@@@@@9144
#CX06,00,X	112358XXXXXXXXXXXXX

Currency Symbols

The amount field type codes (A1, A2, A3, AB, LB, S1, and S2) can have a monetary symbol (such as \$, •, or £) that prints on the receipt. You can enter the monetary symbol either manually or have it determined automatically by the

Device Handler process. To enter the symbol manually, type the appropriate symbol after each occurrence of the field type code. Any text after the code is considered part of the monetary symbol until the next field marker is encountered.

The monetary symbol that is used can be determined automatically by the use of a table in the N50NAMS File. For the symbol to be determined automatically, you must enter CC\$ after the field type code. The symbol is determined by the currency code table (rcpt_money_sym_tbl_g). This table contains an entry for each monetary system for which you want a special symbol to be printed. Each entry in the table contains the country code, the number of characters required in the sequence to send the special symbols to the receipt printer (including escape codes), the number of characters taken up on the receipt as a result of the sequence, and the 10-character sequence that prints the proper monetary symbol. If a country code is encountered that is not in the table, the 3-character ISO currency code is used (for example, USD for US dollars). If neither the country code nor the ISO currency code exist in the table, no symbol is printed.

Transaction Receipt Header

The receipt header is printed at the top of the receipt for all financial transactions. The Device Handler process uses the header receipt templates (i.e., HDR receipt type in the RCPT extended memory table) to format the receipt header for all transactions. The process uses the receipt group identifier, the language indicator, the receipt type, and the device type from the Receipt File (RCPT) extended memory table to find the corresponding header receipt template.

Once this step is completed, the Device Handler process moves to the first character of the header receipt template and uses the field markers and field type codes described earlier in this section to scan the receipt template to determine the receipt text.

The following is a representation of the default entry in the header receipt template (receipt type HDR) in the RCPT extended memory table for the Device Handler process.

Transaction Type	Receipt Type	Default RCPT Extended Memory Table Template Data
General Header	HDR	#DA #T1 #TM>>> #CN>> #TL>>>

Transaction Receipt Trailer

A trailer or footer can also be added to a receipt to contain marketing information, or other information you want to print at the bottom of all financial receipts. You can choose not to use a trailer by not having a TRLR receipt type.

So that the trailer is not printed multiple times on one receipt, the trailer is ignored if the cardholder is chaining transactions and you have the TRANSACTIONS ON RECEIPT field on ATD screen 8 set to a value of 0, for multiple transactions per receipt.

An example of the trailer receipt (receipt type TRLR) is not provided, because it is specific to each institution and no default template is included with the RCPT.

Financial Receipt Samples

The following pages describe the default receipt templates provided in the RCPT. Default templates are provided in English only regardless of the language indicator associated with template (e.g., default receipt templates for language indicators 0 and 1 are configured in English). You must modify the appropriate template if you want another language.

The following pages show default financial receipts for the following transactions:

- Withdrawal from checking, no surcharge
- Withdrawal from checking, with surcharge
- Deposit to checking
- Transfer from checking to savings
- Savings account inquiry
- Credit payment from checking
- Payment enclosed
- Message enclosed
- Log only
- PIN change
- Denied financial transactions

Withdrawal from Checking, No Surcharge

The receipt type WD is used for withdrawal transactions. The NCR default receipt template is shown below, followed by an example of the default receipt for a withdrawal from checking account transaction that does not have an added surcharge fee. Note that in the case of a partial dispense, the actual amount dispensed is printed, but no balance information is printed.

```
@RECORD NO.      #SQ>
?IF FEES
#TT              #S1$>
@FEE             #S2$>
@TOTAL          #A1$>
?ENDIF
?IF NOFEES
#TT              #A1$>
?ENDIF
@FR  #FR>>
#XL#LB$>
#XA#AB$>>
```

```
1              A C I   D E M O
2
3      DATE      TIME   TERM
4      MM/DD/YY  HH:MM  S1AN5070
5
6      1234567000000003-003
7
8      330 SOUTH 108TH AVE
9
10     RECORD NO.              570
11     WITHDRAWAL              $20.00
12     FR      013333
13     LEDGER BAL              $3412.00
14     AVAIL BAL              $1760.00
```

```
1      2
123456789012345678901234567
```

Withdrawal from Checking, with Surcharge

The following is an example of the NCR default receipt format for a withdrawal from checking account transaction that has an added surcharge fee. Note that this receipt is produced using the same template as the example on the previous page. Also note that in the case of a partial dispense, the actual amount dispensed is printed, but no balance information is printed.

1	A C I	D E M O
2		
3	DATE	TIME TERM
4	MM/DD/YY	HH:MM S1AN5070
5		
6	123456700000003-003	
7		
8	330 SOUTH 108TH AVE	
9		
10	RECORD NO.	571
11	WITHDRAWAL	\$20.00
12	FEE	\$1.00
13	TOTAL	\$21.00
14	FR 013333	
15	LEDGER BAL	\$3412.00
16	AVAIL BAL	\$1760.00

1	2
123456789012345678901234567	

Deposit to Checking

The receipt type DP is used for deposit transactions. The NCR default receipt template is shown below, followed by an example of the default receipt for a deposit to checking account transaction.

```
@RECORD NO.      #SQ>
@DEPOSIT         #A1$>
@TO   #TO>>
#LB$>
#AB$>
```

```
1           A C I   D E M O
2
3      DATE      TIME   TERM
4      MM/DD/YY  HH:MM  S1AN5070
5
6      1234567000000003-003
7
8      330 SOUTH 108TH AVE
9
10     RECORD NO.      572
11     DEPOSIT         $100.00
12     TO      013333
13     LEDGER BAL      $3512.00
14     AVAIL BAL      $1760.00
```

1 2
123456789012345678901234567

Transfer from Checking to Savings

The receipt type TR is used for transfer transactions. The NCR default receipt template is shown below, followed by an example of the default receipt for a transfer from checking to savings account transaction.

```
@RECORD NO.      #SQ>
@TRANSFER        #A1$>
@FR   #FR>
@TO   #TO>
```

1	A C I D E M O		
2			
3	DATE	TIME	TERM
4	MM/DD/YY	HH:MM	S1AN5070
5			
6	123456700000003-003		
7			
8	330 SOUTH 108TH AVE		
9			
10	RECORD NO.	573	
11	TRANSFER	\$50.00	
12	FR	013333	
13	TO	113333	

12

1

2

123456789012345678901234567

Savings Account Inquiry

The receipt type INQ is used for inquiry transactions. The NCR default receipt template is shown below, followed by an example of the default receipt for a savings account inquiry transaction.

```
@RECORD NO.      #SQ>
@INQUIRY>
@FR  #FR>
#LB$>
#AB$>
```

```
1          A C I   D E M O
2
3      DATE      TIME  TERM
4      MM/DD/YY  HH:MM S1AN5070
5
6      1234567000000003-003
7
8      330 SOUTH 108TH AVE
9
10     RECORD NO.          574
11     INQUIRY
12     FR      113333
13     LEDGER BAL          $1595.00
14     AVAIL BAL          $1310.00
```

1 2
123456789012345678901234567

Credit Payment from Checking

The receipt type PY is used for payment transactions. The NCR default receipt template is shown below, followed by an example of the default receipt for a credit account payment from a checking account.

@RECORD NO. #SQ>
@PAYMENT #A1\$>
@FR #FR>
@TO #TO>

1	A C I D E M O		
2			
3	DATE	TIME	TERM
4	MM/DD/YY	HH:MM	S1AN5070
5			
6	123456700000003-003		
7			
8	330 SOUTH 108TH AVE		
9			
10	RECORD NO.	575	
11	PAYMENT	\$100.00	
12	FR	013333	
13	TO	313333	

1

2

123456789012345678901234567

Payment Enclosed

The receipt type PYEN is used for payment enclosed transactions. The NCR default receipt template is shown below, followed by an example of the default receipt for a payment placed in the envelope depository.

@RECORD NO. #SQ>
@PAYMT ENCL #A1\$>

1	A C I D E M O		
2			
3	DATE	TIME	TERM
4	MM/DD/YY	HH:MM	S1AN5070
5			
6	123456700000003-003		
7			
8	330 SOUTH 108TH AVE		
9			
10	RECORD NO.	576	
11	PAYMENT ENCL	\$100.00	

1

2

123456789012345678901234567

Message Enclosed

The receipt type MGEN is used for message enclosed transactions. The NCR default receipt template is shown below, followed by an example of the default receipt for a message from the cardholder to the institution placed in the envelope depository.

@RECORD NO. #SQ>
@MSG ENCL>

1	A C I D E M O		
2			
3	DATE	TIME	TERM
4	MM/DD/YY	HH:MM	S1AN5070
5			
6	123456700000003-003		
7			
8	330 SOUTH 108TH AVE		
9			
10	RECORD NO.	577	
11	MSG ENCL		

1

2

123456789012345678901234567

Log Only

The receipt type OL is used for log only transactions. The NCR default receipt template is shown below, followed by an example of the default receipt for a log only message.

```
@RECORD NO.      #SQ>
@TYPE #OL @LOG ONLY MSG>
```

```
1          A C I   D E M O
2
3      DATE      TIME  TERM
4      MM/DD/YY  HH:MM S1AN5070
5
6      1234567000000003-003
7
8      330 SOUTH 108TH AVE
9
10     RECORD NO.      578
11     TYPE 1 LOG ONLY MSG
```

1 2
123456789012345678901234567

PIN Change

The receipt type PC is used for PIN change transactions. The NCR default receipt template is shown below, followed by an example of the default receipt for a PIN change.

@RECORD NO. #SQ>
@PIN CHANGE>

1	A C I D E M O		
2			
3	DATE	TIME	TERM
4	MM/DD/YY	HH:MM	S1AN5070
5			
6	123456700000003-003		
7			
8	330 SOUTH 108TH AVE		
9			
10	RECORD NO.	579	
11	PIN CHANGE		

1

2

123456789012345678901234567

Denied Financial Transactions

The Device Handler process converts receipt templates for denied transactions in the same manner as approved transactions with one exception. In addition to the receipt templates, a response code text table is used to associate one line of receipt text per ATM response code. The response code text table is described later in this section.

Also note that if a PIN is entered incorrectly, the transaction is denied, but no receipt is produced and the cardholder is prompted to reenter the PIN.

The receipt type DENY is used for denied financial transactions. The NCR default receipt template is shown below, followed by an example of the default receipt for a denied withdrawal transaction.

```
@RECORD NO.      #SQ>
#TT>>
@TRANSACTION DECLINED>
#RC>
#RT>
```

```
1          A C I   D E M O
2
3      DATE      TIME  TERM
4      MM/DD/YY  HH:MM S1AN5070
5
6      1234567000000003-003
7
8      330 SOUTH 108TH AVE
9
10     RECORD NO.      580
11     WITHDRAWAL
      TRANSACTION DECLINED
      51
      EXPIRED CARD
```

1 2
123456789012345678901234567

Response Code Text Table for Financial Receipts

The response code text table is a separate table in the N50NAMS and is used in conjunction with the #RT (Response Text) field type code when that particular field type code is used in a receipt template for denied financial transactions.

The response code text table provides additional receipt text for denied transaction receipts. This text gives the cardholder an explanation for the denied transaction and is in addition to the text provided in the receipt template.

Mapping Internal Response Codes

When the Device Handler process encounters a #RT field type code in a receipt template, the process scans the response code text table and scans for a match on the internal response code that it generates or that it receives from the Authorization process. The following table illustrates how BASE24-atm response codes are mapped to entries in the response code text table. Refer to the *BASE24-atm Transaction Processing Manual* for complete descriptions of the BASE24-atm response codes.

Note: Transactions can be declined by the Device Handler process before they are sent to the Authorization process. In this case, the Device Handler process generates its own response code and there is no STM response code to map to.

BASE24-atm Response Code	Response Code Text Table Entry Number
50	11
51	12
52	13
53	1
54	24
55	10
56	10
57	8
58	17

BASE24-atm Response Code	Response Code Text Table Entry Number
59	21
60	16
61	3 (normal)
61	4 (fast cash)
61	142 (deposit with cash back)
62	2
63	34
64	15
65	29
66	33
67	142
68	24
69	19
70	24
71	20
72	9
73	24
74	24

Mapping Response Codes to Receipt Text

The response code text table includes one line of receipt text for each response code. Each line of text can be configured in up to three languages. Text for each language can be defined in three groups of up to 24 characters each. The line of text from the table is printed on the denied transaction receipt at the location of the #RT field type code in the receipt template.

To locate the correct information in the response code text table, the Device Handler process performs the following:

1. Locates the applicable response code entry in the response code text table.
2. When a match is made on the response code, determines which portion of the entry to send to the ATM as receipt print data by using the language indicator. If the language indicator is 0, the Device Handler process uses the first 24 characters of the response code entry as receipt print data. If the language indicator is 1, the Device Handler process scans past the first 24 characters of the response code entry and uses the second group of 24 characters of the entry as receipt print data. If the language indicator is 2, the Device Handler process scans past the first two sets of 24 characters of the response code entry and uses the third group of 24 characters of the entry as receipt print data.

Default Entries for Denied Transactions

Default entries for the response code text table are provided in the N50NAMS.

All of the default entries are in English; however, up to three languages can be defined for the NCR terminals. For example, the table for invalid PIN text can be set up in the following way:

Language	Text
First	Invalid PIN
Second	*Invalid PIN
Third	%Invalid PIN

The first language has no prefix; the second language uses an asterisk (*) as a prefix; the third language uses a percent symbol (%) as a prefix.

The default entries used for denied transactions are shown below.

The numbers used in the table correspond to the response codes for the NCR ATM. In the following table, only default English text is shown; however, a total of three languages are available for use.

Response Code	Initial Message Text
1	INVALID PIN
2	PIN TRIES EXCEEDED
3	ENTER LESSER AMOUNT
4	UNABLE TO PROCESS AMT
8	TRANSACTION NOT ALLOWED
9	UNABLE TO PROCESS
10	INVALID TRAN OR ACCOUNT
11	UNAUTHORIZED CARD USAGE
12	EXPIRED CARD
13	INVALID CARD
15	INVALID CREDIT CARD AMT
16	USES LIMIT EXCEEDED
17	INSUFFICIENT FUNDS
19	NO MATCHING SHARING GRP
20	CONTACT BANK
21	INSUFFICIENT FUNDS
24	UNABLE TO PROCESS
29	NO STATEMENT INFO
31	UNABLE TO DISPENSE
32	UNABLE TO DEPOSIT
33	STATEMENT INFO NOT AVAIL

Response Code	Initial Message Text
34	DAILY WD LIMIT REACHED
38	UNABLE TO PROCESS
140	UNABLE TO ACCEPT CHECKS
141	UNABLE TO DISPENSE COINS
142	DEP MUST EXCEED CASH
144	NO CHECK ENTERED
147	UNABLE TO CASH CHECK
175	COURTESY AMT MISMATCH
176	COURTESY AMT READ ERROR
177	INVALID DOCUMENT
178	ZONE VERIFY FAILURE
179	RETRY COURTESY AMOUNT
180	TRAN COMPL DESPITE ERR
181	INVALID COIN AMOUNT
182	TRAN COMPL DOCUMENT JAM
200	SVC SIGNATURE CHK FAILED

Statement Print Transaction Receipts

Cardholders have the option of performing statement print transactions at NCR ATMs. The format of this receipt is determined by the host computer linked to the ATM.

Statement Print Journal Entry

When processing statement print transactions, a journal entry receipt is printed only if the print option field in the TRF extended memory table indicates that a receipt should be printed on the receipt and audit (or journal) printer. This header is printed on the journal printer only. Enter the value A in the printer type key field on the RCPT screen to indicate that this receipt type is printed on the journal printer.

The format of the body of the statement receipt is determined by the host computer linked to the ATM. The total number of characters of print data for all the blocks combined in one Transaction Reply message cannot exceed 500 characters. If this number is exceeded, the Device Handler process generates an event message and no receipt is printed.

The following is a representation of the default entry in the statement header journal entry receipt template (receipt type STMT) in the RCPT extended memory table for the Device Handler process.

Transaction Type	Receipt Type	Default RCPT Extended Memory Table Template Data
Statement Print Journal Entry	STMT	#DA #T1 #TM> #CN> @RECORD NO. #SQ> @STATEMENT PRINT>

Note: The field type codes used in the template above are described in the previous topic, “Field Type Codes.”

Statement Print Header

The header on the consumer receipt for a statement print transaction can come from the host or from the Receipt Table (RCPT) extended memory table. To configure a template in the RCPT for a statement print header on the consumer receipt, use the receipt type STMT and the printer type *. If this information is not in the RCPT extended memory table, the header information from the host is used.

Sample Statement Print Receipt

The following is an example of a statement print receipt generated at an NCR ATM. Note that the exact format of this receipt can differ because the host formats this receipt.

```
1          A C I   D E M O
2
3  DATE          TIME          TERM
4  LAST STMT DATE MM/DD/YY
5  J. CUSTOMER          CURRENT DATE MM/DD/YY
6  123 MAIN STREET
7  ANYTOWN, U.S.A.          CHECKING ACCOUNT
8  45678                      #01-234567-89
9  -----
10 DATE          DESCRIPTION          AMOUNT
11 MM/DD/YY DIRECT DEPOSIT          $900.00
12 MM/DD/YY CHECK #2304          $25.00-
13 MM/DD/YY ATM 5655 W/D #1030          $50.00-
14 MM/DD/YY ATM 4322 W/D #1333          $50.00-
15 MM/DD/YY DEPOSIT ATM #2327          $500.00
16 MM/DD/YY CHECK #5343          $100.00-
17 MM/DD/YY CHECK #1443          $25.00-
18 MM/DD/YY ATM 1223 W/D #0023          $50.00-
19 MM/DD/YY DEPOSIT HOME OFFICE          $900.00
20 MM/DD/YY PAYMENT VENDOR #122          $250.34-
21 MM/DD/YY ATM 5344 W/D #1054          $100.00-
22 MM/DD/YY ATM 3244 W/D #4530          $150.00-
```

```
1          2          3          4
1234567890123456789012345678901234567890
```


Administrative Transaction Receipts

The BASE24-atm Terminal Data Dynamic File—general data (ATDD1) accumulates transaction totals as transactions are completed for each ATM. These totals are used in balancing the ATM and monitoring the cash supply. They can also be printed at the ATM using terminal balancing and terminal subtotaling administrative transactions. All balancing totals are accumulated from the last time the ATM was balanced and the totals were reset to zero.

In the RCPT, the default administrative transaction receipt format has 25 machine print lines and 40 characters per line.

Administrative Receipt Descriptions

Administrative transaction receipts are printed on the same type of receipt as financial receipts. The following table contains the position of the line and its description. The fields described below are totals accumulated in the ATDD1.

Line or Field Label	Description
1–2	A space reserved for the name and address of the institution along with space for an institution's free-format marketing message. The NCR ATM printer can utilize preprinted receipt paper or print these lines itself.
3	DATE TIME TERM
4	The date of the transaction, the time the transaction occurred, and the first 8 digits of the 16-digit terminal ID of the NCR ATM.
5	Blank
6	The card number that was used to initiate the transaction.
7	Blank
8	ATM location; obtained from the ATD.
9	Blank
C1 STR	CASSETTE 1 STARTING AMOUNT — Total cash in cassette 1 at the beginning of the balancing period.

Line or Field Label	Description
C1 INC	CASSETTE 1 CASH INCREASE — Total increase to cash in cassette 1 by administrative transactions during this balancing period.
C1 DEC	CASSETTE 1 CASH DECREASE — Total decrease to cash in cassette 1 by administrative transactions during this balancing period.
C1 OUT	CASSETTE 1 CASH OUT — Total amount of cash dispensed from cassette 1.
C1 END	CASSETTE 1 CASH END — Sum of starting amount, cash increases, cash decreases, and cash dispensed in cassette 1 for this balancing period.
C2 STR	CASSETTE 2 STARTING AMOUNT — Total cash in cassette 2 at the beginning of the balancing period.
C2 INC	CASSETTE 2 CASH INCREASE — Total increase to cash in cassette 2 by administrative transactions during this balancing period.
C2 DEC	CASSETTE 2 CASH DECREASE — Total decrease to cash in cassette 2 by administrative transactions during this balancing period.
C2 OUT	CASSETTE 2 CASH OUT — Total amount of cash dispensed from cassette 2.
C2 END	CASSETTE 2 CASH END — Sum of starting amount, cash increases, cash decreases, and cash dispensed in cassette 2 for this balancing period.
C3 STR	CASSETTE 3 STARTING AMOUNT — Total cash in cassette 3 at the beginning of the balancing period.
C3 INC	CASSETTE 3 CASH INCREASE — Total increase to cash in cassette 3 by administrative transactions during this balancing period.
C3 DEC	CASSETTE 3 CASH DECREASE — Total decrease to cash in cassette 3 by administrative transactions during this balancing period.

Line or Field Label	Description
C3 OUT	CASSETTE 3 CASH OUT — Total amount of cash dispensed from cassette 3.
C3 END	CASSETTE 3 CASH END — Sum of starting amount, cash increases, cash decreases, and cash dispensed in cassette 3 for this balancing period.
C4 STR	CASSETTE 4 STARTING AMOUNT — Total cash in cassette 4 at the beginning of the balancing period.
C4 INC	CASSETTE 4 CASH INCREASE — Total increase to cash in cassette 4 by administrative transactions during this balancing period.
C4 DEC	CASSETTE 4 CASH DECREASE — Total decrease to cash in cassette 4 by administrative transactions during this balancing period.
C4 OUT	CASSETTE 4 CASH OUT — Total amount of cash dispensed from cassette 4.
C4 END	CASSETTE 4 CASH END — Sum of starting amount, cash increases, cash decreases, and cash dispensed in cassette 4 for this balancing period.
DEP0	Number of deposits made through the envelope depository.
DEP1	Number of deposits made through the commercial (e.g., Securomatic) depository.
MSG	Number of messages to the institution.
PAY	Number of payment transactions.
TOT	Total number of envelopes in the envelope depository.
CRD	CARD — Number of cards expected to be in the capture card bin in the ATM.

Line or Field Label	Description
SUB/BAL/ ADJ	An abbreviation of the administrative transaction type that was performed. SUB = PRINT MACHINE SUBTOTALS BAL = BALANCE TERMINAL ADJ = MID-POINT ADJUSTMENT

Sample Administrative Receipt

The following is an example of the NCR default ATM receipt format for an administrative midpoint adjustment transaction. This format is used for all administrative transactions.

```

1          A C I   D E M O
2
3      DATE          TIME   TERM
4      MM/DD/YY    HH:MM S1AN5070
5
6      1234567000000003-003
7
8      330 SOUTH 108TH AVE
9
10     C1 STR    10500   C1 INC    10000
11     C1 DEC         0   C1 OUT     100
12     C1 END    20460
13     C2 STR    49980   C2 INC         0
14     C2 DEC         0   C2 OUT     760
15     C2 END    49220
16     C3 STR         0   C3 INC         0
17     C3 DEC         0   C3 OUT         0
18     C3 END         0
19     C4 STR         0   C4 INC         0
20     C4 DEC         0   C4 OUT         0
21     C4 END         0
22     DEPO  55  DEP1  00
23     MSG   00  PAY   00
24     TOT   55  CRD   00  ADJ
25

```

```

1          2
12345678901234567890123456789

```

Administrative Receipt Templates

The administrative receipt is split into three sections. The top of the receipt is a header, which can contain any information you want. The body of the receipt contains cassette totals for hoppers 1 through 4. The bottom of the receipt contains depository totals.

There are one or more receipt templates for each part of the administrative receipt. The Device Handler process converts receipt templates for administrative transactions in the same manner as approved and denied financial transactions. However, administrative receipts require their own set of field type codes for the receipt templates. The field markers used are the same as those used for approved and denied financial transactions.

Field Type Code Descriptions

The following tables describe the standardized field type codes that are used in receipt templates for administrative receipts. Each field type code corresponds to a predetermined piece of fixed-length data. The data is retrieved from the ATD and the N50NAMS. Data names are shown in parentheses following each field type code description.

Cassette Field Type Code Descriptions

The following field type codes are used in the portion of the administrative receipt that includes cassette totals.

Code	Description
AT	The administrative transaction type (e.g., subtotals, adjustment or totals).
BV	The denomination of the bills in the cassette. This code specifies the hopper contents value and whether the hopper contains cash or noncurrency items. (ATDD1.ATDD1-CORE.HOPR <i>i</i> .BILL-VAL where <i>i</i> refers to cassettes 1 through 4).
CD	The number of bills removed from the cassette by midpoint adjustment during the current balancing period multiplied by the denomination of the bills in the cassette. This field type code has a 13-byte length. (ATDD1.ATDD1-CORE.HOPR <i>i</i> .CASH-DECR $\times$ ATDD1.ATDD1-CORE.HOPR <i>i</i> .BILL-VAL where <i>i</i> refers to cassettes 1 through 4).

Code	Description
CE	The current number of bills in the cassette multiplied by the denomination of the bills in the cassette. This field type code has a 13-byte length. (ATDD1.ATDD1-CORE.HOPR <i>i</i> .CASH-END $\times$ ATDD1.ATDD1-CORE.HOPR <i>i</i> .BILL-VAL where <i>i</i> refers to cassettes 1 through 4).
CI	The number of bills added to the cassette by midpoint adjustment during the current balancing period multiplied by the denomination of the bills in the cassette. This field type code has a 13-byte length. (ATDD1.ATDD1-CORE.HOPR <i>i</i> .CASH-INCR $\times$ ATDD1.ATDD1-CORE.HOPR <i>i</i> .BILL-VAL where <i>i</i> refers to cassettes 1 through 4).
CO	The number of bills dispensed from the cassette or removed from the cassette for cardholder withdrawals during the current balancing period multiplied by the denomination of the bills in the cassette. This field type code has a 13-byte length. (ATDD1.ATDD1-CORE.HOPR <i>i</i> .CASH-OUT $\times$ ATDD1.ATDD1-CORE.HOPR <i>i</i> .BILL-VAL where <i>i</i> refers to cassettes 1 through 4).
CS	The number of bills in the cassette at the beginning of the current balancing period of the terminal multiplied by the denomination of the bills in the cassette. This field type code has a 13-byte length. (ATDD1.ATDD1-CORE.HOPR <i>i</i> .BEG-CASH $\times$ ATDD1.ATDD1-CORE.HOPR <i>i</i> .BILL-VAL where <i>i</i> refers to cassettes 1 through 4).
HN	The cassette number.
RA	The response text for denied administrative transactions (located in the admin_resp_text_g table in the N50NAMS). This code is used with denied administrative transaction receipts. The printed receipt text depends on the value of ATM.N50.TRAN-RESP-CDE. The Device Handler process searches the admin_resp_text_g table in the N50NAMS to find a match on the response code. The corresponding text is then moved to the print buffer. The response text is configurable and is described in the topic “Response Code Text Table for Administrative Receipts,” later in this section.

Note: Each cassette field type code (i.e., CS, CI, CD, CO, CE, BV) can be followed with a variable length monetary symbol in the receipt template to identify the monetary unit used for the transaction. The following tags are used to describe the contents of the cassettes:

C 00 = Cash
C 01 = Coin
N 02 = Noncurrency item or travelers check

Currency Symbols

The amount field type codes (BV, CD, CE, CI, CO and CS) can have a monetary symbol (such as \$, •, or £) that prints on the receipt. You can enter the monetary symbol either manually or have it determined automatically by the Device Handler process. To enter the symbol manually, type the appropriate symbol after each occurrence of the field type code. Any text after the code is considered part of the monetary symbol until the next field marker is encountered.

The monetary symbol that is used can be determined automatically by the use of a table in the N50NAMS File. For the symbol to be determined automatically, you must enter CC\$ after the field type code. The symbol is determined by the currency code table (rcpt_money_sym_tbl_g). This table contains an entry for each monetary system for which you want a special symbol to be printed. Each entry in the table contains the country code, the number of characters required in the sequence to send the special symbols to the receipt printer (including escape codes), the number of characters taken up on the receipt as a result of the sequence, and the 10-character sequence that prints the proper monetary symbol. If a country code is encountered that is not in the table, the 3-character ISO currency code is used (for example, USD for US dollars). If neither the country code nor the ISO currency code exist in the table, no symbol is printed.

Depository Field Type Code Descriptions

The following field type codes are used in the portion of the administrative receipt that includes depository totals.

Code	Description
AC	The total amount of commercial deposits accepted since the terminal was last balanced (ATDD1.ATDD1-CORE.AMT-CMRCL-DEP).

Code	Description
AD	The unverified amount of deposits accepted since the terminal was last balanced (ATDD1.ATDD1-CORE.AMT-DEP).
AK	The unverified amount of checks cashed and received since the terminal was last balanced (ATDD1.ATDD1-CORE.AMT-CHK).
AP	The unverified amount of payments enclosed received since the terminal was last balanced (ATDD1.ATDD1-CORE.AMT-PAY).
CK	The total number of checks cashed since the terminal was last balanced (ATDD1.ATDD1-CORE.NUM-CHK).
CR	The total number of cards retained since the terminal was last balanced (ATDD1.ATDD1-CORE.CRDS-RET).
D0	The number of deposits received since the terminal was last balanced (ATDD1.ATDD1-CORE.NUM-DEP).
D1	The number of commercial deposits received since the terminal was last balanced (ATDD1.ATDD1-CORE.NUM-CMRCL-DEP).
EN	The sum of the following values: number of deposits, number of commercial deposits, number of message envelopes, number of payment envelopes, and number of checks cashed since the terminal was last balanced.
MG	The total number of message to financial institution envelopes received since the terminal was last balanced (ATDD1.ATDD1-CORE.NUM-MSG).
PY	The total number of payment envelopes received since the terminal was last balanced (ATDD1.ATDD1-CORE.NUM-PAY).

Note: Each depository amount field type code can be followed with a variable length monetary symbol, which identifies the monetary unit used for the transaction.

Administrative Templates

The administrative header does not have a default receipt template in the RCPT. If you create a receipt type of ADHD, this receipt type is used only for administrative transactions. If the ADHD receipt type does not exist, the HDR receipt type is used for both financial and administrative transactions.

The following are representations of the default entries in the RCPT for administrative receipts.

Transaction Type	Receipt Type	Default RCPT Template Data
Administrative transaction denied	ADDN	#E^4@ADMIN TRANSACTION DECLINED> #E^4@RA>>
Cassette Totals	ADHO	#E^4@C#HN @STR#CS\$> #E^4@C#HN @INC#CI\$@ C#HN @DEC#CD\$> #E^4@C#HN @OUT#CO\$@ C#HN @END#CE\$>
Depository Totals (for terminal subtotalling, terminal balancing, and midpoint adjustment)	ADTO	#E^4@DEP0#D0 @DEP1#D1 @CHK #CK> #E^4@MSG #MG @PAY #PY> #E^4@TOT #EN @CRD #CR #AT>

Response Code Text Table for Administrative Receipts

This response code text table is a separate table in the N50NAMS and is used in conjunction with the #RA (Response Text Administrative) field type code when that particular field type code is used in an administrative receipt template for denied transactions.

The response code text table for denied administrative transactions provides additional receipt text for denied transactions. This text gives the service person an explanation for the denied transaction and is in addition to the text provided in the receipt template.

When the Device Handler process encounters a #RA field type code in a receipt template, the process moves to the administrative response code text table and scans for a match on the internal response code it generates or that it receives from the Authorization process. From this point on, the Device Handler process uses the administrative response code text table in the same manner as the response code text table for financial transactions. This processing is described earlier in this section. The following entries are the default entries in the response code text table for denied administrative transactions. All of the default entries are in English; however, up to three languages can be defined for the NCR terminals.

Response Code	Initial Message Text
4	UNABLE TO PROCESS AMT
26	ATD BILL VALUE ZERO
27	INVALID AMT FOR HOPPER
30	ATM ALREADY BALANCED
36	TRANSACTION CANCELLED
143	CPLF PRINT RQST DENIED

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A: Statement Print Data Compression

This appendix describes the statement print data compression feature. This feature is available for use with statement print transaction requests initiated at NCR 5XXX-series ATMs that are supported by the BASE24-atm NCR 5XXX Device Handler process.

This appendix includes the following information:

- A description of data compression and its purpose
- A description of BASE24-atm support of data compression
- An analysis of the impacts of data compression
- BASE24-atm data compression standards
- Statement print message specifications
- Configuration requirements for the statement print data compression feature

Data Compression on Statement Print Transactions

Data compression involves reducing, or compressing, the amount of transaction data that is carried in an ATM transaction message. Reducing the amount of transmitted data results in a more efficient use of processing resources. Data compression is accomplished by reducing the number of characters and spaces transmitted within an individual transaction.

The greatest benefits of data compression can be achieved by reducing the amount of data transmitted that is required to print a statement. An ATM statement print transaction provides the cardholder with an account statement containing a detailed listing of the cardholder's account transactions for a specified period of time. Statements are printed on either receipt-size paper using a 40-column format or, when so equipped, a larger statement printed on paper that accepts an 80-column format. The number of items listed on the receipt depends on the amount of transaction activity that has taken place on the account during the specified time and on the capability of the host to store and send such data. Statement print transactions usually contain more lines of information than financial transactions.

The additional amount of information contained in a statement print transaction requires that a series of messages be sent back and forth between the BASE24-atm system and the host. These messages consist of request messages from the BASE24-atm system to the host and response messages from the host to the BASE24-atm. The number of request and response messages sent between BASE24 system and the host depends on the amount of transaction activity on the account and on the capacity of the host for sending statement information. As a result, compared to financial transactions, statement print transactions take longer to send, require increased use of communications lines, and place a heavier burden on transaction processing facilities.

Forms of Data Compression

Two techniques are used to compress the data in statement print transactions. These techniques are referred to as *space compression* and *text compression*. The benefits of data compression can be achieved by using either space compression or text compression or by using a combination of both techniques. Implementing both techniques results in the greatest efficiency. These two techniques are discussed below.

Space Compression

Space compression reduces the number of blank spaces and repetitive characters transmitted in a statement print transaction message. Space compression uses special characters to represent multiple spaces, blank lines, and repetitive characters contained in the statement print transaction message as it is carried through the ATM transaction processing system.

For example, in a statement print transaction, the first line of information can contain ten blank spaces between two items of information. When the statement print transaction message is sent, a space compression symbol is substituted for the ten blank spaces. Using space compression symbols to represent blank spaces and repetitive characters reduces the amount of data transmitted in a statement print message.

Text Compression

Text compression reduces the amount of descriptive information (for example, headings and labels) transmitted in a statement print transaction message. Text compression uses special characters to represent the descriptive text contained in the statement print message.

For example, one line of text in a statement print transaction might be displayed as AVAILABLE BALANCE. When the statement print transaction message is sent, a text compression symbol is substituted for the words AVAILABLE BALANCE. One text compression symbol replaces two words in the message. Using text compression symbols to represent descriptive text reduces the amount of data sent in a statement print message. You can define the wording of text messages for the type of transaction and printer used.

BASE24-atm Statement Print Data Compression Support

The BASE24-atm product establishes data compression standards that allow compressed data to be carried in a statement print transaction message. Data compression standards consist of a set of reserved characters used for space and text compression. These reserved characters are referred to as *format codes*.

BASE24-atm data compression standards affect the statement print transaction message at two separate points in the statement print transaction flow. During statement print transaction processing, the response to a statement print transaction request initiated at an ATM is sent from the host computer to the BASE24-atm system. In turn, the BASE24-atm product recodes, reformats, and sends the statement print data to the ATM device.

Note: Compression information sent from the host computer to the BASE24-atm system is in a different format than the data compression information sent from the BASE24-atm system to the ATM.

BASE24-atm Data Compression Message Format

BASE24-atm standardized data compression format codes are used to carry statement print messages from the host to the BASE24-atm system. That is, the statement print message format sent from the host to the BASE24-atm system contains the set of reserved characters established by the BASE24-atm system to achieve space and text compression.

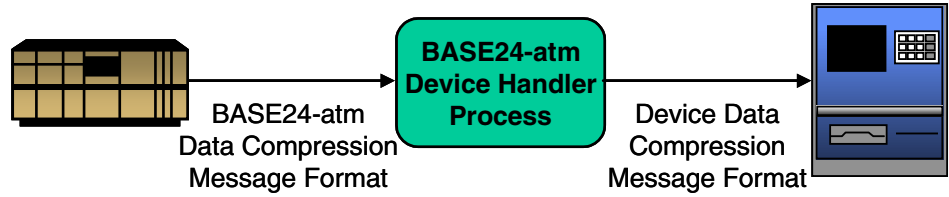
Device Data Compression Message Format

ATM devices do not support the standardized BASE24-atm data compression message format used between the host and the BASE24 system. Therefore, before the compressed statement print message from the host can be sent to an ATM, the BASE24-atm system converts the standardized BASE24-atm data compression format codes into a device-specific data compression message format supported by the ATM. The BASE24-atm Device Handler process converts the BASE24-atm data compression message format into the device-specific data compression message format used by the ATM.

Note: The device-specific data compression message format for the NCR 5XXX ATMs supports space compression only; it does not support text compression.

Use of Data Compression Message Formats

The following diagram is a conceptual illustration of the data compression message formats used during two segments of the statement print transaction flow.



Data Compression Impacts

One of the primary objectives of implementing the BASE24-atm statement print data compression feature is to decrease the number of messages required to generate a statement print transaction. Decreasing the number of statement print messages required to complete a transaction can result in significant savings to system resources.

The performance gains that can be achieved by implementing the BASE24-atm statement print data compression feature are directly related to the extent that space and text compression standards are used, and the number of statement print transactions an institution typically processes during a given period of time.

Reducing the number of messages to process a statement print transaction can impact the following areas:

- BASE24-atm processing. Fewer messages mean less interprocess traffic for statement print transactions.
- ATM response time. Improved response time means faster service for the cardholders who have requested statement print transactions.
- Communications lines usage. Fewer messages per transaction mean more efficient use of communications lines between the host, the BASE24 system, and the ATM.

The extent to which any of the above factors is affected is based on the number of statement print transactions the institution typically handles as well as the extent to which data compression is implemented.

BASE24-atm Data Compression and Device Data Compression

Two methods of data compression are used for statement print transaction flows. One method is used between the host and the BASE24-atm system. The other is used between the BASE24-atm Device Handler process and the ATM device.

Comparison of Data Compression Formats

The following table provides an example of how the BASE24-atm data compression format differs from the NCR 5XXX device data compression format.

Data Compression Method	Maximum Number of Compressed Spaces	Space Compression Symbol	Text Compression
BASE24-atm format (Host to BASE24)	35	[^] x	Supported
NCR 5XXX device format (BASE24 to Device)	15	Shift Out control character (i.e., an ASCII code) and ?	Not supported

BASE24 space compression symbols and their use are explained later in this appendix. Refer to the *NCR NDC+ Programmer's Reference Manual* for more information about NCR 5XXX space compression symbols.

Relationship Between BASE24-atm Data Compression Standards and Device Data Compression

Because there is a difference between the BASE24-atm data compression format (host to BASE24) and the data compression format used by the ATM devices (BASE24 to device), the BASE24-atm Device Handler process provides the connecting link between the two formats so that one format is converted to the other.

When the Device Handler process receives statement print messages from the host that contain text compression symbols, the BASE24 conversion process uses information in the N50NAMS to convert the received symbols into predetermined text used by the ATM.

N50NAMS File

The N50NAMS is an edit file that contains a series of tables used by the Device Handler processes. You can edit these tables to allow customization of specific types of information displayed on ATM receipts or screens. At the time the Device Handler process is compiled, the information in the N50NAMS is incorporated into the Device Handler object file.

N50NAMS Description

The N50NAMS contains a table with a series of identifiers that cross reference BASE24-atm data compression symbols. The Device Handler process uses these identifiers as a key to convert BASE24-atm compression symbols into a format that the designated ATM device can use to print statement print receipts.

Editing the N50NAMS

The statement print transaction description table (`ncr_stmt_prnt_tran_descr`) in the N50NAMS must be configured to support the BASE24-atm statement print data compression feature. This table allows you to define receipt text for statement print transactions. The Device Handler process uses this table to convert compressed text data into the actual receipt text displayed at the ATM.

To set up the ASCII hexadecimal index and its values in the statement print transaction description table, edit the N50NAMS. When setting up the statement print transaction description table, it is important to list the entries for each index code by frequency of use. This enables the Device Handler process to scan the index codes in the table more quickly.

After the statement print transaction description table has been built, the Device Handler process must be recompiled.

Statement Print Transaction Description Table

The statement print transaction description table in the N50NAMS supports the BASE24-atm statement print data compression feature. The Device Handler process uses information in this table to convert compressed text in the statement print transaction message from the BASE24-atm data compression format into a format that the designated ATM device can use to print statement print receipts.

ASCII Hexadecimal Index

The statement print transaction description table is built using an ASCII hexadecimal index with values ranging from 00 to FF. Each value used in the index must correspond to a BASE24-atm text compression symbol used by the host. For example, if an index code value of 00 is established and defined in the statement print transaction description table, the corresponding BASE24-atm text compression symbol [00 can be used by the host. The symbol [is a table lookup character, as explained later in this section. The compressed text in the message, represented by the symbol [00, is converted into the text string defined for index code value 00 in the statement print transaction description table.

You determine the value for each of the hexadecimal index codes for the primary, secondary and tertiary languages and establish the text associated with the index codes used in the statement print transaction description table. By editing the statement print transaction description table in the N50NAMS, message text and hexadecimal index codes are linked; the hexadecimal values defined for each of the messages determines the text that is printed on receipts or statements.

Using the ASCII Hexadecimal Index

The following description illustrates how the ASCII hexadecimal index is used. In this example, the value of index code 00 in the statement print transaction description table has been defined as CURRENT BALANCE. Whenever the Device Handler process receives the text compression symbol [00 in a statement print message from the host, the process finds the corresponding index value in the statement print transaction description table. In this case, since 00 has been defined as CURRENT BALANCE, the Device Handler process translates the text compression symbol [00 into the text CURRENT BALANCE before sending the statement print message to the ATM device. The ATM receipt or statement printer then prints CURRENT BALANCE on the receipt or statement form.

The following table shows how ASCII hexadecimal index values can be defined in the statement print transaction description table. You can define the statement text that equates to any given hexadecimal value.

Index Values (Hexadecimal)	Example of Institution-Defined ATM Statement Text
00	CURRENT BALANCE
01	BEGINNING BALANCE
02	AVAILABLE BALANCE
0A	RETURNED CHECK-NSF

Setting Up the ASCII Hexadecimal Index

You can establish up to 256 hexadecimal index codes in the statement print transaction description table. The length of the text for each of the index values is limited to a maximum of 25 ASCII characters and spaces for each hexadecimal value.

When setting up values for index codes, it is important to list entries by frequency of use; the most frequently used text messages should be included in the entries at the beginning (hexadecimal 00) of the table. If CURRENT BALANCE is the text displayed most frequently on printed statements, then the first index code defined in the statement print transaction description table should be CURRENT BALANCE. In this way, the Device Handler process can scan the statement print transaction description table more efficiently when converting BASE24-atm text compression symbols to ATM text.

Text Compression Conversion

For ATM devices that do not support text compression, such as the NCR 5XXX-series devices, the Device Handler process must convert BASE24-atm text compression symbols used in the statement print response message before sending the message to the ATM. This conversion is accomplished by using BASE24-atm data compression symbols called table lookup characters.

Table Lookup Characters

The standard BASE24-atm data compression format codes carried in the statement print message from the host to the Device Handler process include two text compression symbols called the table lookup character ([) and the suppress table lookup character (]). These lookup character symbols are used to compress text as the statement print message is carried from the host to the Device Handler process. The ATM device does not support the use of these table lookup character symbols; therefore, the Device Handler process must convert these symbols before sending the message on to the ATM device.

When the Device Handler process receives the statement print transaction message from the host, it matches the lookup character symbols in the message to the index codes in the ASCII hexadecimal index in the statement print transaction description table in the N50NAMS. The Device Handler process converts the table lookup characters into the text defined for each of the index codes before sending the statement print message to the ATM device.

The table lookup symbols consist of a bracket character ([or]) followed by two hexadecimal digits from the ASCII hexadecimal index. The lookup symbol is represented as [*hh* or]*hh* where *hh* is a two-digit hexadecimal value. The table lookup symbol is a cross reference to the ASCII hexadecimal index in the statement print transaction description table in the N50NAMS. For example, the table lookup symbol might take the following forms: [01,]99, or [FF.

Use of the @ Symbol with Table Lookup Characters

The @ symbol is used in conjunction with the table lookup character ([) or the suppress table lookup character (]) to determine whether blanks are displayed or suppressed when transmitting and formatting ATM statement printer information. The @ symbol is placed after specific table entries in the statement print transaction description table.

When the table lookup character ([) is used to access a table entry, @ symbols placed after a table entry are converted to blank spaces on the statement receipt. The @ symbols are converted to blank spaces up to the maximum table width of one line.

When the suppress table lookup character (]) is used to access an entry, blanks represented by an @ symbol are suppressed beginning with the first @ symbol after the table entry. Suppressing blanks enables the host to eliminate the blank space after the receipt text message and insert cardholder-specific data immediately after the text on the statement receipt. Cardholder-specific data to be inserted is included in the statement data response message sent from the host.

Using the table lookup characters in combination with the @ symbols in a table entry makes it possible to display one table entry in the statement print transaction description table in different ways.

The following table illustrates how the same table entry can be displayed in different ways on the ATM statement printer receipt.

Table entries shown and character count are for illustration purposes. Numbers shown represent cardholder check numbers and are included for illustration purposes.

Text Compression Symbol Used		Statement Print Transaction Table Entry	ATM Statement Printer Receipt Text
Lookup Character	[	RET CK # @@@@@@@@@@@@@@@@	RET CK # 5600
Suppress Lookup Character	]	RET CK # @@@@@@@@@@@@@@@@	RET CK #5600
Lookup Character	[	RET CK #	RET CK # 1675
Suppress Lookup Character	]	RET CK #	RET CK # 1675

Example of Text Compression Conversion

The following is an example of how the Device Handler converts text compression symbols into statement print receipt text using the statement print transaction description table in the N50NAMS.

Statement Print Transaction Description Table

You edit the statement print transaction description table in the N50NAMS to establish ASCII hexadecimal index codes and specify the values for each of the index codes in the table.

Sample entries used in a statement print transaction description table are shown in the following table.

Index Values (Hexadecimal)	Institution-Defined ATM Statement Text
00	CURRENT BALANCE
01	BEGINNING BALANCE
02	AVAILABLE BALANCE
0A	RETURNED CHECK-NSF

In this example, the Device Handler process uses the values in the statement print transaction description table to convert text compression symbols received in the statement print message from the host.

Table Lookup Character

The Device Handler process receives table lookup symbols in the statement print message received from the host.

Using the index entries shown in the above table, the Device Handler process converts the table lookup symbol [01 as follows:

[01 = BEGINNING BALANCE

Conversion

Using the index entries shown in the statement print transaction description table above, the following table illustrates how the Device Handler process converts compressed text data received in the statement print message from the host.

Index Value	Format Code	Data Received by BASE24 from the Host	Data Sent to the ATM by Device Handler process
00	[	[00	CURRENT BALANCE
01	[	[01	BEGINNING BALANCE
02	[	[02	AVAILABLE BALANCE
0A	[	[0A	RETURNED CHECK-NSF

BASE24-atm Data Compression Format Codes

The BASE24-atm product uses a standardized set of format codes that allows a host to send a compressed statement print message to the BASE24-atm system.

These codes can be placed in the statement print response messages sent by the host to the BASE24-atm system. The host does not have to use all of the data compression format codes. These format codes can be used to achieve space compression or text compression or both. However, the greatest savings are attained when both forms of compression are used.

The following table identifies and describes the BASE24-atm data compression format codes used for space and text compression.

Format Code	Definition
{	Line Feed Symbol — A left brace ({}) indicates the end of a line of statement data. This symbol must be placed at a column position less than or equal to the value of the PRNT-SIZ field provided in the statement print request message.
}	Page Feed Symbol — A right brace ({}) indicates the last line of a page in a multiple-page receipt. This symbol must be at a line position less than or equal to the value specified in the LINES/PAGE field on BASE24-atm Terminal Data files (ATD) screen 7. This value is not provided for the host in the statement print request message
^x	Compression Symbol — The space compression symbol is a caret (^) followed by the variable <i>x</i> , where <i>x</i> indicates the number of spaces that have been compressed. Valid values for <i>x</i> are 1 through 9 and A through Z for a maximum of 35 spaces. The values A through Z indicate the number of spaces according to alphabetic order: A = 10, B = 11, etc.
> <i>znn</i>	Repeat Symbol — The repeat symbol is a right angle bracket (>) followed by the variable <i>znn</i> , where <i>z</i> indicates the character to be repeated and <i>nn</i> is the number of times it is to be repeated.

Format Code	Definition
<i>[hh</i>	Table Lookup Symbol — The table lookup symbol is a left bracket ([) followed by the variable <i>hh</i> , where <i>hh</i> (hexadecimal value) is a two-digit cross-reference into the alpha hexadecimal index in the N50NAMS statement print transaction description table. Using the table lookup symbol ([), all @ symbols used in a specific table entry are changed to blanks up to the maximum line length of 35.
<i>]hh</i>	Suppress Table Lookup Symbol — The suppress table lookup symbol is a right bracket (]) followed by the variable <i>hh</i> , where <i>hh</i> (hexadecimal value) is the two-digit cross-reference into the alpha hexadecimal index in the N50NAMS statement print transaction description table. Using the suppress table lookup symbol (]), blanks are suppressed at the first @ symbol found in a specific table entry.
<i>\nn</i>	Increment Date Symbol — The increment date symbol, or date calculation, is a backslash (\) followed by the variable <i>nn</i> , where <i>nn</i> indicates the number of days that should be added to the date in the STMT.LAST-STMT-DAT field in the STM. This format code tells the Device Handler process to calculate a date using the STMT.LAST-STMT-DAT field in the STM. The format of this date is based upon the value of the DATE-DISPLAY-TYPE param in the Logical Network Configuration File (LCONF). This calculated date is printed on the statement receipt where this symbol occurred.
<i>?nnn</i>	Ignore Text Symbol — The ignore text symbol is composed of a question mark followed by a three-character (decimal) variable <i>nnn</i> , where <i>nnn</i> indicates the number of characters the Device Handler process is to ignore in the statement print data from the host. This symbol, if used, can be included in the first position of the statement data to indicate that the specified number of characters in the statement print data are not to be used as statement data, but are echoed back to the host as is. The purpose of this symbol is to allow the host to set aside a certain number of characters in the statement data for tracking purposes. Use of this symbol requires setting the IGNORE-TXT field on ATD screen 7 to a value of 1.

Application of Data Compression

The following discussion provides examples of how data compression standards work when applied to statement print transaction data. This is illustrated by showing a sample statement print receipt and applying data compression standards to information in the receipt. The receipt is used as a representation of the effect data compression has on the information contained in the statement print message.

The examples show how data compression affects the statement print message format in the following two segments of the transaction flow:

- From the host to the BASE24-atm Device Handler process
- From the BASE24-atm Device Handler process to the ATM device

Also included are descriptions of the data compression symbols used for selected lines of the statement print receipt.

Sample Statement Print Receipt

An example of an ATM statement print receipt appears below. This receipt represents the statement print transaction information before data compression techniques are applied.

1	ACCOUNT UPDATE SINCE MOST RECENT STATEMENT			
2	INFORMATION FROM mm/dd/yy TO mm/dd/yy hh:mm xM.			
3				
4	ACCOUNT NUMBER - 3333331111123456			
5				
6	DATE	TRANSACTION	DEBIT	CREDIT
7				
8	mm/dd/yy	BEGINNING BALANCE		1,703.70
9				
10	mm/dd/yy	RETURNED CHECK-NSF		22.97
11	mm/dd/yy	RETURNED CHECK-NSF		56.41
12	mm/dd/yy	CHECK WITHDRAWAL #5654	60.90	
13	mm/dd/yy	WITHDRAWAL BIG BUCKS ATM	20.00	
14	mm/dd/yy	DEPOSIT BIG BUCKS ATM		840.76
15	mm/dd/yy	DEPOSIT BIG BUCKS ATM		840.76
16	mm/dd/yy	DEPOSIT		1,000.00
17	mm/dd/yy	WITHDRAWAL BIG BUCKS ATM	10.00	
18	mm/dd/yy	CHECK WITHDRAWAL #5653	20.00	
19	mm/dd/yy	CHECK WITHDRAWAL #5660	247.82	
20	mm/dd/yy	WITHDRAWAL BIG BUCKS ATM	40.00	
21	mm/dd/yy	STOP PAYMENT CHARGE	10.00	
22	mm/dd/yy	WITHDRAWAL BIG BUCKS ATM	50.00	
23	mm/dd/yy	WITHDRAWAL BIG BUCKS ATM	30.00	
24	mm/dd/yy	DEPOSIT BIG BUCKS ATM		7.62
25	mm/dd/yy	CHECK WITHDRAWAL #5658	71.19	
26	mm/dd/yy	CHECK WITHDRAWAL #5659	3.00	
27	mm/dd/yy	CHECK WITHDRAWAL #5661	31.99	
28	mm/dd/yy	CHECK WITHDRAWAL #5662	1,012.50	
29	mm/dd/yy	WITHDRAWAL BIG BUCKS ATM	20.00	
30	mm/dd/yy	CHECK WITHDRAWAL #5665	32.70	
31	mm/dd/yy	CHECK WITHDRAWAL #5664	69.96	
32	mm/dd/yy	DEPOSIT		1,222.33
33	mm/dd/yy	CHECK WITHDRAWAL #5663	61.50	
34	mm/dd/yy	WITHDRAWAL BIG BUCKS ATM	75.00	
35	mm/dd/yy	WITHDRAWAL BIG BUCKS ATM	5.00	
36	mm/dd/yy	WITHDRAWAL BIG BUCKS ATM	40.00	
37	mm/dd/yy	CHECK WITHDRAWAL #5667	78.29	
38	mm/dd/yy	CHECK WITHDRAWAL #5668	89.69	
39	mm/dd/yy	CHECK WITHDRAWAL #5669	21.50	
40	mm/dd/yy	WITHDRAWAL BIG BUCKS ATM	5.00	
41	mm/dd/yy	ACCOUNT UPDATE FEE	1.00	
42	*****			
43				
44	mm/dd/yy	CURRENT BALANCE		1,865.71
45				
46	mm/dd/yy	AVAILABLE BALANCE		1,965.71

Sample Statement Print Transaction Description Table

The following illustration shows the statement print transaction description table used for the statement print transaction represented by the receipt shown on the previous page. You can modify the text used in this table to suit the needs of the institution.

ASCII	
Hexadecimal	
Index	Table Description
	1 2
	1234567890123456789012345
01	"BEGINNING BALANCE "
02	"AVAILABLE BALANCE "
03	"CURRENT BALANCE "
04	"ACCOUNT UPDATE FEE "
05	"WITHDRAWAL BIG BUCKS ATM "
06	"CHECK WITHDRAWAL @@@@@@@@ "
07	"STOP PAYMENT CHARGE "
08	"DEPOSIT BIG BUCKS ATM "
09	"DEPOSIT "
0A	"RETURNED CHECK-NSF "
94	"DEBIT CREDIT@@@@ "
95	"DATE TRANSACTION@@@@ "
96	"ACCOUNT NUMBER - @@@@@@@@ "
97	"INFORMATION FROM @@@@@@@@ "
98	" RECENT STATEMENT@@@@@@@@ "
99	"ACCOUNT UPDATE SINCE MOST "

Note: Entries between ASCII hexadecimal index values 0A and 94 are not shown in this table because they are not used in the statement illustration.

Message Data — Host to BASE24-atm

The example below illustrates how the statement print transaction data looks when it is transmitted from the host to the Device Handler process. This example applies BASE24-atm space and text compression standards to the statement print transaction data contained in the statement print receipt shown previously.

Stmt Line	Host Compressed Data Format	Line Length	Position	Message Number
1	^G[99]98{	9	1-9	1
2	^E]97\00 TO \23 hh:mm xM.{	26	10-35	
3	{	1	36	
4	4]963333331111123456{	22	37-58	
5	{	1	59	
6	^6]95^O]94{	11	60-70	
7	{	1	71	
8	^6\nn [01^O1,703.70{	21	27-92	
9	{	1	93	
10	^6\nn [0A^R22.97{	18	94-111	
11	^6\nn [0A^R56.41{	18	112-129	
12	^6\nn]06#5654^D60.90{	23	130-152	
13	^6\nn [05^B20.00{	18	153-170	
14	^6\nn [08^Q840.76{	19	171-189	
15	^6\nn [08^Q840.76{	19	190-208	
16	^6\nn [09^O1,000.00{	21	209-229	
17	^6\nn [05^B10.00{	18	230-247	
18	^6\nn]06#5653^D20.00{	23	248-270	
19	^6\nn]06#5660^C247.82{	24	271-294	
20	^6\nn [05^B40.00{	18	295-312	
21	^6\nn [07^B10.00{	18	313-330	
22	^6\nn [05^B50.00{	18	331-348	
23	^6\nn [05^B30.00{	18	1-18	2
24	^6\nn [08^S7.62{	17	19-35	
25	^6\nn]06#5658^D71.19{	23	36-58	
26	^6\nn]06#5659^E3.00{	22	59-80	
27	^6\nn]06#5661^D31.99{	23	81-103	
28	^6\nn]06#5662^A1,012.50{	26	104-129	
29	^6\nn [05^B20.00{	18	130-147	
30	^6\nn]06#5665^D32.70{	23	148-170	
31	^6\nn]06#5664^D69.96{	23	171-193	
32	^6\nn [09^O1,222.33{	21	194-214	
33	^6\nn]06#5663^D61.50{	23	215-237	
34	^6\nn [05^B75.00{	18	238-255	
35	^6\nn [05^C5.00{	17	256-272	
36	^6\nn [05^B40.00{	18	273-290	
37	^6\nn]06#5667^D78.29{	23	291-313	
38	^6\nn]06#5668^D89.69{	23	314-336	
39	^6\nn]06#5669^D21.50{	23	337-359	

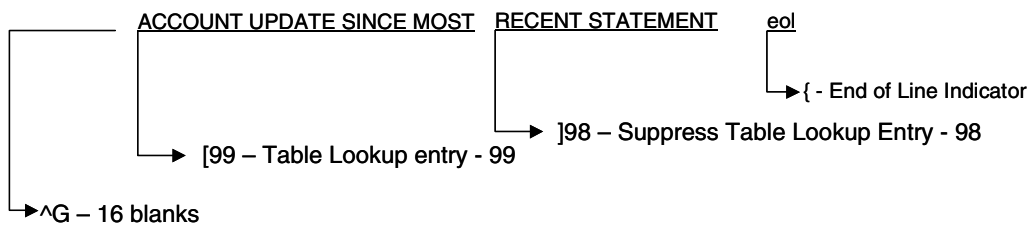
Stmt Line	Host Compressed Data Format	Line Length	Position	Message Number
40	^6\nn [05^C5.00{	17	19-35	3
41	^6\nn [04^C1.00{	17	36-52	
42	>*78{	5	53-57	
43	{	1	58	
44	^6\nn [03^O1,815.71{	21	59-79	
45	{	1	80	
46	^6\nn [02^O1,915.71}	21	81-101	

Applying Data Compression

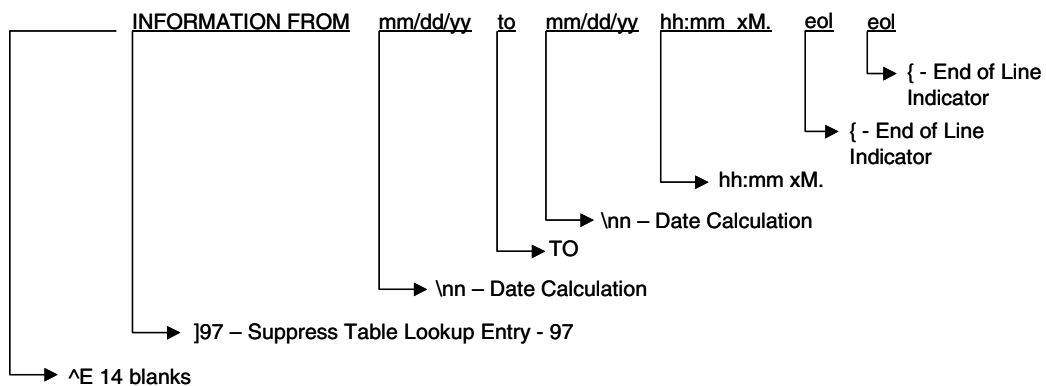
Using the sample statement print receipt and sample statement print transaction description table shown previously, the following examples describe how data compression symbols are used to compress the data in the statement print receipt.

The following pages show the data compression symbols sent in the statement print message for lines 1, 2, 3, 4, 6, 7, 8, 9, 10, 12, and 42 of the sample statement print receipt.

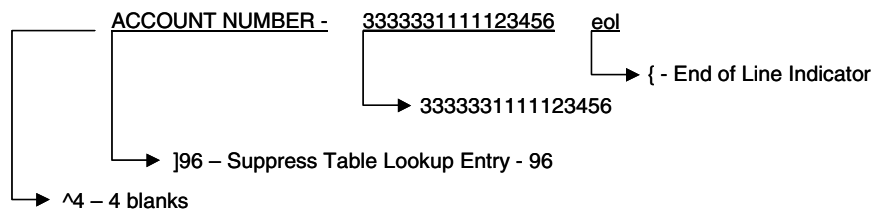
LINE 1



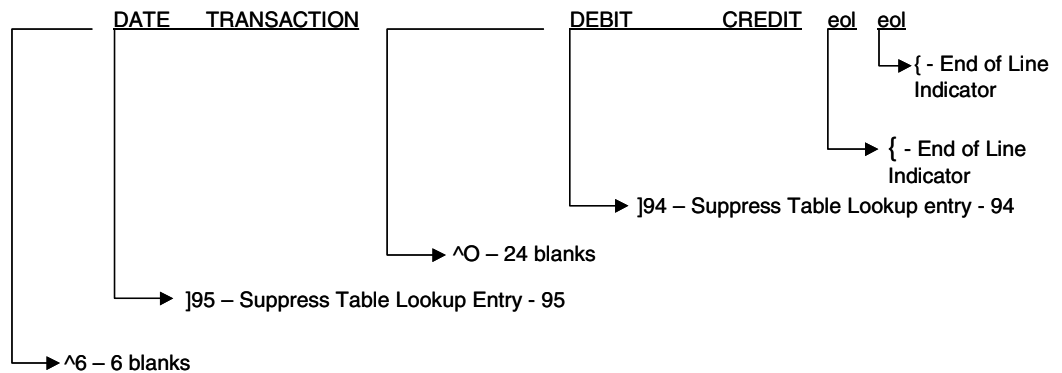
LINES 2 and 3



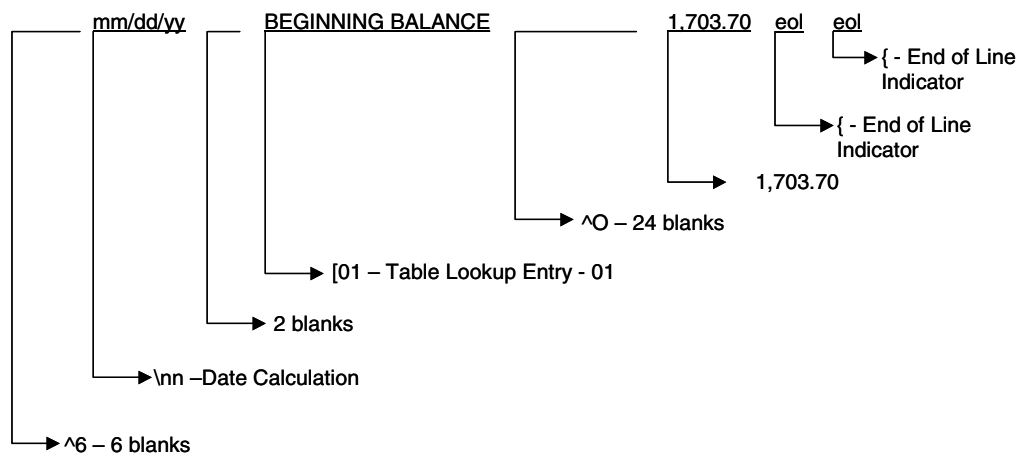
LINE 4



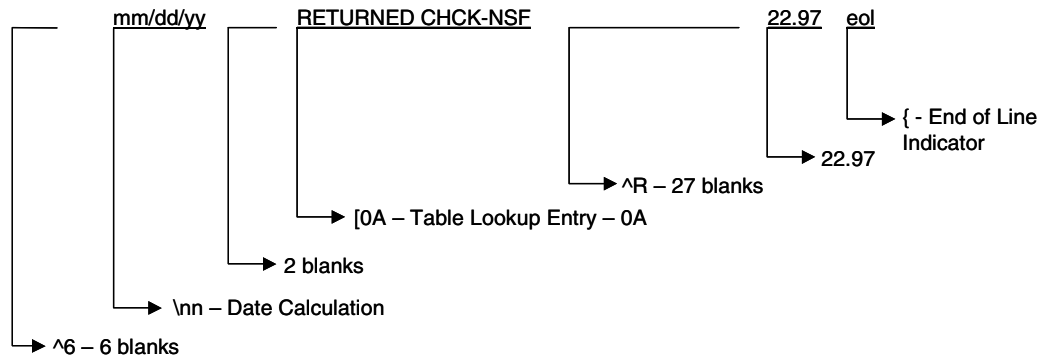
LINES 6 and 7



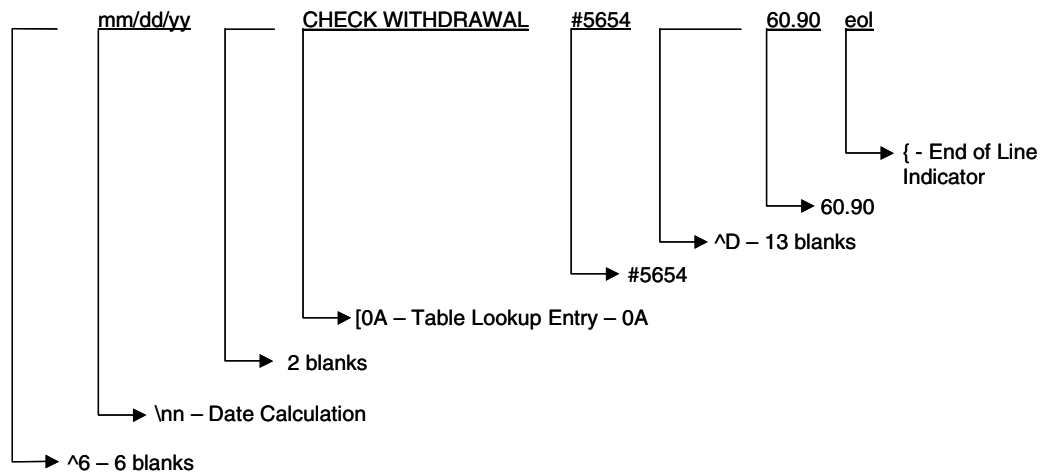
LINES 8 and 9



LINE 10



LINE 12



LINE 42



Message Data — BASE24-atm to the ATM Device

The example below illustrates how the message data looks when it is transmitted from the Device Handler process to the ATM device. BASE24-atm space compression standards have been used in this example for illustration purposes. The space compression symbols actually used vary from one device to another.

Stmt Line	Native Mode Compressed Data Format	Line Length	Position	Msg No.
1	^G ACCOUNT UPDATE SINCE MOST RECENT STATEMENT {	49	1-49	1
2	^EINFORMATION FROM mm/dd/yy TO mm/dd/yy hh:mm xM.{	50	50-99	
3	{	1	100	
4	^4ACCOUNT NUMBER - 3333331111123456{	36	101-136	
5	{	1	137	
6	^6DATE^6TRANSACTION^F^9DEBIT^ACREDIT{	36	138-173	
7	{	1	174	
8	^6mm/dd/yy BEGINNING BALANCE^F^F 1,703.70{	44	175-218	
9	{	1	219	
10	^6mm/dd/yy RETURNED CHECK-NSF^F^F^422.97{	42	220-261	
11	^6mm/dd/yy RETURNED CHECK-NSF^F^F^456.41{	42	262-303	
12	^6mm/dd/yy CHECK WITHDRAWAL #5654^F 60.90{	43	304-346	
13	^6mm/dd/yy WITHDRAWAL BIG BUCKS ATM^C20.00{	44	347-390	
14	^6mm/dd/yy DEPOSIT BIG BUCKS ATM^F^F840.76{	44	391-434	
15	^6mm/dd/yy DEPOSIT BIG BUCKS ATM^F^F840.76{	44	435-478	
16	^6mm/dd/yy DEPOSIT^F^F 1,000.00{	34	1-34	2
17	^6mm/dd/yy WITHDRAWAL BIG BUCKS ATM^C10.00{	44	35-78	
18	^6mm/dd/yy CHECK WITHDRAWAL #5653^F 20.00{	43	79-121	
19	^6mm/dd/yy CHECK WITHDRAWAL #5660^F247.82{	43	122-164	
20	^6mm/dd/yy WITHDRAWAL BIG BUCKS ATM^B40.00{	44	165-208	
21	^6mm/dd/yy STOP PAYMENT CHARGE^B10.00{	39	209-247	
22	^6mm/dd/yy WITHDRAWAL BIG BUCKS ATM^B50.00{	44	248-291	
23	^6mm/dd/yy WITHDRAWAL BIG BUCKS ATM^B30.00{	44	292-335	
24	^6mm/dd/yy DEPOSIT BIG BUCKS ATM^S7.62{	40	336-375	
25	^6mm/dd/yy CHECK WITHDRAWAL #5658^F 71.19{	43	376-418	
26	^6mm/dd/yy CHECK WITHDRAWAL #5659^F 3.00{	43	419-461	
27	^6mm/dd/yy CHECK WITHDRAWAL #5661^F 31.99{	43	1-43	3
28	^6mm/dd/yy CHECK WITHDRAWAL #5662^F1,012.50{	45	44-88	
29	^6mm/dd/yy WITHDRAWAL BIG BUCKS ATM^C20.00{	45	89-133	
30	^6mm/dd/yy CHECK WITHDRAWAL #5665^F 32.70{	43	134-176	
31	^6mm/dd/yy CHECK WITHDRAWAL #5664^F 69.96{	43	177-219	
32	^6mm/dd/yy DEPOSIT^F^F^A1,222.33{	34	220-253	
33	^6mm/dd/yy CHECK WITHDRAWAL #5663^F 61.50{	43	254-296	
34	^6mm/dd/yy WITHDRAWAL BIG BUCKS ATM^C75.00{	44	297-340	

Stmt Line	Native Mode Compressed Data Format		Line Length	Position	Msg No.
35	^6mm/dd/yy	WITHDRAWAL BIG BUCKS ATM^D5.00{	43	341-383	3
36	^6mm/dd/yy	WITHDRAWAL BIG BUCKS ATM^C40.00{	44	384-427	
37	^6mm/dd/yy	CHECK WITHDRAWAL #5667^F 78.29{	43	428-470	
38	^6mm/dd/yy	CHECK WITHDRAWAL #5668^F 89.69{	43	1-43	4
39	^6mm/dd/yy	CHECK WITHDRAWAL #5669^F 21.50{	43	44-86	
40	^6mm/dd/yy	WITHDRAWAL BIG BUCKS ATM^D5.00{	43	87-129	
41	^6mm/dd/yy	ACCOUNT UPDATE FEE^F^41.00{	39	130-168	
42	***** ... **{		79	169-247	
43	{		1	248	
44	mm/dd/yy	CURRENT BALANCE^F^F 1,815.71{	42	249-290	
45	{		1	291	
46	mm/dd/yy	AVAILABLE BALANCE^F^F 1,915.71{	44	292-335	

Statement Print Messages

Statement print transactions routed through the BASE24-atm system to the host do so in a series of transaction messages. These messages consist of request messages to the host and response messages from the host. Because of the amount of information carried in the statement print transaction, several request and response messages can be sent between the BASE24-atm system and the host to complete a single transaction.

BASE24-atm Statement Print Message Types

Transaction types used by the BASE24-atm product for statement print messages are as follows:

Type	Description
0200	Financial Transaction Request — A request to authorize a transaction.
0215	Statement Print Transaction Response — A response carrying the statement information for printing.
0205	Statement Print Transaction Request — A request for additional statement information for a transaction in progress.

BASE24-atm Statement Print Transaction Processing

The following steps describe the processing for statement print transactions being sent from the BASE24-atm system to the host.

1. The BASE24-atm system receives a request for statement information, which is sent to the host as a 0200 request message.
2. The host processes the 0200 message and returns a 0215 response message to the BASE24-atm system. The 0215 message carries the statement information.
3. If additional pages of statement information are required, the BASE24-atm system sends a 0205 request message to the host requesting additional information.

-
4. The host processes the 0205 message and returns a 0215 message carrying additional statement information.

This processing continues until all of the statement information has been sent.

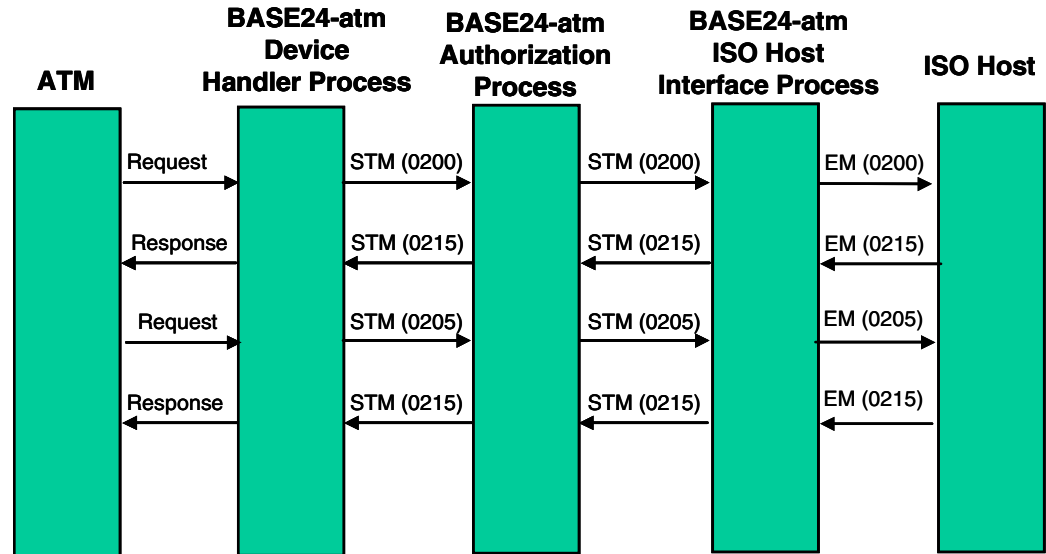
ISO Statement Print Messages

The BASE24 external message is based on the standard external message developed by the International Organization for Standardization (ISO) and is used to route messages between a host and the BASE24-atm system. The BASE24-atm Standard Internal Message (STM) is used to route messages between BASE24-atm processes. The BASE24-atm ISO Host Interface process provides the message interface between the BASE24-atm system and the host.

The fields used to carry data compression information are described elsewhere in this appendix.

For detailed information on the BASE24 external message (identified in the following diagram by the code EM), refer to the ***BASE24 External Message Manual***. For information on the BASE24-atm Standard Internal Message (STM) and the BASE24-atm ISO Host Interface process, refer to the ***BASE24-atm Transaction Processing Manual*** and the ***BASE24 ISO Host Interface Manual***, respectively.

The following diagram illustrates the message flow when the BASE24-atm system sends a statement print transaction to an ISO host. Although this diagram shows only two request and response cycles, each statement print transaction is different and may require a different number of cycles to complete.



ISO Statement Print Data Elements

Statement print data is carried in two elements in the BASE24 external message. Data element S-126 (BASE24-atm Additional Data) carries the request information needed for the statement print transaction request messages (0200 and 0205) from BASE24-atm to the host. Data element S-125 (BASE24-atm Account Indicator/Statement Print Data) carries the statement print transaction response data (0215) from the host to the BASE24-atm system.

Data Element S-126

BASE24-atm uses data element S-126 (Additional Data) to carry BASE24 message tokens. The Statement Print token (token 02) must be added to data element S-126 to include the statement print information in the external message to the host. For outgoing messages, the tokens included in data element S-126 are specified by records in the Token File (TKN). For information on configuring tokens to be included in outgoing external messages, refer to the ***BASE24 Tokens Manual***. For information on the structure of data element S-126, refer to the ***BASE24 External Message Manual***.

Data Element S-125

Data element S-125 (BASE24-atm Account Indicator/Statement Print Data) carries the statement print transaction response (0215) data from the host to the BASE24-atm system. Data element S-125 must be echoed back with the statement print transaction response data from the host. Data element S-125 is returned to the host by BASE24-atm in subsequent requests (0205) if the statement print transaction requires more than one request to the host. For information on the structure of data element S-125, refer to the ***BASE24 External Message Manual***.

Data Compression Fields

The statement print token contains the FRMT (format) field and the PRNT-SIZ (Printer Size) field. The FRMT field indicates to the host if the Device Handler process supports the BASE24-atm statement print data compression feature. The PRNT-SIZ field indicates the maximum number of characters per line supported by the Device Handler process. Information for the FRMT and PRNT-SIZ fields is taken from the ATD. These fields are displayed on the NCR-specific ATD screen 7.

The FRMT and PRNT-SIZ fields are loaded into the Statement Print token from the BASE24-atm Standard Internal Message (STM).

For information on the ATD, refer to the ***BASE24-atm Files Maintenance Manual***.

Setting Up the ATD

The BASE24-atm Terminal Data files (ATD) contain one record for each terminal in the logical network. These records define the characteristics and functionality of the terminal.

The ATD must be updated for the terminals that support the BASE24-atm statement print data compression feature. Changes to the ATD are necessary to specify if the terminal supports data compression and if the host uses the ignore text option.

The following fields on NCR 5XXX screen 7 are used for statement print data compression: LEFT COLUMN, RIGHT COLUMN, FRMT, and IGNORE-TXT. Refer to the ***BASE24-atm Files Maintenance Manual*** for more information about NCR 5XXX ATD screen 7 and complete descriptions of these fields.

Setting Up the EMF

External Message File (EMF) records specify which data elements are included in the BASE24 external message for incoming and outgoing messages. This file is used with the ISO-based external message format.

The following EMF settings are required for data elements S-125 and S-126 to support the BASE24-atm statement print data compression feature. These are the default settings for 0200, 0205, and 0215 messages.

PRODUCT #	MSG TYPE	Data Element S-126 Set To	Data Element S-125 Set To
01 (ATM)	0200	C	C (Outgoing)
01 (ATM)	0205	C	C
01 (ATM)	0215	C	M

If anything other than these values is used for elements S-125 or S-126, the values must be updated to support the BASE24-atm statement print data compression feature.

For more detailed information on updating the EMF, refer to the ***BASE24 Base Files Maintenance Manual***.

Configuring Token Processing

The Token File (TKN) is used to configure token processing. It specifies the following information:

- The tokens that get logged to the BASE24-atm Transaction Log File (TLF) and the Interchange Log Files (ILFs).
- The tokens that get extracted from the transaction log files by the Super Extract process. Extracting allows the token information to be processed by the host.
- The tokens that get sent to the host with each transaction in the ISO-based external message.

The Statement Print token contains the FRMT field which indicates whether the BASE24-atm statement print data compression feature is supported. The Token File (TKN) must be modified to ensure that this token gets sent from the BASE24 system to the ISO Host Interface process or the ISO BASE24 Interchange (BIC) Interface process in the external message and to ensure that this token gets logged to the appropriate transaction log files.

Refer to the ***BASE24 Tokens Manual*** for complete information on the Statement Print token (token ID 02) and on using the TKN to specify tokens to be sent in the external message. Refer to the ***BASE24 Base Files Maintenance Manual*** for detailed field descriptions for the TKN and the HCF.

B: Digital Audio Service

The NDC+ Digital Audio Service (DAS) enables the ATM to provide certain transaction response information, such as balance and account information in a voice response, in addition to or in place of a printed receipt. DAS uses a template similar to receipt templates to format variable audio response messages.

Note: No default DAS receipt templates are provided in the RCPT. If you want DAS support, you must create the appropriate DAS templates.

The Receipt File and DAS Templates

Configuring DAS audio receipts is similar to the configuring of printed receipts. DAS records are created and stored in the Receipt File (RCPT). To create a DAS record, the PRNTR TYPE field of the RCPT must be set to a value of D.

The format for DAS records are made up of DAS field type codes and associated audio file numbers that create the messages used in the voice response by the ATM. Like standard receipt field type codes, the field marker # precedes each DAS field type code.

DAS Field Type Codes

DAS field type codes correspond to predetermined pieces of data that the Device Handler process retrieves from the BASE24-atm Standard Internal Message (STM), the BASE24-atm Terminal Data files (ATD), or the N50NAMS. The field type code can be followed by an optional three-digit audio file number. The field code WV, described in the following table, requires the three-digit audio file number because it does not have any transactional data associated with it. The WV field type code is used only to specify a particular audio file.

Note: The audio messages “dollars” and “and cents” are automatically supplied by the Device Handler process whenever an amount (such as available balance, ledger balance, transaction amount, or surcharge amount) is specified. If the currency at the ATM is not U.S. dollars, the audio file numbers in the `ncr_das_rsrvd_tbl` in the N50NAMS must be changed and the code recompiled.

The following table describes the field type codes that are used in DAS templates. The variable *nnn* is an audio file number in the range of 000 through 999. This number is required for the *WVnnn* field type code and optional for all others. Data names are included in parentheses after each field type code description to indicate the specific data field from which the information is taken.

Code	Description
A1 <i>A1nnn</i>	The transaction amount (STM.RQST.AMT-1), for instance, the withdrawal amount or deposit amount. In the case of a withdrawal with a surcharge fee, A1 contains the withdrawal amount plus the surcharge. This total is deducted from the cardholder’s account. The audio messages “dollars” and “and cents” are automatically supplied by the Device Handler process.
A2 <i>A2nnn</i>	The transaction amount (STM.RQST.AMT-2). The audio messages “dollars” and “and cents” are automatically supplied by the Device Handler process.
A3 <i>A3nnn</i>	The transaction amount (STM.RQST.AMT-3). The audio messages “dollars” and “and cents” are automatically supplied by the Device Handler process.

Code	Description
AB ABnnn	The available balance (STM.RQST.AMT-3). The available balance voice response is used only if a response code 0 is returned from the BASE24-atm Authorization process and the CUSTOMER BALANCE INFO field on Institution Definition File (IDF) screen 13 and the CUSTOMER BALANCE INFORMATION field on ATD screen 8 indicate that Amount 3 (STM.RQST.AMT-3) is to be displayed. The audio messages “dollars” and “and cents” are automatically supplied by the Device Handler process.
AM AMnnn	The transaction amount (STM.RQST.AMT-1 + STM.RQST.AMT-2 for split transactions, STM.RQST.AMT-1 for other transactions). This code is used with the BASE24-atm self service banking (SSB) Enhanced ATM Application only. The audio messages “dollars” and “and cents” are automatically supplied by the Device Handler process.
CN CNnnn	The card number from the ATM card (ATM.N50.CRD-NUM) and the member number (STM.RQST.MBR-NUM). To provide the voice response for the card number only, see the CX code below.
CXll,rr CXnnn, ll, rr	The card number from the ATM card (ATM.N50.CRD-NUM). The variables <i>ll</i> and <i>rr</i> are used in truncation of the card number, where <i>ll</i> is the number of digits starting from the left that are provided and <i>rr</i> is the number of digits starting from the right that are provided. If the variables are not present, the card number is provided in the voice response in its entirety or for a maximum of the first 15 digits in the number. Refer to the topic “Account Number Truncation” later in this section for more information about using these variables to control truncation for the voice response of a card number.
FA FAnnn	The <i>from</i> account type (STM.FROM-ACCT-TYP). The Device Handler process, using the das_acct_tbl_g table in the N50NAMS, matches the transaction <i>from</i> account type to a specific reserved audio file number. The message in the corresponding audio file is then played.

Code	Description
FR ll,rr FR nnn,ll , rr	The <i>from</i> account number (STM.RQST.FROM-ACCT). The variables ll and rr are used in truncation of the card number, where ll is the number of digits starting from the left that are provided and rr is the number of digits starting from the right that are provided. If the variables are not present, the card number is provided in the voice response in its entirety or for a maximum of the first 15 digits in the number. Refer to the topic “Account Number Truncation” later in this section for more information about using these variables to control truncation for the voice response of an account number.
LB LB nnn	The ledger balance (STM.RQST.AMT-2). The ledger balance is voiced only if a response code of 0 is returned from the BASE24-atm Authorization process and the CUSTOMER BALANCE INFO field on Institution Definition File (IDF) screen 13 and the CUSTOMER BALANCE INFORMATION field on ATD screen 8 indicate that Amount 2 (STM.RQST.AMT-2) is to be displayed. The audio messages “dollars” and “and cents” are automatically supplied by the Device Handler process.
MB MB nnn	Cardholder member number (STM.RQST.MBR.NUM).
NC NC nnn	Non–Currency Dispense item description. The Device Handler process searches the <code>das_ncd_tbl_g</code> in the N50NAMS to find a match on the hopper contents. The message in the corresponding audio file is then played. This code is used with the Non–Currency Dispense add-on product only.
ND ND nnn	The net deposit amount (STM.RQST.AMT-1 minus STM.RQST.AMT-2). This code is used with the SSB Device Handler Extension only. The audio messages “dollars” and “and cents” are automatically supplied by the Device Handler process.
NP NP nnn	The net deposit amount for a partial dispense transaction (STM.RQST.AMT-1 minus STM.RQST.AMT-3). This code is used with the SSB Device Handler extension only. The audio messages “dollars” and “and cents” are automatically supplied by the Device Handler process.

Code	Description
RT RT _{nnn}	The response message associated with a denied transaction. The specific message depends on the value of ATM.N50.TRAN-RESP-CDE. The Device Handler process searches the das_resp_txt_tbl_g table in the N50NAMS to find a match on the response code. The message in the corresponding audio file is then voiced.
S1 S1 _{nnn}	In the case of a withdrawal with a surcharge fee, this code is the amount to be dispensed, without including the surcharge amount. (STM.RQST.AMT-1 minus TRAN-FEE) The audio messages “dollars” and “and cents” are automatically supplied by the Device Handler process.
S2 S2 _{nnn}	In the case of a withdrawal with a surcharge fee, this code is the amount of the surcharge (TRAN-FEE, from the Surcharge Data token). The audio messages “dollars” and “and cents” are automatically supplied by the Device Handler process.
TA TA _{nnn}	The <i>to</i> account type (STM.TO-ACCT-TYP). The Device Handler process, using the das_acct_tbl_g table in the N50NAMS matches the transaction <i>to</i> account type to a specific reserved audio file number. The corresponding audio file is then played.
TOLL, <i>rr</i> TO _{nnn} , <i>ll,rr</i>	The <i>to</i> account number (STM.RQST.TO-ACCT). The variables <i>ll</i> and <i>rr</i> are used in truncation of the card number, where <i>ll</i> is the number of digits starting from the left that are provided and <i>rr</i> is the number of digits starting from the right that are provided. If the variables are not present, the card number is provided in the voice response in its entirety or for a maximum of the first 15 digits in the number. Refer to the topic “Account Number Truncation” later in this section for more information about using these variables to control truncation for the voice response of an account number.
TT TT _{nnn}	The transaction type (STM.TRAN-CDE). The Device Handler process using the tran_tbl_g table in the N50NAMS, matches the transaction type to a specific audio file number. The appropriate audio file is then voiced.
WV _{nnn}	The Audio Wave field type code. This code is used to indicate when a specific message is to be voiced. The number corresponds to the audio file that contains the appropriate message to be played in the voice response. This code is required for audio receipts.

Account Number Truncation for Voice Responses

Three of the field type codes listed in the previous discussion have additional variables that allow the account number or card number voice response to be truncated. This feature prevents a cardholder's entire account number from being played.

The three field type codes are CX (card number without member number), FR (*from* account number), and TO (*to* account number). The two variables that can be appended to these field type codes are *ll*, indicating the number of digits starting from the left that are provided and *rr* indicating the number of digits starting from the right that are provided. If these variables are not present, the card or account number is played in its entirety or until a maximum of 15 digits have been voiced. The ATM truncates the rightmost digits of any card or account number greater than 15 digits. The NDC+ DAS feature has a limit of 15 digits in a voice response.

Truncating one or more of the numbers does not affect how the number is stored internally; the complete number is still shown on the Transaction Log File (TLF) and reports.

The following examples use a 19-digit primary account number (PAN) of 1123581321345589144.

Field Type Code in DAS Receipt Template	PAN as Voice Response
#CX00,04	9144
#CX06,04	112358 (pause) 9144
#CX06,00	112358
#CX444,00,04 (where 444 is the audio file containing the message "The last four digits of your account number are")	The last four digits of your account number are 9144
#CX555,06,04 (where 555 is the audio file containing the message "The first six characters in your card number and the last four characters of your card number are")	The first six characters in your card number and the last four characters of your card number are 112358 (pause) 9144

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C: Dispensing Bills and Noncurrency Items

This appendix describes the algorithms the BASE24-atm NCR 5XXX Device Handler process uses to dispense bills and noncurrency items. The methods include the least bills and configurable dispense algorithm methods. A third method, the thresholds method, is used with the BASE24-atm self-service banking (SSB) add-on product, and is described in detail in the ***BASE24-atm NCR 5XXX SSB Manual***.

This appendix also describes the Non-Currency Dispense add-on product.

Bill Dispensing Methods

You can configure the BASE24-atm NCR 5XXX Device Handler process to dispense notes in three different ways:

- Least bills method
- Thresholds method
- Configurable dispense algorithm method

The thresholds method is used with the BASE24-atm self-service banking (SSB) add-on product. Refer to the ***BASE24-atm NCR 5XXX SSB Manual*** for more information about the thresholds method of configurable bill dispensing.

The DISPENSE ALGORITHM field on BASE24-atm Terminal Data files (ATD) NCR 5XXX screen 7 enables you to select the bill dispensing method for your terminal. Refer to the ***BASE24-atm Files Maintenance Manual*** for more information about this field.

Least Bills Method

The least bills method is used with the standard BASE24-atm product. Using the least bills method, the NCR Device Handler process calculates the hopper bill counts from high to low bill denomination for bill dispensing. The Device Handler process dispenses as many bills as it can from the highest denomination. When the amount left to be dispensed is lower than the highest denomination, the Device Handler process calculates the number of bills to be dispensed from the hopper with the next highest denomination. For example, a transaction for \$375 (U.S. dollars) at a terminal with hoppers containing \$5, \$20, \$50, and \$100 would be dispensed as follows:

- Three \$100 bills
- One \$50 bill
- One \$20 bill
- One \$5 bill

To configure the least bills method, set the DISPENSE ALGORITHM field on ATD NCR 5XXX screen 7 to a value of 0. The least bills method is the default bill dispensing method for the BASE24-atm product. If the ATM cannot dispense the

requested amount using one of the other methods, the least bills method is used. The Non–Currency Dispense add-on product can use the least bills method to dispense noncurrency items.

Note: In situations where the requested amount cannot be dispensed because one or more hoppers is out of service or because the overall limit of 40 bills is exceeded, the Device Handler process returns a suggested amount to the customer displaying one amount lower than was requested and one amount higher than was requested.

Configurable Dispense Algorithm Method

The configurable dispense algorithm method gives an institution flexibility in its hopper configuration. The algorithm uses start and stop amounts and the maximum number of bills allowed to calculate the correct number of bills to dispense. This method is used when the DISPENSE ALGORITHM field on ATD NCR 5XXX screen 7 is set to a value of 2.

The *start dispense amount* is the minimum amount needed before the ATM starts dispensing bills from the hopper. For example, if a terminal contains \$5 (U.S. dollars) bills in hoppers 1 and 2 and \$50 bills in hoppers 3 and 4, hopper 4 could be configured with a start amount of \$200. This means the device does not dispense bills out of hopper 4 unless the amount to be dispensed is \$200 or more. For amounts less than \$200, the device could dispense \$50 bills from hopper 3.

The *stop dispense amount* is the amount at which the ATM stops dispensing from the hopper. When the amount to be dispensed is less than or equal to this amount, the ATM proceeds to the next hopper. Using the hopper values from the previous example, you could set a \$100 stop dispense amount on hopper 4. This means that when the amount to be dispensed is \$100 or less, the ATM would stop dispensing bills from hopper 4 and start dispensing bills from hopper 3. For a \$200 transaction, the ATM dispenses \$100 from hopper 4 and the remaining \$100 from the other hoppers.

The maximum number of bills is the number of bills at which the ATM stops dispensing bills from a hopper.

If you have purchased the BASE24-atm Non–Currency Dispense add-on product and one or more hoppers contains nonvalue items, you can use the configurable dispense algorithm *nonvalue dispense amount* to specify, by nonvalue hopper, the amount at which the ATM dispenses a nonvalue item. For example, you could configure one hopper to dispense a coupon for transactions of \$50 or more and a

second hopper to dispense an advertisement for transactions of \$20 or more. Refer to the topic “Non–Currency Dispense Add-on Product” later in this appendix for more information about dispensing cash value and nonvalue items.

If the algorithm fails (that is, the limits set by the start dispense amount, stop dispense amount and maximum number of bills prevent the ATM from dispensing the requested amount), the Device Handler process defaults to the least bills method.

Configuring the BASE24-atm Terminal Data Files (ATD)

Amounts for the configurable dispense algorithm are defined using the BASE24-atm Terminal Data files (ATD).

The DISPENSE ALGORITHM field on ATD NCR 5XXX screen 7 must be set to a value of 2 (start and stop amounts) for the configurable dispense algorithm.

ATD NCR 5XXX screen 13 is a device-specific screen. You can use this screen to define the configurable dispense algorithm.

The START DISPENSE AMT field contains the start dispense amount. Entering a value in this field is optional.

The STOP DISPENSE AMT field contains the stop dispense amount. Entering a value in this field is optional.

The MAX BILLS field indicates the maximum number of bills per hopper. Note that the maximum number of bills dispensed for a transaction cannot exceed 40.

If you have purchased the Non–Currency Dispense add-on product and you plan to use the configurable dispense algorithm, you can use the NON-VALUE DISPENSE field to specify the amount at which a nonvalue item is dispensed. This field is used only when the Non–Currency Dispense add-on product and the configurable dispense algorithm are used in combination. Refer to the topic “Non–Currency Dispense Add-On Product” later in this section for more information about dispensing noncurrency items.

For more information about the ATD, refer to the ***BASE24-atm Files Maintenance Manual***.

Device Handler Processing

The Device Handler process uses the bill value, start dispense amounts, stop dispense amounts, and maximum number of bills to compute the bill mix for a withdrawal transaction. The Device Handler process starts with hopper 4 and works its way down. If the ATM cannot dispense bills from the hopper (that is, the checks described below fail), the Device Handler process resumes calculations with the next hopper. If the Device Handler process runs out of hoppers before the entire transaction amount is dispensed, the process defaults to the least bills method.

The Device Handler process performs the following steps starting with hopper 4:

1. Verifies that the hopper is operational, the ATD hopper contents value is equal to the local contents value, the bill value is greater than 0, and the amount to be dispensed is greater than or equal to the hopper's start dispense amount. If the conditions are not met, the Device Handler process proceeds to the next hopper and performs the same checks until the conditions are met.
 - Once the conditions are met, processing continues with step 2.
 - If the conditions are not met with the final hopper, the Device Handler process defaults to the least bills method.
2. Compares the amount to be dispensed to the stop dispense amount for the hopper.
 - If the amount to be dispensed is less than or equal to the stop dispense amount, the Device Handler process proceeds to the next hopper and starts again with step 1.
 - If the amount to be dispensed is greater than the stop amount, processing continues with step 3.
3. Compares the amount to be dispensed to the bill value.
 - If the amount to be dispensed is less than the bill value, the Device Handler process proceeds to the next hopper and starts again with step 1.
 - If the amount to be dispensed is greater than or equal to the bill value, processing continues with the step 4.
4. Compares the number of bills dispensed from the hopper to the hopper's maximum number of bills allowed.
 - If the number of bills dispensed is greater than or equal to the hopper's maximum, the Device Handler process proceeds to the next hopper and starts again with step 1.

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- If the number of bills dispensed is less than the hopper's maximum number of bills, processing continues with step 5.
5. Compares the number of bills to be dispensed to the maximum number of bills allowed for the device.
 - If the number of bills to be dispensed is greater than or equal to the device's maximum, the Device Handler process proceeds to the next hopper and starts again with step 1.
 - If the number of bills to be dispensed is less than the device's maximum, the Device Handler process continues with step 6.
 6. Calculates that the hopper can dispense a bill, and subtracts the bill value from the transaction amount, giving the new amount to be dispensed. The Device Handler process also updates the bill count for the hopper and the device.
 7. Repeats steps 2, 3, and 4 until it reaches the full transaction amount, the stop amount, or maximum number of bills.
 - If the stop amount or maximum bills is reached, the Device Handler process resumes calculations with the next highest hopper, starting with step 1.
 - If there are no more hoppers available, the Device Handler process defaults to the least bills method.

Currency Dispense Algorithm Examples

The following is an example of how the Device Handler process calculates the bill mix using the configurable dispense algorithm. The table below shows a sample ATM hopper configuration in U.S. currency.

	Hopper 1	Hopper 2	Hopper 3	Hopper 4
Bill Value	\$5	\$20	\$20	\$100
Start dispense amount	0	20	100	300
Stop dispense amount	1	19	89	101
Max bills	6	6	4	6

Using this configuration and the configurable dispense algorithm, the Device Handler process would calculate the number of bills as shown in the table below. Descriptions of the transactions follow the table.

Transaction Amount	Hopper 1	Hopper 2	Hopper 3	Hopper 4
\$15	3 bills	0 bills	0 bills	0 bills
\$20	0 bills	1 bill	0 bills	0 bills
\$50	2 bills	2 bills	0 bills	0 bills
\$100	0 bills	4 bills	1 bill	0 bills
\$150	2 bills	3 bills	4 bills	0 bills
\$250*	2 bills	0 bills	2 bills	2 bills
\$300	0 bills	4 bills	1 bill	2 bills
\$425	1 bill	1 bill	0 bills	4 bills
\$500	0 bills	4 bills	1 bills	4 bills
\$750	2 bills	3 bills	4 bills	6 bills

* The configurable dispense algorithm fails, and the Device Handler process defaults to the least bills method. Refer to the transaction description below for more processing information.

\$15 Transaction. The transaction amount is less than the start dispense amount for hoppers 4, 3, and 2. The Device Handler process calculates 3 bills from hopper 1.

\$20 Transaction. The transaction amount is less than the start dispense amount for hoppers 4 and 3. The transaction amount is equal to the start dispense amount, greater than the stop dispense amount, and equal to the bill value for hopper 2, so one bill can be dispensed from the hopper.

\$50 Transaction. The transaction amount is less than the start dispense amount for hoppers 4 and 3, so the Device Handler process begins calculating the bill mix with hopper 2. The transaction amount is greater than the start dispense amount, stop dispense amount, and bill value for the hopper, so bills can be dispensed from the hopper. After hopper 2 dispenses \$40 (two bills), the amount remaining to be dispensed (\$10) is less than the stop dispense amount, so the Device Handler begins calculating the amount to be dispensed from the next hopper. The remaining \$10 is dispensed as two bills from hopper 1.

\$100 Transaction. The transaction amount is less than the start dispense amount for hopper 4, so the Device Handler process begins calculating the bill mix with hopper 3. The transaction is equal to the start dispense amount and greater than the stop dispense amount and bill value for that hopper. The Device Handler process calculates that \$20 (one bill) can be dispensed from hopper 3. After that the amount to be dispensed (\$80) is less than the \$89 stop dispense amount. The Device Handler process begins calculating the bill mix with hopper 2. Four bills (\$80) can be dispensed from hopper 2.

\$150 Transaction. The transaction amount is less than the start dispense amount for hopper 4, so the Device Handler process begins calculating the bill mix using hopper 3. The transaction amount is greater than the start dispense amount, stop dispense amount and bill value for hopper 3, so the Device Handler process begins calculating the bill mix from that hopper. The first \$80 (four bills) can be dispensed from hopper 3. After that, the amount remaining to be dispensed (\$70) is less than the \$89 stop dispense amount for that hopper. The Device Handler process begins calculating using hopper 2. The next \$60 (three bills) can be dispensed from hopper 2. After that, the amount remaining to be dispensed (\$10) is less than the stop dispense amount for that hopper. The Device Handler process begins calculating using hopper 1. The remaining amount can be dispensed with 2 bills from hopper 1.

\$250 Transaction. The transaction amount is less than the start dispense amount for hopper 4, so the Device Handler process begins calculating the bill mix with hopper 3. The transaction amount is greater than the start dispense amount, stop dispense amount, and bill value for hopper 3, so the Device Handler process begins

calculating the bill mix from that hopper. The Device Handler process calculates that the first \$80 can be dispensed from hopper 3. Although the remaining amount (\$170) is greater than the stop dispense amount, the maximum bills limit has been reached. The Device Handler process resumes calculations with hopper 2. The next \$120 (six bills) can be dispensed from hopper 2. The maximum number of bills is reached, so the Device Handler process must resume calculations with hopper 1. The \$50 dollars remaining to be dispensed cannot be dispensed from hopper 1 because there is a six bill maximum. The algorithm fails, and the Device Handler defaults to the least bills method.

\$300 Transaction. The transaction amount is equal to the start dispense amount and greater than the stop dispense amount and bill value for hopper 4, so the Device Handler process begins calculating the bill mix. The first \$200 can be dispensed from hopper 4 (two bills). After this, the amount remaining to be dispensed (\$100) is less than the \$101 stop dispense amount. The Device Handler process resumes calculations with hopper 3. The amount remaining to be dispensed is equal to the start dispense amount and greater than the stop dispense amount and the bill value for that hopper. Hopper 3 can dispense \$20 (1 bill). After this, the amount to be dispensed (\$80) is less than the \$89 stop dispense amount, so the Device Handler process resumes calculations with hopper 2. Hopper 2 can dispense the remaining \$80 (4 bills).

\$425 Transaction. The transaction amount is greater than the start dispense amount, stop dispense amount, and bill value for hopper 4, so the Device Handler process begins calculating the bill mix with hopper 4. The first \$400 dollars can be dispensed from hopper 4. After this, the amount remaining to be dispensed (\$25) is less than the \$101 stop dispense amount, so the Device Handler process resumes calculations with hopper 3. The amount to be dispensed is less than the start dispense amount (\$100) for hopper 3, so no bills can be dispensed from hopper 3. The Device Handler process resumes calculations with hopper 2. The amount to be dispensed is greater than the start dispense amount, stop dispense amount, and bill value for hopper 2. One bill (\$20) can be dispensed from hopper 2. The amount remaining (\$5) is less than the stop dispense amount of hopper 2, so the Device Handler process resumes calculations with hopper 1. The remaining \$5 can be dispensed with one bill from hopper 1.

\$500 Transaction. The transaction amount is greater than the start dispense amount, stop dispense amount, and bill value for hopper 4, so the Device Handler process begins calculating the bill mix. The first \$400 dollars (four bills) can be dispensed from hopper 4. After this, the amount remaining to be dispensed (\$100) is less than the \$101 stop dispense amount, so the Device Handler process resumes calculations with hopper 3. \$20 (1 bill) can be dispensed from hopper 3. After

this, the amount remaining to be dispensed (\$80) is less than the \$89 stop dispense amount for hopper 3. The Device Handler process resumes calculations with hopper 2. Hopper 2 can dispense the remaining \$80 (4 bills).

\$750 Transaction. The transaction amount is greater than the start dispense amount, stop dispense amount, and bill value for hopper 4, so the Device Handler process begins calculating the bill mix. The first \$600 (six bills) can be dispensed from hopper 4. This is the maximum that can be dispensed from the hopper. The Device Handler process resumes calculations with hopper 3. \$80 dollars (4 bills) can be dispensed from hopper 3. After this, the amount remaining to be dispensed (\$70) is less than the \$89 stop dispense amount. The Device Handler process resumes calculations with hopper 2. \$60 can be dispensed from hopper 2. After this, the amount remaining to be dispensed (\$10) is less than the stop dispense amount for the hopper. The Device Handler process resumes calculations with hopper 1. The remaining \$10 can be dispensed from hopper 1 with two bills.

Non-Currency Dispense Add-On Product

The BASE24-atm Non-Currency Dispense add-on product enables an institution to dispense noncurrency items through its ATMs. Noncurrency items can include cash value items, such as travelers checks, telephone cards, postage stamps, and transportation tickets, or nonvalue items, such as coupons or advertisements.

An institution can configure hoppers for up to four noncurrency items at an ATM. This is the maximum number of note hoppers for an NCR 5XXX ATM.

Processing Non-Currency Dispense Transactions

The BASE24-atm system treats a Non-Currency Dispense transaction the same as it does a cash withdrawal or cash advance transaction. When the NCR 5XXX Device Handler process receives a Non-Currency Dispense transaction, it adds the Non-Currency Dispense token to the end of the Standard Internal Message (STM). The token carries the information necessary for the transaction to be processed by the BASE24-atm system.

The NCR 5XXX Device Handler process performs the following functions on a Non-Currency Dispense transaction:

- Adds the Non-Currency Dispense token to the STM.
- Computes the total cost of the purchase, and places the value in the STM. This value is later displayed to the cardholder on the ATM screen. This is done for quantity-based transactions (for example, purchasing postage stamps) but not amount-based transactions (that is, traveler's checks).
- Checks thresholds configured for dispensing nonvalue items.

The Device Handler process searches the internal operations code table when it receives a request message. If the operations code indicates that the cardholder has requested a noncurrency item, the Device Handler process moves the value in the NCD ROUTING GROUP field on ATD screen 1 to the RQST.RTE-GRP field of the STM instead of the value from the ALT ROUTING GROUP field on ATD screen 1.

Refer to the *BASE24-atm Transaction Processing Manual* for more information about processing cash withdrawal and cash advance transactions.

Non–Currency Dispense Configuration

The Non–Currency Dispense add-on product can be used with the BASE24-atm NCR 5XXX Device Handler process. The following NCR and BASE24 files contain fields needed for the Non–Currency Dispense add-on product:

- NN50NAMS
- Acquirer Processing Code File (APCF)
- BASE24-atm Terminal Data files (ATD)
- Card Prefix File (CPF)
- Cardholder Authorization File (CAF)
- External Message File (EMF)
- Institution Definition File (IDF)
- Logical Network Configuration File (LCONF)
- Receipt File (RCPT)
- Surcharge File (SURF)
- Terminal Receipt File (TRF)
- Usage Accumulation File (UAF)

Modifying the NN50NAMS

The NN50NAMS is an edit file that contains a series of tables used by the BASE24-atm NCR 5XXX Device Handler process for Non–Currency Dispense transactions. You can edit these tables to customize specific information displayed on ATM receipts.

The noncurrency description table in the NN50NAMS (ncd_item_descr_g) supports the Non–Currency Dispense enhancement. The Device Handler process uses information in this table to map noncurrency items to receipt descriptions.

Modifying the Acquirer Processing Code File

The Acquirer Processing Code File (APCF) contains one record for each combination of acquirer transaction profile, message category, and ISO processing code supported by the BASE24-atm system. The APCF also supports the use of wildcard characters for the acquirer transaction profile and message category, so

that one APCF record can cover many combinations. Each acquirer transaction profile defines a set of transactions supported by a terminal or group of terminals. Each APCF record contains a code indicating if and how the transaction is allowed at the terminal.

APCF screen 2 contains fields that enable you to specify the types of Non-Currency Dispense transactions that can be performed at your ATMs. The transactions are grouped by processing code. For each processing code (that is, transaction type and account types), you can specify the circumstances under which a transaction is allowed.

Valid values are as follows:

- 0 = Transaction is not allowed
- 1 = Transaction is allowed within the county
- 2 = Transaction is allowed within the state
- 3 = Transaction is allowed nationally
- 4 = Transaction is allowed internationally

For more information about the APCF screens, refer to the ***BASE24 Base Files Maintenance Manual***.

Modifying the BASE24-atm Terminal Data Files

The BASE24-atm Terminal Data files (ATD) contains configuration information for each individual terminal in the logical network. The fields described below have specific Non-Currency Dispense requirements. These ATD fields are described in detail in the ***BASE24-atm Files Maintenance Manual***.

ATD Screen 1

ATD screen 1 contains information about the Device Handler process and Authorization process associated with the terminal and sharing information. The NCD ROUTING GROUP field specifies the routing group number used to route Non-Currency Dispense transactions initiated by cardholders who are outside the logical network.

ATD Screen 2

You can use ATD screen 2 to specify the acquirer transaction profile for Non–Currency Dispense transaction types of transactions available to on-us and not-on-us cardholders at a particular terminal.

ATD Screens 4 and 5

ATD screens 4 and 5 enable you to define hopper contents. You can define up to four noncurrency dispense hoppers on a NCR 5XXX ATM. (This is the maximum number of note hoppers for these ATMs.)

Hoppers containing cash are configured first, followed by noncurrency hoppers. The CONTENTS field for noncurrency hoppers is set to a value of 02 through 10. Values 02 through 10 are defined in the COBNAMES file, with the value 02 reserved for traveler's checks. Within a contents type, the value in the BILL VALUE field is entered from lowest to highest. For example, if your ATMs carry \$20 traveler's checks and \$50 traveler's checks, the hopper with the \$20 traveler's checks would be configured first. When configuring the BILL VALUE field for nonvalue items, use the value 0.

ATD Screen 7

ATD NCR 5XXX screen 7 is a device-specific screen. The screen includes the NON-VALUE ACCTS fields used to indicate the type or types of accounts for which a nonvalue item is dispensed. A value of Y in the CK, SV, or CC field indicates that a nonvalue item can be dispensed for a checking account transaction, savings account transaction, or credit account transaction, respectively.

Note that this screen also includes the DISPENSE ALGORITHM field. You can use any algorithm with the Non–Currency Dispense add-on product.

ATD Screen 13

ATD NCR 5XXX screen 13 is a device-specific screen. You can use this screen to define nonvalue item dispense amounts.

The NON-VALUE DISPENSE field contains the amount at which nonvalue items are dispensed when using the configurable dispense algorithm. Using this field and the configurable dispense algorithm enables the institution to set a nonvalue dispense amount for each hopper that contains a nonvalue item.

Setting Card Prefix File and Cardholder Authorization File Usage Limits

Non-Currency Dispense transactions are processed in the same manner as cash withdrawal transactions, with the Authorization process checking limits in the Card Prefix File (CPF), Cardholder Authorization File (CAF), and the Usage Accumulation File (UAF). Your institution can include Non-Currency Dispense transactions in the limits set for ATM transactions, or it can set limits specifically for Non-Currency Dispense transactions.

You can set Non-Currency Dispense limits in the CPF and CAF for general noncurrency items and two specific noncurrency items. Hopper content codes specify the two noncurrency items for which to set limits. The *general* noncurrency items include any noncurrency item dispensed by an ATM other than those specified by content codes. For example, you could define a set of limits for the amount of general noncurrency items, a set of limits specifically for the amount of traveler's checks, and a third set of limits for the amount of event tickets. Within the content categories, you can specify limits for cash withdrawals and credit withdrawals. Within the cash withdrawal category, you can set limits for the total amount of transactions and the total amount of offline transactions.

CPF screen 5 contains Non-Currency Dispense limits set at the card prefix level. CAF screen 9 contains Non-Currency Dispense limits and activity for individual cardholders. Fields on these screens are grouped by hopper contents. During authorization processing, the Authorization process checks the HOPR-CONTENTS field of the Non-Currency Dispense token. If the value in the HOPR-CONTENTS field is equal to one of the content codes, the Authorization process uses limits specific to that content code field. If the value of the HOPR-CONTENTS field is not equal to one of the content codes, the Authorization process uses the set of limits defined for all other noncurrency items. Refer to the ***BASE24 Tokens Manual*** for more information about the Non-Currency Dispense token.

Fields are arranged in a similar manner on UAF screen 5. This screen contains system-protected fields showing activity for the usage period.

For more information about the CPF, CAF, and UAF, refer to the ***BASE24 Base Files Maintenance Manual***.

Modifying the External Message File

When configuring the External Message File (EMF) you can change the default value of data element S-116 (BASE24-from host maintenance CAF Non-Currency Dispense) from a value of C (conditional) to a value of M (mandatory) if your institution requires that this element be present in messages to and from the From Host Maintenance process. If the element is not included in the message, you can set the value to a blank.

Modifying the Institution Definition File

The ACQUIRER TRANSACTION PROFILE field on Institution Definition File (IDF) screen 9 enable you to specify transaction profiles for Non-Currency Dispense transactions. The processing codes for the profiles are defined in the Acquirer Processing Code File (APCF). For more information about IDF screen 9, refer to the *BASE24 Base Files Maintenance Manual*.

Modifying the Logical Network Configuration File

The Logical Network Configuration File (LCONF) contains two params for the Non-Currency Dispense add-on product.

The ATM-DH-NON-VALUE-THRESHOLD param should be present at initialization if an ATM is configured to dispense nonvalue items and is not using the configurable dispense algorithm. This param sets a minimum whole dollar withdrawal amount for determining whether a nonvalue item should be dispensed along with the requested cash or noncurrency item. This param is used when the DISPENSE ALGORITHM field on ATD NCR 5XXX screen 7 is set to a value of 0 or 1. If the DISPENSE ALGORITHM field is set to a value of 2, this param is ignored.

The N50-NCD-CRD-EJECT-OPTION param is an optional param that indicates whether the terminal ejects the card before, after, or at the same time it dispenses the noncurrency item. If the parameter is not present, the NCR 5XXX Device Handler process uses a default value indicating that the terminal should dispense the noncurrency item before ejecting the card.

Refer to the *BASE24 Logical Network Configuration File Manual* for more information about the ATM-DH-NON-VALUE-THRESHOLD and N50-NCD-CRD-EJECT-OPTION params.

Modifying the Receipt File

You can add receipt templates to the Receipt File (RCPT) for Non-Currency Dispense transactions. Receipt condition codes NCD and NONCD that are used with the IF and ENDIF directives allow noncurrency data to be displayed on withdrawal transaction receipts. The field type code NC displays the noncurrency description text defined in the NN50NAMS.

The receipt type WD (withdrawal) is used for a Non-Currency Dispense transaction. A sample receipt template as defined in the RCPT is shown below.

```
@RECORD NO.    #SQ>
?IF NONCD
@WITHDRAWAL#A1$>
?ENDIF
?IF NCD
@PURCHASE#A1$>
#NC>>
?ENDIF
@FR #FR>>
#XL#LB$>
#XA#AB$>>
```

The above template creates the following receipt for a cash withdrawal transaction:

1	A C I D E M O		
2			
3	DATE	TIME	TERM
4	MM/DD/YY	HH:MM	S1AN5070
5			
6	123456700000003-003		
7			
8	330 SOUTH 108TH AVE		
9			
10	RECORD NO.	570	
11	WITHDRAWAL	\$20.00	
12	FR	013333	
13			
14	LEDGER BAL	\$3412.00	
15	AVAIL BAL	\$1760.00	

1 2
123456789012345678901234567

The same template creates the following receipt for a Non-Currency Dispense purchase transaction (in this case traveler's checks).

1	A C I D E M O		
2			
3	DATE	TIME	TERM
4	MM/DD/YY	HH:MM	S1AN5070
5			
6	123456700000003-003		
7			
8	330 SOUTH 108TH AVE		
9			
10	RECORD NO.	577	
11	PURCHASE	\$50.00	
12	TRAVELERS CHECKS		
13			
14	FR	013333	
15			
16	LEDGER BAL	\$3362.00	
17	AVAIL BAL	\$1710.00	

1	2
123456789012345678901234567	

See section 5 for more information about condition values and field type codes.

Modifying the Surcharge File

The Surcharge File (SURF) contains information regarding surcharging. If you want to assess a surcharge for Non-Currency Dispense transactions, set the NON-CURRENCY DISPENSE field on SURF screen 3 to a value of 1. This enables the Authorization process to assess the surcharge as configured in the SURF. If you do not want to assess a surcharge for Non-Currency Dispense transactions, set the value in this field to 0.

Refer to the *BASE24 Base Files Maintenance Manual* for more information about configuring the SURF.

Modifying the Terminal Receipt File

Terminal Receipt File (TRF) screen 2 contains fields that enable you to specify the receipt options for approved and denied Non–Currency Dispense transactions at your ATMs. Each terminal receipt profile defines a set of receipt options supported by a terminal or a group of terminals. Each TRF record contains a code indicating if and how the transaction is allowed at the terminal. For more information about the TRF screens, refer to the ***BASE24-atm Files Maintenance Manual***.

D: NCR 5XXX Dial-Up Device Handler Extension

With the saturation of ATMs, the financial industry demands cost-effective methods of distributing ATMs to new locations. Because a dial-up ATM runs on a standard business line, it is a less expensive alternative to a leased line, especially in areas that do not see much consumer traffic.

An institution can support the use of dial-up ATMs using the BASE24-atm NCR 5XXX Dial-up Device Handler extension.

The NCR 5XXX Dial-Up Device Handler extension can be used with transaction surcharging and the BASE24-atm Non-Currency Dispense add-on product.

This appendix describes the NCR 5XXX Dial-Up Device Handler extension. It includes the following topics:

- Supported protocols
- Dial-up message header
- Device Handler processing
- Files maintenance considerations
- Network control commands

Supported Protocols

Dial-up ATMs are physically connected to the BASE24 system using standard communication protocols (refer to section 1 for more information on the protocols supported by the NCR 5XXX Device Handler process) in conjunction with the Visa II protocol.

If the dial-up device uses the Async Visa II protocol, the TYPE attribute of the corresponding DEVICE object in the XPNET system must be set to a value of 255. This suppresses the automatic disconnect function for connection complete messages. A value other than 255 retains the automatic disconnect function.

Dial-Up Message Header

The request message from a dial-up ATM is distinguished from other request messages by the dial-up header, which precedes the standard message header. The format of the dial-up header is shown below, followed by descriptions of its fields:

Position	Field Length	Field Name
1–2	2 numeric characters	Device Type
3–9	7 numeric characters	BIN
10–21	12 alphanumeric characters	Terminal ID
22	1 hexadecimal character	Field Separator
23–24	2 alphabetic characters	Message Type

Device Type — A code indicating the Visa II device. This field is always set to a value of 50.

BIN — Communication and routing information. This information is used in routing the message to the BASE24 system. The Device Handler process does not use this information.

Terminal ID — A value that identifies the NCR 5XXX terminal. Positions 3 through 8 of this field contain the six-digit machine number entered locally at the ATM. When a message comes into the BASE24 system, the Device Handler process reads the BASE24-atm Terminal Data Static File—general data (ATDS1) record using this terminal ID instead of the BASE24 station name from the XPNET process.

Field Separator — A period (.) representing a field separator character (hexadecimal 1C).

Message Type — A two-character code representing the type of inbound message. Valid values are as follows:

CC = Connect continuation
 CP = Communication complete
 RQ = Transaction request message

Device Handler Processing

The following pages discuss request and response message processing. They also describe how the Device Handler process handles surcharge and Non–Currency Dispense transactions.

Request Message Processing

When the Device Handler process receives a message, it checks for the dial-up header. If the message does not start with the dial-up header, the Device Handler process reads the BASE24-atm Terminal Data Static File—general data (ATDS1) using the BASE24 station name from the XPNET process. If the message starts with the dial-up header, the Device Handler process reads the ATDS1 with the terminal ID from the header but saves the BASE24 station name, so that the response can be sent back to the correct terminal.

The ATM can send the following types of messages:

- Request message (e.g., cardholder transaction request)
- Communication complete message indicating that a transaction is complete or that a disconnect command was sent
- Connect continuation message used to maintain a connection with the BASE24 system when the current sequence of messages is not complete (e.g., downloading the ATM)

In addition, the ATM can connect when its internal *I'm Alive* timer has expired (indicating that no messages have been sent for a period of time). In the case of an event, such as an indicator from the tamper sensor or a card reader fault, the ATM connects to the BASE24 system and sends an unsolicited status message.

Response Message Processing

When responding to the ATM, the Device Handler process uses the BASE24 station name from the XPNET process to send a message back to the correct ATM.

When the Device Handler is done processing a request and there are no messages to be sent to the ATM, it sends a disconnect command to the ATM. The format of this command is the normal terminal command format with a command code of F. If the ATM has another message to send to the BASE24 system, the message is

sent. If the ATM does not have any other messages when it receives the disconnect command, it sends a request message with the Message Type field of the dial-up header set to a value of CP.

Failed Messages

In the case of a message failure, the Device Handler process reads the ATDS1 with the terminal ID from the XPNET process. For message failure type 106 with the message flag turned off (i.e., the MSG FAIL field on ATD screen 7 is set to a value of 0), the Device Handler process generates an event message stating that the transaction was not reversed. If the MSG FAIL field on ATD screen 7 is set to a value of 1, the Device Handler process reverses all failed messages. If the MSG FAIL field is set to a value of 2, the Device Handler process waits for the next request and then determines whether to reverse the transaction based on the status of the last transaction.

Surcharge Processing

The NCR 5XXX Dial-Up Device Handler extension supports ATM surcharging. An institution using the surcharge application in a dial-up environment can choose for each ATM the interactive message sequence of the standard BASE24-atm processing or a noninteractive message sequence.

Interactive Surcharge Processing

Standard surcharge processing uses an interactive message sequence. In this environment, the cardholder selects a withdrawal transaction, and it is sent to the Authorization process, which calculates the surcharge. A message containing the surcharge amount is presented to the cardholder to allow him or her to agree to the charge or terminate the transaction. When the cardholder agrees to the surcharge, the transaction is sent to the authorization process again for approval.

Institutions using the surcharging interactive message sequence must use load group 307 for its screens, states, tables and parameters. (Surcharging load group 304 is used in leased line environments.) Load group 307 allows a clean disconnect sequence if the cardholder decides not to continue the transaction after being presented with the surcharge notification. This helps the institution avoid a long connection time.

Noninteractive Surcharge Processing

Using the surcharging interactive message sequence can result in long dial-up times for the institution. To decrease dial-up time, an institution can choose a noninteractive message sequence. The NCR 5XXX Dial-Up Device Handler extension enables an institution to define a maximum surcharge amount that can be charged for a transaction at the terminal. This maximum amount is displayed to the cardholder during the initial transaction sequence. The actual amount the cardholder is charged might be less than the maximum amount shown during transaction selection. The maximum surcharge amount is sent to the ATM during any download that contains the surcharge notification screen. Because there is only one communication sequence, the dial-up time is shorter.

When the Device Handler process receives the surcharge notification message (9901) from the Authorization process, it determines whether noninteractive surcharging is in use (that is, the DOWNLOAD SURCHARGE AMT field on ATD screen 7 contains a nonzero value, and the value has been downloaded to the ATM). If noninteractive surcharging is in use, and the actual surcharge is less than or equal to the maximum amount, the request is sent to the Authorization process without presenting a surcharge notification screen to the cardholder.

Refer to the *BASE24-atm Transaction Processing Manual* for a message flow diagram of a noninteractive surcharge transaction.

Institutions using the surcharging noninteractive message sequence must use load group 305 for withdrawal transactions and load group 306 for fast cash transactions.

Non–Currency Dispense Processing

The NCR 5XXX Dial-Up Device Handler extension supports dispensing noncurrency items using the Non–Currency Dispense add-on product. An institution using the Non–Currency Dispense add-on product in a dial-up environment can choose the interactive message sequence of the standard BASE24-atm processing or a noninteractive message sequence.

Interactive Non–Currency Dispense Processing

Standard Non–Currency Dispense processing uses an interactive message sequence. In this environment, the cardholder selects a noncurrency dispense transaction, and it is sent to the Authorization process, which calculates the

transaction amount. A message containing the transaction amount is presented to the cardholder to allow him or her to agree to the amount or terminate the transaction.

Institutions using the surcharging interactive message sequence must use load group 501 for its screens, states, tables and parameters. (Non–Currency Dispense load group 500 is used in leased line environments.) Load group 501 allows a clean disconnect sequence if the cardholder decides not to continue the transaction after being presented with the transaction amount. This helps the institution avoid a long connection time.

Noninteractive Non–Currency Dispense Processing

Using the Non–Currency Dispense interactive message sequence can result in long dial-up times for the institution. To decrease dial-up time, an institution can choose a noninteractive message sequence.

Institutions using the Non–Currency Dispense noninteractive message sequence must use load group 502. States in group 502 connect the state flow to the noninteractive surcharging state flow.

Files Maintenance Considerations

The following NCR and BASE24 files contain fields used with the NCR 5XXX Dial-up Device Handler extension:

- Logical Network Configuration File (LCONF)
- BASE24-atm Terminal Data files (ATD)
- NCR Configuration File (CNFG5070)
- Dial-up Configuration File (DIALUP)

In addition, this subsection describes system configuration using text commands issued from a network control facility.

Logical Network Configuration File

The Logical Network Configuration File (LCONF) contains one assign and several params for the NCR 5XXX Dial-up Device Handler extension.

The ATM-ADCF assign contains the fully qualified file name of the ATM Dial-Up Command File (ADCF). The ADCF queues commands to a dial-up ATM until it initiates a connection with the BASE24 system.

The ATM-DH-MAX-QUEUED-CMD param specifies the maximum number of commands that can be queued to a terminal. This param can be any integer value greater than zero.

The ATM-DH-MAX-LOAD-CMD param specifies the number of ADCF LOAD commands that can be processed by the Device Handler process at one time.

The ATM-DH-DYN-TBL-RETN-PRD-DIAL-UP param specifies the number of seconds the Device Handler process waits for the next leg of a multiple-leg transaction before the dynamic data for a terminal connected to a dial-up line can be removed from the dynamic data table. A value of zero disables this processing.

The ATM-DH-SUSPECT-RVSL-ON-IM-ALIVE param determines whether to generate a suspect reversal when an I'm Alive message is received while the Device Handler process is expecting the completion message from the previous transaction.

Refer to the *BASE24 Logical Network Configuration File Manual* for more information about the assign and params.

BASE24-atm Terminal Data Files

BASE24-atm Terminal Data files (ATD) NCR 5XXX screen 7 is a device-specific screen. This screen contains the following fields used with the NCR 5XXX Dial-Up Device Handler extension:

- The DIAL-UP field indicates whether the terminal is a dial-up ATM.
- The MSG FAIL field indicates failed-message processing.
- If surcharging is supported at the terminal, the DOWNLOAD SURCHARGE AMT field specifies the surcharge amount to be downloaded to the device. This amount is displayed to the cardholder during noninteractive surcharge processing. Refer to the topic, “Surcharging” in this appendix for more information on noninteractive surcharge processing.

If you are using load groups for interactive surcharge processing, set the DOWNLOAD SURCHARGE AMT field to the value 0. If you are using load groups for noninteractive surcharge processing, set the DOWNLOAD SURCHARGE AMT field to a nonzero value. The nonzero value must be greater than or equal to any possible surcharge value as calculated in the Surcharge File (SURF).

Note: It is important that the DOWNLOAD SURCHARGE AMT field is set with the same rules as the load groups. For example, if you are using load groups for interactive surcharge processing, and the DOWNLOAD SURCHARGE AMT field contains a nonzero value, a surcharge is applied to the transaction but the cardholder is not presented with a surcharge notification screen. Likewise, if you are using load groups for noninteractive surcharge processing and the DOWNLOAD SURCHARGE AMOUNT field is set to a value of zero, the transaction is processed as if interactive surcharging is in effect, except that the cardholder is presented with both the noninteractive surcharge screen and the interactive surcharge screen.

When you update the DOWNLOAD SURCHARGE AMT field, an ATD flag is set to indicate that the field has been changed and must be downloaded to the ATM.

In addition, the INACTIVITY TIME-OUT (WORKDAY), INACTIVITY TIME-OUT (WEEKEND/HOLIDAY), and NON-PEAK INACTIVITY TIME-OUT fields on this screen are used to identify ATMs that are not active because of communication problems. They should be set to values greater than the longest anticipated disconnect period for a dial-up ATM.

Refer to the *BASE24-atm Files Maintenance Manual* for more information on ATD NCR 5XXX screen 7.

NCR Configuration File

The NCR Configuration File (CNFG5070) enables the institution to maintain ATM screen displays, state table entries, FIT table data, configuration parameters, and timers. The following load groups are used for dial-up processing with surcharging or the Non-Currency Dispense add-on product. Load groups are entered in the GROUP NUMBER field of the NCR 5XXX Configuration File Menu.

- 305 = The load group for noninteractive surcharge processing for withdrawal and deposit transactions.
- 306 = The load group for noninteractive surcharge processing for fast cash transactions. This load group must be loaded on top of group 305.
- 307 = The load group for dial-up interactive surcharge processing.
- 308 = The group for dial-up ATM noninteractive surcharge processing (includes support for withdrawal, envelope deposit, SSB check deposit, and deposit with cash back transactions). This group must be loaded with groups 0 and 106.
- 309 = The group for dial-up ATM noninteractive surcharge processing (includes support for withdrawal, envelope deposit, SSB check deposit, deposit with cash back, and split deposit transactions). This group must be loaded with groups 0, 106, and 110.*
- 310 = The load group for surcharging with surcharge interactive response notification processing for withdrawal transactions.
- 315 = The load group for dial-up noninteractive surcharge processing for withdrawal and deposit transactions. Canceled transactions are logged to the TLF.
- 317 = The group for dial-up interactive surcharging and logging canceled transactions to the TLF.
- 318 = The group for dial-up noninteractive surcharging and logging canceled transactions to the TLF (includes support for withdrawal, envelope deposit, SSB check deposit, and deposit with cash back transactions). This group must be loaded with groups 0 and 106.
- 319 = The group for dial-up noninteractive surcharging and logging canceled transactions to the TLF (includes support for withdrawal, envelope deposit, SSB check deposit, and deposit with cash back transactions). This group must be loaded with groups 0, 106, and 110.
- 501 = The load group for dial-up Non-Currency Dispense processing.
- 502 = The load group for dial-up Non-Currency Dispense noninteractive surcharge processing. This load group must be loaded on top of group 501.
- 610 = The EMV Dial-Up Device Handler group for surcharge interactive response notification processing.
- 611 = The EMV Dial-Up Device Handler group for noninteractive surcharge processing.

- 612 = The EMV Dial-Up Device Handler group for noninteractive surcharge processing for fast cash transactions.
- 613 = The EMV Dial-Up Device Handler group for processing Non–Currency Dispense transactions.
- 614 = The EMV Dial-Up Device Handler group for processing noninteractive Non–Currency Dispense transactions.

* If the ATM supports split deposit transactions but not SSB check deposit or deposit with cash back transactions, you can load group 309 with only groups 0 and 110 (and not 106).

Refer to section 4 for more information about the CNFG5070.

To use surcharge interactive response notification processing with a dial-up ATM (i.e., using load group 310), you must set the ATM-FEE-INTRACTV-RESP-NTFY param to a value of Y. If you are not using surcharge interactive response notification processing (i.e, using load groups other than 310) and the ATM-FEE-INTRACTV-RESP-NTFY param is present, it must be set to a value of N.

Dial-Up Configuration File

The Dial-Up Configuration File (DIALUP) is a data file held on the NCR NDC+ ATM OS/2 hard drive. It defines NCR NDC+ configuration options pertaining to the Visa II dial-up protocol. The file is built using a DOS-based utility, (TABU.EXE), supplied by NCR. The utility produces the DIALUP from an ASCII text file.

For more information about this file, see the *NCR 56XX ATM Dial-Up Interface Specification*.

Network Control Commands

Because the Device Handler process cannot initiate a call to a dial-up ATM to execute network control commands, it queues and stores commands in the ATM Dial-Up Command File (ADCF) until the ATM connects to the BASE24 system.

When a request initiating a connection is received, the Device Handler process checks the ADCF. If there are no stored commands, the request is processed. Otherwise, the transaction request is declined while the ATM closes and the ADCF commands are processed. If the ATM is already connected to the BASE24 system and a command is received, the command is queued to the ADCF and processed after all preceding commands in the file are processed. The commands stored in the ADCF are executed using first in, first out (FIFO) processing, except when priority commands are generated programmatically (for example, initial loads started by the Device Handler process as a result of receiving ATM unsolicited power failure status messages).

The maximum number of commands that can be queued to a terminal is specified in the ATM-DH-MAX-QUEUED-CMD param. When the number of commands in the queue surpasses the limit specified by the param, the new command is dropped and an event message is generated indicating the command was dropped.

The maximum number of simultaneous downloads to multiple terminals that can be queued by the Device Handler process at one time is specified in the ATM-DH-MAX-LOAD-CMD param. Once this limit is reached, the Device Handler process retains the load command in the ADCF. A counter is increased any time a queued load message is sent from the Device Handler process and decreased when the command has been processed.

You can list the commands in the ADCF by entering the LISTQ command from a network control facility. This command causes the Device Handler process to list all records from the ADCF for the specified terminal ID. Each command is listed in an event message on the network log.

To clear the queue for a terminal you can enter the CLEARQ command from a network control facility. This command causes the Device Handler process to delete all records from the ADCF for a specified terminal ID. Each command is listed in event messages on the network log before it is deleted. The number of commands pending for that terminal is then set to a value of 0.

The LISTQ and CLEARQ commands cannot be entered from the Device Control Terminal (DCT).

Refer to the *BASE24 Text Command Reference Manual* for more information about the LISTQ and CLEARQ commands.

ACI Worldwide, Inc.

E: Distributing Master Keys and Transaction Keys

This appendix describes add-on products that manage master keys and transaction keys. The BASE24-atm uses transaction keys, also known as working keys, to perform specific cryptographic operations, such as encrypting a message authentication code (MAC) or a PIN. A *master key* is used to encrypt the transaction key.

The Automated Key Distribution System add-on product enables you to remotely load a master key to the ATM. The Dynamic Key Management add-on product enables you to manage transaction keys.

Note that the Automated Key Distribution System add-on product manages master keys only, and the Dynamic Key Management add-on product manages transaction keys only.

Automated Key Distribution System Add-On Product

The BASE24-atm Automated Key Distribution System add-on product enables an institution to remotely load a unique master key to an ATM. In the past, master keys, which are used to encrypt all other keys, were loaded manually at ATMs. Using the BASE24-atm Automated Key Distribution System add-on product, you can load the master key to the ATM from the BASE24-atm system. In addition, the key is randomly generated, ensuring a unique key for all ATMs in the network.

This topic describes the BASE24-atm Automated Key Distribution System add-on product, its configuration, and its processing.

Overview

The ATM master key (also known as the key exchange key or A-key) is the cryptographic key used to encrypt all other keys (e.g., PIN encryption key, MAC key) that are exchanged between the BASE24-atm system and an ATM.

Master keys are by nature long term, and, once in place, are usually left unchanged unless they are compromised. Without the BASE24-atm Automated Key Distribution System add-on product, a new key must be manually loaded into the ATM if the key is ever compromised.

The BASE24-atm Automated Key Distribution System add-on product enables an institution to remotely load a unique and secure master key to an ATM using the Rivest-Shamir-Adleman (RSA) public key scheme.

Before the master key is downloaded, the BASE24-atm system and the Encrypting PIN Pad (EPP) on the NCR 5XXX ATM must first authenticate each other and exchange their public keys. (The key mode also can be set at this time if it had not been set when the ATM was in supervisor mode.) The EPP is a tamper-proof component of the NCR 5XXX ATM consisting of a PIN pad, a security module, data encryption algorithms, and the cryptographic keys. The Transaction Security Services process provides an enhanced interface between the BASE24-atm system and the host security module.

Refer to the *BASE24 Transaction Security Services Processing Guide* for more information about the role of the Transaction Security Services process in automated key distribution.

Configuration Considerations

The following Hardware and Logical Network Configuration File (LCONF) configuration considerations must be addressed for the use of the BASE24-atm Automated Key Distribution System add-on product.

In addition, you must address the Transaction Security Services process and hardware security module implementation requirements. Refer to the ***BASE24 Transaction Security Services Processing Guide*** for more information about these requirements.

NCR Software and Hardware Requirements

Signature-based processing: The NCR 5XXX ATM must be running a minimum of NCR NDC+ version 7.0 software. The ATM also must be equipped with an NCR encrypting PIN pad (EPP). Note that NCR's enhanced BAPE software that mimics triple DES encryption does not support the RSA algorithm in loading initial keys.

Enhanced Signature-based processing: The Basic Protocol was found to present PCI security failures. NCR addressed these audit gaps and released the Enhanced RKL Protocol as a result, designed for EPP firmware version 7.xx. This version is fully compliant with the logical security requirements of the new PCI standards, and contains both Basic and Enhanced protocols so migration to the Enhanced Protocol can be done at the desired pace. The NCR ATM must be running Aprta Advance version 3.04 or Aprta Edge version 2.05 software (or later).

Certificate-based processing: The Diebold ATM must be running Aprta Edge release 2.04.00 software. The ATM also must be equipped with a Diebold encrypting PIN pad (EPP).

Wincor Nixdorf's shared specification signature- and certificate-based processing: The ATM (whether it is a Wincor, NCR, Diebold, or another vendor's device) must be running Wincor Nixdorf's ProCash/NDC firmware version 13.00 (or greater) or firmware that is compatible with Wincor Nixdorf's shared RKL specification, version 1.5.

Hardware Security Module Requirements

The Automated Key Distribution System add-on product supports Atalla Key Block (AKB) and the Thales e-Security (Racal) security modules.

Protocol Requirements

NCR ATMs support the extended message lengths required by the Automated Key Distribution System add-on product using the following protocols:

- TC500 Async
- TC500 Sync
- SNA (LU0)/SDLC
- SNA/X.25 (QLLC)
- X.25 Lap6/X.21 bis
- TCP/IP

Logical Network Configuration File

The following Logical Network Configuration File (LCONF) assigns and params must be configured to use the Automated Key Distribution System add-on product. For full descriptions of these assigns and params, refer to the ***BASE24 Logical Network Configuration File Manual***.

Assigns. The following assigns must be configured in the LCONF to use the Automated Key Distribution System add-on product.

SECURE-DEV1	The PPD name of the primary Transaction Security Services process, rather than a hardware security module. This assign is not specific to the Automated Key Distribution System add-on product.
SECURE-DEV2	The PPD name of the secondary Transaction Security Services process, rather than a hardware security module. This assign is not specific to the Automated Key Distribution System add-on product.
SECURE-DEV-xx	The PPD name of one of 20 Transaction Security Services processes accessed on a rotating basis, where <i>xx</i> is any two characters. This assign is not specific to the Automated Key Distribution System add-on product.

Params. The following params must be configured in the LCONF to use the Automated Key Distribution System add-on product.

ATM-DH-AKDS-TSS-TIMER	The number of seconds the NCR 5XXX Device Handler process waits for a reply from the Transaction Security Services process. If the Device Handler process has not received a reply before the timer expires, it aborts the load. Valid values are 10 through 600. This param is specific to the Automated Key Distribution System add-on product.
ATM-DH-N50-AKDS-SERIAL-NUM-VRFY	A code indicating whether to verify the EPP serial number against a list of approved numbers maintained in the Transaction Security Services (TSS) database. This verification occurs during the authentication and public key exchange sequence of Automated Key Distribution System processing.
SECURE-DEV-TYP	Specifies the security device or process the BASE24 security utilities communicate with directly. This param must be set to the value TSS to indicate the use of the Transaction Security Services process instead of a hardware security module. This param is not specific to the Automated Key Distribution System add-on product.

Terminology

The following terminology is used in describing the Automated Key Distribution add-on product:

Base64 Encoding — A method of encoding data used by Diebold to convert binary data into encoded characters. A base64 algorithm can produce a 4 to 3 ratio of encoded bytes to binary bytes.

Base94 Encoding — A method of encoding data used by NCR to convert binary data into the 94 displayable ASCII characters, ranging from hex 20 (space) to hex 7E (tilde). Using these characters, a base94 algorithm can produce a 5 to 4

ratio of encoded bytes to binary bytes. Given the block size (256 bytes for RSA-encrypted data) and message length constraints, NCR has determined that other encoding methods are unsuitable for transmitting RSA blocks.

Certificate — An electronic document binding some pieces of information together, such as a user's identity and public key. Certificate Authorities (CAs) provide certificates.

Certificate Authority (CA) — An organization that creates certificates and signatures. The CA might be an ATM manufacturer, a financial institution or a third party.

Encrypting PIN Pad (EPP) — A tamper-proof ATM component comprised of a PIN pad, a security module to perform cryptographic processing, data encryption algorithms (e.g., DES, Triple DES, MAC, RSA), and the cryptographic keys. All cryptographic keys are stored in the EPP. All cryptographic functions are performed by the EPP.

Encryption Key Change (EKC) — An NDC message that is used to load cryptographic data to the EPP, request cryptographic information from the EPP, and perform other functions.

Encryptor Initialization Data (EID) — A solicited NDC message that is returned by the ATM in response to an extended encryption key change (EEKC) message.

Extended Encryption Key Change (EEKC) — An NDC message that is used to load cryptographic data to the EPP, request cryptographic information from the EPP, and perform other functions. The ATM responds with an encryptor initialization data (EID) message.

Host Security Module (HSM) — A security device that acts as a peripheral to the host computer. It performs cryptographic processing in a physically secure environment on behalf of the host.

Key — A string of bits widely used in cryptography, enabling users to encrypt and decrypt data. A key can be used to perform other mathematical operations, as well. Given a cipher, a key determines the mapping of the plaintext to the ciphertext. Also known as the A-Key, Key Exchange Key (KEK), and Terminal Master Key (TMK).

Key Verification Value (KVV) — The NCR equivalent of BASE24's check digits.

Master key — A key used to encrypt and decrypt the working keys that are downloaded from the host to the ATM. Also known as the A-Key, Key Exchange Key (KEK), and Terminal Master Key (TMK).

Out-of-Band — Describes a function that is performed offline, rather than via online transaction processing. For example, manually loading keys at an ATM is a function that is performed out-of-band.

Private key — One-half of a cryptographic key pair used in a public key algorithm (RSA), the other half being the Public Key. The private key is uniquely associated with an entity and is not available publicly. It is used to decrypt data that was encrypted using the matched public key. It is also used to create signatures that can only be verified using the matched public key. There is a mathematical relationship between the private key and the public key, but it is virtually impossible to derive one from the other. Also known as Secret Key.

Public key — One half of a cryptographic key pair used in the RSA public key algorithm (the other half being the secret key). The public key is uniquely associated with an entity and is available publicly. It is used to encrypt data that can only be decrypted using the matched secret key. It also is used to verify signatures that were created with the matched secret key. There is a mathematical relationship between the public key and the secret key, but it is virtually impossible to derive one from the other. In the Automated Key Distribution System add-on product, the BASE24 system, the EPP, and NCR have public keys.

Public Key Infrastructure (PKI) — A security infrastructure whose services are implemented and delivered using public key concepts and techniques.

Remote Key Loading (RKL) — Wincor Nixdorf's keying process using RSA or public cryptography.

RSA — A public key algorithm based on the factoring problem. RSA stands for Rivest-Shamir-Adleman, the developers of the RSA public key algorithm.

Secret key — One half of a cryptographic key pair used in the RSA public key algorithm (the other half being the public key). The secret key is uniquely associated with an entity and is not available publicly. It is used to decrypt data

that was encrypted using the matched public key. It also is used to create signatures that only can be verified using the matched public key. There is a mathematical relationship between the secret key and the public key, but it is virtually impossible to derive one from the other. In the Automated Key Distribution add-on product, the BASE24 system, the EPP, and NCR have secret keys.

Signature — A method of preventing data from being changed during transmission and of preventing the sender from being impersonated.

Working key — A category of keys used to perform specific cryptographic operations. This key is encrypted under the Master Key prior to being downloaded from the Host to the ATM. PIN encryption keys, MAC keys, and PIN verification keys are all examples of Working Keys. Also known as the B-Key.

Basic Signature-Based Processing

The following topics describe the steps the BASE24-atm NCR 5XXX Device Handler process follows during Basic Protocol for automated key distribution using signatures. Automated key distribution is performed in phases as follows:

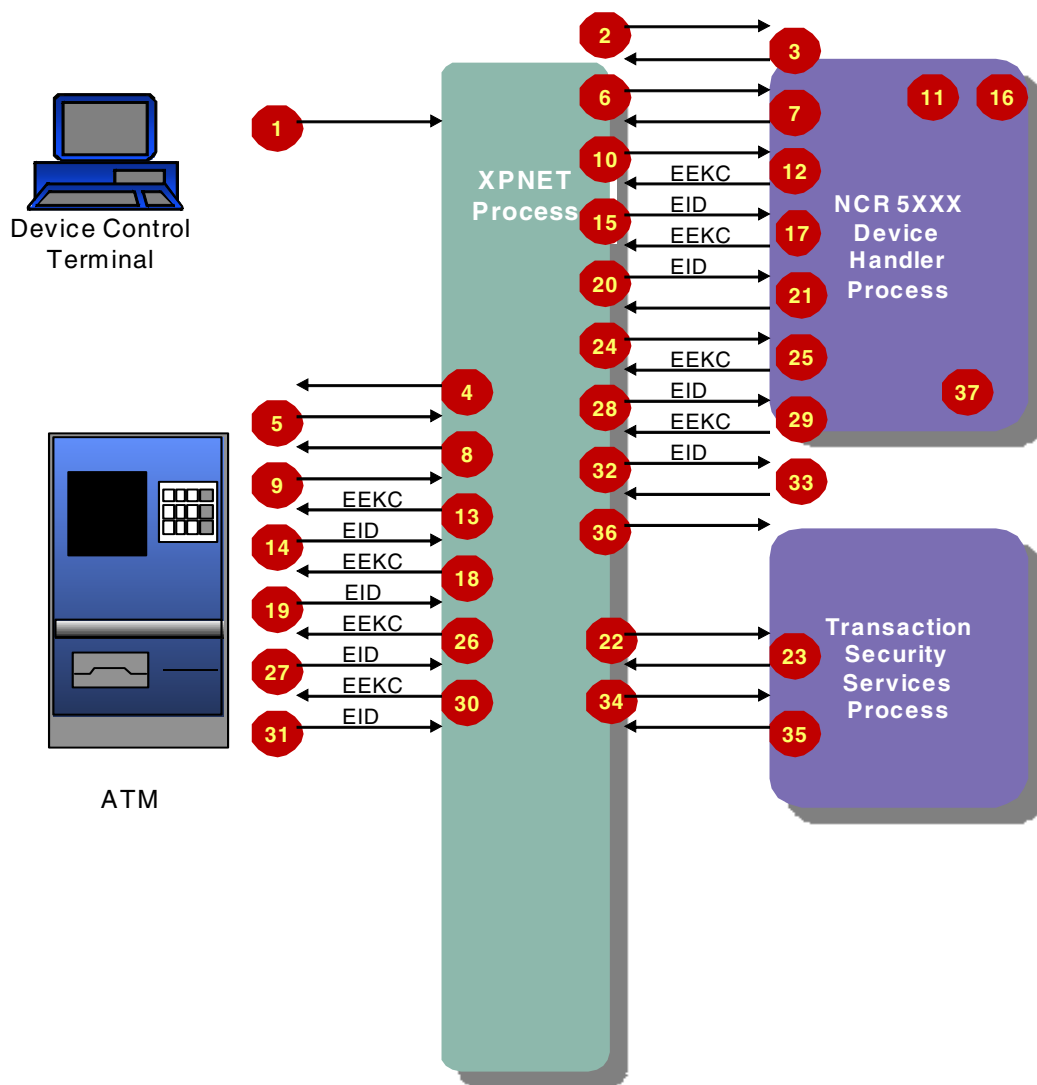
Phase	Description
1	Generation of the NCR RSA key pair.
2	Manufacture of the NCR Encrypting PIN Pad (EPP).
3	Generation of the BASE24 RSA key pair.
4	Authentication and public key exchange.
5	Master key download.

The processing described in this appendix is for phases 4 and 5. Phases 1 through 3 are described in the ***BASE24 Transaction Security Services Processing Guide***. This appendix also includes message flows for the Load Key Mode command. This command sequence is performed after authentication and public key exchange. The key mode is loaded only if it was not set while the ATM was in supervisor mode.

Authentication and Public Key Exchange Using Basic Signatures

The following message flow illustrates the processing that takes place during authentication and public key exchange (phase 4 of automated key distribution processing) which utilizes signatures. At this stage in the exchange, the BASE24 system and the EPP have established the keys and signatures required for automated key distribution. (Refer to the *BASE24 Transaction Security Services Processing Guide* for the message flow for establishing keys and signatures.) In the following steps, the BASE24 system and the EPP authenticate each other and exchange public keys. These steps are typically performed only once, when the EPP is initially deployed.

Refer to the *BASE24 Transaction Security Services Processing Guide* for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps	Processing
1, 2	An operator initiates a request for a public key exchange. In this example, the LOAD PUBLIC KEY command is sent from the Device Control Terminal (DCT), but a request can also be made from a network control facility.
3, 4	The BASE24-atm NCR 5XXX Device Handler process sends an operational command message to the ATM requesting a shutdown.
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends a configuration request to the ATM.
9, 10	The ATM sends a configuration response message to the Device Handler process.
11	The Device Handler process searches the configuration response message for the value 7 or 8 in the encryptor type field and the TERMINAL LINE PROTOCOL field on NCR 5XXX screen 7 of the ATD for the protocol value. An encryptor type of 7 or 8 and a protocol value other than B, C, or S indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is stopped and the ATM is restarted. If the ATM is capable of processing Automated Key Distribution System commands, the Device Handler process continues with the next check. The Device Handler checks the software ID and release number in the configuration response message to determine if the ATM is capable of sending the encryptor capabilities and state. If the software ID begins with G531 (Apra Advance) but the release number is less than 3.02, or if the software ID begins with V351 (Apra Edge) but the release number is less than 2.04.00, the ATM is not capable of sending the encryptor capabilities and state, nor does it support certificates; therefore, the Device Handler process continues with step 17. Otherwise, the Device Handler process continues with step 12.

Steps	Processing
12, 13	The Device Handler process sends an extended encryption key change (EEKC) message to request the encryptor capabilities and state.
14, 15	The ATM responds with an encryptor initialization data (EID) message.
16	The Device Handler process checks the Remote Key Scheme and Certificate State provided in the encryptor initialization data (EID) message. If the Remote Key Scheme is “00” (none) or “03” (signature + certificate), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “00” (not ready/not supported) or “02” (certificate secondary), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “01” (certificate primary), the Device Handler process proceeds with the load using the certificate-based method; refer to the “Authentication and Public Key Exchange Using Certificates” topic in this manual for more information. If the Remote Key Scheme is “01” (signature), the Device Handler process continues with step 17.
17, 18	The Device Handler process sends an extended encryption key change (EEKC) message to request the encrypting PIN pad (EPP) serial number.
19, 20	The ATM responds with an encryptor initialization data (EID) message. The message contains the eight-byte EPP serial number and the base94-encoded 320-byte signature of the serial number.
21, 22	The Device Handler process decodes the base94-encoded signature and sends a nowaited request to the Transaction Security Services process to verify the decoded signature.

Steps	Processing
23, 24	The Transaction Security Services process verifies the signature using the NCR public key and stores the serial number of the EPP in its database. The Transaction Security Services process returns a response to the Device Handler process. Included in the response is the BASE24 public key and the signature of the public key.
25, 26	The Device Handler process encodes the BASE24 public key and its signature using the base94 algorithm. The Device Handler process downloads the encoded BASE24 public key and signature to the ATM EPP by sending an extended encryption key change (EEKC) message.
27, 28	The ATM EPP verifies the signature using the NCR public key. The BASE24 public key is stored in the EPP. The ATM responds to the Device Handler process with an encryptor initialization data (EID) message indicating the key was loaded successfully.
29, 30	The Device Handler process sends an extended encryption key change (EEKC) message to the ATM requesting the EPP public key.
31, 32	The ATM responds with an encryptor initialization data (EID) message containing the EPP public key and the signature of the public key. Each are 320 bytes and base94 encoded.
33, 34	The Device Handler process decodes the base94-encoded EPP public key and signature and sends a nowaited request to the Transaction Security Services process to verify the decoded signature.
35, 36	The Transaction Security Services process verifies the signature using the NCR public key and stores the EPP public key in its database. It also generates a message authentication code (MAC) of the EPP public key and stores it in its database. The Transaction Security Services process sends a response to the Device Handler process.
37	The Device Handler process receives the response. At this point the host and EPP have successfully authenticated each other and exchanged their public keys.

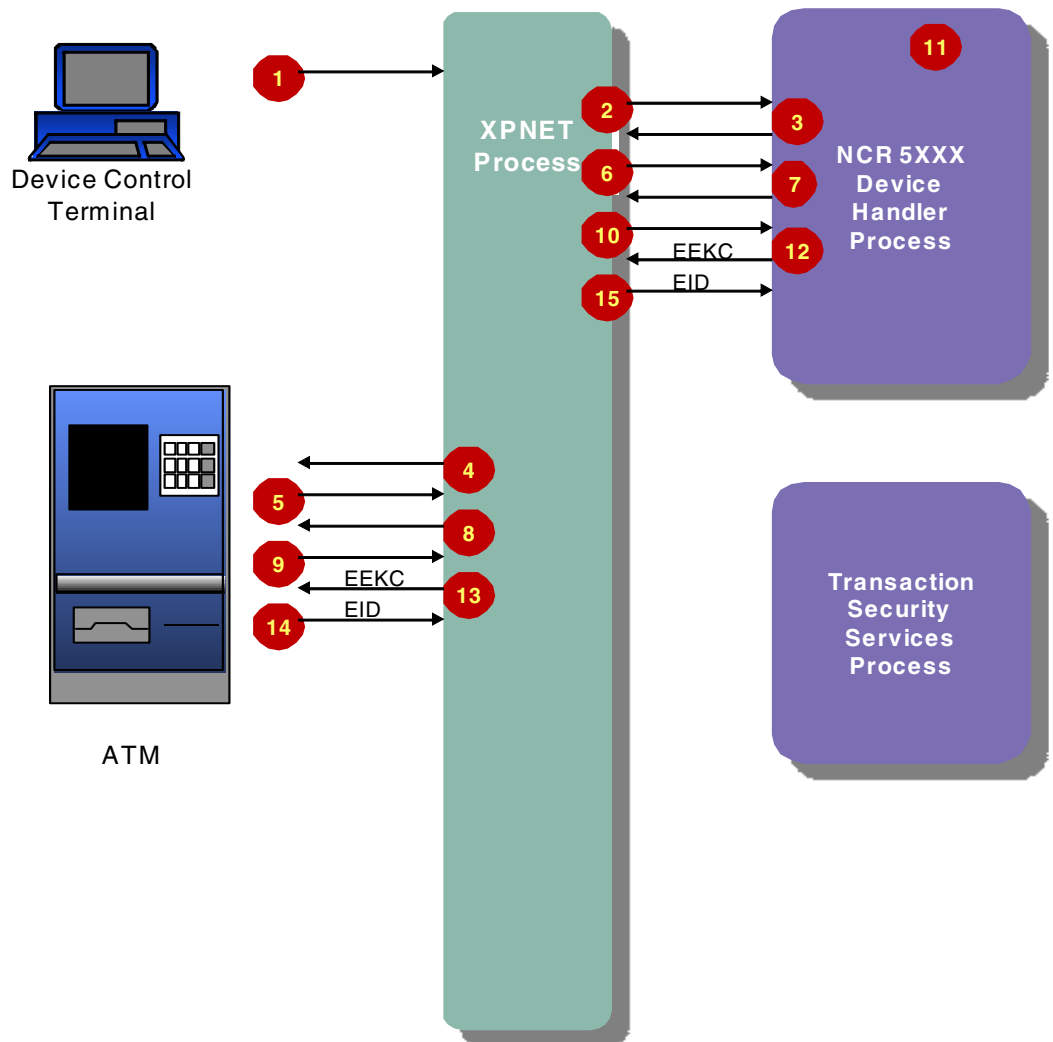
Downloading the Key Entry Mode Using Basic Signatures

In addition to downloading keys, the Automated Key Distribution System can be used to change the key entry mode (for example, going from single-length to double-length keys). This is accomplished by entering the LOAD ENTRY KEY MODE command from the DCT or the 9500LKEYMODE command from a network control facility.

Note: Changing the key entry mode deletes all the DES keys in the EPP as well as the BASE24 public key and the master key. To reduce the chances of accidentally changing the mode and deleting the keys, the Load Key Entry Mode command is accepted only when the authentication and public key exchange sequence has been performed since the last power up, ensuring that BASE24 has the ability to restore the keys. If this sequence has not been performed since the last power-up, the EPP rejects the command.

The download key entry mode command also can be used to initially set the key mode. Changing the key entry mode switches the encryptor into EPP mode. For example, if the EPP was installed before the Automated Key Distribution System add-on product was implemented (i.e., it is running in BAPE emulation mode, which does not support the Automated Key Distribution System), the encryptor's keys remain preserved. When the Automated Key Distribution System is implemented, the EPP must be enabled. If the EPP is installed with NDC+ 7.00 software and is running in BAPE emulation mode when the key entry mode is changed, it is also reprogrammed as an EPP and moves out of BAPE emulation mode. Note that you also can load the key entry mode at the ATM when the ATM is in supervisor mode.

After the key entry mode has been successfully loaded, the authentication and public key exchange sequence and the master key download sequence must be performed. The following message flows illustrate the processing that takes place when the key entry mode is downloaded.



Steps

1, 2

Processing

An operator initiates a request to download the key entry mode. In this example, the LOAD KEY ENTRY MODE command is entered from the Device Control Terminal (DCT), but a request can also be made from a network control facility.

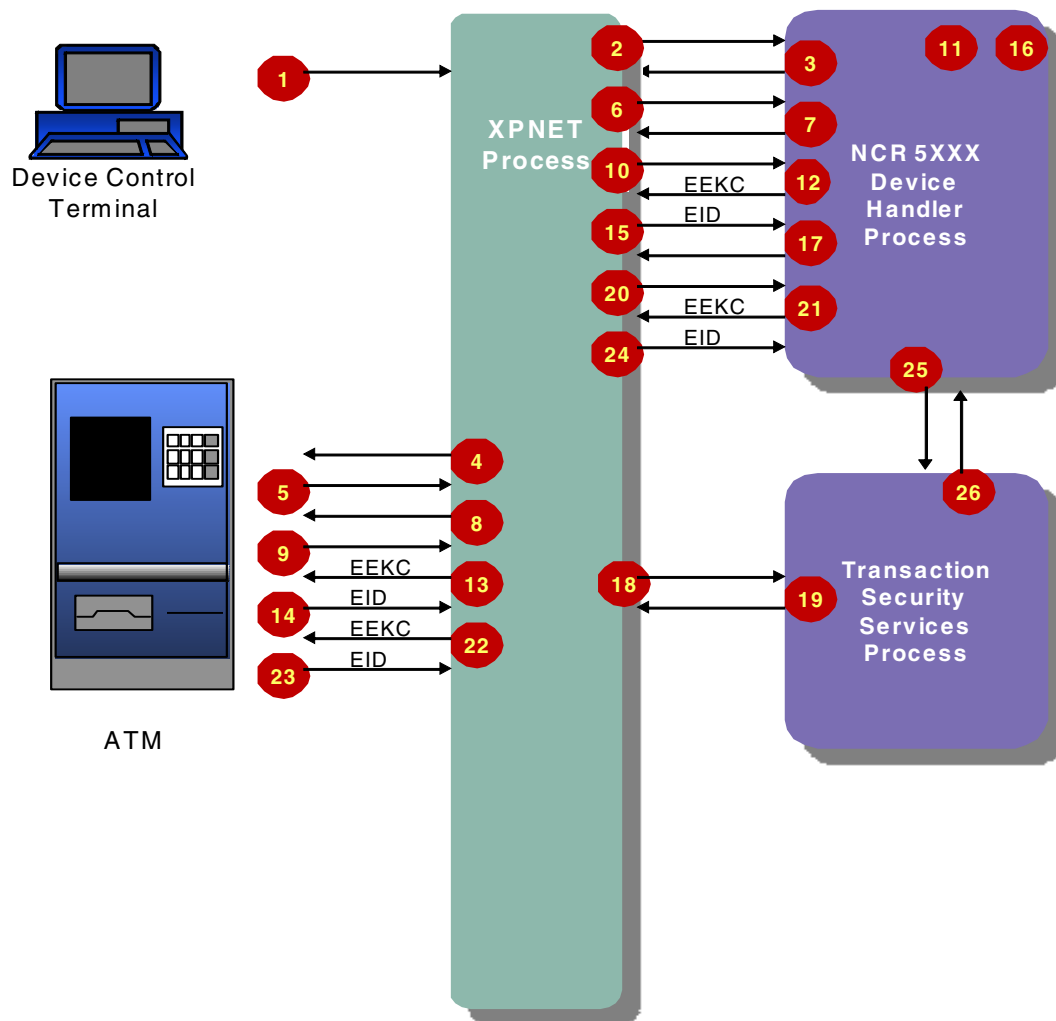
Note: When this command is performed, all keys in the encryptor are cleared and further operation of the device is prevented until the keys have been reloaded.

Steps	Processing
3, 4	The BASE24-atm NCR 5XXX Device Handler process sends a terminal command message to the ATM requesting a shutdown.
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends a configuration request to the ATM.
9, 10	The ATM sends a configuration response message to the Device Handler process.
11	The Device Handler process searches the configuration response message for the value 7 or 8 in the encryptor type field and the TERMINAL LINE PROTOCOL field on NCR 5XXX screen 7 of the ATD for the protocol value. An encryptor type of 7 or 8 and a protocol value other than B, C, or S indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is stopped and the ATM is restarted. If the ATM is capable of processing Automated Key Distribution System commands, the Device Handler process continues with the next check. The Device Handler checks the software ID and release number in the configuration response message to determine if the ATM is capable of sending the encryptor capabilities and state. If the software ID begins with G531 (Aptra Advance) but the release number is less than 3.02, or if the software ID begins with V351 (Aptra Edge) but the release number is less than 2.04.00, the ATM is not capable of sending the encryptor capabilities and state, nor does it support certificates. The Device Handler process continues with the basic scheme documented previously, otherwise, the Device Handler process continues with step 12.
12, 13	The Device Handler process sends an extended encryption key change (EEKC) message to download the key entry mode to the EPP.
14, 15	The EPP accepts the key entry mode. The ATM responds with an encryptor initialization data (EID) message indicating the key entry mode was accepted.

At this point the key entry mode at the EPP has been modified, and all keys in the EPP have been deleted, including the BASE24 public key and the master key (the EPP public and secret key pair and the NCR public key are not deleted). The authentication and public key exchange must be performed, followed by the master key download.

Downloading the Master Key Using Basic Signatures

The following message flow illustrates the processing that takes place during the master key download phase (phase 5). At this stage in the processing, the BASE24 system and the EPP have authenticated each other and exchanged public keys. Refer to the ***BASE24 Transaction Security Services Processing Guide*** for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps

Processing

- 1, 2 An operator initiates a request to download the master key. In this example, the LOAD MASTER KEY command is entered from the Device Control Terminal (DCT), but a request can also be made from a network control facility.
- 3, 4 The BASE24-atm NCR 5XXX Device Handler process sends an operational command message to the ATM requesting a shutdown.
- 5, 6 The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.

Steps	Processing
7, 8	The Device Handler process sends a configuration request to the ATM.
9, 10	The ATM sends a configuration response message to the Device Handler process.
11	The Device Handler process searches the configuration response message for the value 7 or 8 in the encryptor type field and the TERMINAL LINE PROTOCOL field on NCR 5XXX screen 7 of the ATD for the protocol value. An encryptor type of 7 or 8 and a protocol value other than B, C, or S indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is stopped and the ATM is restarted. If the ATM is capable of processing Automated Key Distribution System commands, the Device Handler process continues with the next check. The Device Handler checks the software ID and release number in the configuration response message to determine if the ATM is capable of sending the encryptor capabilities and state. If the software ID begins with G531 (Aptra Advance) but the release number is less than 3.02, or if the software ID begins with V351 (Aptra Edge) but the release number is less than 2.04.00, the ATM is not capable of sending the encryptor capabilities and state, nor does it support certificates; therefore, the Device Handler process continues with step 17. Otherwise, the Device Handler process continues with step 12.
12, 13	The Device Handler process sends an extended encryption key change (EEKC) message to request the encryptor capabilities and state.
14, 15	The ATM responds with an encryptor initialization data (EID) message.

Steps	Processing
16	The Device Handler process checks the Remote Key Scheme and Certificate State provided in the encryptor initialization data (EID) message. If the Remote Key Scheme is “00” (none) or “03” (signature + certificate), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “00” (not ready/not supported) or “02” (certificate secondary), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “01” (certificate primary), the Device Handler process proceeds with the load using the certificate-based method; refer to the “Downloading the Master Key Using Certificates” topic in this manual for more information. If the Remote Key Scheme is “01” signature, the Device Handler process continues with step 17.
17, 18	The Device Handler process sends a nowaited request to the Transaction Security Services process to generate a master key.
19, 20	The Transaction Security Services process generates the master key, which is encrypted under the EPP public key, and a signature for the master key. It stores the master key in its database and returns a response to the Device Handler process.
21, 22	The Device Handler process encodes the master key and the signature using the base94 algorithm. The Device Handler process downloads the encoded master key and signature to the EPP by sending an extended encryption key change (EEKC) message.
23, 24	The EPP verifies the signature using the BASE24 public key and decrypts the master key using the EPP secret key. The master key is stored in the EPP. The ATM responds to the Device Handler process with an encryptor initialization data (EID) message. The message contains the six-byte key verification value (KVV) for the master key just loaded.

Steps	Processing
25	The Device Handler process sends a waited request to the Transaction Security Services process to verify the key verification value returned by the ATM. Note that this step does not go through the XPNET process.
26	The Transaction Security Services process verifies the key verification value by comparing it to the master key check digits stored in its database. The Transaction Security Services process returns a response to the Device Handler process. Note that this step does not go through the XPNET process.

The BASE24 system and the EPP have the master key stored in their databases. The working keys (e.g., PIN encryption or MAC keys) can now be downloaded to the EPP.

Note: A new financial institution table (FIT) must be downloaded to the ATM if the encrypted PIN verification key in the financial institution table is not set to blanks. The encrypted PIN verification key must be translated from encryption under the old master key to encryption under the new master key.

Load All Keys Using Basic Signatures

The NCR 5XXX Device Handler process enables you to download all keys (i.e., public key, master key, MAC encryption key, and PIN encryption key) with one command.

After an operator initiates a request to download all keys, the BASE24-atm NCR 5XXX Device Handler process shuts down the ATM, determines whether the ATM supports the Automated Key Distribution System add-on product, and determines the ATM's remote key scheme (signature or certificate), as described in the previous message flows.

If the ATM is capable of processing Automated Key Distribution System commands, the Device Handler process downloads the following RSA and DES keys in this order:

1. Public key
2. Master key
3. MAC encryption key (if the MAC SUPPORT TYPE field on ATD screen 9 is set to the value 3)

-
4. PIN encryption key (if the PIN ENCRYPT TYPE field on ATD screen 9 is set to the value 01, 03, 04, or 05)

If the ATM is not capable of processing Automated Key Distribution System commands, the Device Handler process downloads only the following DES keys (in this order):

1. MAC encryption key (if the MAC SUPPORT TYPE field on ATD screen 9 is set to the value 3)
2. PIN encryption key (if the PIN ENCRYPT TYPE field on ATD screen 9 is set to the value 01, 03, 04, or 05)

The public key and master key are downloaded to the ATM as described earlier in this appendix. The MAC encryption and PIN encryption keys are downloaded as described in the ***BASE24 Transaction Security Manual***.

The ATM is reopened at the end of the Load All Keys command.

Note that this command is supported only in the Automated Key Distribution System extension of the NCR 5XXX Device Handler process. Customers who have not purchased the add-on product are not able to use this command.

Enhanced Signature-Based Processing

The following topics describe the steps the BASE24-atm NCR 5XXX Device Handler process follows during Enhanced Protocol for automated key distribution using signatures. Automated key distribution is performed in phases as follows:

Phase	Description
1	Generation of the NCR RSA key pair.
2	Manufacture of the NCR Encrypting PIN Pad (EPP).
3	Generation of the BASE24 RSA key pair.
4	Authentication and public key exchange.
5	Master key download.

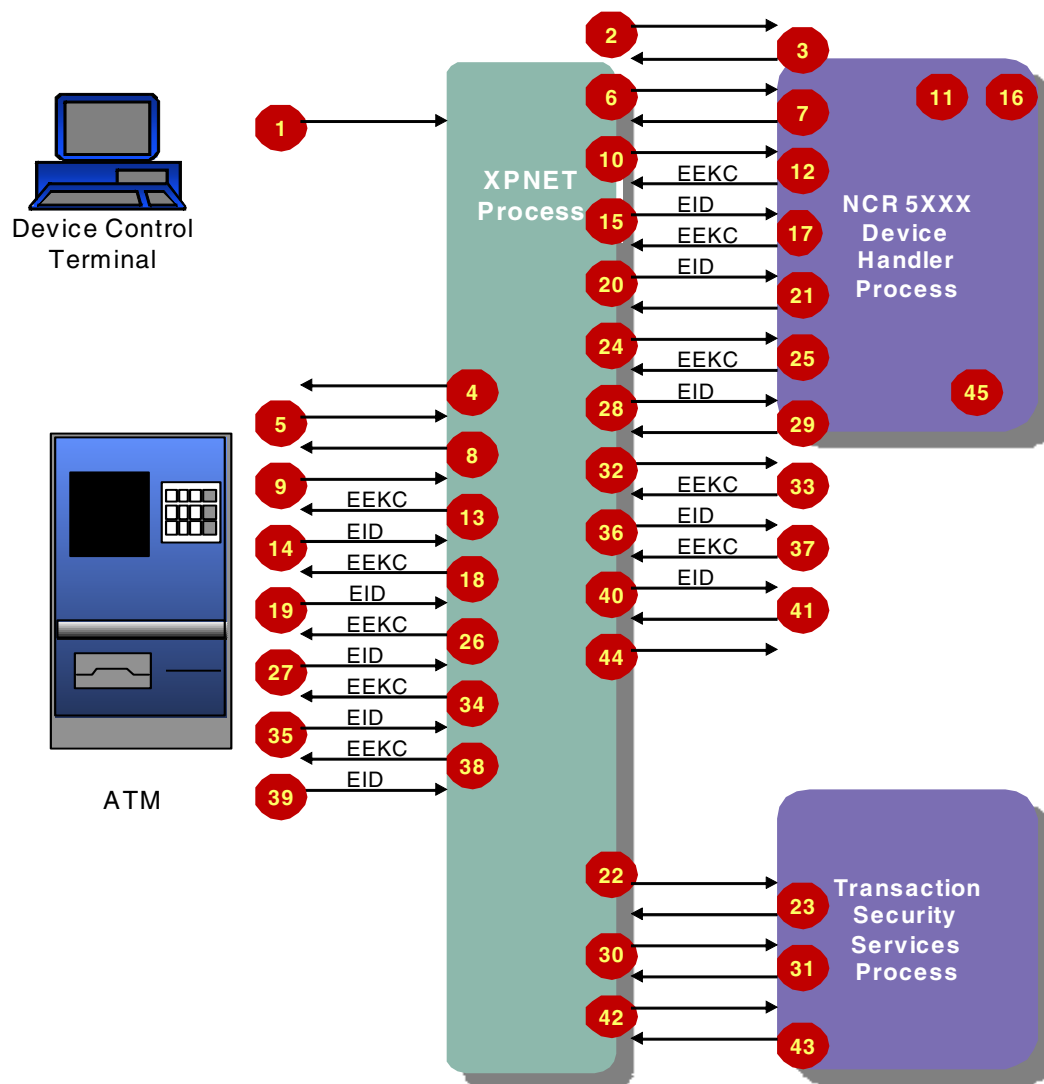
The processing described in this appendix is for phases 4 and 5. Phases 1 through 3 are described in the ***BASE24 Transaction Security Services Processing Guide***. This appendix also includes message flows for the Load Key Mode command.

This command sequence is performed after authentication and public key exchange. The key mode is loaded only if it was not set while the ATM was in supervisor mode.

Authentication and Public Key Exchange Using Enhanced Signatures

The following message flow illustrates the processing that takes place during Enhanced Protocol authentication and public key exchange (phase 4 of automated key distribution processing) which utilizes signatures. At this stage in the exchange, the BASE24 system and the EPP have established the keys and signatures required for automated key distribution. (Refer to the ***BASE24 Transaction Security Services Processing Guide*** for the message flow for establishing keys and signatures.) In the following steps, the BASE24 system and the EPP authenticate each other and exchange public keys. These steps are typically performed only once, when the EPP is initially deployed. If these steps are to be repeated, the BASE24 Public keys must first be deleted from the EPP by performing the Delete Public Key command.

Refer to the ***BASE24 Transaction Security Services Processing Guide*** for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps

Processing

- 1, 2 An operator initiates a request for a public key exchange. In this example, the LOAD PUBLIC KEY command is sent from the Device Control Terminal (DCT), but a request can also be made from a network control facility.
- 3, 4 The BASE24-atm NCR 5XXX Device Handler process sends an operational command message to the ATM requesting a shutdown.

Steps	Processing
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends a configuration request to the ATM.
9, 10	The ATM sends a configuration response message to the Device Handler process.
11	The Device Handler process searches the configuration response message for the value 7 or 8 in the encryptor type field and the TERMINAL LINE PROTOCOL field on NCR 5XXX screen 7 of the ATD for the protocol value. An encryptor type of 7 or 8 and a protocol value other than B, C, or S indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is stopped and the ATM is restarted. If the ATM is capable of processing Automated Key Distribution System commands, the Device Handler process continues with the next check. The Device Handler checks the software ID and release number in the configuration response message to determine if the ATM is capable of sending the encryptor capabilities and state. If the software ID begins with G531 (Apra Advance) but the release number is less than 3.02, or if the software ID begins with V351 (Apra Edge) but the release number is less than 2.04.00, the ATM is not capable of sending the encryptor capabilities and state, nor does it support certificates. The Device Handler process continues with the basic scheme documented previously, otherwise, the Device Handler process continues with step 12.
12, 13	The Device Handler process sends an extended encryption key change (EEKC) message to request the encryptor capabilities and state.
14, 15	The ATM responds with an encryptor initialization data (EID) message.

Steps	Processing
16	The Device Handler process checks the Remote Key Scheme and Certificate State provided in the encryptor initialization data (EID) message. If the Remote Key Scheme is “00” (none) or “03” (signature + certificate), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “00” (not ready/not supported) or “02” (certificate secondary), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “01” (certificate primary), the Device Handler process proceeds with the load using the certificate-based method; refer to the “Authentication and Public Key Exchange Using Certificates” topic in this manual for more information. If the Remote Key Scheme is “01” (signature), the Device Handler process continues with the basic scheme documented previously. If the Remote Key Scheme is “04” (enhanced signature), the Device Handler process continues with step 17.
17, 18	The Device Handler process sends an extended encryption key change (EEKC) message to request the encrypting PIN pad (EPP) serial number.
19, 20	The ATM responds with an encryptor initialization data (EID) message. The message contains the base94-encoded 320-byte signature of the serial number.
21, 22	The Device Handler process decodes the base94-encoded signature and sends a nowaited request to the Transaction Security Services process to verify the decoded signature.
23, 24	The Transaction Security Services process verifies the signature using the NCR public key and stores the serial number of the EPP in its database. The Transaction Security Services process returns a response to the Device Handler process. Included in the response is the BASE24 Root Public key and its signature.

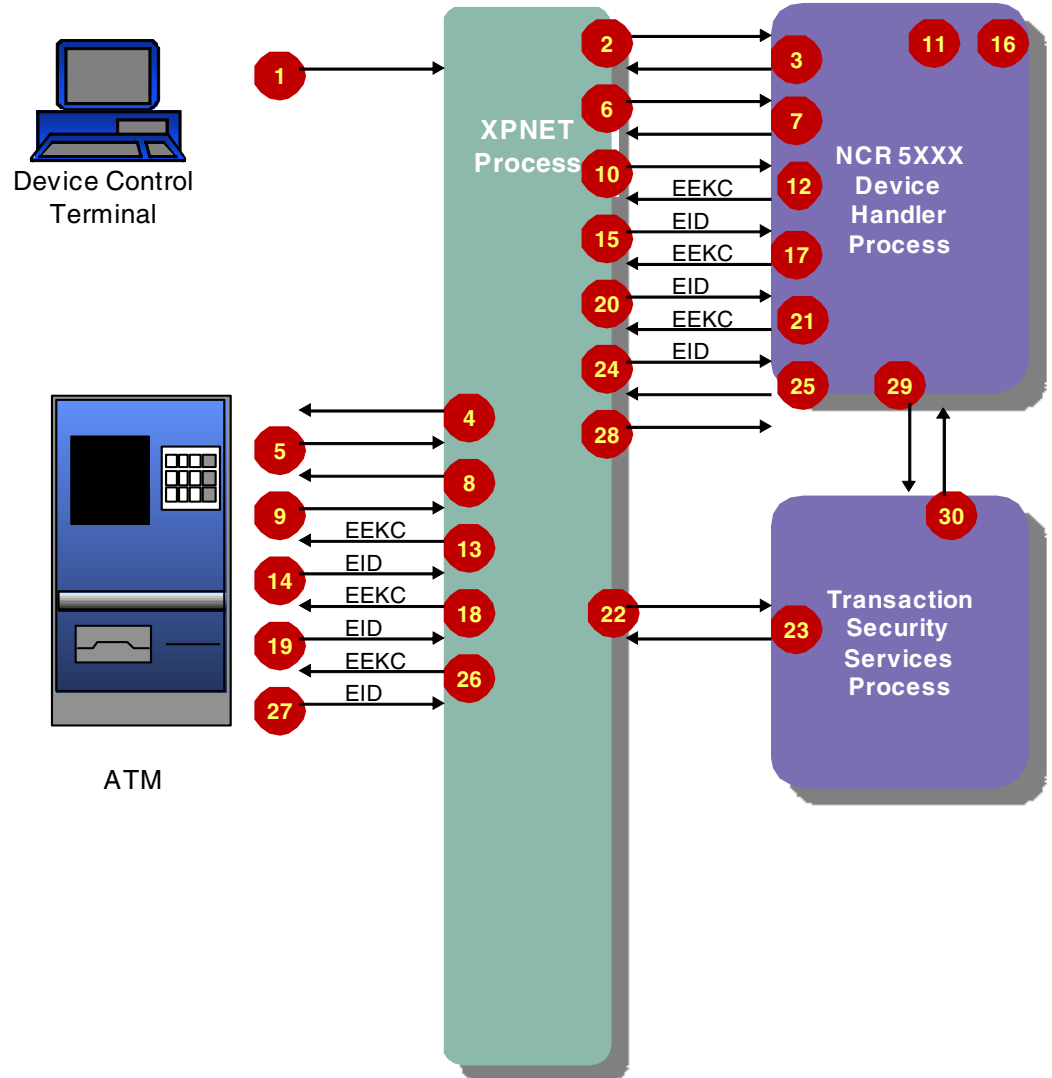
Steps	Processing
25, 26	The Device Handler process encodes the BASE24 Root Public key and its signature using the base94 algorithm. The Device Handler process downloads the encoded BASE24 Root Public key and signature to the ATM EPP by sending an extended encryption key change (EEKC) message.
27, 28	The ATM EPP verifies the signature using the NCR public key. The BASE24 Primary Root Public key is stored in the EPP. The ATM responds to the Device Handler process with an encryptor initialization data (EID) message indicating the key was loaded successfully.
29, 30	The Device Handler process sends a nowaited request to the Transaction Security Services process to retrieve the Host Primary Public key.
31, 32	The Transaction Security Services process returns a response to the Device Handler process. Included in the response is the BASE24 Primary Public key and the signature of the Primary Public key.
33, 34	The Device Handler process encodes the BASE24 Primary Public key and its signature using the base94 algorithm. The Device Handler process downloads the encoded BASE24 Primary Public key and signature to the ATM EPP by sending an extended encryption key change (EEKC) message.
35, 36	The ATM EPP verifies the signature using the Host Root Public key. The BASE24 Primary Public key is stored in the EPP. The ATM responds to the Device Handler process with an encryptor initialization data (EID) message indicating the key was loaded successfully.
37, 38	The Device Handler process sends an extended encryption key change (EEKC) message to the ATM requesting the EPP public key.
39, 40	The ATM responds with an encryptor initialization data (EID) message containing the EPP public key and the signature of the public key. Each are 320 bytes and base94 encoded.

Steps	Processing
41, 42	The Device Handler process decodes the base94-encoded EPP public key and signature and sends a nowaited request to the Transaction Security Services process to verify the decoded signature.
43, 44	The Transaction Security Services process verifies the signature using the NCR public key and stores the EPP public key in its database. It also generates a message authentication code (MAC) of the EPP public key and stores it in its database. The Transaction Security Services process sends a response to the Device Handler process.
45	The Device Handler process receives the response. At this point the host and EPP have successfully authenticated each other and exchanged their public keys.

Master Key Download Using Enhanced Signatures

The following message flow illustrates the processing that takes place during the Master key download phase (phase 5). At this stage in the processing, the BASE24 system and the EPP have authenticated each other and exchanged public keys.

Refer to the *BASE24 Transaction Security Services Processing Guide* for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps	Processing
1, 2	An operator initiates a request to download the master key. In this example, the LOAD MASTER KEY command is entered from the Device Control Terminal (DCT), but a request can also be made from a network control facility.
3, 4	The BASE24-atm NCR 5XXX Device Handler process sends an operational command message to the ATM requesting a shutdown.
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends a configuration request to the ATM.
9, 10	The ATM sends a configuration response message to the Device Handler process.
11	The Device Handler process searches the configuration response message for the value 7 or 8 in the encryptor type field and the TERMINAL LINE PROTOCOL field on NCR 5XXX screen 7 of the ATD for the protocol value. An encryptor type of 7 or 8 and a protocol value other than B, C, or S indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is stopped and the ATM is restarted. If the ATM is capable of processing Automated Key Distribution System commands, the Device Handler process continues with the next check. The Device Handler checks the software ID and release number in the configuration response message to determine if the ATM is capable of sending the encryptor capabilities and state. If the software ID begins with G531 (Aptra Advance) but the release number is less than 3.02, or if the software ID begins with V351 (Aptra Edge) but the release number is less than 2.04.00, the ATM is not capable of sending the encryptor capabilities and state, nor does it support certificates; therefore, the Device Handler process continues with step 17. Otherwise, the Device Handler process continues with step 12.

Steps	Processing
12, 13	The Device Handler process sends an extended encryption key change (EEKC) message to request the encryptor capabilities and state.
14, 15	The ATM responds with an encryptor initialization data (EID) message.
16	The Device Handler process checks the Remote Key Scheme and Certificate State provided in the encryptor initialization data (EID) message. If the Remote Key Scheme is “00” (none) or “03” (signature + certificate), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “00” (not ready/not supported) or “02” (certificate secondary), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “01” (certificate primary), the Device Handler process proceeds with the load using the certificate-based method; refer to the “Downloading the Master Key Using Certificates” topic in this manual for more information. If the Remote Key Scheme is “01” (signature), the Device Handler process continues with the basic scheme documented previously. If the Remote Key Scheme is “04” (enhanced signature), the Device Handler process continues with step 17.
17, 18	The Device Handler process sends an extended encryption key change (EEKC) message to request a random number from the ATM.
19, 20	The ATM responds with an encryptor initialization data (EID) message containing the random number.
21, 22	The Device Handler process sends a nowaited request to the Transaction Security Services process with the random number to generate a master key.

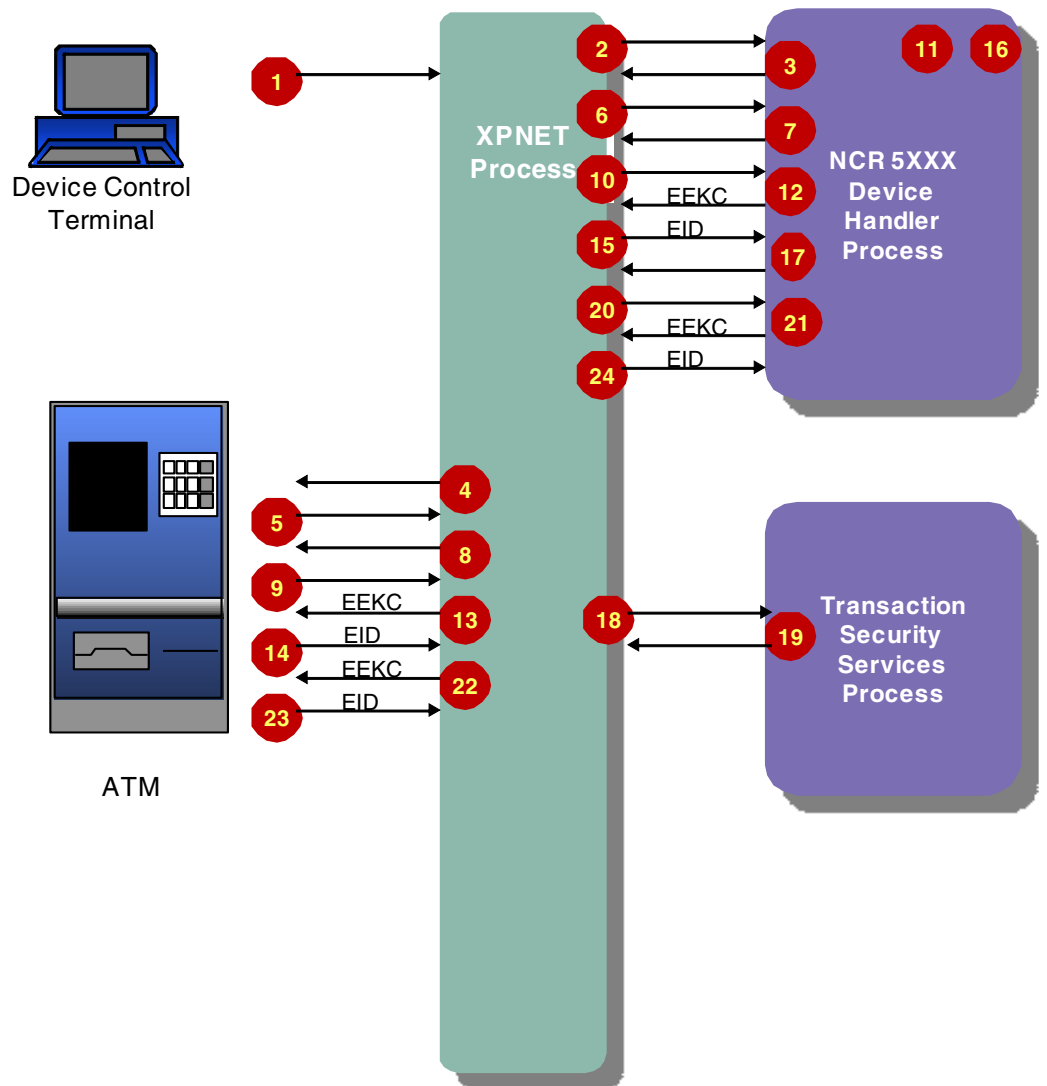
Steps	Processing
23, 24	The Transaction Security Services process generates the master key, which is encrypted under the EPP public key, and a signature for the master key. It stores the master key in its database and returns a response to the Device Handler process.
25, 26	The Device Handler process encodes the master key and the signature using the base94 algorithm. The Device Handler process downloads the encoded master key and signature to the EPP by sending an extended encryption key change (EEKC) message.
27, 28	The EPP verifies the signature using the BASE24 public key and decrypts the master key using the EPP secret key. The master key is stored in the EPP. The ATM responds to the Device Handler process with an encryptor initialization data (EID) message. The message contains the six-byte key verification value (KVV) for the master key just loaded.
29	The Device Handler process sends a waited request to the Transaction Security Services process to verify the key verification value returned by the ATM. Note that this step does not go through the XPNET process.
30	The Transaction Security Services process verifies the key verification value by comparing it to the master key check digits stored in its database. The Transaction Security Services process returns a response to the Device Handler process. Note that this step does not go through the XPNET process.

The BASE24 system and the EPP have the Master key stored in their respective databases. The working keys (e.g., PIN encryption or MAC keys) can now be downloaded to the EPP.

Note: A new financial institution table (FIT) must be downloaded to the ATM if the encrypted PIN verification key in the financial institution table is not set to blanks. The encrypted PIN verification key must be translated from encryption under the old master key to encryption under the new master key.

Delete Public Keys Using Enhanced Signatures

This command deletes the BASE24 Root Public key or BASE24 Primary Public key from the EPP. Keys must be reloaded before transactions from the ATM can be authorized. If the BASE24 Root Public key is deleted, the EPP automatically deletes the BASE24 Primary Public key as well.



Steps	Processing
1, 2	An operator initiates a request for a public key exchange. In this example, the DELETE PUBLIC KEY command is sent from the Device Control Terminal (DCT), but a request can also be made from a network control facility.
3, 4	The BASE24-atm NCR 5XXX Device Handler process sends an operational command message to the ATM requesting a shutdown.
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends a configuration request to the ATM.
9, 10	The ATM sends a configuration response message to the Device Handler process.
11	The Device Handler process searches the configuration response message for the value 7 or 8 in the encryptor type field and the TERMINAL LINE PROTOCOL field on NCR 5XXX screen 7 of the ATD for the protocol value. An encryptor type of 7 or 8 and a protocol value other than B, C, or S indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is stopped and the ATM is restarted. If the ATM is capable of processing Automated Key Distribution System commands, the Device Handler process continues with the next check. The Device Handler checks the software ID and release number in the configuration response message to determine if the ATM is capable of sending the encryptor capabilities and state. If the software ID begins with G531 (Aptra Advance) but the release number is less than 3.02, or if the software ID begins with V351 (Aptra Edge) but the release number is less than 2.04.00, the ATM is not capable of sending the encryptor capabilities and state, nor does it support certificates. The Device Handler process continues with the basic scheme documented previously, otherwise, the Device Handler process continues with step 12.

Steps	Processing
12, 13	The Device Handler process sends an extended encryption key change (EEKC) message to request the encryptor capabilities and state.
14, 15	The ATM responds with an encryptor initialization data (EID) message.
16	The Device Handler process checks the Remote Key Scheme and Certificate State provided in the encryptor initialization data (EID) message. If the Remote Key Scheme is “00” (none) or “03” (signature + certificate), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “00” (not ready/not supported) or “02” (certificate secondary), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “01” (certificate primary), the Device Handler process proceeds with the load using the certificate-based method; refer to the “Authentication and Public Key Exchange Using Certificates” topic in this manual for more information. If the Remote Key Scheme is “01” (signature), the Device Handler process aborts the load as this command is only valid for Enhanced AKDS. If the Remote Key Scheme is “04” (enhanced signature), the Device Handler process continues with step 17.
17, 18	The Device Handler process sends a nowaited request to the Transaction Security Services process to retrieve the key and signature of the Host Primary Public key or the Host Root Public key to be deleted.
19, 20	The Transaction Security Services process returns a response to the Device Handler process containing the key and its signature.
21, 22	The Device Handler process encodes the public key and the signature using the base94 algorithm. The Device Handler process downloads the encoded public key and signature to the EPP by sending an extended encryption key change (EEKC) message.

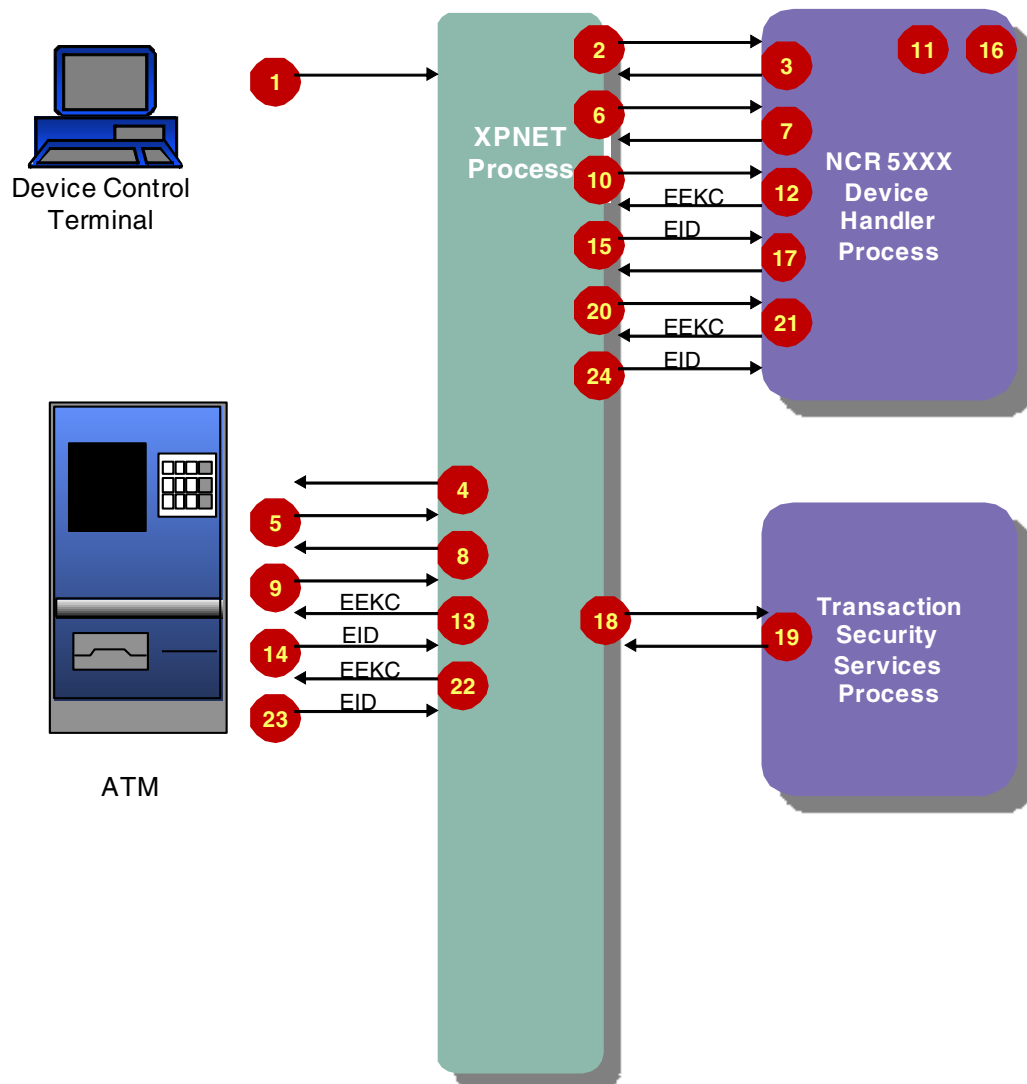
Steps

Processing

- 23, 24 The ATM deletes the appropriate key(s) and responds with an encryptor initialization message (EID) indicating the key has been deleted.

Load Primary Public Key Using Enhanced Signatures

This command loads the BASE24 Primary Public key, and can be used after the BASE24 Primary Public key has been deleted.



Steps	Processing
1, 2	An operator initiates a request for a public key exchange. In this example, the LOAD PRIMARY PUBLIC KEY command is sent from the Device Control Terminal (DCT), but a request can also be made from a network control facility.
3, 4	The BASE24-atm NCR 5XXX Device Handler process sends an operational command message to the ATM requesting a shutdown.
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends a configuration request to the ATM.
9, 10	The ATM sends a configuration response message to the Device Handler process.
11	The Device Handler process searches the configuration response message for the value 7 or 8 in the encryptor type field and the TERMINAL LINE PROTOCOL field on NCR 5XXX screen 7 of the ATD for the protocol value. An encryptor type of 7 or 8 and a protocol value other than B, C, or S indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is stopped and the ATM is restarted. If the ATM is capable of processing Automated Key Distribution System commands, the Device Handler process continues with the next check. The Device Handler checks the software ID and release number in the configuration response message to determine if the ATM is capable of sending the encryptor capabilities and state. If the software ID begins with G531 (Aptra Advance) but the release number is less than 3.02, or if the software ID begins with V351 (Aptra Edge) but the release number is less than 2.04.00, the ATM is not capable of sending the encryptor capabilities and state, nor does it support certificates. The Device Handler process continues with the basic scheme documented previously, otherwise, the Device Handler process continues with step 12.

Steps	Processing
12, 13	The Device Handler process sends an extended encryption key change (EEKC) message to request the encryptor capabilities and state.
14, 15	The ATM responds with an encryptor initialization data (EID) message.
16	The Device Handler process checks the Remote Key Scheme and Certificate State provided in the encryptor initialization data (EID) message. If the Remote Key Scheme is “00” (none) or “03” (signature + certificate), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “00” (not ready/not supported) or “02” (certificate secondary), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “01” (certificate primary), the Device Handler process proceeds with the load using the certificate-based method; refer to the “Authentication and Public Key Exchange Using Certificates” topic in this manual for more information. If the Remote Key Scheme is “01” (signature), the Device Handler process aborts the load as this command is only valid for Enhanced AKDS. If the Remote Key Scheme is “04” (enhanced signature), the Device Handler process continues with step 17.
17, 18	The Device Handler process sends a nowaited request to the Transaction Security Services process to retrieve the Host Primary Public key and signature.
19, 20	The Transaction Security Services process returns a response to the Device Handler process containing the key and its signature.
21, 22	The Device Handler process encodes the public key and the signature using the base94 algorithm. The Device Handler process downloads the encoded public key and signature to the EPP by sending an extended encryption key change (EEKC) message.

Steps**Processing**

- | | |
|--------|--|
| 23, 24 | The ATM verifies the signature, stores the key in the EPP, and responds with an encryptor initialization message (EID) indicating the key has been loaded. |
|--------|--|

Load Master and Communication keys Using Enhanced Signatures

The NCR 5XXX Device Handler process enables you to download all keys (i.e. Master key, MAC encryption key, and PIN encryption key) with one command. After an operator initiates a request to download all keys, the BASE24-atm NCR 5XXX Device Handler process shuts down the ATM, determines whether the ATM supports the Automated Key Distribution System add-on product, and identifies the ATM's remote key scheme (signature or certificate), as described in the previous message flows.

If the ATM is capable of processing Automated Key Distribution System Enhanced Protocol commands, the Device Handler processes download the following RSA and DES keys in this order:

1. Downloads the Master key
2. Downloads the MAC encryption key (if the MAC SUPPORT TYPE field on ATD screen 9 is set to the value 3)
3. Downloads the PIN encryption key (if the PIN ENCRYPT TYPE field on ATD screen 9 is set to the value 01, 03, 04, or 05)

If the ATM is not capable of processing enhanced Automated Key Distribution System commands, the Device Handler aborts the load as this command is only valid for Enhanced AKDS. The Master key is downloaded to the ATM as described earlier in this appendix. The MAC encryption and PIN encryption keys are downloaded as described in the ***BASE24 Transaction Security Manual***.

The ATM is reopened at the end of the Load Master/Com Keys command.

Note that this command is supported only in the Automated Key Distribution System extension of the NCR 5XXX Device Handler process. Customers who have not purchased the add-on product are not able to use this command.

Load All Keys Using Enhanced Signatures

The NCR 5XXX Device Handler process enables you to download all keys (i.e. Public keys, Master key, MAC encryption key, and PIN encryption key) with one command. After an operator initiates a request to download all keys, the BASE24-atm NCR 5XXX Device Handler process shuts down the ATM, determines whether the ATM supports the Automated Key Distribution System add-on product, and identifies the ATM's remote key scheme (signature or certificate), as described in the previous message flows.

If the ATM is capable of processing Automated Key Distribution System Enhanced Protocol commands, the Device Handler processes download the following RSA and DES keys in this order:

1. Deletes the Host Root Public Key and Host Primary Public Key
2. Downloads the Host Public keys
3. Downloads the Master key
4. Downloads the MAC encryption key (if the MAC SUPPORT TYPE field on ATD screen 9 is set to the value 3)
5. Downloads the PIN encryption key (if the PIN ENCRYPT TYPE field on ATD screen 9 is set to the value 01, 03, 04, or 05)

If the ATM is not capable of processing enhanced Automated Key Distribution System commands, the Device Handler process downloads only the following DES keys in this order:

1. MAC encryption key (if the MAC SUPPORT TYPE field on ATD screen 9 is set to the value 3)
2. PIN encryption key (if the PIN ENCRYPT TYPE field on ATD screen 9 is set to the value 01, 03, 04, or 05)

The Public key and Master key are downloaded to the ATM as described earlier in this appendix. The MAC encryption and PIN encryption keys are downloaded as described in the ***BASE24 Transaction Security Manual***.

The ATM is reopened at the end of the Load All Keys command.

Note that this command is supported only in the Automated Key Distribution System extension of the NCR 5XXX Device Handler process. Customers who have not purchased the add-on product are not able to use this command.

Certificate-Based Processing

The following topics describe the processing steps the BASE24-atm NCR 5XXX Device Handler process takes in automated key distribution using certificates. Automated key distribution is performed in phases as follows:

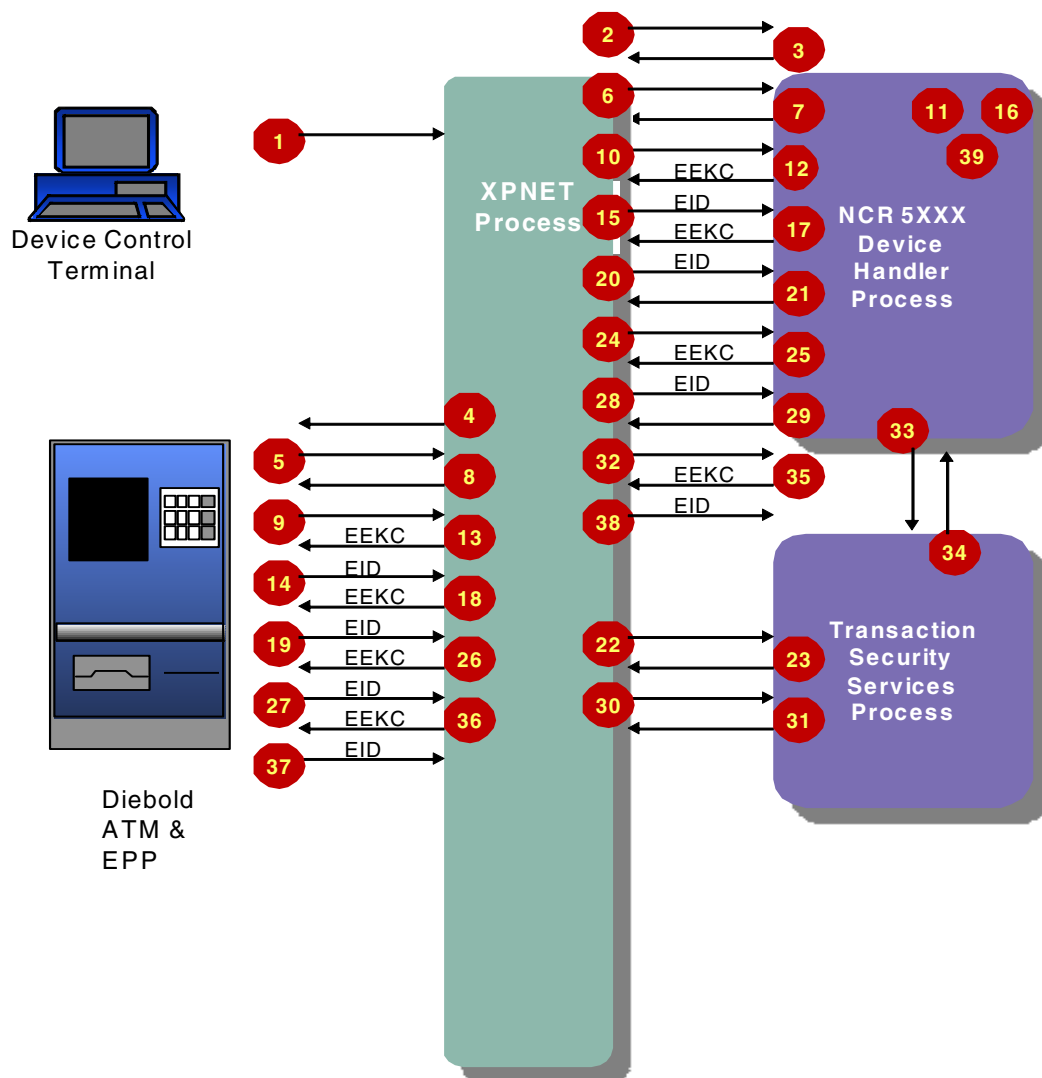
Phase	Description
1	Generation of the Diebold RSA key pair.
2	Manufacture of the Diebold Encrypting PIN Pad (EPP).
3	Generation of the BASE24 RSA key pair.
4	Authentication and public key exchange using certificates.
5	Master key download using certificates.

The processing described in this appendix is for phases 4 and 5. Phases 1 through 3 are described in the ***BASE24 Transaction Security Services Processing Guide***. This appendix also includes messages flows for the Load Key Mode command. This command sequence is performed after authentication and public key exchange. The key mode is loaded only when the key mode was not set while the ATM was in supervisor mode.

Authentication and Public Key Exchange Using Certificates

The following message flow illustrates the processing that takes place during authentication and public key exchange (phase 4 of automated key distribution processing) which utilizes certificates. At this stage in the exchange, the BASE24 system and the EPP have established the keys and signatures required for automated key distribution. (Refer to the ***BASE24 Transaction Security Services Processing Guide*** for the message flow for establishing keys and signatures.) In the following steps, the BASE24 system and the EPP authenticate each other and exchange public keys. These steps are typically performed only once, when the EPP is initially deployed.

Refer to the ***BASE24 Transaction Security Services Processing Guide*** for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps	Processing
1, 2	An operator initiates a request for a public key exchange. In this example, the LOAD PUBLIC KEY command is sent from the Device Control Terminal (DCT), but a request can also be made from a network control facility.
3, 4	The BASE24-atm NCR 5XXX Device Handler process sends a terminal command message to the ATM requesting a shutdown.
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends a configuration request to the ATM.
9, 10	The ATM sends a configuration response message to the Device Handler process.
11	The Device Handler process searches the configuration response message for the value 7 or 8 in the encryptor type field and the TERMINAL LINE PROTOCOL field on NCR 5XXX screen 7 of the ATD for the protocol value. An encryptor type of 7 or 8 and a protocol value other than B, C, or S indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is stopped and the ATM is restarted. If the ATM is capable of processing Automated Key Distribution System commands, the Device Handler process continues with the next check. The Device Handler checks the software ID and release number in the configuration response message to determine if the ATM is capable of sending the encryptor capabilities and state. If the software ID begins with G531 (Apra Advance) but the release number is less than 3.02, or if the software ID begins with V351 (Apra Edge) but the release number is less than 2.04.00, the ATM is not capable of sending the encryptor capabilities and state, nor does it support certificates. Therefore, the Device Handler process proceeds with the load using the signature-based method; refer to the “Downloading the Master Key Using Signatures” topic in this manual for more information. Otherwise, the Device Handler process continues with the next step.

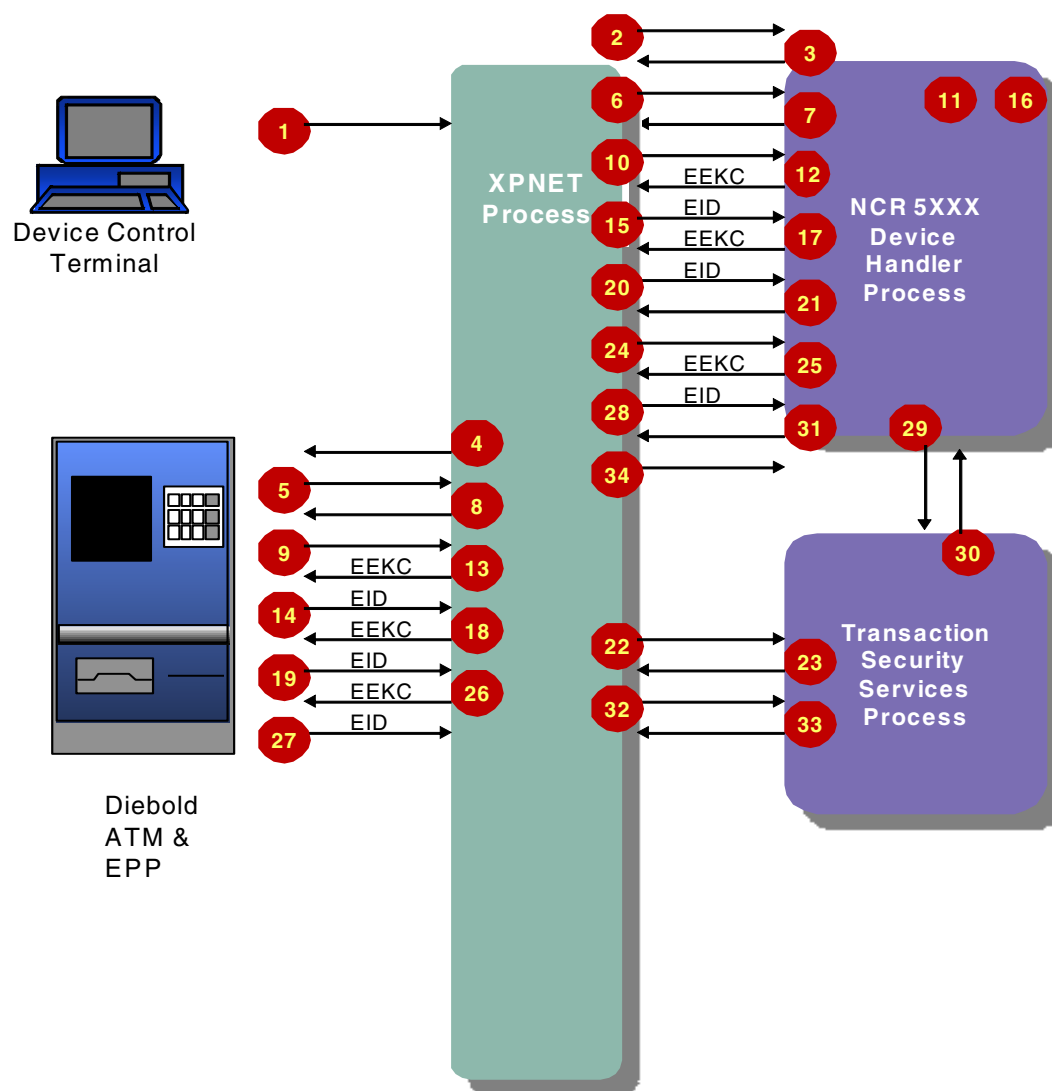
Steps	Processing
12, 13	The Device Handler process sends an extended encryption key change (EEKC) message to request the encryptor capabilities and state.
14, 15	The ATM responds with an encryptor initialization data (EID) message.
16	The Device Handler process checks the Remote Key Scheme and Certificate State provided in the encryptor initialization data (EID) message. If the Remote Key Scheme is “00” (none) or “03” (signature + certificate), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “00” (not ready/not supported) or “02” (certificate secondary), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “01” signature, the Device Handler process proceeds with the load using the signature-based method; in which case you will refer to the “Authentication and Public Key Exchange Using Signatures” topic in this manual for more information. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “01” (certificate primary), the Device Handler process continues with step 17.
17, 18	The Device Handler process sends an extended encryption key change (EEKC) message to request the encrypting PIN pad (EPP) public encryption key certificate.
19, 20	The ATM responds with an encryptor initialization data (EID) message.
21, 22	The Device Handler process base94-decodes the EPP public encryption key certificate and sends a nowaited request to the Transaction Security Services process to verify the decoded certificate.
23, 24	The Transaction Security Services process verifies the EPP public encryption key certificate and stores it in its database. The Transaction Security Services process returns a nowaited response to the Device Handler process.

Steps	Processing
25, 26	The Device Handler process sends an extended encryption key change (EEKC) message to request the EPP public verification key certificate.
27, 28	The ATM responds with an encryptor initialization data (EID) message.
29, 30	The Device Handler process base94-decodes the EPP public verification key certificate and sends a nowaited request to the Transaction Security Services process to verify the decoded certificate.
31, 32	The Transaction Security Services process verifies the EPP public verification key certificate and stores it in its database. The Transaction Security Services process returns a nowaited response to the Device Handler process.
33	The Device Handler process sends a waited request to the Transaction Security Services process to retrieve the BASE24 public verification key certificate.
34	The Transaction Security Services process returns a waited response to the Device Handler process. Included in the response is the BASE24 public verification key certificate.
35, 36	The Device Handler process base94-encodes the public verification key certificate and downloads the encoded certificate to the EPP by sending an extended encryption key change (EEKC) message.
37, 38	The ATM EPP verifies the BASE24 public verification key certificate. The certificate key is stored in the EPP. The ATM responds to the Device Handler process with an encryptor initialization data (EID) message indicating the certificate was loaded successfully.
39	The Device Handler process receives the response. At this point the host and EPP have successfully authenticated each other and exchanged their public keys.

Downloading the Master Key Using Certificates

The following message flow illustrates the processing that takes place during the master key download phase (phase 5). At this stage in the processing, the BASE24 system and the EPP have authenticated each other and exchanged public keys.

Refer to the *BASE24 Transaction Security Services Processing Guide* for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps

Processing

- 1, 2 An operator initiates a request to download the master key. In this example, the LOAD MASTER KEY command is entered from the Device Control Terminal (DCT), but a request can also be made from a network control facility.
- 3, 4 The BASE24-atm NCR 5XXX Device Handler process sends a terminal command message to the ATM requesting a shutdown.
- 5, 6 The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.

Steps	Processing
7, 8	The Device Handler process sends a configuration request to the ATM.
9, 10	The ATM sends a configuration response message to the Device Handler process.
11	The Device Handler process searches the configuration response message for the value 7 or 8 in the encryptor type field and the TERMINAL LINE PROTOCOL field on NCR 5XXX screen 7 of the ATD for the protocol value. An encryptor type of 7 or 8 and a protocol value other than B, C, or S indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is stopped and the ATM is restarted. If the ATM is capable of processing Automated Key Distribution System commands, the Device Handler process continues with the next check. The Device Handler checks the software ID and release number in the configuration response message to determine if the ATM is capable of sending the encryptor capabilities and state. If the software ID begins with G531 (Aptra Advance) but the release number is less than 3.02, or if the software ID begins with V351 (Aptra Edge) but the release number is less than 2.04.00, the ATM is not capable of sending the encryptor capabilities and state, nor does it support certificates. Therefore, the Device Handler process proceeds with the load using the signature-based method; refer to the “Downloading the Master Key Using Signatures” topic in this manual for more information. Otherwise, the Device Handler process continues with the next step.
12, 13	The Device Handler process sends an extended encryption key change (EEKC) message to request the encryptor capabilities and state.
14, 15	The ATM responds with an encryptor initialization data (EID) message.

Steps	Processing
16	The Device Handler process checks the Remote Key Scheme and Certificate State provided in the encryptor initialization data (EID) message. If the Remote Key Scheme is “00” (none) or “03” (signature + certificate), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “00” (not ready/not supported) or “02” (certificate secondary), the load is stopped and the ATM is restarted. If the Remote Key Scheme is “01” signature, the Device Handler process proceeds with the load using the signature-based method; in which case you will refer to the “Downloading the Master Key Using Signatures” topic in this manual for more information. If the Remote Key Scheme is “02” (certificate) and the Certificate State is “01” (certificate primary), the Device Handler process continues with step 17.
17, 18	The Device Handler process sends an extended encryption key change (EEKC) message to request the ATM’s random number.
19, 20	The ATM responds with an encryptor initialization data (EID) message.
21, 22	The Device Handler process decodes the ATM’s random number and sends a nowaited request to the Transaction Security Services process to generate a master key.

Steps	Processing
23, 24	The Transaction Security Services process generates the master key and stores it in its database. It then constructs a master key block containing the BASE24 distinguishing identifier, the master key, and the key identifier; and then goes on to encrypt the master key block using the EPP public encryption key. The Transaction Security Services process then generates BASE24's random number and constructs a key load packet containing BASE24's random number, the ATM's random number, the ATM distinguishing identifier, and the encrypted master key block. The Transaction Security Services signs the entire key load packet using BASE24's private signing key. Finally, the Transaction Security Services process returns the signed key load packet in a nowaited response to the Device Handler process.
25, 26	The Device Handler process base94-encodes the signed key load packet and downloads it to the EPP by sending an extended encryption key change (EEKC) message.
27, 28	The EPP verifies the signature on the key load packet using the BASE24 public verification key. In addition, the EPP checks the ATM's distinguishing identifier and verifies that the ATM's random number in the key load packet matches the one it sent in steps 19 and 20. The EPP then decrypts the encrypted master key block using the EPP's private decryption key and checks BASE24's distinguishing identifier. If all of these checks are successful, the EPP accepts the master key and stores it according to the key identifier. The EPP creates an acknowledgement packet containing the random number for BASE24 and the ATM, as well as BASE24's distinguishing identifier. The EPP signs the entire acknowledgement packet using the EPP's private signing key. Finally, the EPP sends the signed acknowledgement packet and a six-byte key verification value (KVV) for the master key that was just loaded to the Device Handler in an encryptor initialization data (EID) message.
29	The Device Handler process sends a waited request to the Transaction Security Services process to verify the key verification value (KVV) returned by the ATM.

Steps	Processing
30	The Transaction Security Services process verifies the key verification value (KVV) by comparing it to the master key check digits that are stored in its database. The Transaction Security Services process returns a response to the Device Handler process.
31, 32	The Device Handler process base94-decodes the signed acknowledgement packet and sends a nowaited request to the Transaction Security Services process to verify the decoded and signed acknowledgement packet.
33, 34	The Transaction Security Services process verifies the EPP's signature on the acknowledgement packet using the EPP's public verification key. In addition, the Transaction Security Services process checks BASE24's distinguishing identifier and verifies that BASE24's random number in the acknowledgement packet matches the one it sent in steps 23 and 24. The TSS process goes on to check that the ATM's random number in the acknowledgement packet matches the one received in steps 19 and 20. Finally, TSS returns a nowaited response message to the Device Handler.

The BASE24 system and the EPP have the master key stored in their databases. The working keys (e.g., PIN encryption or MAC keys) can now be downloaded to the EPP.

Note: A new financial institution table (FIT) must be downloaded to the ATM if the encrypted PIN verification key in the financial institution table is not set to blanks. The encrypted PIN verification key must be translated from encryption under the old master key to encryption under the new master key.

Load All Keys Using Certificates

The NCR 5XXX Device Handler process enables you to download all keys (i.e., public key, master key, MAC encryption key, and PIN encryption key) with one command.

After an operator initiates a request to download all keys, the BASE24-atm NCR 5XXX Device Handler process shuts down the ATM, determines whether the ATM supports the Automated Key Distribution System add-on product, and determines the ATM's remote key scheme (signature or certificate), as described in the previous message flows.

If the ATM is capable of processing Automated Key Distribution System commands, the Device Handler process downloads the following RSA and DES keys in this order:

1. Public key
2. Master key
3. MAC encryption key (if the MAC SUPPORT TYPE field on ATD screen 9 is set to the value 3)
4. PIN encryption key (if the PIN ENCRYPT TYPE field on ATD screen 9 is set to the value 01, 03, 04, or 05)

If the ATM is not capable of processing Automated Key Distribution System commands, the Device Handler process downloads only the following DES keys (in this order):

1. MAC encryption key (if the MAC SUPPORT TYPE field on ATD screen 9 is set to the value 3)
2. PIN encryption key (if the PIN ENCRYPT TYPE field on ATD screen 9 is set to the value 01, 03, 04, or 05)

The public key and master key are downloaded to the ATM as described earlier in this appendix. The MAC encryption and PIN encryption keys are downloaded as described in the ***BASE24 Transaction Security Manual***.

The ATM is reopened at the end of the Load All Keys command.

Note that this command is supported only in the Automated Key Distribution System extension of the NCR 5XXX Device Handler process. Customers who have not purchased the add-on product are not able to use this command.

Wincor Nixdorf's Shared Specification Certificate-Based Processing

The following topics describe the processing steps the BASE24-atm NCR 5XXX Device Handler process takes in automated key distribution using certificates for Diebold terminals running firmware that is compatible with Wincor Nixdorf's RKL specification running in Diebold 912 emulation mode. Automated key distribution is performed in phases as follows:

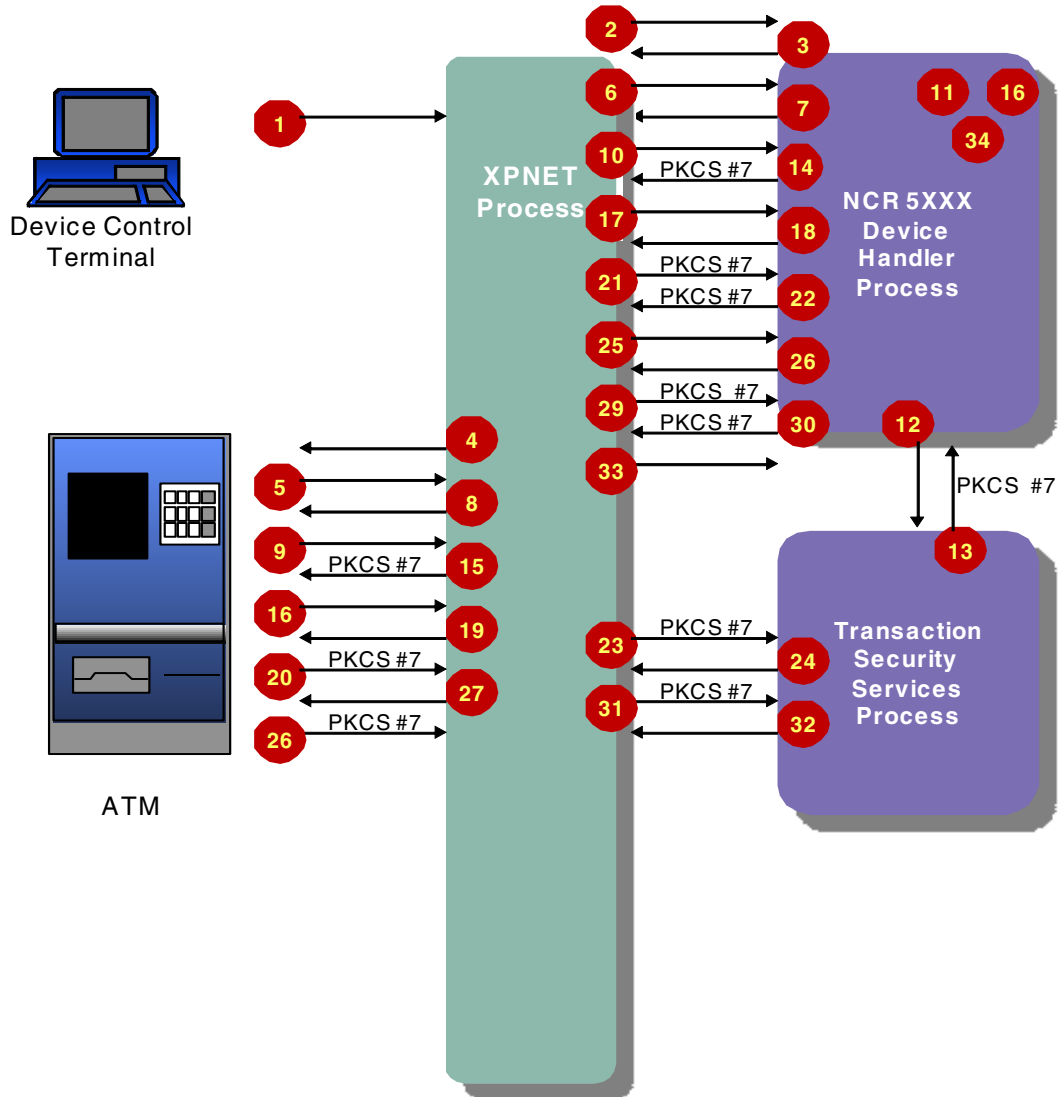
Phase	Description
1	Initial registration and certification for the Diebold Encrypting PIN Pad (EPP).
2	Initial registration and certification for the BASE24 system.
3	Initial certificate exchange.
4	Remote key transport.

The processing described in this appendix is for phases 3 and 4. Phases 1 and 2 are described in the ***BASE24 Transaction Security Services Processing Guide***.

Initial Certificate Exchange Using Certificates

The following message flow illustrates the processing that takes place during the initial certificate exchange (phase 3 of automated key distribution processing). At this stage in the exchange, the BASE24 system and the EPP have established the keys and certificates required for automated key distribution. (Refer to the ***BASE24 Transaction Security Services Processing Guide*** for a message flow example for establishing keys and certificates.) In the following steps, the BASE24 system and the EPP authenticate each other and exchange public keys. These steps are typically performed only once, when the EPP is initially deployed.

Refer to the *BASE24 Transaction Security Services Processing Guide* for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps	Processing
1, 2	An operator initiates a request for a public key exchange. In this example, the LOAD PUBLIC KEY command is sent from the Device Control Terminal (DCT), but a request also can be made from a network control facility.
3, 4	The BASE24-atm NCR 5XXX Device Handler process sends a terminal command message to the ATM requesting a shutdown.
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends a terminal command message, "Send RKL information, EPP capabilities" to the ATM.
9, 10	Provided there is AKDS firmware running at the ATM, the ATM responds to the Device Handler process with a solicited status message. Otherwise a command reject will be received.
11	The Device Handler process searches the signature/certificate indicator for the value of 2 (certificate). A value of 2 indicates that the ATM supports the Automated Key Distribution add-on product. If the ATM is not capable of processing Automated Key Distribution commands, the load is discontinued and a startup message is sent to the ATM. If the value is not found (a value of 0 is present), this phase is not performed and the ATM is restarted. If the value is found, the Device Handler process continues with step 12.
12	The Device Handler process sends a message to the Transaction Security Services to request the BASE24 primary certificate containing the BASE24 public signature verification key.
13	The Transaction Security Services process responds to the Device Handler process with the length of the Public Key Cryptography Standards (PKCS) #7 message and the PKCS #7 message that contains the BASE24 primary certificate containing the BASE24 public signature verification key.
14, 15	The Device Handler process base64-encodes the PKCS #7 message and downloads it to the ATM EPP.

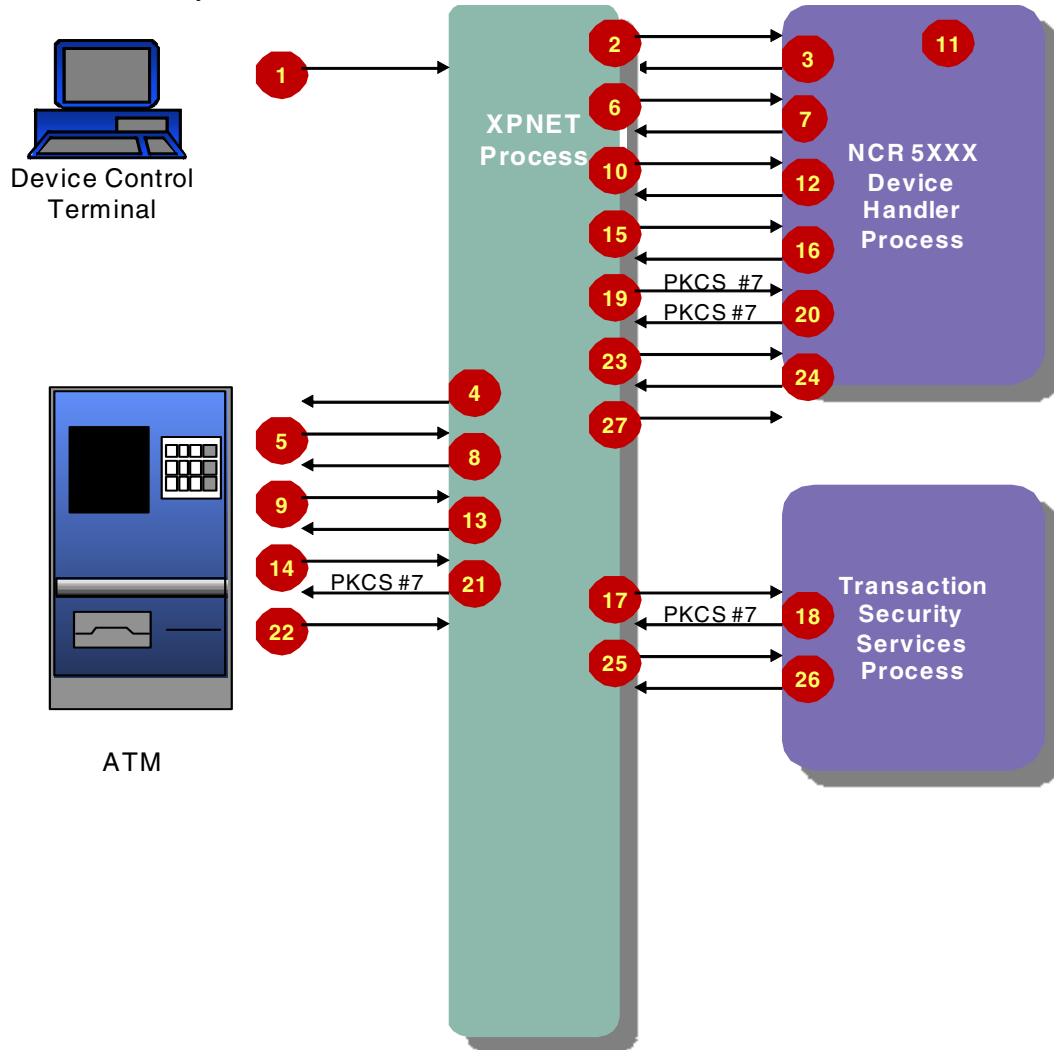
Steps	Processing
16, 17	The ATM EPP verifies the BASE24 primary certificate using the certificate authority's primary public signature verification key. The BASE24 public signature verification key is stored in the EPP. The ATM responds to the Device Handler process with a solicited status message indicating the key was loaded successfully.
18, 19	The Device Handler process requests the EPP public encryption/decryption key by sending a Terminal Command message, "Send RKL information, EPP encryption/decryption Public Key certificate" to the ATM.
20, 21	The ATM responds with a solicited status message. The solicited status message contains a base64-encoded PKCS #7 cryptographic message, which contains the EPP primary certificate which in turn contains the EPP public enciphering key.
22, 23	The Device Handler process base64-decodes the PKCS #7 message and sends a nowaited request to the Transaction Security Services process to verify the decoded certificate.
24, 25	The Transaction Security Services process verifies the EPP primary certificate using the certificate authority's primary public signature verification key and stores the EPP public enciphering key in its database. The Transaction Security Services process sends a nowaited response to the Device Handler process.
26, 27	The Device Handler process receives the nowaited response message. After processing the response, the Device Handler process requests the EPP public signature verification key by sending a Terminal command message, "Send RKL information, EPP signature verification Public Key certificate" to the ATM.
28, 29	The ATM responds with a solicited status message. The solicited status message contains a base64-encoded PKCS #7 cryptographic message, which contains the EPP primary certificate which in turn contains the EPP public signature verification key.
30, 31	The Device Handler process base64-decodes the PKCS #7 message and sends a nowaited request to the Transaction Security Services process to verify the decoded certificate.

Steps	Processing
32, 33	The Transaction Security Services process verifies the EPP primary certificate using the certificate authority's primary public signature verification key and stores the EPP public signature verification key in its database. The Transaction Security Services process sends a nowaited response message to the Device Handler process.
34	The Device Handler process receives the message and processes the response. At this point the BASE24 system and EPP have successfully authenticated each other and exchanged their public keys.

Remote Key Transport

The following message flow illustrates the processing that takes place during the remote key transport phase (phase 4). At this stage in the processing, the BASE24 system and the EPP have authenticated each other and exchanged public keys.

Refer to the *BASE24 Transaction Security Services Processing Guide* for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps

Processing

- | | |
|------|---|
| 1, 2 | An operator initiates a request to download the master key. In this example, the LOAD MASTER KEY command is entered from the Device Control Terminal (DCT), but the command also can be made from a network control facility. |
| 3, 4 | The BASE24-atm NCR 5XXX Device Handler process sends a terminal command message to the ATM requesting a shutdown. |

Steps	Processing
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends a terminal command message, “Send RKL information, EPP capabilities” to the ATM.
9, 10	Provided there is AKDS firmware running at the ATM, the ATM responds to the Device Handler process with a solicited status message. Otherwise a command reject will be received.
11	The Device Handler process searches the signature/certificate indicator for the value of 2 (certificate). A value of 2 indicates that the ATM supports the Automated Key Distribution add-on product. If the ATM is not capable of processing Automated Key Distribution commands, the load is discontinued and a startup message is sent to the ATM. If the value is not found (a value of 0 is present), this phase is not performed and the ATM is restarted. If the value is found, the Device Handler process continues with step 12.
12, 13	The Device Handler process sends a terminal command to the ATM indicating “Send RKL information, Start key exchange, create random number.”
14, 15	The ATM responds with a solicited status message containing the EPP’s random nonce.
16, 17	The Device Handler process sends a nowaited request to the Transaction Security Services process to generate a master key.

Steps	Processing
18, 19	The Transaction Security Services process generates the master key and stores it in its database. It then constructs a key block containing BASE24's distinguishing identifier, the master key, and the key identifier and encrypts the key block using the EPP public enciphering key. The Transaction Security Services process then constructs a key token containing a data block of BASE24's random nonce, the EPP's random nonce, the EPP's distinguishing identifier and the encrypted key block. The Transaction Security Service process then signs the data block using BASE24's private signature generation key. Finally, the Transaction Security Services process puts the entire key token in the PKCS #7 message and returns it to the Device Handler process.
20, 21	The Device Handler process base64-encodes the PKCS #7 message containing the enciphered and signed key block. The Device Handler process downloads the encoded PKCS #7 message to the EPP.
22, 23	The EPP verifies the signature on the key token using the BASE24 public signature verification key. It also checks the EPP's distinguishing identifier in the key token, and verifies that the EPP's random nonce in the key token is the same as the one sent in step 9. The EPP decrypts the key block using the EPP private deciphering key. The EPP also checks BASE24's distinguishing identifier. If all checks are successful, the master key is stored in the EPP according to the key identifier. The EPP then constructs a key token containing a data block of BASE24's random nonce, the EPP's random nonce, and BASE24's distinguishing identifier. The EPP then signs the data block using the EPP private signature verification key. Finally, the EPP puts the entire key token in the PKCS #7 cryptographic message and sends it in a solicited status message to the Device Handler process.
24, 25	The Device Handler process base64-decodes the PKCS #7 message and sends a nowaited request to the Transaction Security Services process to verify the key token.

Steps**Processing**

26, 27

The Transaction Security Services process verifies the signature on the key token using the EPP public signature verification key. It also checks BASE24's distinguishing identifier and checks that the received BASE24 random nonce agrees with the one sent in step 13. In addition it checks that the received EPP random nonce is the same as the one received in step 12. The Transaction Security Services process returns a nowaited response to the Device Handler process.

The BASE24 system and the EPP have the master key stored in their databases. The working keys (e.g., PIN encryption or MAC keys) can now be downloaded to the EPP.

Note: A new financial institution table (FIT) must be downloaded to the ATM if the encrypted PIN verification key in the financial institution table is not set to blanks. The encrypted PIN verification key must be translated from encryption under the old master key to encryption under the new master key.

Load All Keys Using Certificates

The NCR 5XXX Device Handler process enables you to download all keys (i.e., public key, master key, MAC encryption key, and PIN encryption key) with one command.

After an operator initiates a request to download all keys, the BASE24-atm NCR 5XXX Device Handler process sends a terminal command message to the ATM requesting a shutdown, as described in the previous message flows. The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.

The Device Handler process sends a Terminal command message "Send RKL information, EPP Capabilities" to the ATM. Provided there is AKDS firmware running at the ATM, the ATM responds to the Device Handler process with a solicited status message. Otherwise, a command reject will be received.

The Device Handler process searches the signature/certificate indicator for the value of 2 (certificate). A value of 2 indicates that the ATM supports the Automated Key Distribution add-on product. If the ATM supports the Automated Key Distribution System add-on product, the Device Handler process downloads the following RSA and DES keys (in this order):

1. Public key
2. Master key
3. MAC encryption key (if the MAC SUPPORT TYPE field on ATD screen 9 is set to the value 3)
4. PIN encryption key (if the PIN ENCRYPT TYPE field on ATD screen 9 is set to the value 01, 03, 04, or 05)

If the ATM does not support the Automated Key Distribution add-on product, the Device Handler process downloads only the following DES keys (in this order):

1. MAC encryption key (if the MAC SUPPORT TYPE field on ATD screen 9 is set to the value 3)
2. PIN encryption key (if the PIN ENCRYPT TYPE field on ATD screen 9 is set to the value 01, 03, 04, or 05)

The public key and master key are loaded to the ATM as described earlier in this appendix. The MAC encryption and PIN encryption keys are loaded as described in the ***BASE24 Transaction Security Services Processing Guide***.

Note that this command is supported only in the Automated Key Distribution System extension of the NCR 5XXX Device Handler process. Customers who have not purchased the add-on product are not able to use this command.

Wincor Nixdorf's Shared Specification Signature-Based Processing

The following topics describe the processing steps the BASE24-atm NCR 5XXX Device Handler process takes in automated key distribution for Wincor Nixdorf terminals running in NCR NDC+ emulation mode or other vendor's terminals

running firmware that is compatible with Wincor Nixdorf's RKL specification, running in NCR NDC+ emulation. Automated key distribution is performed in phases as follows:

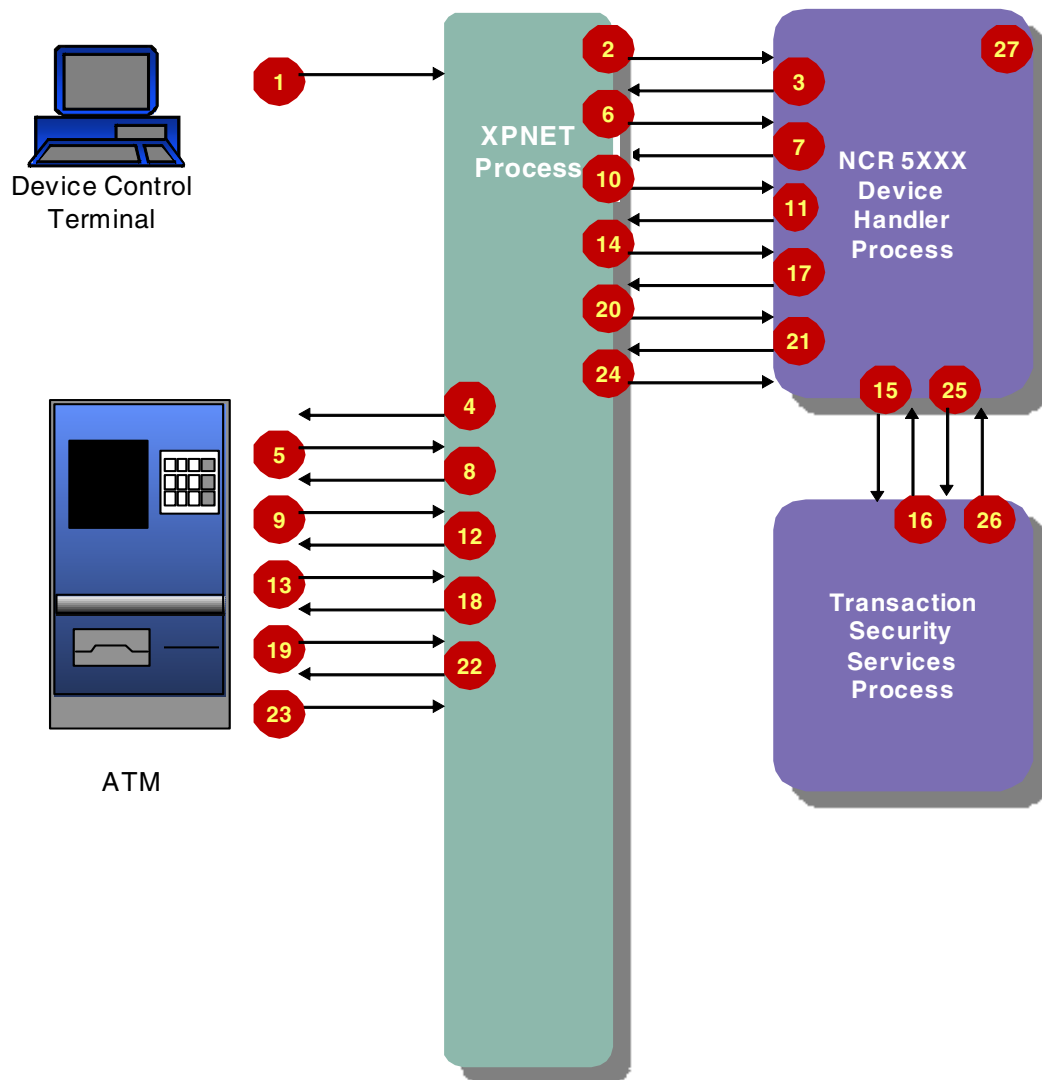
Phase	Description
1	Generation of the Wincor Nixdorf RSA key pair.
2	Manufacture of the vendor's encrypting pin pad (EPP).
3	Generation of the BASE24 RSA key pair.
4	Authentication and public key exchange.
5	Download of the master key.

The processing described in this appendix is for phases 4 and 5. Phases 1 through 3 are described in the ***BASE24 Transaction Security Services Processing Guide***.

Authentication and Public Key Exchange for Wincor Nixdorf

The following message flow illustrates the processing that takes place during authentication and public key exchange (phase 4 of automated key distribution processing). At this stage in the exchange, the BASE24 system and the EPP have established the keys and signatures required for automated key distribution. (Refer to the ***BASE24 Transaction Security Services Processing Guide*** for the message flow for establishing keys and signatures.) In the following steps, the BASE24 system and the EPP authenticate each other and exchange public keys. These steps are typically performed only once, when the EPP is initially deployed.

Refer to the ***BASE24 Transaction Security Services Processing Guide*** for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



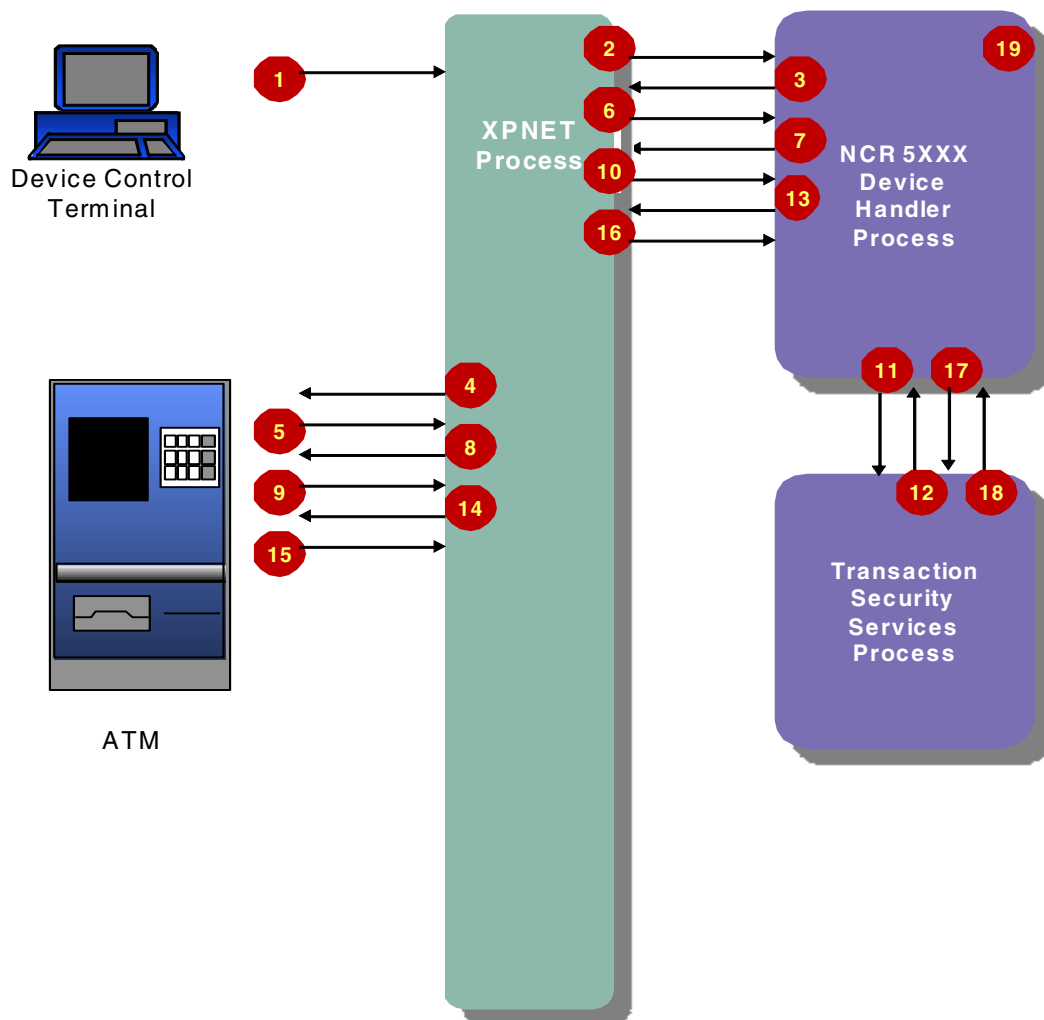
Steps	Processing
1, 2	An operator initiates a request for a public key exchange. In this example, the LOAD PUBLIC KEY command is sent from the Device Control Terminal (DCT), but a request can also be made from a network control facility.
3, 4	The BASE24-atm NCR 5XXX Device Handler process sends a terminal command message to the ATM requesting a shutdown.
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends a terminal command message, “Send RKL information, EPP capabilities” to the ATM.
9, 10	Provided there is AKDS firmware running at the ATM, the ATM responds to the Device Handler process with a solicited status message. Otherwise, a command reject will be received.
11, 12	The Device Handler process searches the signature/certificate indicator for the value 1 or 3. A value of 1 or 3 indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is discontinued and a startup message is sent to the ATM. If the ATM is capable of processing Automated Key Distribution System commands, the Device Handler process requests the encrypting PIN pad (EPP) serial number and signature using a terminal command message, “Send RKL information, EPP serial number and signature” to the ATM.
13, 14	The ATM responds to the Device Handler process with a solicited status message.
15	The Device Handler process decodes the base64-encoded signature and sends a request to the Transaction Security Services process to verify the decoded signature.

Steps	Processing
16	The Transaction Security Services process verifies the signature using the Wincor Nixdorf public key and stores the serial number of the EPP in its database. The Transaction Security Services process returns a response to the Device Handler process. Included in the response is the BASE24 public key and the signature of the public key.
17, 18	The Device Handler process encodes the BASE24 public key and its signature using the base64 algorithm. The Device Handler process downloads the encoded BASE24 public key and signature to the ATM EPP by sending an encryption key change (EKC) message.
19, 20	The ATM EPP verifies the signature using the Wincor Nixdorf public key. The BASE24 public key is stored in the EPP, and then the ATM responds with a solicited status message.
21, 22	The Device Handler process requests the EPP public key by sending a terminal command message, “Send RKL information, EPP Public Key and Signature” to the ATM.
23, 24	The ATM responds to the Device Handler process with a solicited status message.
25	The Device Handler process decodes the base64-encoded EPP public key and signature and sends a request to the Transaction Security Services process to verify the decoded signature.
26	The Transaction Security Services process verifies the signature using the Wincor Nixdorf public key and stores the EPP public key in its database. Then, the TSS process sends a response to the Device Handler process.
27	The Device Handler process receives the response. At this point the host and EPP have successfully authenticated each other and exchanged their public keys.

Downloading the Master Key for Wincor Nixdorf

The following message flow illustrates the processing that takes place during the master key download phase (phase 5). At this stage in the processing, the BASE24 system and the EPP have authenticated each other and exchanged public keys.

Refer to the *BASE24 Transaction Security Services Processing Guide* for information about the interaction among the Transaction Security Services process, the Transaction Security Services (TSS) database, and the hardware security module.



Steps

Processing

- 1, 2 An operator initiates a request to download the master key. In this example, the LOAD MASTER KEY command is entered from the Device Control Terminal (DCT), but a request can also be made from a network control facility.
- 3, 4 The BASE24-atm NCR 5XXX Device Handler process sends a terminal command message to the ATM requesting a shutdown.

Steps	Processing
5, 6	The ATM shuts down and responds to the Device Handler process with a solicited status message, indicating the ATM is ready.
7, 8	The Device Handler process sends a terminal command message, “Send RKL information, EPP capabilities” to the ATM.
9, 10	Provided there is AKDS firmware running at the ATM, the ATM responds to the Device Handler process with a solicited status message. Otherwise, a command reject will be received.
11	The Device Handler process searches the signature/certificate indicator for the value 1 or 3. A value of 1 or 3 indicates that the ATM supports the Automated Key Distribution System add-on product. If the ATM is not capable of processing Automated Key Distribution System commands, the load is discontinued and a startup message is sent to the ATM. The Device Handler process sends a request to the Transaction Security Services process to generate a master key.
12	The Transaction Security Services process generates the master key, which is encrypted under the EPP public key, and a signature for the master key. It stores the master key in its database and returns a response to the Device Handler process.
13, 14	The Device Handler process encodes the master key and the signature using the base64 algorithm. The Device Handler process downloads the encoded master key and signature to the EPP by sending an encryption key change (EKC) message.
15, 16	The EPP verifies the signature using the BASE24 public key and decrypts the master key using the EPP secret key. The master key is stored in the EPP. The ATM responds to the Device Handler process with a solicited status message.
17	The Device Handler process sends a request to the Transaction Security Services process to verify the key verification value returned by the ATM.
18	The Transaction Security Services process verifies the key verification value by comparing it to the master key check digits stored in its database. The Transaction Security Services process returns a response to the Device Handler process.

Steps**Processing**

- | | |
|----|--|
| 19 | The BASE24 system and the EPP have the master key stored in their databases. The working keys (e.g., PIN encryption or MAC keys) can now be downloaded to the EPP. |
|----|--|

Note: A new financial institution table (FIT) must be downloaded to the ATM if the encrypted PIN verification key in the financial institution table is not set to blanks. The encrypted PIN verification key must be translated from encryption under the old master key to encryption under the new master key.

Load All Keys Command for Wincor Nixdorf

When you enter the Load All Keys command (entered as LOADALLKEYS or LALLKEYS), the NCR 5XXX Device Handler process closes the specified ATM and sends a configuration request message. This message requests the ATM to send remote key loading (RKL) information and EPP capabilities to the BASE24 system. If the ATM does not have an AKDS-capable EPP installed, it rejects this request. If the ATM responds, the NCR 5XXX Device Handler process checks the Signature/Certificate status in the response. If the status is not 1 (Signature) or 3 (Signature and certificate), the process aborts the load, generates one or more event messages, and reopens the ATM.

If the ATM supports AKDS, the NCR 5XXX Device Handler process sequentially performs the following for the Load All Keys command:

1. Performs phase 4, which is to authenticate and exchange public keys.
2. Performs phase 5, which is to download the master key.
3. Downloads the MAC encryption key if the MAC SUPPORT TYPE field on NCR 5XXX screen 9 of the BASE24-atm Terminal Data files (ATD) is set to a value of 3 for the ATM.
4. Downloads the PIN encryption key if the PIN ENCRYPT TYPE field on NCR 5XXX screen 9 of the BASE24-atm Terminal Data files (ATD) is set to a value of 01, 03, 04, or 05 for the ATM.
5. Reopens the ATM.

If the ATM does not support AKDS, the NCR 5XXX Device Handler process sequentially performs only steps 3 through 5 above for the Load All Keys command.

Load Commands

You can issue a command to download the public key, the master key, or all keys to the ATM using the Device Control Terminal (DCT) or a network control facility.

The Automated Key Distribution System add-on product supports both leased-line and dial-up modes of the NCR 5XXX Device Handler process. When using the NCR 5XXX Dial-Up Device Handler extension, load commands are written to the ATM Dial-Up Command File (ADCF). The ADCF queues commands to a dial-up ATM until the ATM initiates a connection with the BASE24-atm Device Handler process. Refer to appendix D for more information about the NCR 5XXX Dial-Up Device Handler extension.

Command Syntax

The Load Primary Public, Load Public, Load Master, Load Master/Com, Load All Keys, and Load Key Entry Mode commands can be entered from a network control facility. The following paragraphs describe the commands and their syntax. Refer to the *BASE24 Text Command Reference Manual* for more information about these commands.

9500LPRIPUBLIC or 9500LOADPRIPUBLIC

The 9500LPRIPUBLIC or 9500LOADPRIPUBLIC command initiates a request to authenticate the encrypted PIN pad (EPP) and download the BASE24 Primary Public key to the EPP of the named ATM.

Syntax

9500LPRIPUBLIC *terminal-name*

9500LOADPRIPUBLIC *terminal-name*

terminal-name — The name of the ATM where the Primary Public key is loaded.

9500LPUBLIC or 9500LOADPUBLIC

The 9500LPUBLIC or 9500LOADPUBLIC command initiates a request to authenticate the encrypted PIN pad (EPP) and download the BASE24 public key to the EPP of the named ATM.

Syntax

9500LPUBLIC *terminal-name*

9500LOADPUBLIC *terminal-name*

terminal-name — The name of the ATM where the public key is loaded.

9500LMASTER or 9500LOADMASTER

The 9500LMASTER or 9500LOADMASTER command initiates a request to download the master key to the encrypted PIN pad of the named ATM.

Syntax

9500LMASTER *terminal-name*

9500LOADMASTER *terminal-name*

terminal-name — The name of the ATM where the Master key is loaded.

9500LMSTRCOM or 9500LOADMSTRCOM

The 9500LMSTRCOM or 9500LOADMSTRCOM command initiates a request to download the new Master, PIN, and MAC keys to the encrypted PIN pad of the named ATM.

Syntax

9500LMSTRCOM *terminal-name*

9500LOADMSTRCOM *terminal-name*

terminal-name — The name of the ATM where all the keys are loaded.

9500LALLKEYS or 9500LOADALLKEYS

The 9500LOADALLKEYS or 9500LOADALLKEYS command initiates a request to download new public, master, MAC encryption, and PIN encryption keys to the encrypted PIN pad of the named ATM.

Syntax

9500LALLKEYS *terminal-name*

9500LOADALLKEYS *terminal-name*

terminal-name — The name of the ATM where all the keys are loaded.

9500LKEYMODE

The 9500LKEYMODE command initiates a request to download the key entry mode to the encrypted PIN pad (EPP) of the named ATM. This command is used only when the key entry mode is not set through the supervisor mode at the ATM.

Note: This command is not valid for ATMs running firmware that is compatible with Wincor Nixdorf's RKL specification, version 1.5.

Syntax

9500LKEYMODE *terminal-name* *x*

terminal-name — The name of the ATM where the key entry mode is loaded.

x — A code indicating the key entry mode. Valid values are as follows:

- 1 = Single length without XOR
- 2 = Single length with XOR
- 3 = Double length with XOR
- 4 = Double length restricted

Sending a Command Using the DCT

You also can use the Device Control Terminal (DCT) to initiate a Load Primary Public, Load Public, Load Master, Load Master/Com, or Load Key Entry Mode command. The commands are entered using the LOAD PRIMARY PUBLIC KEY, LOAD PUBLIC KEY, LOAD MASTER KEY, LOAD MASTER/COM, LOAD ALL KEYS, or LOAD KEY ENTRY MODE field on the DCT NCR 5XXX Load Commands screen. Refer to the ***BASE24 Device Control Manual*** for more information on these fields and the NCR 5XXX Load commands screen.

BASE24-atm Dynamic Key Management Add-On Product

The BASE24-atm Dynamic Key Management add-on product enables an institution to determine when PIN and MAC keys need to be changed. The BASE24-atm Dynamic Key Management add-on product shuts down the ATM and initiates commands to change the keys when predetermined thresholds are exceeded.

This topic describes the BASE24-atm Dynamic Key Management add-on product and its configuration.

Overview

The BASE24-atm system uses transaction keys to secure PINs and perform message authentication in the BASE24 network. Transaction keys should be changed regularly to maintain proper security. The standard BASE24-atm NCR 5XXX Device Handler process monitors transactions that include message authentication codes (MACs) and shuts down an ATM when a predetermined number of consecutive MAC errors is exceeded. In this case the ATM's MAC key must be changed. An operator must then issue a command to load the key and reopen the ATM. The operator also must manually monitor and load PIN keys.

The BASE24-atm Dynamic Key Management add-on product automates the key management process. When the NCR 5XXX Device Handler process determines that the keys must be changed (based on criteria set at the process level), it issues commands to automatically load the keys and reopen the ATM.

Determining When Keys are Changed

The standard BASE24-atm NCR 5XXX Device Handler process uses the ATM-DH-CONS-MAC-ERR-LMT Logical Network Configuration (LCONF) param to determine when a MAC key may have been compromised. This param specifies the number of consecutive number of MAC errors that can occur before the Device Handler process shuts down the ATM.

The BASE24-atm Dynamic Key Management add-on product allows you to set thresholds for PIN and MAC keys at the process level. The following table lists the thresholds that can initiate a transaction key change and the associated LCONF param in which the threshold is set:

Threshold	Param
The number of PIN verifications exceeds a predetermined limit.	ATM-DH-DKM-PIN-TXN-LMT
The number of MAC verifications and MAC generations exceed a predetermined limit.	ATM-DH-DKM-MAC-TXN-LMT
The number of PIN verification failures exceeds a predetermined limit.	ATM-DH-DKM-PIN-ERR-LMT
The number of MAC verification and MAC generation failures exceeds a predetermined limit.	ATM-DH-DKM-MAC-ERR-LMT
The number of consecutive MAC failures exceeds a predetermined limit.	ATM-DH-CONS-MAC-ERR-LMT
A specified time interval has passed since the last MAC and PIN key change.	ATM-DH-DKM-KEY-CHNG-INTRVL

The counters and timestamp that correspond to these thresholds are stored in the BASE24-atm Terminal Data files (ATD). A timestamp for the last key change initiated by the Dynamic Key Management add-on product is also stored in the ATD.

Refer to the *BASE24 Logical Network Configuration File Manual* for more information about these params.

Changing Keys

Once any of the thresholds for a key change has been reached, the process to load keys to the ATM is automatically initiated and all of the threshold counters and timer are reset. To minimize the time an ATM is down, both the PIN key and MAC key are loaded at the same time. The Dynamic Key Management key change timestamp also is updated.

Refer to the ***BASE24 Transaction Security Manual*** for information about transmitting transaction keys to the device.

F: Multiple Currency Dispense Add-On Feature

This appendix describes the BASE24-atm Multiple Currency Dispense add-on feature. This enables NCR 5XXX ATMs to dispense cash in multiple currencies.

Note: This function is for dispensing cash only. The BASE24-atm Multiple Currency add-on product for authorization, which supports the authorization of multiple currencies, can work with this product but is not required. Refer to the ***BASE24-atm Multiple Currency Support Manual*** for more information about the BASE24-atm Multiple Currency add-on product for authorization.

This appendix documents the cash dispensing portion of BASE24-atm multiple currency functions only. Refer to the ***BASE24-atm Multiple Currency Support Manual*** for information about authorization and currency conversion processing of multiple currency transactions.

Overview

The BASE24-atm Multiple Currency Dispense add-on feature enables an institution to dispense cash in a single currency or up to four different currencies from a single ATM.

Currency Types

BASE24-atm supports transaction processing in multiple currencies by defining currency types and allowing these to be used in data files and financial messages within BASE24. Some of these currency types are briefly defined in the following list. The *BASE24-atm Multiple Currency Support Manual* describes these currency types in more detail.

Transaction Currency — The currency in which the transaction is delivered to BASE24 from a device, a host, or an interchange. This is the key currency for multiple currency processing.

Acquirer Institution Currency — The default currency for the institution that acquires the transaction.

Issuer Institution Currency — The default currency for the institution that is the issuer for the transaction.

Account Currency — The designated currency for the account or accounts to which the transaction applies.

Base Currency — The currency through which all currency conversions take place. All exchange rates are defined in the Base currency. This currency is selected at installation and it is unlikely to change.

Original Currency — The currency used at the ATM at which the transaction originates. If the transaction is acquired by BASE24, Original currency is the same as the Transaction currency. However, if the transaction enters BASE24 through an interchange, the Original currency may be different than any other currency known to BASE24.

Terminal Currency — The currency for the ATM at which the transaction is performed.

Logical Network Currency — Logical network currency is maintained in the Multiple Currency product primarily to provide compatibility with standard BASE24 processing for a number of subsidiary functions.

Supported Transactions

The BASE24-atm Multiple Currency Dispense add-on feature supports cash withdrawals from checking, savings, and credit accounts.

Support with Other BASE24 Products

The BASE24-atm Multiple Currency add-on feature is supported by the Surcharge option. The BASE24-atm Non-Currency Dispense add-on product is compatible with the BASE24-atm Multiple Currency Dispense add-on feature (i.e., they can exist in the same environment), but BASE24-atm Non-Currency Dispense transactions cannot occur in more than a single currency. Only cash withdrawals can be done in more than a single currency.

Support is not provided for the Fraud Control option or the BASE24-atm self-service banking application.

Account Balance Information

When the Authorization process processes a Multiple Currency Dispense transaction, it converts the account currency to the Transaction currency. To display the account balance on the receipt and on the screen, the Device Handler process converts the amount given in the Transaction currency to an amount in the Account currency. Because of the multiple currency conversions, the account balance that appears on the receipt and on the ATM screen may differ slightly from the actual account balance.

Exchange Rate Notification

Institutions can optionally include an exchange rate notification step in multiple currency dispense transactions. If the value of the ATM-DH-XCHG-RATE-NTFY param is set to a value of Y, the cardholder is presented the rate of exchange and is prompted to confirm or cancel the transaction. If the cardholder cancels the transaction, the BASE24-atm NCR 5XXX Device Handler process sends a reversal message to the BASE24-atm Authorization process. Refer to the topic

“Processing” for more information about exchange rate notification and the topic “Configuring the Multiple Currency Dispense Add-On Feature” for more information about the ATM-DH-XCHG-RATE-NTFY param.

Exchange rate notification can exist in the same environment as response notification surcharging; however, only one interactive response message can be presented to the cardholder during a transaction. If both types of interactive response scenarios occur in the same transaction, the surcharge response notification takes precedence, and the exchange rate screen is not displayed.

Processing

Any process performing multiple currency processing converts the Transaction currency into the various currencies specified by the currency type using the Currency Conversion utilities. Currency conversion uses exchange rates held against the user-selected Base currency and follows appropriate conversion rules. Currency conversion processing uses the triangulation method, in which conversion between two currencies uses exchange rates held against the Base currency.

Currency conversion processing is discussed in the *BASE24-atm Multiple Currency Support Manual*.

Exchange rates are configured at the network level in the Exchange Rate File (ERF). The ERF contains one record for each currency (except the Base currency) that is used by the BASE24 network. The Base currency in multiple currency processing is the single currency against which exchange rates for all other currencies are expressed. All exchange rates held in the ERF are in respect to the BASE currency to a specific currency rate. That is, the value by which the amount must be multiplied or divided to obtain the equivalent.

The exchange rate information is stored in and retrieved from the Multiple Currency token. Refer to the *BASE24-atm Tokens Manual* for a detailed description of this token.

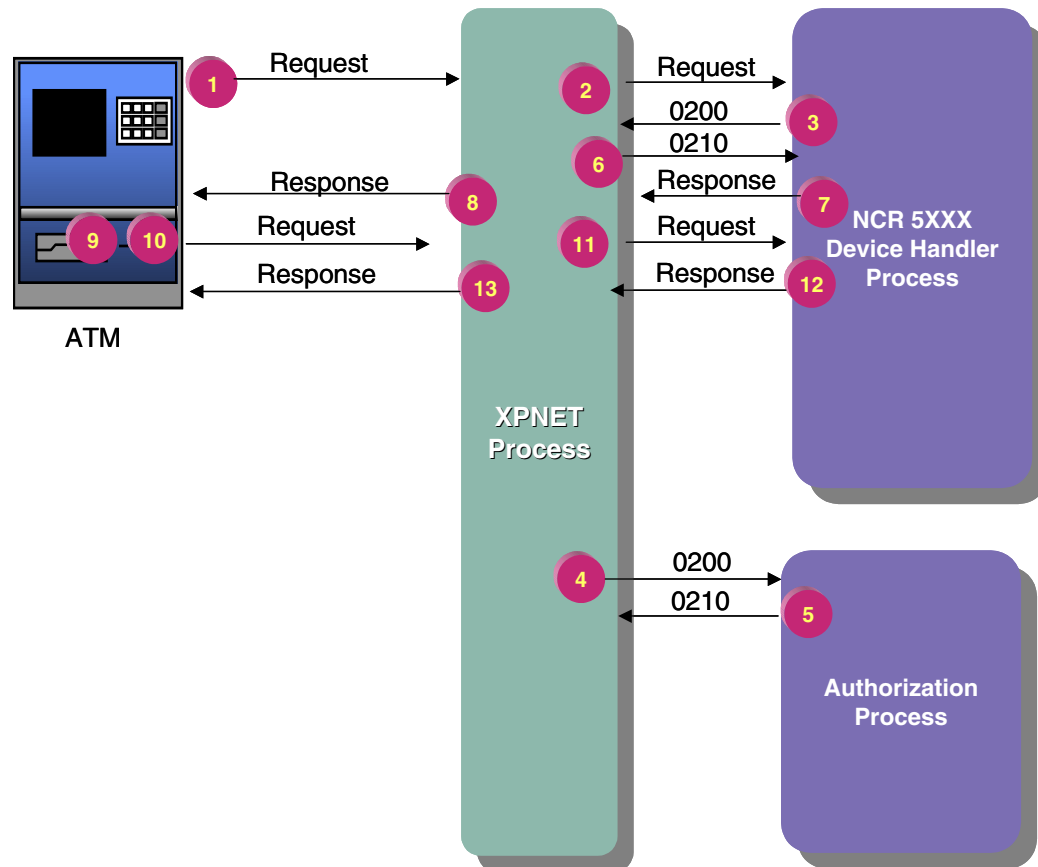
Message Flows

The following message flow diagrams describe the processing that takes place when the institution uses exchange rate notification (that is, the ATM-DH-XCHG-RATE-NTFY param is present and set to a value of Y). With exchange rate notification, the exchange rate is displayed to the cardholder, and the cardholder can confirm or cancel the transaction.

Note: The message flow for a multiple currency dispense transaction without exchange rate notification is the same as the message flow for a withdrawal transaction. Refer to the *BASE24-atm Transaction Processing Manual* for message flows for withdrawal transactions. Message flows for multiple currency dispense transactions that include multiple account selection or surcharges are similar to those in the *BASE24-atm Transaction Processing Manual*, with the exchange rate notification steps described here occurring after the multiple account selection and surcharging steps.

Withdrawal Transaction, Exchange Rate Notification, Cardholder Confirms Exchange Rate

The following diagram shows the message flow of a multiple currency dispense transaction with the cardholder confirming the exchange rate. Descriptions of each step follow the illustration.

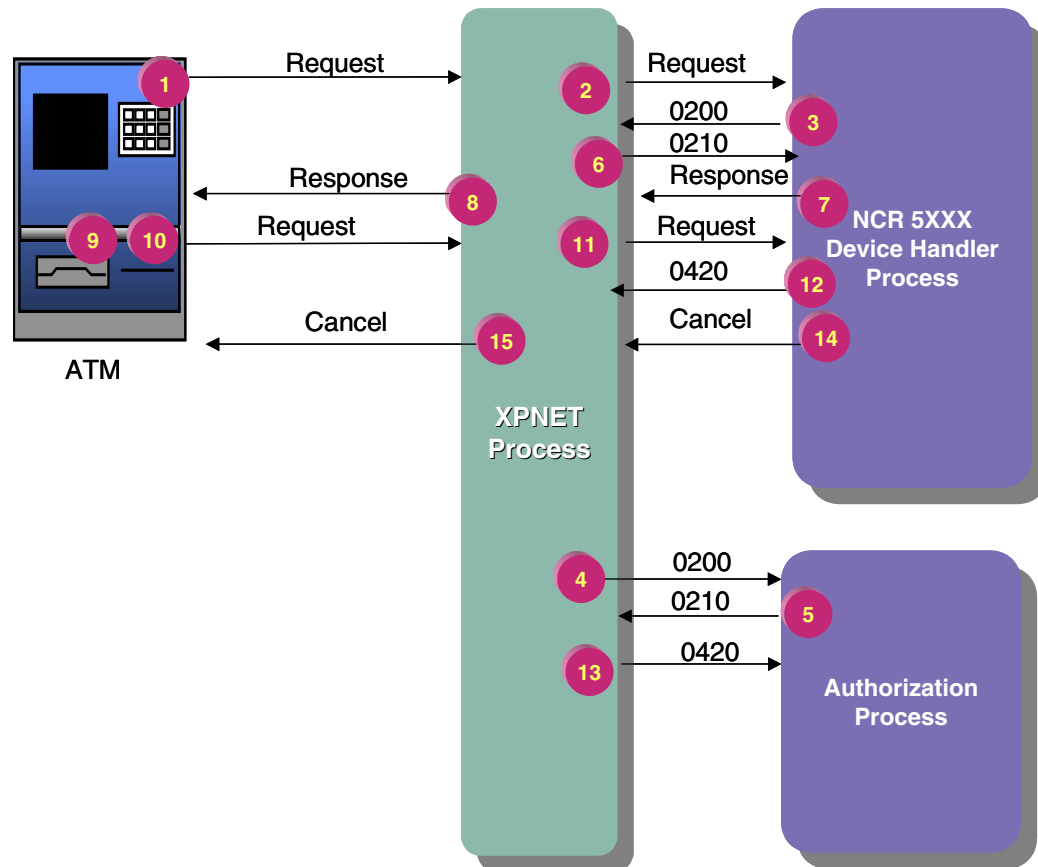


Step	Description
1, 2	The cardholder selects a transaction in a specific currency, the account to be used, and the transaction amount. The ATM sends a native-mode request message to the BASE24-atm NCR 5XXX Device Handler process.
3, 4	The Device Handler process formats a 0200 request message with the multiple currency token and sends it to the BASE24-atm Authorization process.

Step	Description
5, 6	Once the transaction is authorized (by the Authorization process itself or by the host or an interchange), the Authorization process formats a 0210 response message and sends it to the Device Handler process.
7, 8	The Device Handler process formats a native mode response message instructing the ATM to display the exchange rate and prompt the customer for confirmation.
9	The ATM displays the exchange rate and prompts the customer for confirmation. The customer elects to confirm the transaction.
10, 11	The ATM formats a native-mode request and sends it to the Device Handler process.
12, 13	The Device Handler process sends the dispense and print information to the ATM in a native response message.

Withdrawal Transaction, Exchange Rate Notification, Cardholder Cancels Transaction

The following message flow diagram illustrates a transaction in which the cardholder is presented with the exchange rate information and chooses to cancel the transaction. A description of the steps follows the diagram.



Step	Description
1, 2	The cardholder selects a transaction in a specific currency, the account to be used, and the transaction amount. The ATM sends a native-mode request message to the BASE24-atm NCR 5XXX Device Handler process.
3, 4	The Device Handler process formats a 0200 request message with the multiple currency token and sends it to the BASE24-atm Authorization process.

Step	Description
5, 6	Once the transaction is authorized (by the Authorization process itself or by the host or an interchange), the Authorization process formats a 0210 response message and sends it to the Device Handler process.
7, 8	The Device Handler process formats a native mode response message instructing the ATM to display the exchange rate and prompt the customer for confirmation.
9	The ATM displays the exchange rate and prompts the customer for confirmation. The customer elects to cancel the transaction.
10, 11	The ATM formats a native-mode request and sends it to the Device Handler process.
12, 13	The Device Handler process formats a 0420 reversal message which includes the Multiple Currency token and sends it to the Authorization process.
14, 15	The Device Handler process formats a native-mode message to cancel the transaction and sends it to the ATM.

Receipt Processing

The BASE24-atm NCR 5XXX Device Handler process uses the following field type codes to process receipts for multiple currency dispense transactions:

Field Type Code	Description
CV	Equivalent converted amount
EV	Exchange rate value
XC	Converted currency description
XE	Exchange rate description
XS	Selected currency description

The following is an example of a receipt for a transaction in which the currency selected by the cardholder is different than the account or terminal currency. The selected amount, exchange rate, and converted amount were formatted from the XS, EV, and CV field type codes, respectively.

```
1          A C I   D E M O
2
3      DATE          TIME   TERM
4      MM/DD/YY   HH:MM S1AN5070
5
6      123456700000003-003
7
8      330 SOUTH 108TH AVE
9
10     RECORD NO.          570
11     WITHDRAWAL
12     SELECTED AMOUNT     $20.00
13     EXCHANGE RATE       .1000
14     CONVERTED AMOUNT    p200.00
15     FR      1111222233334444
16
17     LEDGER BAL          $3412.00
18     AVAIL BAL           $1760.00
```

1 2
123456789012345678901234567

Note that although the balance information is in the Account currency, the figure may differ slightly from the actual account balance because of multiple currency conversions that have taken place.

Refer to section 5 for detailed information about these field type codes and configuring receipts.

Configuring the Multiple Currency Dispense Add-On Feature

The Multiple Currency Dispense add-on feature can be used with the BASE24-atm NCR 5XXX Device Handler process. The following NCR and BASE24 files contain fields needed for the Multiple Currency Dispense add-on feature.

- N50NAMS
- NCR Configuration File (CNFG5070)
- Logical Network Configuration File (LCONF)
- BASE24-atm Terminal Data files (ATD)
- Receipt File (RCPT)
- Exchange Rate File (ERF)

N50NAMS

The N50NAMS is an edit file that contains a series of tables used by the BASE24-atm NCR 5XXX Device Handler process. You can edit these tables to customize specific information displayed on ATM receipts. The `ncr_mult_crncy_tbl` table in the N50NAMS supports the multiple currency dispense add-on feature. This table allows the receipt information specific to the Multiple Currency Dispense add-on feature to be displayed in multiple languages.

NCR Configuration File (CNFG5070)

The NCR Configuration File (CNFG5070) enables the institution to maintain ATM screen displays, state table entries, FIT table data, configuration parameters, and timers, which are then downloaded to the ATM. These screens, states, etc., are identified by a load group number. The load group number for the Multiple Currency Dispense add-on feature is 901.

States for the Multiple Currency Dispense add-on feature include the following:

State Type	State Number
D—Preset Operations Code Buffer State	092, 093, 094
E—Four FDK Selection State	091
F—Amount Entry State	387
H—Four FDK Information Entry State	386
I—Transaction Request State	187
W—FDK Switch State	032

Refer to section 4 for more information about configuring the CNFG5070.

Logical Network Configuration File (LCONF)

The Logical Network Configuration File (LCONF) links physical file names to the files names as they are referenced in the BASE24 code and contains processing values used throughout the logical network. The LCONF contains one assign and three params that are related to the Multiple Currency Dispense add-on feature.

The ERFEMT assign, which is not specific to the Multiple Currency Dispense add-on feature (but is used by an Multiple Currency product), defines the location of the Exchange Rate File (ERF) extended memory table to be used by this logical network.

The BASE-CURRENCY-CODE param contains the ISO currency code of the Base currency to be used by this logical network. This code must match the Base currency code as set in the COBNAMES File at installation. This param is used by the Multiple Currency Dispense add-on feature and the BASE24-atm Multiple Currency add-on product.

The CURRENCY-CODE param contains the ISO currency code of the primary type of currency the logical network uses. The value in this param must match a value in the COBNAMES file. This param is not specific to the Multiple Currency Dispense add-on feature.

The ATM-DH-XCHG-RATE-NTFY param indicates whether to prompt the cardholder to confirm the exchange rate. This param is specific to the BASE24-atm Multiple Currency Dispense add-on feature.

BASE24-atm Terminal Data Files (ATD)

The BASE24-atm Terminal Data files (ATD) contain configuration information for each terminal in the logical network. The CURRENCIES CONFIGURED FOR SELECTION BY THE CARDHOLDER fields on ATD screen 4 have specific Multiple Currency Dispense requirements. These fields correspond to the currencies displayed to the cardholder at the ATM and must contain valid ISO currency codes from the COBNAMES File.

These ATD fields are described in detail in the *BASE24-atm Files Maintenance Manual*.

Receipt File (RCPT)

The Receipt File (RCPT) contains receipt templates that define the actual text and formats for the various types of receipts. You can configure receipts for the Multiple Currency Dispense add-on feature using the field type codes described in the topic “Receipt Processing” earlier in this section.

Refer to section 5 for more information about configuring the RCPT.

Exchange Rate File (ERF)

The Exchange Rate File (ERF) contains one record for each currency used in the BASE24 system. When using the Multiple Currency Dispense add-on feature, ensure that all currencies used are configured in the ERF. Refer to the *BASE24-atm Multiple Currency Support Manual* for more information about the ERF.

ACI Worldwide, Inc.

G: BASE24-atm Mobile Top-Up Extension

The BASE24-atm Mobile Top-Up Device Handler extension enables cardholders to replenish, or *top up*, their mobile telephone accounts at an ATM. The replenishment transaction is a noncurrency dispense transaction. The institution must purchase both the Non-Currency Dispense add-on product and the Mobile Top-Up Device Handler extension to offer mobile top-up services.

A telecommunications services provider or an individual mobile operator, rather than the institution that owns the ATM or issues cards, is responsible for authorizing the minutes purchase, maintaining the consumer mobile telephone account, and calculating any taxes applied to the transaction. The BASE24-atm product authorizes the funds portion of the transaction. Refer to the topic “Processing Mobile Top-Up Transactions” for more information about routing transactions to the telecommunications services provider.

Cardholders who perform mobile top-up transactions receive either real-time replenishment or voucher replenishment. Real-time replenishment enables the telecommunications services provider or mobile operator to replenish the cardholder’s account as soon as the transaction is authorized. Voucher replenishment provides a stock receipt (i.e., voucher) on the ATM receipt for the top-up portion of the transaction. Voucher information includes a voucher number and expiration date. The cardholder must then use his or her mobile telephone to enter this information for the replenishment to take place. Whether a customer receives real-time replenishment or a voucher depends on the configuration of a host system (either the institution’s host or the telecommunications services provider host). Typically, if the host recognizes the mobile telephone number and carrier, the cardholder receives real-time replenishment. If the host does not recognize the number and carrier, the cardholder receives a voucher printed on the ATM receipt for the top-up portion of the transaction.

Terminology

The following terms appear in this documentation to describe the BASE24-atm Mobile Top-Up Device Handler extension.

Definitions-based model — A transaction processing method for the BASE24-atm Mobile Top-Up Device Handler extension that uses predefined values for mobile operator-supported allowed amounts and telephone time increments.

Funds authorizer — The entity responsible for authorizing the withdrawal, or *funds*, portion of a mobile top-up transaction (e.g., the BASE24-atm Authorization process).

Message-based model — A transaction processing method for the BASE24-atm Mobile Top-Up Device Handler extension that uses a series of messages between the Transaction Context Manager process, the funds authorizer, and the top-up authorizer to verify mobile telephone numbers, the number of minutes, and other purchase information.

Mobile operator — A company that provides application services for the transaction (e.g., an individual mobile telephone company).

Telecommunication service provider — An entity that provides telecommunication services for individual mobile operators.

Top-up authorizer — The entity responsible for authorizing the top-up portion of a mobile top-up transaction. This entity can verify that the mobile telephone number exists, that the account can be replenished, and that the amount or number of minutes requested is valid or may issue a voucher.

Transaction Context Manager process — A BASE24 module that enables the BASE24 system to receive a single request from an endpoint and split the routing between two or more destinations.

Voucher — Information printed on the ATM receipt that indicates mobile telephone time was purchased and a code to enter to activate the replenishment. An ATM generates two receipts for a mobile top-up transaction: one receipt for the funds portion and one for the top-up portion.

Mobile Top-Up Models

Customers can employ one of two models in processing mobile top-up transactions: the definitions-based model or the message-based model.

Definitions-Based Model

The *definitions-based* model uses locally defined values for mobile telephone operators (i.e., the actual service providers), the amount of time that can be purchased, and other information needed to complete the purchase of mobile telephone time. This information is stored in the Mobile Operator File (MOF), a separately licensable BASE24 module, which is read by the Transaction Context Manager process. Customers that employ the definitions-based model must license the Non–Currency Dispense add-on product, the Mobile Top-Up Device Handler extension, the Mobile Top-Up Infrastructure, the Transaction Context Manager product, and the Mobile Top-Up Transaction Context Manager extension.

Because operator information is locally defined, a single message sequence between BASE24 and the mobile operator contains all the information needed to complete the purchase.

Message-Based Model

The *message-based* model of mobile top-up processing uses a series of messages between BASE24 and a telecommunication service provider to gather the information needed to complete the purchase. These messages can include verifying whether the mobile telephone number supplied in the request can accept a replenishment and whether the number of minutes or the amount of the purchase is valid.

Customers that employ the message-based model must license the Non–Currency Dispense add-on product, the Mobile Top-Up Device Handler extension, and the Transaction Context Manager product.

Processing Mobile Top-Up Transactions

A mobile top-up transaction is initiated by the cardholder at the ATM. The cardholder inserts his or her card and selects a noncurrency dispense (withdrawal) transaction, specifying the purchase item as mobile top-up. In addition, the cardholder specifies an amount measured in monetary units (e.g., \$20 US dollars) or time units (e.g., 30 minutes). The cardholder can use the ATM keypad to specify a mobile telephone number. The transaction is processed first by the NCR 5XXX Device Handler process and then the Transaction Context Manager process, which routes the transaction, based on settings in the Split Transaction Routing File (STRF), to the BASE24-atm Authorization process and the mobile operator or telecommunication service provider.

NCR 5XXX Device Handler Process

The NCR 5XXX Device Handler process performs the following functions for a mobile top-up transaction:

- Receives the transaction request from the ATM and appends the necessary tokens to identify the transaction as a mobile top-up transaction
- Forwards customer actions to the Transaction Context Manager process
- Receives information from the Transaction Context Manager process
- Formats and sends native mode messages to the ATM

After the ATM sends the transaction information to the NCR 5XXX Device Handler process, the Device Handler process formats the transaction as a noncurrency dispense transaction and builds the Non-Currency Dispense token (setting the HOPR-CONTENTS field to the value 11), Pre-Pay Top-Up token, and the Transaction Subtype token. Although the mobile top-up transaction is a noncurrency dispense (withdrawal) transaction, the transaction requires additional processing. The Device Handler process uses `mtu_op_key_tbl` table in the T100NAMS file to fill the TXN-SUBTYP field of the Transaction Subtype token.

A mobile top-up transaction message has two processing requirements: the funds portion and the top-up portion (includes the mobile telephone number and all processing information). If the SPLIT-TXN-RTG-DEST param is present, the Device Handler process forwards the transaction to the Transaction Context Manager process specified in the param, which can route the transaction to multiple authorizing entities. If the param is not present, the transaction is declined.

Note that when the SPLIT-TXN RTG-DEST param is present, the Device Handler process always sends transactions to the Transaction Context Manager process and not the BASE24-atm Authorization process.

After the Device Handler process retrieves the destination name from the SPLIT-TXN-RTG-DEST param it adds the Split Transaction Routing token with the name of the Authorization process in the FUNDS-AUTH-DEST field.

When the BASE24-atm system uses the Transaction Context Manager process, all messages between the Device Handler process and the Authorization process are routed through the Transaction Context Manager process.

The BASE24-atm Mobile Top-Up Device Handler extension also supports receiving a 9909 mobile top-up notification message from the Transaction Context Manager process, requesting additional action from the cardholder at the ATM, such as selecting a mobile operator and amount, selecting a new amount, or approving a tax. The content of the 9909 message is based on the value of the Pre-Pay Response token RESP-IND field.

Refer to the ***BASE24 Tokens Manual*** for more information about the Split Transaction Routing token, Pre-Pay Response token, Transaction Subtype token, Pre-Pay Top-Up token and Non-Currency Dispense token.

Transaction Context Manager Process

The Transaction Context Manager process performs routing to the telecommunication service provider and the issuer financial institution. When the Transaction Context Manager process receives a transaction it searches for the Transaction Subtype token. If the token is not present, the transaction is declined. If the token is present, the Transaction Context Manager retrieves the transaction subtype. Using the transaction subtype as the key, the Transaction Context Manager process reads the STRF extended memory table to find the secondary source and the authorizer (primary or secondary) to which it sends the transaction first. The ROUTING HIERARCHY field on the STRF screen specifies the order in which the Transaction Context Manager process sends the transaction message to the destinations.

Splitting the Transaction Routing

The Transaction Context Manager process splits the transaction routing to different destinations. In the case of mobile top-up processing, mobile top-up with funds transaction routing is split between the top-up authorizer and the funds authorizer.

The Transaction Context Manager process uses the routing hierarchy specified in the STRF to determine the order to send the transaction message to the authorizing entities. The Transaction Context Manager process uses the Split Transaction Routing token to keep track of the transaction routing. The Transaction Context Manager process updates the TXN-RESP-IND, the FUNDS-AUTH-STAT, and the SCND-SVC-STAT fields in the Split Transaction Routing token each time it passes the transaction to either the funds authorizer or the top-up authorizer.

Logging to the Transaction Log File (TLF)

The Transaction Context Manager process logs the secondary service (i.e., top-up) transaction information to the Transaction Log File (TLF). This is done prior to sending the final response for the mobile top-up transaction to the Device Handler process. All approved mobile top-up transactions have two transactions logged to the TLF: one for the funds authorization and one for the top-up authorization. In the definitions-based model of mobile top-up processing, the Transaction Context Manager process logs the top-up transaction with a transaction code of 25 (top-up). In the message-based model of mobile top-up processing, the Transaction Context Manager process logs the top-up transaction with a transaction code of 10 (withdrawal). In both models the Transaction Context Manager process logs the value S in the responder field, indicating a secondary service transaction and enables Transaction-based Pricing and ATM reports to process the secondary service transactions appropriately.

Configuration Considerations

The following files contain fields needed for the mobile top-up product. These files are in used in conjunction with the files used in Non–Currency Dispense processing.

- TN50NAMS
- Logical Network Configuration File (LCONF)
- Transaction Code/Subtype Relationship File (TSRF)
- Surcharge File (SURF)
- Receipt File (RCPT)
- Split Transaction Routing File (STRF)
- Mobile Operator File (MOF)

Modifying the TN50NAMS

The TN50NAMS is an edit file that contains a series of tables used by the BASE24-atm NCR 5XXX Mobile Top-Up Device Handler extension for mobile top-up transactions. Customers can edit these tables to customize specific information displayed on ATM receipts.

The transaction type table in the TN50NAMS (mtu_tran_tbl_g) supports the mobile top-up transaction type in receipt printing. The Device Handler process uses information in this table to format receipt descriptions with text specific to mobile top-up transactions, instead of generic withdrawal text.

Also in this file is the operations code table containing a subtype for each mobile top-up authorizer destination.

Modifying the Logical Network Configuration File

The Logical Network Configuration File (LCONF) contains seven params specific to the Mobile Top-Up Device Handler extension.

The SPLIT-TXN-RTG-DEST param specifies the destination of the Transaction Context Manager process. The Transaction Context Manager process is required for split transaction routing (in a mobile top-up transaction, part of the transaction is routed to the third-party telecommunications process and part to the BASE24-atm Authorization process).

The ATM-DH-RCPT-PHONE-NUM-FRMT param specifies whether the telephone number printed on the receipt is formatted in the xxx-xxx-xxxx format.

The CARRIER-VT*identifier* param, where *identifier* is a value in the range of 00–29 representing a voucher type, specifies a four-digit mobile operator ID. This mobile operator ID matches a mobile operator with an entry in the Mobile Operator File (MOF) and enables the mobile operator to be identified for the transaction.

The CARRIER-AMT-VT*identifier* param, where *identifier* is a value in the range of 00–29 representing a voucher type, specifies the six predetermined amounts, in monetary units, supported for this mobile operator. Valid values must be formatted with eight amounts and each amount must be separated by semicolons. If the mobile carrier does not support eight amounts, the remaining values in this param must be set to zeros.

The MOBILE-OPERATOR-ID-VT*identifier* param, where *identifier* is a value in the range of 00–29 representing a voucher type, specifies a mobile operator six-digit IIN, two-digit digit product ID, and the mobile operator name (up to 20 characters) associated with the mobile telephone operator.

The MOBILE-NUM-MIN-LGTH param specifies the minimum length for the mobile telephone number entered by the cardholder.

The MOBILE-NUM-MAX-LGTH param specifies the maximum length for the mobile telephone number entered by the cardholder.

Modifying the Transaction Code/Subtype Relationship File

The Transaction Code Subtype Relationship File (TSRF) contains one record for each transaction subtype supported in the network. You can use the TSRF to set up a transaction subtype and related transaction codes for mobile top-up transactions. The transaction subtype distinguishes the mobile top-up transaction from other withdrawal transactions. You can use the subtype when configuring the Surcharge File (SURF) and the Split Transaction Routing File (STRF).

Modifying the Surcharge File

Some customer contracts with telecommunications services providers prohibit surcharging on mobile top-up transactions at an ATM. If you use the surcharging feature at your ATMs and want to offer mobile top-up transactions without a surcharge, you must configure the Surcharge File (SURF) to prevent surcharges.

Using the transaction subtype defined in the TSRF, you can specify transactions with a subtype do not have a surcharge by setting the MIN TXN, FLAT FEE, and PERCENT FEE to the value 0.00 and the USE MIN/MAX field to a blank.

For more information about the SURF, refer to the *BASE24 Base Files Maintenance Manual*.

Modifying the Receipt File

The Receipt File (RCPT) contains receipt templates. You can add receipt templates for mobile top-up transactions. ACI supplies the PPTU receipt type template for real-time mobile top-up transactions that do not involve a voucher and the VCHR receipt type template for mobile top-up transactions that include a voucher. The VT00 through VT29 receipt types are used in the definitions-based model to provide a unique voucher receipt record for each mobile operator supported.

Note that the ATM prints two receipts for a mobile top-up transaction. The funds portion of the transaction is printed using the same template as a noncurrency dispense (withdrawal) transaction. The receipt for the top-up portion uses the field type codes and templates described here. The ATM always prints the top-up portion first.

Mobile top-up transaction templates can use the following codes specific to the BASE24-atm Mobile Top-Up add-on product:

- TT (print transaction type)
- V0 (print activation number)
- V1 (print stock number)
- V2 (print shelf date, that is the date the minutes expire once activated)
- V3 (print telephone number)
- V4 (print unique transaction number)
- V5 (print telecommunications company marketing message)

-
- V6 (print telecommunications company name)
 - V7 (print carrier telephone number)
 - VA (print mobile telephone balance)
 - VB (print batch ID)
 - VE (print expiration days)
 - VI (print control ID number)
 - VN (print voucher number)
 - VS (print expiration date)
 - VT (print tax amount)
 - VX (print actual amount added to the mobile telephone account)

Note: Care should be taken when configuring the receipt templates as to avoid exceeding the maximum size of the print buffer in the device message.

Refer to section 6 for detailed definitions of these codes and information about configuring the RCPT.

A sample PPTU template as defined in the RCPT is shown below.

```
@MOBILE TOP-UP SUMMARY>>
#V6>
@MOBILE: #V3>
@TOP-UP VALUE: #VX>
@TAXES PAID: #VT>
@UPDATED PREPAID BALANCE: #VA>
@NEW EXPIRATION DATE: #VS>@REF NUM: #V4>>
@CUSOTMER SERVICE: #V7>
#V5
```

The above template creates the following receipt for a mobile top-up transaction that does not require a voucher.

```

1          A C I   D E M O
2
3  DATE          TIME  TERM
4  MM/DD/YY  HH:MM  S1AN5070
5
6  MOBILE TOP-UP SUMMARY
7
8  XYZ TELECOMMUNICATIONS
9  MOBILE:  402-555-5555
10 TOP-UP VALUE:                $20.00
11 TAXES PAID:                  $1.00
12 UPDATED PREPAID BALANCE: 30
13 NEW EXPIRATION DATE:   MM/DD/YY
14 REF NUM: 123456789012345
15
16 402-390-7600
17   USE XYZ TEXT MESSAGING
18   AND WIN

```

1 2 3
12345678901234567890123456789012

A sample VCHR or VTxx, where xx is 00–29, template as defined in the RCPT is shown below.

```

@MOBILE TOP-UP SUMMARY>>
#V6>
@MOBILE: #V3>@TOP-UP VALUE: #VX>@TAXES PAID:   #VT>
@TOP-UP VOUCHER: #VN>@VOUCHER EXPIRY: #VE @DAYS>
@REF: #V4>>
@CUSTOMER SERVICE: #V7>
#V5>>

```

The above template creates the following receipt for a mobile top-up transaction that includes a voucher.

```
1          A C I   D E M O
2
3  DATE          TIME   TERM
4  MM/DD/YY   HH:MM S1AN5070
5
6  MOBILE TOP-UP SUMMARY
7
8  XYZ TELECOMMUNICATIONS
9  MOBILE:  402-555-5555
10 TOP-UP VALUE:                      $20.00
11 TAXES PAID:                        $1.00
12 TOP-UP VOUCHER: 1234567890123456
13 VOUCHER EXPIRY:                      14 DAYS
14 REF: 123456789012345
15
16 402-390-7600
17     USE XYZ TEXT MESSAGING
18           AND WIN
```

1 2 3
12345678901234567890123456789012

Modifying the BASE24-atm Terminal Data Files

Care should be taken when configuring the receipt templates as to avoid exceeding the maximum size of the print buffer in the device message. If you exceed the maximum, you can set the LARGE MESSAGE MODE field on BASE24-atm Terminal Data files NCR 5XXX screen 7 to the value Y to increase the maximum message size to 2048 bytes. Refer to the ***BASE24-atm Files Maintenance Manual*** for more information about using the large message mode.

Modifying the Split Transaction Routing File

The Split Transaction Routing File (STRF) contains one record for each transaction subtype in the BASE24 network that requires split routing. Once you have set up transaction subtypes for mobile top-up transactions in the TSRF, you must create STRF records for each subtype so that the transactions can be routed properly. This file contains the process name of the top-up authorizer for the subtype.

The ROUTING HIERARCHY field specifies the order to which the Transaction Context Manager process routes the transaction for authorization of the funds and top-up portions of the transaction. For the definitions-based model of mobile top-up processing, you can configure sequential routing, with the transaction going to the funds authorizer first by setting this field to the value B0, or you can configure sequential routing with the transaction going to the top-up authorizer first by setting this field to the value B1. For the message-based model of mobile top-up processing, you can configure multiple messages with the transaction going to the top-up authorizer first by setting the field to the value B2.

Refer to the *BASE24 Base Files Maintenance Manual* for more information about this file.

Modifying the Mobile Operator File

The Mobile Operator File (MOF) contains one record for each mobile operator defined to the BASE24 system. This file is used in definitions-based processing only. Refer to the *BASE24 Base Files Maintenance Manual* for more information about the MOF.

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H: NCR 5XXX Aptra Edge Standard NDC Device Handler Extension

NCR 5XXX Aptra Edge Standard NDC support has been developed as an extension to the existing BASE24 NCR 5XXX Device Handler process. This extension makes it possible for institutions to operate both Aptra Edge Standard NDC message format and standard NDC+ devices using a single Device Handler process.

This section describes the impacts of the Standard Aptra Edge NDC Device Handler extension as follows:

- Overview
- Device Control Terminal
- Device Configuration
- Processing

Overview

The Aptra Edge Standard NDC Device Handler extension is responsible for processing transaction request messages that originate from an Aptra Edge ATM. These messages conform to the standard NDC+ message format. However, the Aptra Edge Standard NDC message format software has its own proprietary op codes and next states which differ from those defined in the BASE24 Release 6.0 NCR 5XXX Device Handler. Therefore, when processing Aptra Edge transaction requests and responses, op codes, state and screen numbers, and response codes require appropriate mapping. This mapping, as well as all other Edge-specific processing takes place within the Aptra Edge Standard NDC extension.

In addition, Aptra Edge Standard NDC message format ATMs can be remotely configured using Extensible Markup Language (XML) data language downloads. This is accomplished as part of a regular ATM screen load, using the existing NCR Configuration File (CNFG5070). To remotely customize Aptra Edge terminals, blank customer screens can be painted with specific XML commands that are downloaded to the terminal. These XML commands overlay and override the existing configuration.

The standard NDC+ Device Handler determines the source of the incoming request message, whether from an NDC+ terminal or from an Aptra Edge terminal. The Device Handler makes this determination based on the ATM's software ID.

Device Control Terminal (DCT)

Aptra Edge XML configuration load messages are constructed using the screen/keyboard data access capability of the Device Control Terminal. Institutions can use the DCT screen entry capability to create XML load messages that can be downloaded to the Aptra Edge terminal.

The Aptra Edge Standard NDC message format application is configured using a resident Data Dictionary file and various other XML files. These files contain instructional language defining ATM capability. Institutions can customize their ATMs through modification of the instructions contained within the various files. Downloading modifications to the Data Dictionary and XML files is accomplished using the standard NDC+ screens.

Using the DCT ATM NCR 5XXX Commands screen 45 to obtain access to the configuration file, institutions can construct and gain access to XML configuration load messages. This screen is normally used to construct and access customer screens; however, for Aptra Edge, it can also be used to construct and gain access to XML configuration load messages. The screen input is unrestricted and free-format. It is the responsibility of the operator constructing the XML instructions to ensure the XML instructions are properly formatted.

Refer to the *NCR Aptra Edge Remote Configuration Guide, release 02.03.xx*, January 11, 2006 for more specific information on formatting instructional data.

Device Configuration

The ATM NCR 5XXX Device Handler configuration file (CNFG5070) contains Aptra Edge configuration records. Institutions can construct their own XML load messages depending on their specific circumstances.

Processing

The following describes the processing for the Aptra Edge Standard Device Handler extension for the op codes and next states.

Op Codes

The Aptra Edge Standard NDC message format application has its own proprietary op codes that are different from those defined within BASE24. Request messages originating from an Aptra Edge ATM must have their op codes mapped to internal BASE24-atm Standard Internal Message (STM) transaction codes in the same way that BASE24-atm now maps existing BASE24-atm op codes to STM transaction codes. This mapping takes place within the Aptra Edge Standard NDC Device Handler extension.

Next States

The Aptra Edge Standard device has its own proprietary next states. In the standard NDC+ Device Handler process, the next state is determined by the load group downloaded to the terminal or by the next state being returned in the transaction reply message. In the Aptra Edge terminal, the next state and associated screen verbiage is predetermined by the resident Aptra Edge application software. The Device Handler process maps the appropriate next state, as expected by the Aptra Edge application, when returning the response to the ATM.

For denied transactions, there are many possible next states, depending on the type of transaction involved, the sort of card reader/writer, and whether receipt data is being printed.

The complete list of next denial states and associated screen verbiage for Aptra Edge is included in the ***NCR Aptra Edge Standard NDC Message Format Guidelines.***

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